# AI Strategy Consulting: Foundations to Expert

A structured training programme for financial-services consultants moving into AI strategy, agentic AI and AI business cases

Prepared for an experienced payments and FS consultant preparing for AI consultant and AI strategy interviews

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## Table of Contents

**How to use this document** ..... **5**
- Reading protocol ..... 5
- Labelling conventions used throughout ..... 6
- A note on frameworks and attribution ..... 6
- What was added to the curriculum and why ..... 6

**Module 0 — Diagnostic and Curriculum Map** ..... **7**
- 1. Purpose of this module ..... 7
- 2. The 10-question diagnostic ..... 7
- 3. Self-scoring rubric ..... 8
- 4. Curriculum map ..... 8
- 5. How to use this document ..... 10

**Module 1 — The AI Landscape: Rules Engines, ML, Deep Learning, GenAI and Agentic AI** ..... **11**
- 1. Why a client cares ..... 11
- 2. Concept and mechanism ..... 11
- 3. Frameworks ..... 13
- 4. Worked BFSI example ..... 14
- 5. Real-world case ..... 15
- 6. Consultant language ..... 16
- 7. Common mistakes juniors make ..... 16
- 8. Interview questions with model answers ..... 17
- 9. Self-check quiz ..... 17

**Module 2 — How LLMs Work, at Consultant Depth** ..... **18**
- 1. Why a client cares ..... 18
- 2. Concept and mechanism ..... 18
- 3. Frameworks ..... 22
- 4. Worked BFSI example ..... 22
- 5. Real-world case ..... 23
- 6. Consultant language ..... 23
- 7. Common mistakes juniors make ..... 24
- 8. Interview questions with model answers ..... 24
- 9. Self-check quiz ..... 25

**Module 3 — Agentic AI Mechanics** ..... **26**
- 1. Why a client cares ..... 26
- 2. Concept and mechanism ..... 26

3. Frameworks....30
4. Worked BFSI example....30
5. Real-world case....32
6. Consultant language....33
7. Common mistakes juniors make....33
8. Interview questions with model answers....33
9. Self-check quiz....34

**Module 4 — Data and Tech Stack: Readiness, Vector Stores, LLMOps, Evaluation and Observability..35**

1. Why a client cares....35
2. Concept and mechanism....35
3. Frameworks....38
4. Worked BFSI example....39
5. Real-world case....40
6. Consultant language....40
7. Common mistakes juniors make....41
8. Interview questions with model answers....41
9. Self-check quiz....42

**Module 05 — AI Strategy and Maturity....43**

1. Why a client cares....43
2. Concept and mechanism....43
3. Frameworks....44
4. Worked BFSI example....46
5. Real-world case....48
6. Consultant language....48
7. Common mistakes juniors make....49
8. Interview questions with model answers....49
9. Self-check quiz....50

**Module 06 — Use-Case Discovery....51**

1. Why a client cares....51
2. Concept and mechanism....51
3. Frameworks....53
4. Worked BFSI example....54
5. Real-world case....57
6. Consultant language....57
7. Common mistakes juniors make....58
8. Interview questions with model answers....58
9. Self-check quiz....59

**Module 07 — Use-Case Assessment and Prioritisation....60**

1. Why a client cares....60
2. Concept and mechanism....60
3. Frameworks....62
4. Worked BFSI example....63
5. Real-world case....66
6. Consultant language....67
7. Common mistakes juniors make....67
8. Interview questions with model answers....67

9. Self-check quiz....68
**Module 08 — Cost Factors and Total Cost of Ownership....69**
1. Why a client cares....69
2. Concept and mechanism....69
3. Frameworks....71
4. Worked BFSI example....72
5. Real-world case....75
6. Consultant language....75
7. Common mistakes juniors make....76
8. Interview questions with model answers....76
9. Self-check quiz....77
**Module 09 — The AI Business Case....78**
1. Why a client cares....78
2. Concept and mechanism....78
3. Frameworks....80
4. Worked BFSI example....81
5. Real-world case....85
6. Consultant language....85
7. Common mistakes juniors make....86
8. Interview questions with model answers....86
9. Self-check quiz....87
**Module 10 — Build vs Buy vs Partner, Model Selection and RFP Evaluation....88**
1. Why a client cares....88
2. Concept and mechanism....88
3. Frameworks....91
4. Worked BFSI example....92
5. Real-world case....94
6. Consultant language....94
7. Common mistakes juniors make....94
8. Interview questions with model answers....95
9. Self-check quiz....96
**Module 11 — From Proof of Concept to Scale: Stage Gates and Escaping Pilot Purgatory....97**
1. Why a client cares....97
2. Concept and mechanism....97
3. Frameworks....100
4. Worked BFSI example....100
5. Real-world case....103
6. Consultant language....103
7. Common mistakes juniors make....104
8. Interview questions with model answers....104
9. Self-check quiz....105
**Module 12 — Responsible AI, Regulation and Model Risk for AI in Banking....106**
1. Why a client cares....106
2. Concept and mechanism....106
3. Frameworks....111
4. Worked BFSI example....112

5. Real-world case....114
6. Consultant language....115
7. Common mistakes juniors make....115
8. Interview questions with model answers....116
9. Self-check quiz....117

**Module 13 — AI Operating Model and Talent....118**

1. Why a client cares....118
2. Concept and mechanism....118
3. Frameworks....121
4. Worked BFSI example....122
5. Real-world case....124
6. Consultant language....125
7. Common mistakes juniors make....125
8. Interview questions with model answers....125
9. Self-check quiz....126

**Module 14 — Change Management, Adoption and the KPI Framework....127**

1. Why a client cares....127
2. Concept and mechanism....127
3. Frameworks....130
4. Worked BFSI example....130
5. Real-world case....133
6. Consultant language....133
7. Common mistakes juniors make....133
8. Interview questions with model answers....134
9. Self-check quiz....135

**Module 15 — AI in Managed Services: AI-led Delivery, the Labour-model Shift and Outcome-based Pricing.... 136**

1. Why a client cares....136
2. Concept and mechanism....136
3. Frameworks....139
4. Worked BFSI example....139
5. Real-world case....142
6. Consultant language....142
7. Common mistakes juniors make....143
8. Interview questions with model answers....143
9. Self-check quiz....144

**Module 16 — BFSI Use-case Library....145**

1. Why a client cares....145
2. Concept and mechanism....145
3. Frameworks....152
4. Worked BFSI example....153
5. Real-world case....155
6. Consultant language....155
7. Common mistakes juniors make....155
8. Interview questions with model answers....156
9. Self-check quiz....157

**Module 17 — Public case-study teardowns..... 158**
1. Why a client cares..... 158
2. Concept and mechanism..... 158
3. Frameworks..... 159
4. Worked BFSI example: cross-case synthesis..... 159
5. Real-world case teardowns..... 161
6. Consultant language..... 166
7. Common mistakes juniors make..... 166
8. Interview questions with model answers..... 166
9. Self-check quiz..... 167

**Module 18 — Three end-to-end engagement simulations..... 168**
Purpose and how to use this module..... 168
Simulation A: AI strategy and use-case assessment for a mid-sized bank..... 169
Simulation B: Agentic AI transformation of corporate onboarding (KYB) at a UK Tier-1 bank..... 175
Simulation C: An AI business case for an agentic dispute-resolution platform, defended to a CFO..... 183
How to talk about these simulations in an interview..... 189
Self-check quiz..... 189

**Module 19 — Positioning Your Own Experience as an AI Strategy Consultant..... 191**
1. Why a client cares..... 191
2. Concept and mechanism..... 191
3. Frameworks..... 192
4. Worked BFSI example: your three projects, rewritten..... 192
5. Real-world case..... 198
6. Consultant language..... 198
7. Common mistakes juniors make..... 198
8. Interview questions with model answers..... 199
9. Self-check quiz..... 200

**Module 20 — Interview Arsenal: 40 Questions, 5 Case Drills, 3 Guesstimates and a Full Mock..... 201**
How to use this module..... 201
Part A: Top 40 AI-consulting interview questions with model answers..... 201
Part B: Five AI case drills..... 211
Part C: Three AI guesstimates..... 215
Part D: Full mock interview with grading rubrics..... 216

**Appendix — Answer keys..... 221**

# How to use this document

This document replaces the interactive, one-module-at-a-time format with a complete self-study programme. It covers twenty modules across five phases, four one-page recaps, and an answer-key appendix.

## Reading protocol

1. **Start with Module 0.** Attempt the ten-question diagnostic without looking at the answer key. Score yourself using the rubric. The band tells you whether to read every module in order or to move quickly through Phase 1.

2. **Read one module at a sitting.** Each module follows the same nine-part structure: why a client cares, concept and mechanism, frameworks, worked BFSI example, real-world case, consultant language, common mistakes, interview questions with model answers, and a five-question self-check quiz.
3. **Attempt the quiz before opening the appendix.** Quiz answers are deliberately placed in the appendix at the back, not in the module, so that you test recall rather than recognition. Each answer carries a short explanation and a note on what a manager-level AI consultant would add.
4. **Revise from the recaps.** After Modules 5, 10, 15 and 20 there is a one-page recap built for revision the night before an interview.
5. **Finish with Module 20.** Work the forty questions, the five case drills and the three guesstimates, then sit the mock interview and grade yourself with the rubrics provided.

## Labelling conventions used throughout

Every example, number, client and case carries one of these labels at first mention. Treat them as a contract with you about what is real.

| Label | Meaning | How to use it in an interview |
|---|---|---|
| [PUBLIC — source, year] | Publicly reported fact, with the organisation or publication and year | Cite it, with the source and year, and say if figures may have moved |
| [ILLUSTRATIVE — composite] | A realistic but invented engagement, client or number | Use it as a pattern, never as a client reference |
| [ESTIMATE — assumption] | The author's own assumption, with the assumption stated | Say "my working assumption is" and be ready to flex it |
| [VERIFY — date-sensitive] | Regulation status, model pricing, vendor capability or fast-moving statistic | Check it the week before any interview or client meeting |

## A note on frameworks and attribution

Each framework is attributed to its originator where this is known. Where a framework is generic practice with no clear originator (for example a value-versus-feasibility two-by-two or a weighted scoring model), it is described as "generic, widely used". Where attribution is uncertain, this is stated rather than guessed. In an interview, attributing a framework to the wrong firm is a credibility error that a partner will notice.

## What was added to the curriculum and why

Three additions were made to the twenty-module curriculum as originally specified. Each is small and sits inside an existing module rather than adding a new one.

- **Evaluation design as a first-class topic (Module 4 and Module 11).** Interviewers in 2025–26 increasingly probe how you would know an agent is working. Golden datasets, LLM-as-judge, and numeric gate thresholds are now expected vocabulary, not specialist knowledge.
- **Model-risk classification of GenAI and agentic systems (Module 12).** Banks are being asked by supervisors how LLM-based systems fit into frameworks written for statistical models (SR 11-7, PRA SS1/23). Being able to tier an agentic system under a bank's existing model risk policy is a differentiator.
- **The public reversals (Module 17).** Klarna's re-hiring of humans and Commonwealth Bank of Australia's reversal of AI-linked redundancies are now standard interview material. A strategy consultant must be able to explain why value claims were walked back, not just what was announced.

# Module 0 — Diagnostic and Curriculum Map

## 1. Purpose of this module

Before you study, calibrate. The ten questions below span all five phases of the curriculum. Answer each in writing, in no more than six lines, before you look at the answer key in the appendix. Score yourself honestly using the rubric. The band you land in tells you where to start and how hard to push on each module.

The diagnostic is also a preview of how interviewers probe. Notice that half the questions are not “what is X” but “what would you tell the client” or “defend the number”. That is the register you need to reach.

## 2. The 10-question diagnostic

Answer all ten. Time yourself; a manager-level candidate should finish in 25 to 30 minutes.

1. **Tool choice.** A UK bank wants to automate the decision on whether a disputed card transaction qualifies for a chargeback under scheme rules. The rules are codified, the inputs are structured, and the regulator expects a reproducible outcome. Which class of technology do you recommend (rules engine, classical ML, LLM, agentic AI) and why? Where, if anywhere, would an LLM add value in that process?
2. **LLM mechanics.** Explain what a context window is, why it matters for cost, and why “just put the whole KYC file in the prompt” is usually the wrong design. Mention at least two technical consequences.
3. **Agentic patterns.** Distinguish a supervisor (hierarchical) multi-agent pattern from an orchestrator-worker pattern. Give one BFSI situation where you would choose each.
4. **Evaluation.** A vendor claims their sanctions-alert narrative generator is “95% accurate”. List the three questions you would ask before believing the claim, and name two metrics you would insist on seeing.
5. **Framework attribution.** Which firm is associated with the “10-20-70” rule for AI value, and what do the three numbers refer to? Which firm published “Rewired” and what are its six capabilities?
6. **Prioritisation.** You have scored 24 use cases on value and feasibility. Six land in the “high value, low feasibility” quadrant. What do you do with them, and what is the one common mistake juniors make with this quadrant?
7. **Cost arithmetic.** A KYB refresh process handles 60,000 cases a year. Each case will require an LLM pipeline consuming about 40,000 input tokens and 4,000 output tokens. Using an input price of USD 3 per million tokens and output price of USD 15 per million tokens [ESTIMATE — order of magnitude, VERIFY], what is the annual inference cost? Is inference likely to be the biggest line in the TCO? What is?
8. **Business case.** A CFO looks at your AI business case and says “your benefits assume 40% FTE reduction in year one; nobody achieves that.” Give the two or three adjustments a credible consultant would make to the case and explain how you would present the risk-adjusted view.
9. **Regulation.** Name the four functions of the NIST AI Risk Management Framework. Then: under the EU AI Act, is a model used to assess the creditworthiness of a retail customer high-risk? What does that imply for a bank deploying it?
10. **Positioning.** In two or three sentences, describe your own role on an AI project you have worked on in the language of a strategy consultant rather than an implementer. Then say which of your sentences an interviewer would push on, and why.

## 3. Self-scoring rubric

Score each question 0, 1 or 2 against the model answers in the appendix (answers/00).

| Score | Meaning |
|---|---|
| 0 | Missed the point, or gave a definition where a judgement was required |
| 1 | Correct core, but missing the mechanism, the number, the attribution or the client implication |
| 2 | Would satisfy a manager-level interviewer: structured, correct, and ends with a “so what” |

| Total | Band | What to do |
|---|---|---|
| 0 to 7 | Foundations | Study every module in order. Do every quiz. Expect 60 to 70 hours |
| 8 to 14 | Practitioner | Skim Phase 1, work Phases 2 to 4 carefully, do all mock questions. Expect 40 to 50 hours |
| 15 to 20 | Mock-ready | Go straight to Modules 7, 8, 9, 18, 19 and 20; use the rest as reference. Expect 20 to 25 hours |

Be suspicious of a self-score above 15 before you have read Modules 7 to 9. Most experienced consultants know the vocabulary and under-estimate how badly they defend numbers.

## 4. Curriculum map

Study hours are estimates [ESTIMATE — assumption] for reading, doing the worked example yourself in a spreadsheet, attempting the quiz and the interview questions. Interview weight reflects how often the topic is probed in AI consultant and AI strategy interviews at MBB, Big Four and tech-led consultancies, based on the question patterns compiled in Module 20.

| Module | Phase | Title | Hours | Prerequisites | Interview weight |
|---|---|---|---|---|---|
| 0 | Start | Diagnostic and curriculum map | 1 | None | Low |
| 1 | 1 Foundations | AI landscape: rules, ML, deep learning, GenAI, agentic | 3 | 0 | High |
| 2 | 1 Foundations | How LLMs work at consultant depth | 4 | 1 | High |
| 3 | 1 Foundations | Agentic AI mechanics | 4 | 2 | High |
| 4 | 1 Foundations | Data and tech stack, LLMOps, evaluation | 3 | 2 | Medium |
| 5 | 2 Frameworks | AI strategy and maturity | 3 | 1 | High |
| Recap 1 | | One-page recap of Modules 1 to 5 | 1 | 1 to 5 | |
| 6 | 2 Frameworks | Use-case discovery | 3 | 5 | Medium |

| Module | Phase | Title | Hours | Prerequisites | Interview weight |
|---|---|---|---|---|---|
| 7 | 2 Frameworks | Use-case assessment and prioritisation | 4 | 3, 6 | High |
| 8 | 2 Frameworks | Cost factors and TCO | 4 | 2, 7 | High |
| 9 | 2 Frameworks | Business case | 4 | 8 | High |
| 10 | 2 Frameworks | Build vs buy vs partner, vendor and model selection | 2 | 4, 8 | Medium |
| Recap 2 | | One-page recap of Modules 6 to 10 | 1 | 6 to 10 | |
| 11 | 3 Delivery | PoC to pilot to MVP to scale; pilot purgatory | 3 | 7, 9 | High |
| 12 | 3 Delivery | Responsible AI and regulation | 4 | 3, 4 | High |
| 13 | 3 Delivery | AI operating model and talent | 3 | 5 | Medium |
| 14 | 3 Delivery | Change management, adoption and KPIs | 2 | 11 | Medium |
| 15 | 3 Delivery | AI in managed services | 2 | 8, 13 | Medium |
| Recap 3 | | One-page recap of Modules 11 to 15 | 1 | 11 to 15 | |
| 16 | 4 Industry | BFSI use-case library | 4 | 1, 3, 7 | High |
| 17 | 4 Industry | Public case-study teardowns | 3 | 16 | Medium |
| 18 | 4 Industry | Three end-to-end engagement simulations | 8 | 5 to 16 | High |
| 19 | 5 Interview | Positioning your own experience | 3 | 18 | High |
| 20 | 5 Interview | Interview arsenal: 40 questions, 5 cases, 3 guesstimates, mock | 8 | All | High |
| Recap 4 | | One-page recap of Modules 16 to 20 | 1 | 16 to 20 | |
| | | **Total** | **~80** | | |

## What was added to the curriculum you proposed, and why

- **Evaluation moved earlier and expanded (Module 4).** Interviewers increasingly ask “how do you know it works” before “what does it cost”. Golden datasets and LLM-as-judge are now table stakes.

- **Autonomy levels and failure-mode tables (Module 3).** Regulators and model-risk teams ask for them; most candidates cannot produce one.
- **A CFO-challenge section inside the business case (Module 9) and a dedicated simulation (Module 18c).** Your stated gap is defending numbers; this is where it is drilled.
- **Positioning module split from the arsenal (Module 19).** Your gap is role clarity. It deserves its own module and its own narrative template.
- **Managed services (Module 15) kept, but reframed around labour-model economics**, since this is where tech-led consultancies probe hardest.

## 5. How to use this document

The interactive commands you proposed (NEXT, DEEPER, QUIZ, EXAMPLE, MOCK) are replaced by a reading protocol, because this is a document rather than a conversation.

1. **Read the module** end to end. Do not skip Section 2 (mechanism) even where you know the topic; the terminology is what interviewers listen for.
2. **Rebuild the worked example** in a spreadsheet with your own assumptions. If you cannot reproduce the number, you cannot defend it.
3. **Attempt the quiz** in writing, then check the answer key in the appendix. Score against the "manager-level benchmark" line, not just correctness.
4. **Answer the three interview questions aloud**, timed at two minutes each, before reading the model answers.
5. **After every five modules, read the recap** and write your own one-paragraph summary from memory. Compare.
6. **Return to Module 0** after Module 15 and re-score the diagnostic. The delta is your progress measure.

A note on labels. Every example in this document is tagged [PUBLIC — source, year], [ILLUSTRATIVE — composite] or [ESTIMATE — assumption]. Anything tagged [VERIFY — date-sensitive] must be checked against current sources before you quote it to a client or interviewer. Regulation timelines, model prices and vendor capabilities change quarterly.

# Module 1 — The AI Landscape: Rules Engines, ML, Deep Learning, GenAI and Agentic AI

## 1. Why a client cares

Most failed AI programmes in banks fail at tool selection, not at model quality: a generative model was used where a decision table would do, or a rules engine was stretched over a problem whose inputs vary too much. The client's CRO, COO and CFO each care for a different reason: reproducibility and auditability, run cost and operational risk, and whether the benefit case survives the choice of technology. A consultant who can say "this step is a rules problem, this step is a classification problem, and only this step needs a language model" saves the client millions and earns credibility in the first meeting.

## 2. Concept and mechanism

### 2.1 Five classes of technology, defined by how the decision logic is produced

The classes differ in one fundamental respect: who writes the logic and how it is represented.

| Class | How logic is produced | Representation | Determinism | Typical input |
|---|---|---|---|---|
| Rules engine | Humans encode rules explicitly | Decision tables, decision trees, business rules (DMN, Drools, FICO Blaze) | Fully deterministic | Structured fields |
| Classical ML | Learned from labelled or unlabelled data | Parameters of a statistical model (logistic regression, gradient-boosted trees, random forests, clustering) | Deterministic at inference for a fixed model; learned, not authored | Structured tabular features |
| Deep learning | Learned from large data via multi-layer neural networks | Millions to billions of weights | Deterministic at inference (given fixed weights and no sampling) | Images, audio, sequences, text, graphs |
| Generative AI (LLMs and multimodal models) | Pre-trained on web-scale corpora, then aligned | Transformer weights, 10^9 to 10^12 parameters | Stochastic by default (sampling); can be made near-deterministic with temperature 0 but still not reproducible across versions | Natural language, documents, images, code |
| Agentic AI | Generative model placed in a loop with tools, memory and a planning procedure | Model plus orchestration code plus tool schemas | Stochastic and path-dependent across multiple steps | Goals expressed in language, with access to systems |

### 2.2 Rules engines

A rules engine evaluates explicit conditions against structured data and fires actions. Modern implementations use the Rete algorithm (Forgy, 1982) or sequential evaluation, and are specified in standards such as the OMG Decision Model and Notation (DMN). Their properties are precisely what a regulated bank wants for a decision of record: every outcome traces to a rule, rules are version-controlled, changes go through governance, and the same input always yields the same output.

Their limits are equally precise. Rules do not generalise to inputs not anticipated by the author. Rule sets grow combinatorially as exceptions accumulate (a mature payments-operations rule base of several

thousand rules is common in large banks [ILLUSTRATIVE — composite]). Rules cannot read free text, interpret a scanned document, or weigh weak signals. When the problem has those characteristics, the rules engine is kept for the decision of record and another class of technology supplies the inputs.

## 2.3 Classical machine learning

Classical ML learns a mapping from features to a target from historical data. In banking the dominant algorithms are logistic regression (still the backbone of many regulatory credit scorecards because of monotonicity and explainability), gradient-boosted decision trees (XGBoost, LightGBM, CatBoost) for fraud and propensity, and clustering or anomaly detection for AML segmentation. Key properties:

- **Requires labelled history** for supervised tasks: fraud needs confirmed fraud labels; default prediction needs observed defaults over a performance window.
- **Features are engineered** by humans (velocity counts, ratios, aggregations). Feature engineering is often where most value and most model risk sits.
- **Explainability is tractable.** Coefficients, feature importances, SHAP values (Lundberg and Lee, 2017) and reason codes satisfy adverse-action notice requirements in credit.
- **Model risk management applies squarely.** SR 11-7 (US Federal Reserve and OCC, 2011) and the UK PRA SS1/23 (effective May 2024) were written with these models in mind.

Classical ML is the correct tool when you have structured features, meaningful labelled history, a stable target definition and a need to explain individual decisions. It is a poor tool when the input is unstructured or the target shifts faster than you can re-label.

## 2.4 Deep learning

Deep learning uses neural networks with many layers to learn representations directly from raw or lightly processed data. Convolutional networks for images (cheque and ID document processing), recurrent and later transformer architectures for sequences (transaction sequences, text), and graph neural networks for networks (mule-account detection, beneficial-ownership graphs). In banking before 2022, deep learning was used mostly for perception tasks: OCR, signature verification, document classification, and some fraud sequence models.

Properties the client needs to understand: deep models need large data, are expensive to train, are harder to explain (post-hoc methods only), and can be brittle under distribution shift. Their advantage is handling raw unstructured inputs that no feature engineer could encode.

## 2.5 Generative AI and large language models

A large language model (LLM) is a transformer network (Vaswani et al., 2017) trained to predict the next token over a web-scale corpus, then aligned with instruction tuning and reinforcement learning from human feedback (Ouyang et al., 2022). The consequence of this training objective is the key consultant insight: the model has learned a very good statistical model of language and of much of the knowledge expressed in text, but it has not learned to be correct. It produces plausible continuations. Correctness is an emergent and partial property, which is why grounding (retrieval), constraints (structured outputs, tool calls) and evaluation are non-negotiable in banking deployments.

Three capabilities matter commercially:

1. **Understanding and transforming unstructured text**: summarising, extracting, classifying, translating, drafting. This is where most banking value sits today.
2. **Reasoning in language**: multi-step inference over provided facts, improved by chain-of-thought prompting (Wei et al., 2022) and by “reasoning” model variants that generate internal deliberation tokens before answering.
3. **Code and structured output generation**: producing JSON, SQL, or code that another system executes.

What LLMs are not: they are not databases (they do not reliably recall specific facts), not calculators (arithmetic should be delegated to tools), and not deterministic decision engines.

### 2.6 Agentic AI

An agent is a generative model placed in a loop: it observes a state, decides on an action (often a tool call), observes the result, and continues until a goal is met or a stopping condition triggers. Module 3 covers the mechanics. For tool selection, the defining properties are:

- **Multi-step and path-dependent.** Errors compound across steps; a 95% per-step reliability over 10 dependent steps yields roughly 60% end-to-end success (0.95^10 = 0.60).
- **Acts on systems**, not just on text. This introduces operational risk, authorisation and reversibility questions.
- **Expensive**: multiple model calls per task, often with large contexts.
- **Valuable when the path cannot be scripted in advance**: the next step depends on what was found, and the branching is too wide for a flowchart.

### 2.7 Combining the classes: the production pattern

Almost no production system in banking is one class alone. The dominant pattern [ILLUSTRATIVE — composite of typical architectures] is:

- Perception layer (deep learning or LLM): read documents, emails, call transcripts into structured facts.
- Judgement layer (classical ML or LLM with retrieval): score, classify, recommend.
- Decision of record (rules engine): apply policy deterministically to structured inputs.
- Action layer (agentic or RPA): execute across systems with human gates where irreversibility or regulatory exposure demands them.

The consultant's job is to map each process step to a layer and to justify each boundary.

## 3. Frameworks

### 3.1 Tool-selection decision table [ILLUSTRATIVE — practitioner framework, attribution: composite of common practice; no single originator]

Score the task on each characteristic, then read across.

| Characteristic | Favours rules | Favours classical ML | Favours deep learning | Favours LLM | Favours agentic |
|---|---|---|---|---|---|
| Input structure | Fully structured | Structured features | Raw images, audio, sequences | Free text, documents | Mixed, across systems |
| Logic knowable in advance | Yes, fully | Partly, learnable from history | No, perceptual | Partly, expressed in language | No, path emerges |
| Need for determinism | Absolute (decision of record) | High | Medium | Low to medium (tolerate variance) | Low, with gates |
| Explainability requirement | Rule trace | Reason codes, SHAP | Post-hoc only | Citations, rationale text | Step log, audit trail |
| Labelled history | None needed | Essential | Large volumes | Not needed (zero/few-shot) | Not needed, but evals needed |
| Volume and latency | Millions per second | Millions per day | Thousands per second | Hundreds to thousands per minute | Tens to hundreds per minute |
| Error tolerance | Zero | Low, measurable | Low, measurable | Medium, with | Medium, with |

| Characteristic | Favours rules | Favours classical ML | Favours deep learning | Favours LLM | Favours agentic |
|---|---|---|---|---|---|
| | | | | review | gates and rollback |
| Rate of change in logic | Low (governance per change) | Medium (retrain) | Low | High (re-prompt) | High |

Application steps:

1. Decompose the process into tasks at the level of a single decision or transformation.
2. For each task, score the eight characteristics.
3. Assign the class with the most favourable column. Where two classes tie, choose the simpler and cheaper one (rules before ML, ML before LLM, LLM before agent).
4. Record the boundary: what structured artefact passes from one class to the next.
5. Identify the "decision of record" and insist it is deterministic unless the regulator has explicitly accepted otherwise.

### 3.2 The "simplicity ladder" heuristic

Attribution uncertain; widely used in practice. Always try to solve with the lowest rung that works: rule, then lookup or heuristic, then classical model, then single LLM call, then LLM with retrieval, then agent. Each rung up adds cost, variance and governance burden. In client conversations this is the single most persuasive discipline because it signals you are not selling technology.

### 3.3 Hype-cycle awareness

Gartner's Hype Cycle (Gartner, annual) is useful as a vocabulary, not as evidence. In its 2024 and 2025 editions, generative AI had moved past the peak and agentic AI was near the peak [PUBLIC — Gartner Hype Cycle for AI, 2024 and 2025; VERIFY current placement]. Gartner also predicted that over 40% of agentic AI projects would be cancelled by end 2027 [PUBLIC — Gartner, June 2025; VERIFY]. Use these to frame expectations, not to decide.

## 4. Worked BFSI example

[ILLUSTRATIVE — composite] A UK mid-sized bank's payments and financial-crime operations. We decompose ten common tasks and apply the decision table.

| # | Task | Input structure | Determinism need | Logic knowable | Recommended class | Rationale |
|---|---|---|---|---|---|---|
| 1 | ISO 20022 pacs.008 schema validation and routing | Structured XML | Absolute | Fully | Rules engine | Codified in scheme rulebooks; zero error tolerance |
| 2 | Sanctions name screening against lists | Structured names, fuzzy match | High | Mostly (matching algorithms) | Rules plus classical matching (phonetic, edit distance); ML for alert triage | Decision of record must be reproducible; ML can rank alerts for review |
| 3 | Drafting the sanctions alert disposition narrative | Alert data plus analyst notes | Low | Expressed in language | LLM with retrieval over policy and case data | High-volume text drafting; human approves |
| 4 | Card fraud scoring at | Structured transaction | High, sub-100ms | Learnable from labelled fraud | Classical ML (gradient- | Millions per day, strict |

| # | Task | Input structure | Determinism need | Logic knowable | Recommended class | Rationale |
|---|---|---|---|---|---|---|
| | authorisation | features | | | boosted trees) | latency, labelled history exists |
| 5 | Reading a scanned certificate of incorporation | Image | Medium | Perceptual | Deep learning (document AI) feeding an LLM for field extraction | Perception task; LLM maps fields to schema |
| 6 | Ultimate beneficial owner (UBO) resolution across registries | Mixed: registry APIs, documents, web | Low to medium | Path emerges | Agentic (supervisor pattern) with HITL gate | Next step depends on what each registry returns; branching too wide for rules |
| 7 | Chargeback eligibility under scheme rules | Structured dispute fields | Absolute | Fully | Rules engine | Reason codes and time limits are codified |
| 8 | Classifying a customer's dispute email into reason code | Free text | Medium | Expressed in language | LLM classification with confidence threshold | Unstructured input, tolerable error with fallback to human |
| 9 | Reconciliation break investigation | Structured ledgers plus free-text references | Medium | Partly | Rules for matching; LLM for interpreting references; agent for investigating persistent breaks | Layered: rules first, LLM for the residual, agent for the hard tail |
| 10 | Credit decision for a retail loan | Structured bureau and application data | Absolute, explainable | Learnable, regulated | Classical ML scorecard inside a rules-based policy engine | Adverse-action explainability; EU AI Act Annex III high-risk |

Reading the table, two things stand out for the client. First, only three of ten tasks need a generative model, and only one needs an agent; the "AI programme" is in fact a layered automation programme. Second, the tasks that need agents (6, 9) are the ones with the highest variable cost per case today, because they consume senior analyst time. That is where the benefit case concentrates, and also where governance effort concentrates.

So what for the client: "Your fastest savings are in LLM-assisted drafting and classification at the edges of existing rules-based processes; your largest savings are in two agentic investigations that need a governance design before they need a model."

## 5. Real-world case

[PUBLIC — Google Cloud announcement, June 2023; HSBC public statements; VERIFY current figures] HSBC replaced a rules-based transaction monitoring approach in part with Google Cloud's AML AI (an ML-based "dynamic risk assessment"). Google Cloud reported two to four times more suspicious activity identified and around 60% fewer alerts. The relevant lesson for tool selection is not that ML beat rules; it is that the bank kept rules where regulators required explainable thresholds and introduced a learned model where the rule

base had become a false-positive generator. The ML model's outputs were explained through feature-level reason codes, which allowed model-risk sign-off.

[PUBLIC — Klarna press release, Feb 2024; CEO interviews, May 2025] Klarna's customer-service assistant is the canonical generative-AI case: two-thirds of chats handled by the assistant in its first month, equivalent to about 700 FTE, with an estimated USD 40m profit improvement. In 2025 the CEO said the company had over-rotated on cost and would re-hire humans for quality. For tool selection, the lesson is that the LLM was the right tool for the volume of routine queries, and the wrong tool, unassisted, for the long tail where judgement and empathy drove customer outcomes. A layered design with explicit escalation would have been the strategy consultant's recommendation.

## 6. Consultant language

> *"Let us separate the decision of record from the inputs to that decision. The decision stays deterministic; AI improves the inputs."*

Use when a client or a CRO fears that AI will make regulatory decisions.

> *"We should climb the simplicity ladder: rule, model, language model, agent. Each rung adds cost, variance and governance burden, so we stop at the lowest rung that solves the problem."*

Use in the first architecture discussion to signal you are not selling technology.

> *"Only a minority of steps in this process need a generative model. The programme is a layered automation programme with GenAI at the unstructured edges."*

Use when scoping, to reset expectations set by vendor demos.

> *"Agents earn their cost where the path cannot be scripted in advance. Where it can, an orchestrated workflow with LLM steps is cheaper and more controllable."*

Use when a sponsor wants "agentic" for its own sake.

> *"The compounding of per-step error rates is the central design constraint for agents: ten steps at 95% is 60% end to end."*

Use with technology leaders to justify gates, retries and narrow agent scopes.

## 7. Common mistakes juniors make

- Recommending an LLM for a task that has structured inputs and codified logic, because the demo was impressive.
- Describing classical ML as "legacy" in front of a CRO whose scorecards run on it under regulatory approval.
- Treating "agentic" as a product category rather than an architectural choice with a specific justification (path cannot be scripted).
- Forgetting the decision of record. Every regulated process has one; it must be deterministic or explicitly approved otherwise.
- Ignoring latency: fraud scoring at authorisation has a budget measured in tens of milliseconds, which excludes most LLM designs.
- Quoting hype-cycle positions or analyst failure rates as if they were evidence about this client's process.
- Not asking what labelled history exists before proposing supervised ML.

## 8. Interview questions with model answers

**Q1. When would you not use an LLM?**

Structure: three disqualifiers, then the edge case. I would not use an LLM when the decision must be deterministic and auditable as a decision of record (eligibility, limits, sanctions hits), when the inputs are fully structured and the logic is codified (a rules engine is cheaper, faster and defensible), and when latency budgets are in milliseconds at high volume (authorisation fraud scoring). I would also avoid it where the error mode is unacceptable even at low rates and there is no human review stage. The edge case is where an LLM supplies inputs to such a decision: reading documents, classifying text, drafting narratives. There it is often the best tool, provided the decision itself remains deterministic and the LLM's output is validated before it enters the decision. Example: in a disputes process the chargeback eligibility decision stays in a rules engine; the LLM classifies the customer's email and drafts the letter.

**Q2. Explain the difference between classical ML and generative AI to a CFO.**

Classical ML learns a specific mapping from your historical data to a specific outcome, such as probability of fraud, and gives you a number you can calibrate, back-test and explain with reason codes. It is a measurement instrument. Generative AI is a general-purpose model, trained by a vendor on the world's text, that produces language, code or structured output on request. It is a labour-substitute for cognitive tasks involving unstructured information. The financial consequences differ: classical ML has upfront data and modelling cost and near-zero marginal cost per decision; generative AI has low upfront cost and a meaningful marginal cost per call, so its economics depend on volume and on how much human time each call displaces. The risk profile differs too: classical ML risk is model risk you already govern; generative AI adds hallucination, data leakage and third-party dependency. In a business case I treat them as different asset classes.

**Q3. A client says "we want to be an agentic bank". How do you respond?**

I would affirm the ambition and immediately convert it into a portfolio question. First, which processes have steps whose path cannot be scripted in advance and consume expensive judgement time; those are candidates for agents. Second, which steps are high-volume unstructured text work; those are LLM candidates with better economics and lower risk. Third, which decisions of record must stay deterministic. The result is usually that a small number of agentic use cases carry a large share of value and almost all of the governance burden, and a larger number of LLM-assisted steps carry fast, low-risk savings. I would propose a six to eight week assessment that produces that portfolio, with an agentic-fit score per task and a governance design for the top two agentic candidates. I would also set an autonomy-level ambition with the CRO early, because that constraint shapes every design decision.

## 9. Self-check quiz

1. Name the five classes of technology in this module and, for each, state who or what produces the decision logic.
2. A step in a KYB process has fully structured inputs, codified policy, and a zero-error tolerance. Which class do you recommend, and what role, if any, can an LLM play around it?
3. Calculate the end-to-end success rate of an eight-step agent where each step is 97% reliable. What does the result imply for design?
4. Give two reasons a classical ML scorecard remains the correct tool for retail credit decisions even in 2026, one technical and one regulatory.
5. In the HSBC and Google Cloud AML case, what did the bank keep deterministic and what did it replace with a learned model, and why does that split matter for model-risk sign-off?

# Module 2 — How LLMs Work, at Consultant Depth

## 1. Why a client cares

Every number in an LLM business case, from inference cost to accuracy to latency, is determined by mechanics the client's executives have never seen: tokenisation, context length, retrieval quality, sampling. A consultant who understands the mechanism can tell a CIO why the pilot's cost doubled when the prompt grew, why the assistant answers confidently and wrongly, and why fine-tuning is usually the wrong first move. That understanding is the difference between relaying a vendor's claims and advising.

## 2. Concept and mechanism

### 2.1 Tokens and tokenisation

LLMs do not process words or characters. They process tokens: sub-word units produced by a tokeniser trained on the pre-training corpus, usually with byte-pair encoding (Sennrich et al., 2016) or a variant such as SentencePiece. A token is typically three to four characters of English text, so roughly 750 words per 1,000 tokens [ESTIMATE — common rule of thumb; varies by model and language]. Consequences the client must understand:

- **Pricing is per token**, input and output priced separately, output typically three to five times the input price [ESTIMATE — order of magnitude, VERIFY].
- **Non-English and specialised text costs more.** Code, legal boilerplate, SWIFT MT messages and non-Latin scripts tokenise less efficiently, sometimes two to three times more tokens per character.
- **Numbers tokenise poorly.** A long IBAN or account number may become many tokens, and the model reasons over those fragments, which is one reason arithmetic and exact-match tasks should be delegated to tools.
- **Structured documents carry overhead.** JSON keys, XML tags and table formatting consume tokens without carrying information; flattening a document before sending it can cut cost by 20 to 40% [ESTIMATE — assumption].

### 2.2 The transformer and attention

The architecture is the transformer (Vaswani et al., 2017). Each token is mapped to an embedding vector; the sequence passes through stacked layers, each containing a self-attention block and a feed-forward block. Self-attention computes, for every token, a weighted combination of all other tokens in the context, with the weights derived from learned query, key and value projections. The output is a next-token probability distribution over the vocabulary.

Two properties follow directly and are worth stating precisely to technical clients:

- **Attention is quadratic in sequence length in its naive form.** Doubling the context roughly quadruples attention compute, which is why long contexts cost more and run slower, even where vendors charge linearly. Engineering advances (FlashAttention, grouped-query attention, sparse attention) reduce the constant but not the fundamental scaling.
- **Generation is autoregressive.** Each output token requires a forward pass conditioned on everything before it. Output tokens are therefore slower and more expensive than input tokens; latency is dominated by output length, not input length, once the prompt has been processed.

Modern decoder-only LLMs (the GPT, Claude, Gemini and Llama families) use this architecture with scale: parameter counts from a few billion to over a trillion, trained on trillions of tokens. Scaling laws (Kaplan et al., 2020; Hoffmann et al., 2022) showed that capability improves predictably with compute, data and parameters, which is why frontier models are built by a handful of well-capitalised labs.

## 2.3 Pre-training, alignment and what the model “knows”

Pre-training is next-token prediction over a corpus. The result is a base model with broad knowledge but poor instruction following. Alignment turns it into an assistant: supervised fine-tuning on instruction-response pairs, then reinforcement learning from human feedback, RLHF (Ouyang et al., 2022), or variants such as direct preference optimisation. Alignment makes the model helpful and safe, but also introduces sycophancy and over-confidence, since human raters reward confident answers.

What the model knows is parametric knowledge frozen at its training cut-off, stored implicitly in weights. It is not a lookup table. Retrieval of a specific fact is probabilistic: common facts are reliably reproduced, rare facts are not. This is the technical root of hallucination and the reason bank-specific knowledge must be supplied at inference time, not assumed.

## 2.4 Context window

The context window is the maximum number of tokens the model attends to in one call, including system prompt, conversation history, retrieved documents and the generated output. Frontier models in 2025 offered 128,000 to 1,000,000-plus tokens [PUBLIC — vendor documentation, 2025; VERIFY]. Three consultant-level implications:

1. **Cost scales with what you put in.** Every call re-sends the whole context. A 100,000-token document summarised in a ten-turn conversation costs ten times the document unless prompt caching is used (vendors offer cached-input discounts of 75 to 90% [ESTIMATE — VERIFY]).
2. **Quality degrades with irrelevant length.** The “lost in the middle” effect (Liu et al., 2023) showed models attend less reliably to information in the middle of long contexts. More context is not more accuracy.
3. **Long context is not memory.** Nothing persists between calls unless the application stores and re-sends it. “Memory” in agent systems is an engineering layer (Module 3), not a model property.

## 2.5 Embeddings and vector similarity

An embedding model maps text to a dense vector (typically 256 to 3,072 dimensions) such that semantically similar texts are near each other under cosine similarity or dot product. Embedding models are separate from, and much cheaper than, generative models. They power semantic search, clustering, deduplication and retrieval.

Properties to know: embeddings are model-specific (vectors from two models are not comparable); they are sensitive to chunk length (a 50-word chunk and a 2,000-word chunk of the same document embed differently); and they capture topical similarity better than exact-match semantics, which is why hybrid search (vector plus keyword BM25) outperforms pure vector search on names, codes and identifiers, exactly the content banks care about.

## 2.6 Retrieval-augmented generation (RAG), end to end

RAG (Lewis et al., 2020) grounds the model’s generation in retrieved documents supplied in the prompt. The pipeline has distinct stages, each with design choices and failure modes.

| Stage | What happens | Key design choices | Typical failure |
|---|---|---|---|
| Ingestion | Documents parsed to text | Layout-aware parsing for PDFs and tables; OCR for scans | Tables flattened to nonsense; headers lost |
| Chunking | Text split into retrievable units | Fixed-size with overlap; semantic or structural chunking (by section, clause); parent-child chunks | Chunks split mid-clause; context lost |

| Stage | What happens | Key design choices | Typical failure |
|---|---|---|---|
| Enrichment | Metadata attached | Document type, date, jurisdiction, product, access rights | No access-control metadata, so leakage |
| Embedding and indexing | Vectors stored in a vector index | Embedding model choice; index type (HNSW, IVF); hybrid index | Wrong embedding model for domain |
| Retrieval | Top-k chunks fetched for a query | Query rewriting; hybrid search; re-ranking with a cross-encoder; k and similarity thresholds | Wrong chunks retrieved; right chunk ranked 12th |
| Generation | LLM answers using retrieved context | Prompt instructs to answer only from context; cite sources; abstain if absent | Model ignores context and answers from parametric memory |
| Post-processing | Validation and formatting | Citation checking; faithfulness scoring; structured output | Unchecked claims pass through |

Chunking deserves emphasis because it is where most RAG quality problems originate. Fixed 500-token chunks with 50-token overlap are the default and are wrong for most banking documents, where a policy clause or a KYC procedure step is the natural unit. Structural chunking (split on headings and numbered clauses) with a parent-child scheme (retrieve small child chunks for precision, pass the parent section for context) is usually the right starting design.

Retrieval evaluation is separate from generation evaluation. Measure recall at k (is the correct passage in the top k?), mean reciprocal rank (how high is it?), and context precision (what fraction of retrieved chunks is relevant?). A pipeline with 70% recall at 5 cannot produce better than roughly 70% correct answers no matter how good the generator is.

Variants to know by name: GraphRAG (entity-relationship graphs over documents, useful for beneficial-ownership and counterparty questions), agentic RAG (the model decides when and what to retrieve, iteratively), and long-context RAG (fewer, larger chunks with models that handle long contexts well).

## 2.7 Prompting, RAG and fine-tuning: the decision logic

The three ways to shape model behaviour address different problems. The decision rule most practitioners use [attribution uncertain; widely used in practice]: use prompting to control format and behaviour, RAG to supply knowledge, fine-tuning to teach style or a narrow skill where prompting has demonstrably plateaued.

| Lever | Changes | Cost | Latency effect | Knowledge freshness | When to use |
|---|---|---|---|---|---|
| Prompt engineering (system prompt, few-shot examples, chain-of-thought) | Behaviour and format within a call | Near zero | Adds input tokens | None | Always first |
| RAG | Knowledge available at inference | Engineering plus retrieval infrastructure | Adds retrieval latency and tokens | Real-time, controllable | Whenever bank-specific facts are needed |
| Fine-tuning (full or parameter-efficient such as LoRA, Hu et al., 2021) | Weights; style, format, narrow skill | Training data preparation, compute, re-evaluation | Can reduce prompt length | Frozen at training; must retrain | After prompting plateaus; for consistent style or domain vocabulary; for smaller models to |

| Lever | Changes | Cost | Latency effect | Knowledge freshness | When to use |
|---|---|---|---|---|---|
| | | | | | match frontier on a narrow task |

Fine-tuning does not reliably add facts and is a poor way to add knowledge; it also creates a model the bank must govern under model-risk rules, with its own validation and refresh cycle. Juniors propose fine-tuning first because it sounds rigorous. Partners propose it last.

## 2.8 Hallucination: mechanism and mitigation

Hallucination is the generation of fluent content not supported by the input or by fact. Its mechanism follows from the training objective: the model is optimised for plausible continuation, not truth; alignment rewards confident answers; and parametric recall of rare facts is probabilistic. Three forms matter in banking: fabricated facts (a regulation that does not exist), unfaithful summaries (a correct fact from the wrong document), and fabricated citations or identifiers.

Mitigations, in order of effectiveness in practice [ESTIMATE — practitioner consensus]:

1. **Grounding via RAG with instruction to abstain** when the answer is not in context.
2. **Structured outputs and tool calls** for anything that must be exact: look up the account, calculate the fee, retrieve the clause.
3. **Faithfulness checking** by a second model or rule: every claim must map to a cited span.
4. **Lower temperature** (reduces variance, does not remove hallucination).
5. **Human review** calibrated to risk: sample, exception or full.
6. **Evaluation on a golden set** so the rate is measured, not assumed (Module 4).

Note the thing you cannot do: eliminate hallucination. The client's control framework must assume a non-zero rate and design for detection.

## 2.9 Frontier, mid-tier, small and open models

Frontier models (the largest proprietary models from the major labs) lead on reasoning and breadth. Mid-tier models from the same labs trade some capability for roughly three to ten times lower price [ESTIMATE — VERIFY]. Small models (1 to 10 billion parameters) and open-weight models (Llama, Mistral, Qwen, Gemma families and others) can be self-hosted, fine-tuned and run at very low marginal cost, with capability adequate for classification, extraction and routing tasks.

Selection logic for a bank:

- **Data residency and confidentiality** may require self-hosting or a sovereign cloud region, which favours open-weight models or hyperscaler-hosted proprietary models with contractual zero-retention.
- **Task difficulty** determines the minimum capable tier. Routing and classification rarely need frontier; multi-step reasoning over complex documents usually does.
- **Volume** determines whether the price gap matters. At a million calls a month, a tenfold price difference is a budget line; at ten thousand, it is noise.
- **Model routing**: send easy cases to a small model and escalate hard cases, often cutting cost by half or more while holding quality [ESTIMATE — practitioner experience].

Open-weight does not mean free: inference infrastructure, MLOps, security patching and evaluation are the bank's cost. Module 8 covers this.

## 2.10 Latency and cost trade-offs

Latency has three components: time to first token (prompt processing), generation time (output tokens times per-token time) and orchestration overhead (retrieval, tool calls, network). For a customer-facing chat,

time to first token under one second and streaming output are expected. For back-office batch work, latency is irrelevant and batch APIs with roughly 50% discounts [ESTIMATE — VERIFY] become attractive. Reasoning models that "think" before answering multiply both latency and output cost by a factor that can exceed ten; they are justified for hard judgement tasks, not for drafting letters.

# 3. Frameworks

## 3.1 The "shape the model" ladder (prompt, retrieve, tune)

Described in 2.7. Steps: (1) write the system prompt and test on a golden set; (2) add few-shot examples; (3) add retrieval if the failure is missing knowledge; (4) add structured outputs and tools if the failure is exactness; (5) consider fine-tuning only when the residual failure is style or a narrow skill and you have 500-plus high-quality examples [ESTIMATE — common practitioner threshold].

## 3.2 Token-economics model

A reusable calculation [ILLUSTRATIVE — standard practitioner method]: annual inference cost = volume x (input tokens x input price + output tokens x output price) x (1 + retry and evaluation overhead) x (1 - caching and batching discount). Always show the overhead and discount terms explicitly; juniors omit them and the number is wrong by a factor of two in either direction.

## 3.3 RAG quality decomposition

Attribute end-to-end accuracy to retrieval recall and generation faithfulness separately: accuracy is bounded above by retrieval recall, and the gap between recall and accuracy is a generation problem. This decomposition tells you where to invest and is the first analysis to run when a pilot under-performs.

# 4. Worked BFSI example

[ILLUSTRATIVE — composite] A UK bank's corporate KYC (KYB) team must produce a structured summary of each client's onboarding file: ownership structure, nature of business, source of funds, adverse media and risk rating rationale. Volume: 18,000 new onboardings and 42,000 periodic refreshes a year, 60,000 cases in total. Average file: 35 pages of mixed documents, about 25,000 tokens after parsing.

Design A: whole file in the prompt, frontier model.

| Item | Value | Basis |
|---|---|---|
| Input tokens per case | 25,000 file + 3,000 system prompt and schema = 28,000 | [ESTIMATE] |
| Output tokens per case | 2,500 (structured summary) | [ESTIMATE] |
| Input price | USD 3.00 per million | [ESTIMATE — order of magnitude, VERIFY] |
| Output price | USD 15.00 per million | [ESTIMATE — VERIFY] |
| Input cost per case | 28,000 / 1,000,000 x 3.00 = USD 0.084 | |
| Output cost per case | 2,500 / 1,000,000 x 15.00 = USD 0.0375 | |
| Cost per case | USD 0.1215 | |
| Retry and eval overhead | +30% | [ESTIMATE — one in five cases re-run, plus LLM-as-judge scoring] |
| Annual cost | 60,000 x 0.1215 x 1.30 = USD 9,477 | |

Design B: structural chunking, RAG per section, mid-tier model for extraction and frontier model only for the risk-rating rationale.

| Item | Value | Basis |
|---|---|---|
| Extraction calls per case | 6 sections x 4,000 input tokens x mid-tier USD 0.80 per million = USD 0.0192 | [ESTIMATE — VERIFY pricing] |
| Extraction output | 6 x 400 tokens x USD 3.20 per million = USD 0.0077 | |
| Rationale call (frontier) | 6,000 input x 3.00 + 800 output x 15.00 per million = USD 0.018 + 0.012 = USD 0.030 | |
| Embedding cost | 25,000 tokens x USD 0.10 per million = USD 0.0025 | [ESTIMATE — VERIFY] |
| Cost per case | USD 0.0594 | |
| Overhead | +30% | |
| Annual cost | 60,000 x 0.0594 x 1.30 = USD 4,633 | |

Interpretation. Design B halves inference cost and, more importantly, improves faithfulness because each extraction sees only the relevant section. But the inference saving (about USD 4,800 a year) is immaterial next to the engineering cost of the RAG pipeline (USD 150,000 to 300,000 build [ESTIMATE — assumption]). The real reasons to choose Design B are quality (section-level grounding, citation per field) and model-risk defensibility, not token cost. The business value sits elsewhere: if the summary saves 45 minutes of analyst time per case [ESTIMATE], that is 45,000 hours a year, or roughly 25 FTE at 1,800 productive hours each, which at a loaded cost of GBP 65,000 [ESTIMATE] is about GBP 1.6m of capacity. Inference cost is under 1% of the benefit.

So what for the client: “Token cost is a rounding error here; design for faithfulness and auditability, and let the benefit case rest on analyst hours, with a benefit ramp tied to measured adoption.”

## 5. Real-world case

[PUBLIC — Morgan Stanley and OpenAI announcements, Sept 2023; subsequent press coverage 2024; VERIFY figures] Morgan Stanley’s “AI @ Morgan Stanley Assistant” gives roughly 16,000 financial advisers retrieval-augmented access to the firm’s research and procedures library of around 100,000 documents. Publicly reported design features align with this module: the firm invested heavily in document curation and chunking, built an evaluation process in which advisers and subject experts graded answers before release, and restricted the assistant to answering from the firm’s content. Reported adoption reached around 98% of adviser teams. In 2024 the firm added “Debrief”, which summarises client meetings and drafts follow-up emails. The lesson is that the value came from the retrieval corpus and the evaluation discipline, not from the model, which was a commercially available frontier model.

[PUBLIC — Moffatt v. Air Canada, British Columbia Civil Resolution Tribunal, Feb 2024] A customer-facing chatbot gave incorrect information about bereavement fares; the tribunal held the airline liable for the chatbot’s statements. The mechanism was classic hallucination of policy detail without grounding. For banks the implication is that an ungrounded assistant is a conduct risk, and that “the chatbot said it” is not a defence.

## 6. Consultant language

> *“The model is a statistical model of language, not a system of record. Our design supplies the facts at inference time and validates every claim against them.”*

Use when positioning RAG and groundedness to a risk committee.

> *“We should separate retrieval quality from generation quality. If recall at five is 70%, no prompt will get us above 70%.”*

Use when a pilot under-performs and the team is tuning prompts.

> *"Fine-tuning is the last lever, not the first. It adds a model we have to govern and it does not add knowledge."*

Use when a technology team proposes fine-tuning early.

> *"Inference cost scales with context, so our architecture decision is also a cost decision. Chunk and retrieve rather than stuff the prompt."*

Use in architecture reviews.

> *"We assume a non-zero hallucination rate and design for detection, not for prevention."*

Use with the CRO and internal audit.

## 7. Common mistakes juniors make

- Describing the LLM as "reading" or "understanding" documents without stating the retrieval and grounding design.
- Proposing fine-tuning to add bank knowledge.
- Estimating cost from input tokens only, ignoring output tokens, retries, evaluation calls and multi-turn re-sending.
- Chunking by fixed size and then blaming the model for poor answers.
- Treating long context windows as a reason not to build retrieval.
- Quoting a single "accuracy" number without saying what was measured against what.
- Using a reasoning model for a drafting task and wondering why latency and cost exploded.
- Assuming open-weight means cheap, forgetting infrastructure, MLOps and evaluation.

## 8. Interview questions with model answers

**Q1. Explain RAG to a COO and tell me where it fails.**

RAG means that when the model is asked a question, the system first retrieves the relevant passages from the bank's own documents and gives them to the model alongside the question, with instructions to answer only from those passages and cite them. It lets us use a general-purpose model with bank-specific, current, access-controlled knowledge without retraining anything. It fails in three places. First, ingestion: scanned PDFs and tables parsed badly give the model garbage. Second, retrieval: if the right passage is not in the top results, the answer cannot be right; we measure this as recall. Third, generation: the model may ignore the passages and answer from memory, or blend two passages incorrectly; we measure this as faithfulness. The operational implication is that the programme needs a document-curation workstream and an evaluation harness from day one, and those usually cost more than the model.

**Q2. When would you recommend fine-tuning?**

Rarely, and only after prompting and retrieval have plateaued on a measured golden set. Fine-tuning is appropriate to teach consistent style and format (a house tone for customer letters), to teach a narrow skill the base model performs inconsistently (classifying dispute reasons into our 40 internal codes), or to bring a smaller, cheaper or self-hostable model up to frontier performance on one narrow task, which can matter for cost at very high volume or for data-residency reasons. It is not appropriate for adding knowledge, because the model does not reliably recall fine-tuned facts and the knowledge goes stale. The costs the client often misses are governance costs: a fine-tuned model is a model under SS1/23 or SR 11-7, with validation, monitoring and a refresh cycle every time the base model is deprecated.

**Q3. Our pilot cost three times the estimate. What probably happened?**

Four usual suspects, in order of likelihood. Multi-turn conversations re-send the whole history on every turn, so a ten-turn conversation costs far more than ten single calls; prompt caching mitigates this. The prompt grew: teams add instructions, examples and retrieved chunks until the input is five times the original estimate. Output was under-estimated, and output tokens are three to five times the price of input. And hidden calls: retries, evaluation calls, query rewriting and LLM-as-judge scoring can double call volume. My remediation would be to instrument token usage per call type, cap retrieved context with re-ranking, enable caching, and route simple turns to a smaller model. I would also check whether a reasoning model was used for routine steps.

## 9. Self-check quiz

1. Why are output tokens more expensive and slower than input tokens? Give the architectural reason.
2. A RAG pipeline has recall at 5 of 72% and end-to-end answer accuracy of 61%. Where is the larger problem and what would you change first?
3. State the three distinct problems that prompting, RAG and fine-tuning each solve, and give the one thing fine-tuning does not do well.
4. Recompute the Design A annual cost from Section 4 if the bank adopts prompt caching that cuts the effective price of the 3,000-token system prompt and schema by 90% and leaves everything else unchanged. Is the saving material?
5. Name two chunking strategies and explain why fixed-size chunking is usually wrong for policy documents.

# Module 3 — Agentic AI Mechanics

## 1. Why a client cares

Agentic AI is where the largest benefit claims and the largest operational risks in a bank's AI portfolio meet. An agent does not just produce text; it acts on systems, and errors compound across steps. The COO cares because the labour-substitution case depends on agents completing end-to-end work; the CRO cares because an agent that acts wrongly creates a loss event, not a typo; the regulator cares because accountability for an autonomous action must land on a human. A consultant who can specify the pattern, the autonomy level, the guardrails and the human gates in one coherent design is the one the client trusts with the process.

## 2. Concept and mechanism

### 2.1 Definition: agent = model + tools + memory + planning

An agent is a system in which a language model, placed in a control loop, repeatedly decides what to do next in pursuit of a goal, executes actions through tools, observes results and continues until a termination condition is met. The four components:

- **Model.** The LLM that interprets the goal, the state and the tool results, and emits the next action. Its reasoning quality bounds the agent's reliability.
- **Tools.** Functions the agent can call: search, database query, API call, calculator, code execution, another agent. Each tool is described to the model by a schema (name, purpose, typed parameters). The model emits a structured call; the runtime executes it and returns the result as text or structured data.
- **Memory.** Short-term memory is the context window (the running transcript of actions and observations). Long-term memory is an engineered store: a vector database of past cases, a key-value store of user preferences, a scratchpad file, or a structured state object. Nothing persists between calls unless the application persists it.
- **Planning.** The procedure by which the agent decides the sequence of actions: implicit (the model decides step by step) or explicit (a plan is produced first, then executed, then revised).

The runtime, often called the orchestrator or agent framework (LangGraph, AutoGen, CrewAI, Semantic Kernel, OpenAI Agents SDK, Google ADK, and others [PUBLIC — vendor documentation, 2024 to 2025; VERIFY current landscape]), manages the loop, state, tool execution, retries and termination.

### 2.2 The core loop and its failure arithmetic

The loop is: observe state, reason, select action, execute, observe result, repeat. Termination occurs on goal completion, a maximum step count, a budget limit, an error threshold or a human interrupt.

The failure arithmetic is the central design constraint. If each step succeeds independently with probability p and the task requires n dependent steps, end-to-end success is p^n. At p = 0.95 and n = 10, that is 0.60. At p = 0.98 and n = 20, it is 0.67. The design responses are: fewer steps (narrow scope, deterministic sub-steps), higher per-step reliability (better tools, structured outputs, validation), recovery (self-correction, retries, verification steps that detect and fix errors, which breaks the independence assumption in your favour) and gates (human checkpoints before irreversible actions).

### 2.3 Agentic patterns

**ReAct (Reason and Act)** (Yao et al., 2022). The model interleaves a reasoning trace ("Thought"), an action (tool call) and an observation, in a single loop. Simple, general, and the basis of most single-agent implementations. Weakness: no global plan, so it can wander or loop on long tasks.

**Plan-and-execute.** The model first produces an explicit plan (a list of steps), then an executor carries out each step, with optional re-planning when a step fails or reveals new information. Better on long tasks; more predictable; plans can be reviewed by a human before execution, which is valuable in regulated settings. Related: Reflexion (Shinn et al., 2023), in which the agent critiques its own trajectory and retries; Tree of Thoughts (Yao et al., 2023), which explores multiple reasoning branches.

**Supervisor (hierarchical).** A supervising agent receives the goal, decomposes it, delegates sub-tasks to specialised sub-agents (each with its own prompt, tools and scope), evaluates their returns and decides what to do next. Sub-agents may themselves supervise others. Appropriate when sub-tasks are heterogeneous and the next step depends on judgement about results. This is the pattern used in the KYB engagement in your background, and it is the dominant enterprise pattern for complex case work because scope, prompts and permissions can be set per sub-agent.

**Orchestrator-worker.** A central orchestrator decomposes a task into sub-tasks, dispatches them to workers that are often identical or interchangeable, frequently in parallel, then merges results. Appropriate for homogeneous bulk work: screen 500 names, summarise 80 documents, check 30 fields. The distinction from supervisor is heterogeneity and judgement (supervisor) versus homogeneity and parallelism (orchestrator-worker). In practice systems combine both: a supervisor delegates a bulk sub-task to an orchestrator-worker sub-system.

**Multi-agent collaboration and debate.** Several agents with different roles (drafter, critic, compliance reviewer) exchange messages to converge on an output. Debate patterns (Du et al., 2023) improve factual accuracy on some tasks at a multiple of the cost. Use for high-stakes judgement outputs where a second-opinion structure is worth two to five times the inference cost; do not use for routine drafting.

**Router and workflow patterns.** Many "agentic" deployments are in fact deterministic workflows with LLM steps: a router classifies the input and a fixed pipeline runs. These are cheaper, more testable and often sufficient. The consultant's discipline is to call them what they are.

| Pattern | Decomposition | Parallelism | Judgement between steps | Best for | Main risk |
|---|---|---|---|---|---|
| ReAct single agent | Implicit | No | Yes, local | Short tasks, 3 to 8 steps | Looping, drift |
| Plan-and-execute | Explicit plan | Possible | At re-plan points | Long tasks, reviewable plans | Stale plan |
| Supervisor | By specialism | Possible | Yes, central | Heterogeneous case work (KYB, disputes) | Supervisor becomes bottleneck; delegation errors |
| Orchestrator-worker | By unit of work | High | No | Bulk homogeneous work | Merge errors |
| Collaboration and debate | By role | Low | Yes, dialogic | High-stakes judgement | Cost, latency |
| Router plus workflow | Fixed | Fixed | No | Predictable processes | Mis-labelled as agentic |

## 2.4 Tool calling and function schemas

Tool calling is the mechanism by which an LLM emits a structured request (typically JSON conforming to a declared schema) that the runtime executes. Design principles that determine reliability:

- **Narrow, well-described tools.** A tool named `get_customer_profile(customer_id)` with a one-line description and typed parameters is called correctly far more often than a generic `query_database(sql)`.

- **Idempotency and reversibility flags.** Tools that change state should be marked, so the runtime can require confirmation or a gate.
- **Structured returns.** Tool results returned as compact structured data, not raw dumps, reduce context consumption and mis-reading.
- **Least privilege.** Each agent's tool set and credentials are scoped to its task. A document-reading sub-agent has no payment-initiation tool.
- **Error contracts.** Tools return explicit error types the model can reason about (not found, permission denied, timeout) rather than failing silently.

## 2.5 MCP and agent-to-agent protocols

**Model Context Protocol (MCP)** [PUBLIC — Anthropic, Nov 2024] standardises how an application exposes tools, resources and prompts to a model, in a client-server pattern. Instead of bespoke integration per model and per system, a system publishes an MCP server once and any MCP-compatible agent can use it. For a bank this means a governed catalogue of internal MCP servers (customer data, case management, policy library) with authentication and audit at the server, which is an attractive control point.

**Agent2Agent (A2A)** [PUBLIC — Google, Apr 2025; donated to the Linux Foundation, Jun 2025; VERIFY adoption] standardises how agents discover each other (via "agent cards"), negotiate capabilities and exchange tasks and results across vendors and organisations. Where MCP connects an agent to tools, A2A connects agents to agents. Both are young; treat vendor claims of broad adoption as [VERIFY — date-sensitive].

Consultant implication: protocols shift the integration cost curve and create a new governance surface (who may publish a server, who may call it, what is logged). They belong in the operating model discussion, not just the architecture.

## 2.6 Autonomy levels

There is no single industry standard for AI autonomy levels; several practitioner scales exist, often modelled on the SAE J3016 driving-automation levels [attribution: analogy to SAE levels widely used; specific AI scales vary by vendor and firm]. The scale below is a composite used in practice [ILLUSTRATIVE — composite].

| Level | Name | Who decides | Who acts | Human role | Banking example |
|---|---|---|---|---|---|
| 0 | Manual | Human | Human | Does the work | Analyst reads file, writes summary |
| 1 | Assist | Human | Human | Uses AI output as input | Copilot drafts summary; analyst edits and files |
| 2 | Recommend | AI proposes, human decides | Human | Approves each action | Agent proposes dispute outcome and letter; agent (human) approves |
| 3 | Act with approval gates | AI decides routine, human approves defined classes | AI, gated | Approves exceptions and irreversible actions | Agent resolves low-value disputes; human approves refunds above threshold |
| 4 | Act with oversight | AI | AI | Reviews samples and metrics; intervenes on alerts | Agent resolves within policy; QA samples 5% daily |
| 5 | Fully autonomous | AI | AI | Sets policy only | Not appropriate for regulated decisions of record |

| Level | Name | Who decides | Who acts | Human role | Banking example |
|---|---|---|---|---|---|
| | | | | | today |

How to position with regulators and model-risk teams: state the target level per use case, define the gate criteria for moving from one level to the next (measured accuracy on a golden set over a period, incident-free run, independent validation), and commit to the ability to step down a level on trigger. Regulators respond well to “we start at level 2 and earn level 3 against published criteria”, and badly to “the agent is autonomous”.

## 2.7 Guardrails taxonomy

Guardrails are controls applied around the model. Four layers [ILLUSTRATIVE — composite taxonomy; widely used in practice with varying names]:

| Layer | Purpose | Examples |
|---|---|---|
| Input guardrails | Protect the agent from bad or malicious input | Prompt-injection detection, PII redaction before the model, topic and scope classifiers, document-source allow-lists |
| Output guardrails | Check what the model produces before it is used | Schema validation, faithfulness checks against sources, toxicity and PII filters, confidence thresholds with fallback |
| Action guardrails | Constrain what the agent can do | Tool allow-lists per agent, parameter limits (refund cap), rate limits, dual-control for irreversible actions, sandboxed execution |
| Policy guardrails | Encode business and regulatory rules outside the model | Deterministic rules engine as final check, mandatory disclosures, jurisdictional restrictions, kill switch and step-down procedures |

Prompt injection deserves a specific mention because agents read untrusted content (emails, documents, web pages) and can be instructed by it. Mitigations: treat all retrieved content as data not instructions, separate system and user channels, restrict tools available when processing untrusted content, and require human approval before actions triggered by content from untrusted sources.

## 2.8 Human-in-the-loop design patterns

Human-in-the-loop (HITL) is a design decision about where human judgement sits in the flow, not a slogan. Three patterns, often combined:

- **Approve (pre-action gate).** The agent proposes; a human approves before execution. Highest control, lowest throughput gain. Use for irreversible or high-value actions and during the earning-trust phase.
- **Review-sample (post-action QA).** The agent acts; humans review a statistically meaningful sample (and all flagged cases). Use once accuracy is proven, with sample size set by target confidence and observed error rate.
- **Exception-only.** The agent handles cases within policy and confidence thresholds; everything else routes to a human queue with the agent’s analysis attached. The thresholds are the control.

Design parameters to specify: the gate criteria (what triggers human review), the information presented to the human (the agent’s evidence and rationale, not just the conclusion), the time budget for the human, the feedback loop (human corrections become evaluation data) and the escalation path when the human disagrees.

### 2.9 When agents are the wrong answer

An agent is the wrong tool when the path can be scripted (use a workflow with LLM steps), when the task is a single transformation (use one LLM call), when the decision must be deterministic (use rules), when latency is tight (agents are slow), or when the error cost of an autonomous action exceeds the labour saved. The honest test: write the process as a flowchart. If you can, you do not need an agent; if every attempt produces “it depends on what we find”, you might.

## 3. Frameworks

### 3.1 Agent design canvas [ILLUSTRATIVE — practitioner template; attribution: composite]

For each agent, specify in one page: goal and success criteria; scope (what it does and explicitly does not do); inputs and trusted sources; tools with read/write and reversibility flags; memory (what is stored, where, retention); planning pattern; autonomy level and gate criteria; guardrails by layer; HITL pattern and thresholds; evaluation plan and golden set; observability (what is logged); failure modes and fallbacks; owner.

### 3.2 Autonomy progression ladder

Steps: (1) set the target level with the business and the CRO; (2) start one level below the target; (3) define promotion criteria (for example, 500 cases, accuracy above 97% on sampled review, zero severity-one incidents, independent validation sign-off); (4) define demotion triggers (accuracy drop, incident, model version change); (5) review quarterly.

### 3.3 Failure-mode and effects analysis (FMEA) for agents

FMEA originates in reliability engineering (US military standard MIL-P-1629, 1949; widely adopted in automotive). Applied to agents: list each step, its failure modes, the effect, the detection mechanism and the mitigation; score severity, occurrence and detectability. Model-risk teams recognise the format immediately.

## 4. Worked BFSI example

[ILLUSTRATIVE — composite] Supervisor-led multi-agent system for card payment dispute resolution at a UK bank handling 480,000 disputes a year, of which 60% are “transaction not recognised”, 25% merchant disputes (goods not received, defective, cancelled subscription) and 15% fraud-related.

### 4.1 Target autonomy and scope

Target level 3 for non-fraud disputes under GBP 250 within twelve months; level 2 at launch. Fraud-related disputes excluded from autonomous action (level 2 ceiling) because of regulatory reporting and the customer-harm profile. The chargeback eligibility decision remains in the existing rules engine.

### 4.2 Agent scoping

| Agent | Scope | Tools (R = read, W = write) | Model tier | Autonomy ceiling |
|---|---|---|---|---|
| Supervisor | Receives case, plans, delegates, assembles outcome, routes to HITL | Case management R/W (status only), queue router W | Frontier | Level 3 |
| Intake and classification agent | Parses customer message and channel data; assigns reason code with confidence | Customer message R, reason-code taxonomy R | Mid-tier | n/a (proposes only) |

| Agent | Scope | Tools (R = read, W = write) | Model tier | Autonomy ceiling |
|---|---|---|---|---|
| Evidence agent | Gathers transaction, merchant, device and history data; checks for duplicates and prior disputes | Core banking R, card processor R, merchant directory R | Mid-tier | n/a |
| Merchant-contact agent | Drafts and sends merchant enquiry where scheme rules require pre-chargeback contact; parses response | Merchant messaging W (templated only), inbox R | Mid-tier | Level 3 (templated sends) |
| Policy and eligibility agent | Maps evidence to scheme rule inputs; calls the rules engine; interprets the result | Rules engine R (invoke), scheme rulebook RAG R | Frontier | n/a (rules engine decides) |
| Resolution and communication agent | Drafts the customer letter and the representment package; proposes refund or decline | Letter templates R, refund tool W (capped) | Frontier | Level 3 under GBP 250; level 2 above |
| Compliance critic agent | Independent check of the draft outcome against policy and Consumer Duty guidance; flags issues | Policy RAG R | Frontier | n/a (blocks or passes) |

## 4.3 HITL gates

| Gate | Trigger | Human action | Information presented |
|---|---|---|---|
| G1 Classification | Reason-code confidence below 0.85, or fraud indicators | Confirm or correct reason code | Customer message, proposed code, top-3 alternatives with confidence |
| G2 Evidence gap | Mandatory evidence missing after two retrieval attempts | Source evidence or decide | Checklist of evidence with status |
| G3 Outcome approval | Refund above GBP 250; decline on any case; critic agent flag; vulnerable-customer marker | Approve, amend or reject | Evidence summary, rule trace, draft letter, critic's notes |
| G4 Sample QA | 5% random sample of autonomous closures, daily | Grade outcome and letter | Full trajectory log |

## 4.4 Guardrails

Input: customer messages are treated as data; prompt-injection classifier; PII minimisation to sub-agents. Output: structured schemas for every agent return; faithfulness check of the letter against the evidence; tone and Consumer Duty vocabulary checks. Action: refund tool hard-capped at GBP 250 per case and GBP 5,000 per day per agent instance; merchant messaging restricted to templates; no tool can close a case without the rules-engine result attached. Policy: deterministic eligibility decision; mandatory disclosure text injected by template, not generated; kill switch that reverts all cases to the human queue.

### 4.5 Failure-mode table (abridged FMEA)

| Step | Failure mode | Effect | Detection | Mitigation | Severity (1 to 5) |
|---|---|---|---|---|---|
| Classification | Wrong reason code | Wrong rule path, wrong outcome | Confidence threshold; G1; critic | Threshold 0.85; human confirm below | 4 |
| Evidence | Stale or wrong account data | Wrong eligibility inputs | Data freshness check; cross-source consistency | Retry; G2 | 4 |
| Merchant contact | Agent sends non-templated text or discloses data | Conduct and privacy breach | Template-only tool; output PII filter | Hard restriction; audit log | 5 |
| Eligibility | Model misreads rules-engine output | Wrong communicated outcome | Critic compares letter to rule trace | Structured rule output; critic block | 5 |
| Resolution | Hallucinated policy or deadline in letter | Customer misinformation (Air Canada pattern) | Faithfulness check against templates and evidence | Template-locked clauses; G3 | 5 |
| Refund | Duplicate or excess refund | Financial loss | Idempotency key; cap; reconciliation | Cap; daily limit; G3 above GBP 250 | 4 |
| Supervisor | Infinite loop or runaway steps | Cost, latency | Step and token budget | Hard limits; fallback to human | 2 |
| Any | Prompt injection via customer text | Unauthorised action | Injection classifier; tool restrictions | No write tools while processing untrusted text | 5 |

### 4.6 Economics sketch

Average handling time today: 38 minutes per non-fraud dispute [ESTIMATE]. Target: 70% of non-fraud disputes closed at level 3 with 4 minutes of human time (QA and gates amortised), 30% at level 2 with 15 minutes. Weighted: 0.7 x 4 + 0.3 x 15 = 7.3 minutes, saving about 30 minutes per case on 408,000 non-fraud disputes, roughly 204,000 hours, or about 113 FTE at 1,800 hours [ESTIMATE]. Inference: about 12 model calls per case averaging 6,000 input and 600 output tokens, blended price USD 1.50 input and USD 6 output per million [ESTIMATE — VERIFY], gives about USD 0.15 per case, or USD 72,000 a year. The economics are dominated by labour, and the risk is dominated by the three severity-5 rows above. That is the conversation to have with the client.

## 5. Real-world case

[PUBLIC — Klarna, Feb 2024 and May 2025; VERIFY] Klarna's assistant is often cited as agentic; in its first phase it was a single assistant handling customer-service conversations with tools for refunds, returns and payment queries, operating at high autonomy. The 2025 walk-back, in which the CEO said quality had suffered and humans would be re-hired, is the clearest public illustration of an autonomy level set by cost ambition rather than by measured quality on the long tail. A gated design with exception routing would have preserved most of the savings.

[PUBLIC — Goldman Sachs statements, 2025; VERIFY] Goldman Sachs announced a pilot of Cognition's "Devin" coding agent alongside its own GS AI Assistant, positioning the agent as a supervised "new employee" working alongside engineers, with human review of its output. The framing, agents at an explicit assist-and-review level with humans accountable, is the register banks use publicly.

[PUBLIC — Commonwealth Bank of Australia, Aug 2025; VERIFY] CBA reversed 45 call-centre redundancies after the Finance Sector Union showed the AI voice bot had not reduced call volumes as assumed. The lesson

is that autonomy and benefit assumptions must be validated against measured volumes before the labour case is executed.

## 6. Consultant language

> *"We will specify the autonomy level per use case and earn each step up against published criteria."*

Use with the CRO and in regulatory engagement.

> *"Each sub-agent has a narrow scope, its own tool allow-list and least-privilege credentials. The supervisor can coordinate but cannot pay."*

Use in architecture and security reviews.

> *"The decision of record stays in the rules engine; the agents gather evidence, interpret and communicate."*

Use when positioning agents to risk and compliance.

> *"Our human-in-the-loop design is a set of gates with explicit triggers, not a review of everything. The thresholds are the control."*

Use when an operations head fears either loss of control or no savings.

> *"The failure-mode table is the design artefact model-risk will read first. We build it before we build the agent."*

Use to signal governance maturity to the second line.

## 7. Common mistakes juniors make

- Calling a fixed pipeline with LLM steps "agentic" and inheriting agentic governance scrutiny for no benefit.
- Designing one agent with twenty tools instead of several narrow agents with three each.
- Omitting the failure-mode table and discovering the severity-5 risks in the model-risk review.
- Treating human-in-the-loop as "a human checks everything", which destroys the business case, or "a human is notified", which is not a control.
- Not specifying what persists in memory, where, and for how long (a data-protection question).
- Ignoring prompt injection in any agent that reads customer-supplied text.
- Quoting per-step accuracy instead of end-to-end task success.
- Presenting MCP or A2A adoption as settled when it is still moving.

## 8. Interview questions with model answers

**Q1. Walk me through how you would design an agentic system for KYB refresh.**

I would start from the process, not the technology. KYB refresh decomposes into retrieving the current file, checking registries for ownership and status changes, resolving beneficial owners, running screening and adverse media, assessing whether the risk rating changes, and documenting the decision. The sub-tasks are heterogeneous and the next step depends on what is found, so a supervisor pattern fits: a supervisor plans and delegates to a registry agent, an ownership-resolution agent, a screening agent, an adverse-media agent and a documentation agent, each with a narrow tool set and read-only access except the documentation agent, which writes to the case file. The risk-rating decision itself stays deterministic in the existing policy engine. Autonomy: start at level 2, where the agents assemble the pack and propose the rating and a human

analyst approves; promote low-risk, no-change cases to level 3 after a measured period. Gates on any ownership change, any screening hit, any rating increase and all high-risk segments. I would specify guardrails per layer, a golden set of 300 historical refreshes for evaluation, and a failure-mode table before build. The benefit is analyst hours on the no-change majority; the governance burden concentrates on the exceptions, which is where it belongs.

**Q2. Supervisor versus orchestrator-worker: when and why?**

Supervisor when the sub-tasks differ in nature and the path depends on judgement about intermediate results: case work such as disputes, KYB and complaints. The supervisor carries the judgement and the sub-agents carry specialised scope and permissions, which is also attractive for least-privilege security. Orchestrator-worker when the work is homogeneous and parallelisable: screening a batch of names, summarising a set of documents, extracting the same fields from many forms. The orchestrator decomposes, fans out, and merges; there is little judgement between steps. The trade-offs: supervisors can become bottlenecks and make delegation errors, so you cap depth and log every delegation; orchestrator-worker systems fail at merge time, so you validate merged outputs against schemas. Most real systems nest one inside the other: a supervisor delegates a bulk sub-task to an orchestrator-worker component.

**Q3. How do you make an agentic system acceptable to a bank's model-risk function?**

By speaking their language and giving them their artefacts. First, classify what is a model and what is a control: the LLM is a third-party model under SS1/23 or SR 11-7 principles; the orchestration and guardrails are controls. Second, present the autonomy level and the promotion and demotion criteria, so they see a managed progression rather than a release. Third, deliver a failure-mode and effects analysis with severity scoring and detection mechanisms, which is a format they recognise. Fourth, show the evaluation evidence: a golden set, measured end-to-end task success, hallucination and faithfulness rates, and a monitoring plan with drift triggers. Fifth, confirm that the decision of record remains deterministic and that irreversible actions have human gates with full trajectory logging. Finally, offer independent validation access and a kill switch. In my experience the function's objection is rarely to the technology; it is to the absence of these artefacts.

## 9. Self-check quiz

1. List the four components of an agent and state, for each, the single most important design decision a consultant should specify.
2. An agent completes a 12-step task with 96% per-step reliability. What is the end-to-end success rate, and which two design responses would most improve it?
3. Describe the three HITL patterns and say which you would use for refunds above a threshold, and why.
4. Distinguish MCP from A2A in one sentence each, and name one governance implication of each for a bank.
5. In the disputes design in Section 4, why does the merchant-contact agent have a template-only messaging tool rather than free-text sending? Name the guardrail layer this belongs to.

# Module 4 — Data and Tech Stack: Readiness, Vector Stores, LLMOps, Evaluation and Observability

## 1. Why a client cares

The gap between a convincing demo and a production system in a bank is almost entirely in this module: whether the data can be reached and trusted, whether the system can be measured, and whether it can be run, monitored and changed safely. The CIO cares because this is where the integration budget goes; the CRO cares because “how do you know it works” has to be answered with evidence; the CFO cares because the run cost of evaluation, monitoring and human review is where business cases quietly break. A consultant who can specify an evaluation plan with numeric thresholds is rarer than one who can describe an architecture, and far more valuable.

## 2. Concept and mechanism

### 2.1 Data readiness for generative and agentic AI

Classical ML readiness asked whether labelled, structured history existed. Generative AI readiness asks a different set of questions, because the model brings general capability and the bank supplies context, documents and access.

| Dimension | What to assess | Typical bank finding [ILLUSTRATIVE — composite] |
|---|---|---|
| Availability | Can the documents and data the use case needs be reached programmatically? | Policies in SharePoint, procedures in PDFs, case notes in a legacy system without an API |
| Quality | Are documents current, de-duplicated, versioned, free of contradictions? | Multiple versions of the same procedure; superseded policies still live |
| Structure and parseability | Can PDFs, scans and tables be converted to usable text? | Scanned KYC documents; tables rendered as images |
| Metadata | Do documents carry type, date, jurisdiction, product, owner, classification? | Rarely; must be added during ingestion |
| Access control and lineage | Can retrieval respect entitlements? Can every output be traced to a source version? | Entitlements exist at folder level, not document level |
| Privacy and residency | What personal data is present; where may it be processed? | Customer PII in free-text notes; residency constraints on cloud regions |
| Labelled examples | Do we have historical inputs and correct outputs to build a golden set? | Outcomes exist; rationales rarely recorded |
| Freshness and change rate | How often does the corpus change and who owns updates? | Policy changes monthly; no owner for the AI corpus |

Readiness is scored per use case, not once for the bank, because a disputes assistant and a KYB agent depend on different corpora and systems. The readiness gaps become workstreams in the roadmap and cost lines in the TCO (Module 8).

### 2.2 Vector databases and index types

A vector database stores embeddings with metadata and supports approximate nearest-neighbour (ANN) search. Options fall into three groups [PUBLIC — vendor documentation, 2024 to 2025; VERIFY]: dedicated vector databases (Pinecone, Weaviate, Milvus, Qdrant, Chroma), vector capabilities inside general databases (PostgreSQL with pgvector, Elasticsearch and OpenSearch, MongoDB Atlas, Oracle 23ai, Azure AI Search) and

hyperscaler-managed services. For most banks the pragmatic choice is the vector feature of a database already approved and operated in-house, unless scale or latency demands a specialist.

Index types determine the recall-speed-memory trade-off:

- **Flat (brute force)**: exact search, linear in corpus size; fine below roughly a million vectors.
- **HNSW (Hierarchical Navigable Small World)**: graph-based ANN; high recall, fast queries, high memory; the default in most systems.
- **IVF (inverted file)**: clusters vectors, searches nearest clusters; lower memory, tunable recall.
- **Product quantisation (PQ) and scalar quantisation**: compress vectors to cut memory at some recall cost; combined with IVF or HNSW at scale.

Two design points matter more than the index: metadata filtering (jurisdiction, product, date, entitlement) applied before or alongside vector search, and hybrid search combining vector similarity with lexical (BM25) matching, which is essential for names, codes and identifiers.

## 2.3 LLMOps versus MLOps

MLOps manages the lifecycle of models the bank trains: data pipelines, feature stores, training, versioning, deployment, monitoring for drift and retraining. LLMOps manages the lifecycle of applications built on models the bank mostly does not train. The assets under management differ:

| Asset | MLOps | LLMOps |
|---|---|---|
| Model | Trained in-house; versioned weights | Third-party or open-weight; versioned by vendor; deprecations forced |
| Code | Training and inference code | Prompts, retrieval configuration, tool schemas, orchestration graphs |
| Data | Training sets, features | Corpus, chunks, embeddings, golden sets, trajectories |
| Evaluation | Accuracy, AUC, calibration on hold-out | Task success, faithfulness, hallucination rate, safety, cost, latency on golden sets |
| Monitoring | Data drift, concept drift, performance | Quality drift, prompt and model version changes, cost per task, guardrail triggers, user feedback |
| Change control | Retraining cadence | Prompt change, model upgrade, corpus update, each requiring re-evaluation |

The practical consequence: a prompt change is a release. Banks that treat prompts as configuration rather than code lose control of quality. The LLMOps pipeline should run the evaluation suite on every change to prompt, model version, retrieval configuration or corpus, and block release below thresholds.

## 2.4 Evaluation: the discipline that separates pilots from production

**Golden datasets.** A golden set is a curated collection of inputs with reference outputs or grading criteria, representative of the production distribution including edge cases and known failure types. Sources: historical cases with recorded outcomes, SME-authored examples, synthetic variations. Size: a few hundred items is usually enough to detect meaningful regressions; thousands are needed for fine-grained rates [ESTIMATE — practitioner experience]. The golden set is a governed asset: versioned, owned, refreshed as the process changes, and never used for prompt tuning without a held-out portion.

**Metric families.**

| Family | Metric | What it measures | How measured |
|---|---|---|---|
| Correctness | Exact or fuzzy match; field-level precision and recall; F1 | Did the output match the reference? | Deterministic comparison for structured outputs |

| Family | Metric | What it measures | How measured |
|---|---|---|---|
| Groundedness and faithfulness | Faithfulness score; unsupported-claim rate | Is every claim supported by the provided context? | Claim decomposition then LLM-as-judge or NLI model |
| Hallucination | Fabrication rate | Claims not supported by context or fact | Subset of faithfulness, often reported separately |
| Relevance | Answer relevance; context relevance | Does the answer address the question; was retrieved context on-topic? | LLM-as-judge; human rating |
| Retrieval | Recall at k; MRR; context precision | Did retrieval find the right passages? | Against labelled passage references |
| Task success | End-to-end completion rate | Did the agent achieve the goal? | Programmatic checks plus human review |
| Safety and compliance | Policy violation rate; PII leakage rate; injection success rate | Did guardrails hold? | Red-team sets; automated scanners |
| Operational | Latency p50 and p95; cost per task; steps per task | Is it affordable and responsive? | Telemetry |

**LLM-as-judge.** A separate model grades outputs against a rubric. Cheap and scalable, but biased: judges prefer longer answers, their own style, and the first option in pairwise comparisons; they can be inconsistent. Mitigations: calibrate the judge against a human-labelled subset and report agreement (Cohen's kappa or simple agreement rate), use rubric-based scoring rather than free-form, randomise order, and keep humans on a sampled slice permanently. Frameworks that implement these metrics include RAGAS, DeepEval, promptfoo, OpenAI Evals, LangSmith, Braintrust and the hyperscalers' evaluation services [PUBLIC — open-source and vendor documentation, 2024 to 2025; VERIFY].

**Eval harness design.** An eval harness is the automated pipeline that runs the golden set through the system, computes metrics, compares against thresholds and the previous version, and produces a report. Properties: reproducible (pinned model versions, temperature 0 where possible), fast enough to run on every change, segmented (metrics by case type, product, language), and integrated into release gating. Offline evaluation on the golden set is complemented by online evaluation: sampled production outputs graded by humans and judges, user feedback signals, and guardrail trigger rates.

## 2.5 Hyperscaler platforms

The three major platforms offer model access, hosting for open-weight models, retrieval services, agent frameworks, guardrails and evaluation tooling. Everything below is [PUBLIC — vendor documentation, 2025; VERIFY — capabilities and model availability change quarterly].

| Criterion | Microsoft Azure (Azure OpenAI, Azure AI Foundry) | AWS (Amazon Bedrock, SageMaker) | Google Cloud (Vertex AI) |
|---|---|---|---|
| Proprietary model access | OpenAI models; growing catalogue of others | Anthropic, Amazon Nova, Meta, Mistral, Cohere and others | Gemini; Anthropic and others via Model Garden |
| Open-weight hosting | Yes | Yes | Yes |
| Retrieval service | Azure AI Search | Bedrock Knowledge Bases, OpenSearch | Vertex AI Search, AlloyDB |
| Agent tooling | Agent Service, Semantic Kernel, AutoGen | Bedrock Agents, Strands | Agent Builder, ADK |
| Guardrails | Content Safety | Bedrock Guardrails | Model Armor, safety filters |
| Evaluation | AI Foundry evaluation | Bedrock evaluations | Vertex Gen AI evaluation |

| Criterion | Microsoft Azure (Azure OpenAI, Azure AI Foundry) | AWS (Amazon Bedrock, SageMaker) | Google Cloud (Vertex AI) |
|---|---|---|---|
| Data residency | Regional deployments; EU Data Boundary | Regional; sovereign options | Regional; sovereign controls |
| Typical bank fit | Microsoft 365 estates; OpenAI preference | AWS-first estates; multi-model preference | Data and analytics estates on BigQuery; Gemini preference |

Neutral selection criteria: existing cloud and data-platform estate, model choice and multi-model flexibility, data residency and contractual terms (zero retention, no training on customer data), identity and network integration, pricing including committed-use discounts, exit cost, and regulatory relationships (several banks have public partnerships; see Section 5). Avoid single-platform lock-in at the model layer by using an abstraction layer or gateway.

### 2.6 Observability and tracing

Observability for LLM applications means capturing, for every request: the full trajectory (prompts, retrieved chunks, tool calls and results, model responses, guardrail decisions), token counts and cost, latency per step, model and prompt versions, and user feedback. Tools include LangSmith, Langfuse, Arize Phoenix, Datadog LLM Observability, Weights and Biases Weave and the hyperscaler offerings [PUBLIC — vendor documentation, 2025; VERIFY]. OpenTelemetry conventions for generative AI are emerging as a standard for traces.

For a bank, tracing is also the audit trail: regulators and internal audit will ask to see why the system produced a specific output on a specific date. Retention, access control and PII handling for traces are design decisions, since traces contain customer data.

### 2.7 Gateway and platform layer

Mature programmes introduce an AI gateway: a single service through which all model calls pass, providing authentication, model routing, rate limiting, cost attribution by use case, logging, PII redaction, caching and policy enforcement. It is the control point that makes a portfolio of use cases governable and is where the CoE (Module 13) earns its keep.

## 3. Frameworks

### 3.1 Data readiness scorecard

Score each dimension in 2.1 from 1 (not ready) to 4 (production-ready) per use case; any dimension at 1 is a blocker, dimensions at 2 become enabler workstreams with owners and cost. Aggregate only to communicate; decide on the detail. Attribution: generic practitioner tool.

### 3.2 Evaluation pyramid

Attribution uncertain; widely used in practice. Layer 1, deterministic checks (schema, format, forbidden content, exact-match fields), run on everything, cost nothing. Layer 2, model-graded checks (faithfulness, relevance, rubric scores), run on everything offline and on a sample online. Layer 3, human expert review, run on a calibrated sample and all exceptions. The pyramid's shape is the cost model: broad and cheap at the base, narrow and expensive at the top.

### 3.3 Release gating with thresholds

Define for each metric a release threshold and a regression tolerance versus the previous version. Block release automatically on breach. Thresholds come from the risk appetite for the use case (Module 12), not from what the current system achieves.

# 4. Worked BFSI example

[ILLUSTRATIVE — composite] Evaluation plan for a sanctions-screening alert-narrative generator at a European bank. The system reads the alert (hit details, customer data, transaction context, analyst checklist) and drafts the disposition narrative for the analyst to review and sign. Volume: 220,000 level-one alerts a year, about 96% dispositioned as false positives.

## 4.1 Golden set

- 600 historical alerts with analyst-written narratives and recorded dispositions, stratified: 400 false positives across list types (sanctions, PEP, adverse media) and match types (name, alias, vessel, entity), 150 escalations, 50 true matches. Sampled from the last 18 months across all booking locations.
- 100 SME-authored edge cases: transliteration variants, partial matches on common names, hits on secondary parties, vessels and aircraft, stale list entries.
- 60 adversarial items: alerts whose free-text fields contain instruction-like content (prompt injection), PII that must not appear in the narrative, and contradictory data.
- Held-out 30% never used for prompt iteration.

## 4.2 Metrics and thresholds

| Metric | Definition | Release threshold | Regression tolerance | Method |
|---|---|---|---|---|
| Field correctness | Precision and recall of the six mandatory narrative elements (hit type, matched attributes, discriminating evidence, screening result, rationale, recommended disposition) | Precision 0.98, recall 0.97 | minus 0.01 | Deterministic extraction then comparison |
| Faithfulness | Share of narratives with zero claims unsupported by alert data | 0.97 | minus 0.01 | Claim decomposition, LLM-as-judge calibrated to human kappa 0.8 or higher |
| Unsupported-claim rate | Unsupported claims per 100 narratives | 2.0 or lower | plus 0.5 | Same |
| Disposition agreement | Agreement between recommended and recorded disposition on false-positive items | 0.95 | minus 0.02 | Deterministic |
| True-match sensitivity | Share of the 50 true matches where the narrative recommends escalation | 1.00 (zero misses) | none | Deterministic; any miss blocks release |
| PII leakage | Narratives containing PII beyond the permitted fields | 0 per 1,000 | none | Scanner plus human check |
| Injection resistance | Adversarial items where output follows injected instruction | 0 of 60 | none | Programmatic |
| Analyst edit distance | Median proportion of narrative text changed by analyst before sign- | 0.15 or lower by month three | plus 0.05 | Telemetry |

| Metric | Definition | Release threshold | Regression tolerance | Method |
|---|---|---|---|---|
| | off (online) | | | |
| Latency p95 | Time from alert open to draft available | 8 seconds | plus 2 | Telemetry |
| Cost per narrative | Blended inference cost | USD 0.04 [ESTIMATE — VERIFY] | plus 20% | Telemetry |

### 4.3 Online evaluation and HITL

Every narrative is reviewed by an analyst (level 1 autonomy, “assist”). Additionally, 3% of signed narratives are sampled weekly for second-line QA with a rubric; LLM-as-judge faithfulness is computed on 100% of production output and alerts above the threshold are routed to QA. Analyst edits are captured as feedback and fed into the golden set quarterly after SME review.

### 4.4 Interpreting the thresholds for the client

The zero-miss threshold on true matches is a risk-appetite statement, not a statistical one: a narrative that talks a true match down to a false positive is the failure the regulator will ask about. The faithfulness threshold at 0.97 accepts that roughly one narrative in 33 will contain an unsupported claim before analyst review, which is why the system is at assist level and not above. The business value, analyst time saved per alert, is estimated at 6 minutes of a 22-minute average [ESTIMATE], or roughly 22,000 hours a year, about 12 FTE. The evaluation infrastructure and QA sample cost perhaps 1.5 FTE and USD 120,000 of tooling and engineering a year [ESTIMATE — assumption]. The client should see that ratio explicitly.

So what for the client: “The eval plan is the licence to operate. Without the zero-miss test and the faithfulness threshold, the second line will not sign, and without the online edit-distance metric, you cannot prove the benefit.”

## 5. Real-world case

[PUBLIC — Morgan Stanley and OpenAI, Sept 2023 and later coverage; VERIFY] Morgan Stanley’s adviser assistant is publicly described as having been released only after an evaluation process in which advisers and experts graded answers, with the firm reporting continuous evaluation as part of its governance; the firm’s head of analytics and data described building an evaluation framework before scaling. It is the most-cited example of evaluation discipline in financial services.

[PUBLIC — HSBC and Google Cloud, June 2023; VERIFY] HSBC’s AML AI deployment required the bank to demonstrate model performance, explainability and monitoring to regulators across jurisdictions. The reported 60% reduction in alerts and two to four times increase in identified suspicious activity are, in evaluation terms, a precision improvement at maintained or improved recall. The case shows that evaluation evidence is what unlocks regulatory acceptance of a replacement for rules.

[PUBLIC — Deutsche Bank and Google Cloud, Dec 2020; Lloyds Banking Group and Google Cloud, 2024 to 2025; VERIFY] Several large banks have announced strategic hyperscaler partnerships that include AI platforms, illustrating that platform choice in banking is often made at the enterprise level before any use case is selected. A consultant should establish that constraint in week one.

## 6. Consultant language

> *“The golden set is a governed asset. It has an owner, a version and a held-out portion, and every release runs against it.”*

Use when establishing the evaluation workstream.

> *"We measure faithfulness separately from correctness. A correct answer from the wrong source is a control failure."*

Use with model-risk and compliance.

> *"Every prompt change is a release. The eval harness gates it."*

Use with the technology team and the CoE.

> *"Data readiness is per use case, not per bank. The gaps become enabler workstreams with owners and costs."*

Use in roadmap discussions.

> *"Traces are the audit trail. We design their retention and access with the data-protection office, not after the fact."*

Use with the DPO and internal audit.

## 7. Common mistakes juniors make

- Presenting a single accuracy number without stating the denominator, the reference and the method.
- Building the golden set from the same cases used to tune the prompt.
- Using LLM-as-judge without calibrating it against human labels.
- Treating the vector database choice as the architecture decision when chunking, metadata and hybrid search matter more.
- Assuming the bank's data is ready because it exists.
- Forgetting that traces contain customer data.
- Setting thresholds at whatever the pilot achieved rather than at what the risk appetite requires.
- Comparing hyperscalers on model benchmarks rather than on estate fit, residency, contract terms and exit cost.

## 8. Interview questions with model answers

**Q1. How would you evaluate a GenAI system before putting it in front of customers?**

I would build a layered evaluation. First, a golden set of several hundred representative inputs with reference outputs or grading rubrics, stratified by customer segment, intent and risk, plus adversarial items for prompt injection, PII and out-of-scope requests, with a held-out portion. Second, metrics in families: correctness, faithfulness to the knowledge base, hallucination rate, relevance, safety and policy violations, and operational metrics such as latency and cost. Third, a harness that runs on every change to prompt, model, retrieval or corpus, with release thresholds set by risk appetite, including zero tolerance on specific failures like misstating fees or regulatory terms. Fourth, online evaluation: sampled human review, LLM-as-judge on all output, user feedback, and guardrail trigger rates, with drift alerts. Finally, I would specify the autonomy level and the human-review pattern that the measured error rate justifies. The customer-facing decision is made on evidence from this process, not on a demo.

**Q2. What is the difference between MLOps and LLMOps and why does it matter to a bank?**

MLOps manages models the bank trains: data, features, training, validation, deployment, drift monitoring, retraining. LLMOps manages applications built on models the bank does not train: prompts, retrieval configuration, tool schemas, orchestration, golden sets and the model version supplied by a vendor. It matters for three reasons. Change control: in LLMOps a prompt edit or a vendor model upgrade can change behaviour as much as retraining would, so each must be gated by evaluation. Dependency: vendors

deprecate models on their schedule, so the bank needs a re-evaluation and migration process it does not control the timing of. Governance: model-risk frameworks such as SS1/23 were written for trained models; banks must decide how a third-party foundation model, its prompts and its guardrails map onto model inventory, validation and monitoring. The operating consequence is a platform team that owns the gateway, the eval harness and the release process across use cases.

**Q3. A client asks whether to pick Azure, AWS or Google for GenAI. How do you answer?**

I would reframe from “which is best” to “which fits”. The decision criteria are the existing cloud and data estate, because integration cost dominates; model choice and whether the client wants multi-model flexibility or a single-vendor relationship; data residency and contractual terms, specifically zero retention and no training on the bank’s data; identity, network and security integration; pricing with committed-use discounts; exit cost; and the regulator’s familiarity with the platform in that jurisdiction. In practice banks with Microsoft 365 estates lean to Azure, AWS-first banks to Bedrock for its multi-model catalogue, and banks with Google data platforms to Vertex. I would recommend an AI gateway and abstraction layer so the model choice is not locked, and I would flag that platform capabilities change quarterly, so any comparison is dated the day it is written. The client’s enterprise cloud strategy usually settles this before the AI programme starts, and the consultant’s job is to confirm that constraint early rather than run a beauty parade.

## 9. Self-check quiz

1. List five data-readiness dimensions and give, for each, one typical gap found in a bank.
2. Explain the difference between HNSW and IVF indexes and say why hybrid search matters for sanctions and KYC content.
3. Name three biases of LLM-as-judge and the mitigation for each.
4. In the sanctions eval plan, why is the true-match sensitivity threshold set at 1.00 with no regression tolerance, while faithfulness is set at 0.97? Explain in risk-appetite terms.
5. A proposed GenAI system will cost USD 0.04 per narrative at 220,000 narratives a year, plus 1.5 FTE for QA at GBP 70,000 loaded and USD 120,000 of tooling. Compute the annual run cost in USD (assume GBP 1 = USD 1.27 [ESTIMATE — VERIFY]) and compare it with the 12 FTE benefit at GBP 65,000 loaded. What is the ratio?

# Module 05 — AI Strategy and Maturity

## 1. Why a client cares

A bank CEO does not buy "AI"; they buy a defensible answer to three board questions: where are we versus peers, what are we trying to become, and what do we have to build before any of this pays. Maturity assessment answers the first, ambition setting answers the second, and the capability frameworks in this module answer the third. Get these wrong and the use-case portfolio you build in Modules 6 and 7 has no anchor: you end up with a long list of pilots and no strategy, which is exactly the condition that the pilot-failure statistics describe [PUBLIC — Gartner, Jul 2024; MIT NANDA, Aug 2025, VERIFY].

## 2. Concept and mechanism

### 2.1 What an AI strategy actually is

An AI strategy is a resource-allocation argument. It states which business outcomes the institution will pursue with AI, which domains (end-to-end business areas such as corporate onboarding or collections) it will rewire first, which enabling capabilities it must build in what sequence, and how value will be measured and released into the P&L. Everything else (model choices, vendors, platforms) is a consequence of those decisions, not the strategy itself.

Three structural features distinguish a credible AI strategy from a technology plan:

- **It is domain-led, not use-case-led.** Isolated use cases produce isolated pilots. A domain is large enough to carry a business case, bounded enough to be rewired in 12 to 24 months, and owned by a single accountable executive. McKinsey's "Rewired" (Lamarre, Smaje and Zemmel, 2023) is the clearest published articulation of domain-based transformation [PUBLIC — Wiley, 2023].
- **It sequences enablers against value.** Data products, platform, model risk governance and talent are built to the extent and in the order that the prioritised domains require them, not as a generic "foundation" programme that consumes the first 18 months.
- **It carries a value-release mechanism.** Benefits are tied to budget lines and owners, with leading indicators (adoption, cycle time) and lagging indicators (cost, revenue, loss rates). Module 14 covers the KPI architecture; here the point is that the strategy document must name it.

### 2.2 Maturity: what is being measured

AI maturity is the institution's demonstrated ability to generate value from AI repeatedly, not its count of models or pilots. Most published maturity models converge on five or six dimensions:

| Dimension | What "low" looks like | What "high" looks like |
|---|---|---|
| Strategy and leadership | AI is an IT or innovation-lab topic | CEO-sponsored ambition, domains chosen, value targets in the plan |
| Data | Siloed, poorly governed, no ownership | Reusable data products with owners, lineage, quality SLAs |
| Technology and platform | Point solutions, no shared platform | Shared ML/LLM platform, model registry, reusable components |
| Talent and operating model | Scarce skills, vendor-dependent | Product-and-platform teams, CoE plus embedded squads, clear roles |
| Governance and risk | Ad hoc approvals, no model inventory | Tiered model risk framework covering GenAI, auditable lifecycle |
| Adoption and value tracking | Pilots, anecdotal benefits | Scaled deployments, benefits tracked to P&L, change management funded |

Note that these six dimensions map almost one-to-one onto the “Rewired” six capabilities (section 3.1), which is why that framework doubles as a maturity lens in practice.

### 2.3 Ambition: the choice that drives everything downstream

Ambition is set on two axes: breadth (how much of the institution) and depth (how radically processes change). A useful three-level scale that most practices use in some form:

- **Defend.** Use AI to protect the current franchise: cost efficiency, fraud and financial-crime effectiveness, regulatory compliance. Typical value: 5 to 15 percent of addressable operations cost in the chosen domains. Low disruption to the operating model.
- **Extend.** Use AI to grow the current franchise: relationship manager productivity, personalised pricing and offers, faster credit decisions that win share. Requires new data products and some process redesign.
- **Transform.** Redesign how a domain works around AI, including agentic process execution, and in some cases new propositions (embedded finance, AI-native servicing). This is the “Invent” end of BCG’s framing (section 3.3) and the “Reshape” middle. Requires operating-model change and typically 24 to 36 months.

Ambition must be consistent with maturity. A bank at maturity level 2 declaring a “Transform” ambition across the enterprise is the most common strategic error we see [ILLUSTRATIVE — composite]. The right move is a “Transform” ambition in one or two domains, “Defend” elsewhere, and a maturity-building plan that lets the ambition widen in year two.

### 2.4 The economics that justify the ambition

Any ambition statement should be anchored to an external value estimate and then haircut. McKinsey Global Institute estimated generative AI alone could add USD 200 to 340 billion a year to global banking, roughly 2.8 to 4.7 percent of industry revenues, concentrated in corporate and retail banking operations, software engineering, marketing and risk [PUBLIC — MGI, “The economic potential of generative AI”, Jun 2023]. The consultant’s job is to translate that top-down figure into a bottom-up estimate for one institution: the top-down number tells the board the prize is real; the bottom-up number, built in Modules 7 to 9, tells them what is bankable.

## 3. Frameworks

### 3.1 McKinsey “Rewired”: six capabilities

Source: Eric Lamarre, Kate Smaje and Rodney Zemmel, *Rewired: The McKinsey Guide to Outcompeting in the Age of Digital and AI* (Wiley, 2023) [PUBLIC — 2023]. The book argues that digital and AI transformations fail on capabilities, not ideas, and names six:

1. **Business-led digital roadmap.** Leadership alignment on a small number of domains with quantified value; a roadmap owned by business executives, not the CIO.
2. **Talent.** Build an in-house bench (engineers, data scientists, product owners, designers) rather than relying on vendors; the book argues for a specific talent-win-room approach and a technical-career track.
3. **Operating model.** Product-and-platform teams; agile at scale; business and technology co-located with shared OKRs.
4. **Technology.** Distributed, modular architecture; APIs; cloud; engineering practices (CI/CD, automated testing) that let hundreds of teams ship independently.
5. **Data.** Data products with owners, built to serve the prioritised domains; data governance embedded in delivery, not separate.

6. **Adoption and scaling.** Change management, user adoption, benefits tracking, and the "scaling muscle" to replicate a domain solution across geographies and business units.

How to apply it step by step:

1. Score the client on each capability (1 to 5) using evidence, not interviews alone: count of product teams, cycle time from idea to production, proportion of data with named owners, proportion of pilots that reached production.
2. Identify which capabilities the prioritised domains actually depend on. A KYB rewiring depends heavily on data (entity data, document stores) and governance; a relationship-manager copilot depends more on adoption and talent.
3. Set a target score per capability at the 24-month horizon that is sufficient for the domains in scope, not a generic "all 4s".
4. Turn gaps into enabler workstreams with budgets and owners, sequenced ahead of the domain waves that need them.

## 3.2 BCG 10-20-70

Source: Boston Consulting Group, popularised in its 2023 to 2024 AI publications and client work [PUBLIC — BCG, 2023–24]. The rule states that roughly 10 percent of the effort and value in an AI programme sits in algorithms and models, 20 percent in technology and data, and 70 percent in people, process and business change.

How to use it:

- **As a budget sanity check.** If a client's AI budget is 60 percent model and platform licences and 10 percent change, the programme is shaped to fail. In our experience the 70 percent rarely appears as a budget line at all; it is hidden in business-unit time that nobody has released [ILLUSTRATIVE — composite].
- **As a diagnostic lens.** Ask where the last three pilots stalled. The answer is almost always in the 70: process owners who did not redesign the workflow, controls functions that were not engaged, front-line staff who were not trained.
- **As a narrative device with CFOs.** It reframes the business case from "cost of technology" to "cost of change", which is where the CFO's scepticism actually lives.

The rule is a heuristic, not a measurement. Do not present it as empirical.

## 3.3 BCG Deploy, Reshape, Invent

Source: BCG's generative AI value framing, 2023 to 2024 [PUBLIC — BCG, 2023–24; wording VERIFY as BCG materials vary]. The three plays:

- **Deploy.** Adopt off-the-shelf GenAI tools (copilots, enterprise assistants) for broad productivity gains. Fast, low risk, modest and diffuse value, hard to capture in the P&L.
- **Reshape.** Redesign critical end-to-end functions and processes around AI, for example customer service, underwriting or software development. This is where the measurable value sits for banks.
- **Invent.** Create new products, services or business models that AI makes possible. Highest upside, highest uncertainty.

How to apply: allocate the portfolio of initiatives across the three plays and check the shape. A portfolio that is 90 percent Deploy will show adoption statistics and no P&L. A portfolio that is 50 percent Invent at a bank with maturity level 2 is a venture fund, not a strategy. A common target shape for a mid-sized bank in year one is roughly 30 Deploy, 60 Reshape, 10 Invent by investment [ESTIMATE — assumption based on practice; not a BCG figure].

### 3.4 Maturity models and their attribution

- **Gartner AI Maturity Model.** Five levels, commonly described as Awareness, Active, Operational, Systemic, Transformational [PUBLIC — Gartner, various years; VERIFY level names against the current Gartner publication, as wording has changed over time].
- **MIT Sloan Management Review and BCG AI research.** Annual research series (2017 onward) segmenting companies into Pioneers, Investigators, Experimenters and Passives based on AI understanding and adoption; later reports on organisational learning with AI [PUBLIC — MIT SMR/BCG, 2017–2022]. Useful for benchmarking language; not a scoring tool.
- **Deloitte, Accenture, IBM and others** publish their own AI maturity indices. Attribution for specific level descriptors in client decks is frequently wrong; if you cannot name the source document, present the dimensions as your own practitioner framework and say so.

### 3.5 Structure of an AI strategy document

A board-grade AI strategy has eight parts, and the order matters because each part constrains the next:

1. **Context and ambition.** Where the market is, where the bank is, and the chosen ambition level with a headline value target.
2. **Domain selection.** Three to six domains with a one-page rationale each: value pool, feasibility, strategic fit, regulatory exposure.
3. **Use-case portfolio.** Prioritised and sequenced use cases within each domain (Module 7 output), with wave allocation.
4. **Enablers.** Data, platform, model risk governance, talent and operating model, each with a current-state score, target-state score and investment.
5. **Roadmap.** Waves with milestones, dependencies and decision gates.
6. **Governance and responsible AI.** Decision rights, model risk tiers, regulatory mapping (Module 12).
7. **Economics.** Investment profile, benefits ramp, NPV and payback, sensitivities (Module 9).
8. **Value tracking and change.** KPI tree, benefits ownership, adoption plan (Module 14).

## 4. Worked BFSI example

### 4.1 Setting

[ILLUSTRATIVE — composite] “Northgate Bank”: a mid-sized European universal bank with EUR 45 billion of assets, 1.8 million retail customers, 60,000 SME and corporate clients, 6,500 FTE, cost-to-income ratio of 62 percent, and an operations and technology cost base of EUR 520 million a year. The CEO has asked for an AI strategy ahead of the three-year plan. Twenty-two AI pilots have run in the past 30 months; four are in production, none has a tracked P&L benefit.

### 4.2 Maturity assessment

Scoring is 1 (nascent) to 5 (leading), evidence-based. The rubric is a practitioner composite built on the six “Rewired” capabilities [PUBLIC — Lamarre, Smaje, Zemmel, 2023, for the dimensions; scores ILLUSTRATIVE].

| Dimension | Score | Evidence gathered | Target (24 months) | Gap |
|---|---|---|---|---|
| Business-led roadmap | 2 | AI owned by Innovation Lab; no domain owners; no value targets | 4 | 2 |
| Talent | 2 | 14 data scientists, 3 ML engineers, 0 LLM engineers; 80 percent of pilot build by | 3 | 1 |

| Dimension | Score | Evidence gathered | Target (24 months) | Gap |
|---|---|---|---|---|
| | | vendors | | |
| Operating model | 2 | Project-based funding; no persistent product teams for AI; 9-month idea-to-pilot cycle | 4 | 2 |
| Technology | 3 | Cloud landing zone live; one approved LLM gateway; no model registry; no eval harness | 4 | 1 |
| Data | 2 | Customer and entity data in 4 systems; no data product owners; KYB documents unstructured in 2 repositories | 4 | 2 |
| Adoption and scaling | 1 | 4 of 22 pilots in production; no benefits tracking; no change budget | 3 | 2 |
| **Average** | **2.0** | | **3.7** | **1.7** |

Interpretation for the CEO: the bank is a “Level 2, Active” institution in Gartner terms [PUBLIC — Gartner, VERIFY], with a technology platform slightly ahead of its organisation. The binding constraints are operating model, data and adoption, not technology. That finding changes the investment mix: more money into data products and change, less into additional platform licences.

## 4.3 Ambition setting

| Ambition option | Scope | Indicative value (annual, run-rate, year 3) | Investment (3 years) | Maturity required |
|---|---|---|---|---|
| Defend | Operations efficiency and financial crime across the bank | EUR 35 to 50 million | EUR 40 to 55 million | 3.0 |
| Extend | Defend plus RM and adviser productivity, personalised pricing | EUR 60 to 85 million | EUR 70 to 95 million | 3.5 |
| Transform (two domains) | Extend plus agentic rewiring of corporate onboarding and disputes | EUR 85 to 120 million | EUR 100 to 140 million | 3.7 in those domains |

[ESTIMATE — assumptions] Value is built bottom-up from the cost base: operations and technology cost EUR 520 million, of which roughly EUR 310 million is people cost in operations, servicing, financial crime and credit operations. “Defend” assumes 10 to 15 percent efficiency on that EUR 310 million (EUR 31 to 47 million) plus EUR 4 to 6 million of fraud and financial-crime loss reduction. “Transform” adds EUR 25 to 35 million from faster corporate onboarding (revenue pulled forward and abandonment reduced) and dispute cost per case falling by roughly 40 percent.

Recommendation: “Transform” in corporate onboarding (KYB) and disputes, “Defend” elsewhere, with a maturity-building plan that moves data and operating model from 2 to 4 in 18 months. The so-what for the client: the ambition is not constrained by the technology they already bought; it is constrained by the

organisation they have not yet built, and the investment case should be weighted accordingly, close to the 10-20-70 shape.

### 4.4 Investment shape check against 10-20-70

| Category | Client's draft budget | Recommended | Comment |
|---|---|---|---|
| Models and algorithms (licences, inference, model build) | 35 percent | 12 percent | Over-weighted; tiering and caching cut inference cost (Module 8) |
| Technology and data (platform, integration, data products) | 45 percent | 28 percent | Data products are the real need, not more platform |
| People, process and change (redesign, training, adoption, controls engagement) | 20 percent | 60 percent | Largest under-investment; where pilots have been dying |

[ILLUSTRATIVE — composite; the 10-20-70 reference is PUBLIC — BCG, 2023–24]

## 5. Real-world case

**DBS Bank, Singapore** [PUBLIC — DBS annual reports and press, 2023–2025; figures VERIFY]. DBS is the most frequently cited example of a bank that treated AI as an enterprise capability rather than a set of pilots. Publicly reported features of its approach:

- A decade-long digital transformation preceding GenAI, including a shift to a platform operating model and heavy in-house engineering, which is the "Rewired" argument in practice.
- Roughly 800 AI and machine learning models across about 350 use cases, with economic value from AI reported to be heading above SGD 1 billion in 2025 [PUBLIC — DBS, 2024–25, VERIFY].
- Governance built around a "PURE" framework (Purposeful, Unsurprising, Respectful, Explainable) for responsible data use, pre-dating GenAI.
- In February 2025, DBS said it expected around 4,000 temporary and contract roles to be reduced over three years as AI took on more work, while continuing to hire in AI-related roles [PUBLIC — Reuters and others, Feb 2025, VERIFY].

What the case teaches: the economic value DBS reports followed years of investment in data, platform and operating model, consistent with the 70 percent of 10-20-70. The 2025 workforce announcement also shows the reputational dimension of ambition: a "Transform" ambition is a workforce strategy as much as a technology strategy, and the communications plan belongs in the strategy document.

Contrast case: **Klarna** [PUBLIC — Klarna press release, Feb 2024; CEO interviews, May 2025, VERIFY]. Klarna's 2024 announcement that its AI assistant did the work of about 700 agents was followed in 2025 by the CEO acknowledging that cost had been over-prioritised relative to service quality and that human agents would be re-hired. The lesson for ambition setting: a "Transform" ambition without a quality floor in the KPI set over-reaches, and the walk-back is public.

## 6. Consultant language

> *"We should set ambition by domain, not for the enterprise. Transform where the value pool and your data allow it, defend everywhere else, and widen the ambition as the capabilities mature."*

Use when a CEO wants a single enterprise-wide ambition statement and you need to make it executable.

> *"Your binding constraint is not the platform you have bought; it is the operating model and data ownership you have not built yet. That is where the next euro should go."*

Use after a maturity assessment shows technology ahead of organisation.

> *"Twenty-two pilots and four in production is not a technology problem; it is a scaling-capability problem. The strategy has to fund the scaling muscle explicitly."*

Use when presenting the pilot-to-production diagnostic.

> *"I would anchor the board on the external value estimate, then show them the bottom-up bankable number, and be clear about the gap between the two."*

Use when a board paper quotes McKinsey or BCG headline figures without a bottom-up build.

> *"The budget shape tells me how this programme will end. If change and process redesign are 20 percent of spend, the pilots will stall for the same reasons the last ones did."*

Use when reviewing a draft AI investment case.

## 7. Common mistakes juniors make

- **Confusing a use-case list with a strategy.** Thirty use cases ranked by a 2x2 is an output of Module 7, not an AI strategy. The strategy is the ambition, the domains, the enablers and the value mechanism.
- **Scoring maturity from interviews alone.** Executives rate their own function 3.5 out of 5 almost regardless of reality. Score on evidence: production rates, cycle times, data ownership coverage, proportion of spend on change.
- **Misattributing frameworks.** Presenting "Deploy, Reshape, Invent" as McKinsey, or the six capabilities as BCG, is a credibility-ending error in an MBB interview. If unsure, say "widely used in practice, attribution uncertain".
- **Letting technology maturity set the ambition.** A bank with a strong cloud platform and weak operating model will still fail at "Transform" if the ambition is set off the platform.
- **Treating 10-20-70 as data.** It is a heuristic for shaping budgets and conversations. Quoting it as an empirical finding invites the one question you cannot answer.
- **Ignoring the workforce narrative.** Any ambition above "Defend" has headcount implications. If the strategy document is silent on this, the unions, the regulator and the press will fill the silence.

## 8. Interview questions with model answers

**Q1. A mid-sized bank CEO says: "We want to be an AI-first bank in three years." How do you respond?**

I would first unpack what "AI-first" means in terms of outcomes, because the phrase can hide three very different ambitions: defending the cost base, extending the franchise, or transforming how domains run. Then I would test the ambition against maturity. I would run a rapid six-dimension assessment, built on the six capabilities in McKinsey's "Rewired" [PUBLIC — 2023]: business-led roadmap, talent, operating model, technology, data, and adoption and scaling. For a typical mid-sized bank I would expect scores around 2 out of 5 on operating model, data and adoption, which means an enterprise-wide "Transform" ambition in three years is not credible. My recommendation would be a "Transform" ambition in two or three domains where the value pool and data readiness justify it, for instance corporate onboarding and disputes, with a "Defend" ambition elsewhere and an explicit enabler plan to lift the weak dimensions. I would anchor the prize on the external estimate, for example MGI's USD 200 to 340 billion a year for banking [PUBLIC — Jun 2023], and then present a bottom-up bankable number, typically a fraction of the pro-rata top-down figure. Finally, I would shape the investment close to BCG's 10-20-70 heuristic [PUBLIC — BCG, 2023–24] so that change and process redesign are funded, because that is where previous pilots will have failed.

**Q2. How do you distinguish an AI strategy from an AI roadmap, and why does the distinction matter to a board?**

A roadmap is a sequence of initiatives with dates. A strategy is the set of choices that make that sequence defensible: which outcomes, which domains, which ambition level, which enablers in what order, and how value gets released and tracked. The distinction matters because boards approve capital against choices, not against lists. If I present a roadmap without the choices, the board cannot evaluate trade-offs; if I present the choices, the roadmap follows almost mechanically. In practice I structure the strategy document in eight parts, ambition, domains, use-case portfolio, enablers, roadmap, governance, economics, value tracking, and the roadmap is one part of eight. A second reason the distinction matters is accountability: a roadmap is owned by a programme office; a strategy assigns value to domain owners, which is what makes benefits bankable.

**Q3. What is BCG's 10-20-70 and how would you use it in a client conversation?**

It is BCG's heuristic that roughly 10 percent of the effort and value in an AI programme is in algorithms, 20 percent in technology and data, and 70 percent in people, process and business change [PUBLIC — BCG, 2023–24]. I use it in three ways. First, as a budget diagnostic: if a client's AI spend is dominated by licences and platform, the shape predicts stalled pilots. Second, as a post-mortem lens: when asking why previous pilots did not scale, the answer almost always sits in the 70, in process redesign and controls engagement that never happened. Third, as a reframing device with the CFO: it moves the conversation from "what does the technology cost" to "what does the change cost", which is the more honest question. The caveat I always add is that it is a heuristic, not a measurement, and I would never present it as empirical evidence.

## 9. Self-check quiz

1. Name the six capabilities in McKinsey's "Rewired" and state which two a corporate onboarding (KYB) rewiring depends on most, with one sentence of rationale each.
2. A bank's AI budget is 50 percent model and platform licences, 35 percent integration and data, 15 percent change and training. Using the 10-20-70 heuristic, what is the diagnosis and what is the single most important reallocation?
3. Distinguish BCG's "Deploy", "Reshape" and "Invent" plays and give one BFSI example of each.
4. Northgate Bank's maturity average is 2.0 with technology at 3 and adoption at 1. A vendor proposes a EUR 12 million enterprise GenAI platform upgrade. Write the two-sentence response you would give the CEO.
5. Calculation: a bank has EUR 400 million of operations people cost. A "Defend" ambition assumes 8 to 12 percent efficiency; a "Transform" ambition in two domains covering EUR 90 million of that cost assumes 35 percent efficiency in those domains plus 8 percent elsewhere. Compute the annual run-rate value range for each ambition.

# Module 06 — Use-Case Discovery

## 1. Why a client cares

The quality of an AI portfolio is fixed at discovery. If the long list is built from vendor demos and executive anecdotes, prioritisation in Module 7 simply ranks the wrong things well. Clients care because discovery done properly produces something they almost never have: a task-level map of where effort, error and delay actually sit in a domain, which is reusable for operations improvement regardless of whether AI is the answer. It also produces the baseline that the business case will later need (Module 9), which is why discovery and baselining should be designed together.

## 2. Concept and mechanism

### 2.1 Discovery is decomposition, not brainstorming

Generative and agentic AI act on tasks, not on departments. A department-level statement such as "AI for KYC" is unscoreable: it conflates tasks with radically different automation potential (document classification is highly automatable; MLRO approval of a high-risk client is not). Discovery therefore decomposes a domain to the level at which a single task has a definable input, output, decision logic and error profile, and only then asks what AI pattern fits.

Four levels are standard in process architecture practice (the L1 to L4 convention is from process-classification frameworks such as APQC's Process Classification Framework; the application to AI discovery is practitioner convention) [PUBLIC — APQC PCF, ongoing; application ILLUSTRATIVE]:

| Level | Name | Example in corporate onboarding | Typical count per domain |
|---|---|---|---|
| L1 | Domain or value chain | Corporate client onboarding (KYB) | 1 |
| L2 | Stage or sub-process | Document collection; entity verification; screening | 6 to 10 |
| L3 | Task or activity | Build ownership structure and identify UBOs | 20 to 40 |
| L4 | Step | Open registry extract; map shareholder percentages; flag layers above 25 percent | 100 to 300 |

AI fit is assessed at L3, and sometimes L4 when an L3 task mixes automatable and judgement-heavy steps. Business cases are built at L2 or L1, because benefits only materialise when a whole stage changes.

### 2.2 The value-chain walk

A value-chain walk is a structured traverse of the domain from trigger to outcome, with the people who do the work, capturing for each L3 task:

- **Input and output artefacts** (documents, system records, decisions).
- **Volume and frequency** (cases per month, items per case).
- **Handle time** (touch time, not elapsed time) and **cycle time** (elapsed).
- **Rework and error rate** (proportion of cases sent back, root causes).
- **Systems touched** and the mode of access (API, screen, manual re-keying).
- **Decision logic** (deterministic rules, documented policy, tacit judgement).
- **Controls** (maker-checker, sampling, second-line sign-off) and their regulatory basis.

The walk is done on real cases, not on the documented process. The gap between the two is itself a finding: in corporate onboarding the documented process typically has 15 to 20 steps and the observed process 25 to 40, with the difference made up of chasing, re-keying and discrepancy resolution [ILLUSTRATIVE — composite].

## 2.3 Pain-point heatmaps

A heatmap scores each L3 task on a small number of pain dimensions and renders the result so that concentration of pain is visible. The standard dimensions:

| Dimension | Question | Scale anchor (5) |
|---|---|---|
| Volume | How many times a month is this performed? | Top quintile of tasks in the domain |
| Effort | Handle time per occurrence multiplied by volume | Top quintile of total hours |
| Error and rework | How often is output wrong or sent back? | Above 20 percent rework |
| Delay | Share of end-to-end cycle time this task causes, including waiting | Above 20 percent of cycle time |
| Customer or colleague friction | Does the task generate complaints, chasing, escalations? | Frequent, measurable |

The heat score is a simple sum (maximum 25) or a weighted sum where the client's objective justifies weighting (a cycle-time-driven ambition weights Delay; a cost-driven ambition weights Effort). Heat is not AI fit. A task can be hot and unsuitable for AI (MLRO approval) or cool and ideal (document classification at low volume). The heatmap tells you where value is; the fit assessment in Module 7 tells you whether AI can reach it.

## 2.4 Ideation workshops

Workshops turn the heatmap into candidate use cases. They are necessary because the people who do the work know constraints and workarounds that no process document records, and because ownership of a use case starts in the room. The design that works:

- **Participants (10 to 14).** Process owner, three to five front-line operators and checkers, a second-line compliance representative, a data or platform architect who knows the system landscape, a product owner, two consultants (one facilitating, one capturing).
- **Pre-work.** Participants receive the heatmap and the decomposed task list 48 hours ahead; the facilitator pre-seeds four to six "provocation" use cases based on patterns seen elsewhere so the room has something concrete to react to.
- **Agenda (half day).** Thirty minutes on AI patterns in plain terms (the five tools from Module 1 and the agent patterns from Module 3, no vendor demos); forty-five minutes on the heatmap with the room correcting scores; ninety minutes of ideation per hot zone using prompt cards; forty-five minutes of clustering and first-cut fit discussion; thirty minutes assigning sponsors to each cluster.
- **Prompt cards.** Each card forces a pattern on a task: "What if the system drafted this and a human approved it?" (copilot); "What if an agent executed this end to end and only exceptions came to a human?" (agentic with HITL); "What would need to be true for this to be fully automated?" (constraint surfacing); "What does the checker actually check, and could it be checked by a rule?" (rules-versus-model).
- **Outputs.** A raw idea list (typically 40 to 80 items for one domain), a first clustering, named sponsors per cluster, and a parking lot of non-AI improvements, which is often worth more than the AI list and should be handed to the operations team.

### 2.5 Building and cleaning the long list

Raw ideas are not use cases. Cleaning involves five operations:

1. **Normalise the unit.** Each use case is written as "pattern + task + outcome", for example "Agent that unwinds ownership structures from registry and client documents and produces a UBO schedule for checker review". Ideas phrased as technologies ("use an LLM") or outcomes ("faster onboarding") are rewritten or discarded.
2. **De-duplicate.** Merge ideas that act on the same task with the same pattern. Keep variants that differ in autonomy level as options within one use case, not as separate use cases.
3. **Cluster into domains and stages.** Map each use case back to its L2 stage. A stage with four or more use cases is a candidate for a "reshape" of the whole stage rather than point solutions.
4. **Separate AI from non-AI.** Rules, workflow and data fixes go to the operations improvement backlog. Keeping them in the AI long list inflates the AI business case and misleads prioritisation.
5. **Attach the baseline.** Pull the volume, handle time and error data for the underlying task from the value-chain walk onto the use case. A use case without a baseline cannot be scored on value and goes back to discovery.

A good long list for one domain has 10 to 15 use cases after cleaning, each with a one-paragraph description, pattern, autonomy option, sponsor and baseline.

### 2.6 Where discovery goes wrong structurally

- **Top-down only.** Executive interviews produce the use cases executives have read about. The heatmap is the corrective.
- **Bottom-up only.** Front-line ideation without an ambition statement produces 80 micro-automations and no strategy. The ambition from Module 5 is the filter.
- **Vendor-led.** A platform vendor's catalogue is a sales document; its "use cases" are its features.
- **No baseline.** Discovery that does not capture volumes and times produces a long list that cannot be prioritised or costed.

## 3. Frameworks

### 3.1 Process decomposition (L1 to L4)

Attribution: process-level conventions derive from process-classification frameworks such as the APQC PCF [PUBLIC — APQC, ongoing] and from business process management practice; the use of L3 as the AI assessment unit is practitioner convention, attribution uncertain. Steps:

1. Confirm the L1 domain boundary (trigger, outcome, exclusions) with the sponsor.
2. Draft L2 stages from documentation; validate with the process owner.
3. Build L3 tasks from observed cases (five to ten real cases walked end to end), not from documentation.
4. Decompose to L4 only where an L3 task mixes automatable and judgement steps.
5. Capture the seven attributes in section 2.2 per L3 task.

### 3.2 Pain-point heatmap

Attribution: generic operations-improvement tool, widely used; no single originator. Steps:

1. Choose four to five pain dimensions aligned to the client's ambition.
2. Score each L3 task 1 to 5 per dimension using data where available and operator consensus where not; mark which is which.
3. Compute heat; render as a grid of stages by tasks with colour intensity.

4. Identify hot zones (contiguous tasks with heat 17 or higher out of 25).
5. Record the non-AI fixes revealed, as a separate output.

### 3.3 Ideation workshop with pattern prompt cards

Attribution: practitioner design; the pattern cards draw on the agent patterns documented in Module 3. Steps are in section 2.4.

### 3.4 Use-case canvas

Attribution: adapted from Osterwalder and Pigneur's Business Model Canvas structure (2010) [PUBLIC — 2010]; the AI-specific fields are practitioner convention. One page per use case, fields: task and stage; current baseline (volume, handle time, cycle time, error rate); AI pattern and autonomy level; inputs and data sources; outputs and consumers; human-in-the-loop design; controls affected; value hypothesis (which lever, Module 9); key risks; sponsor; open questions. The canvas is the hand-off artefact into Module 7.

## 4. Worked BFSI example

### 4.1 Setting

[ILLUSTRATIVE — composite] Northgate Bank (from Module 5) selected corporate onboarding (KYB) as a "Transform" domain. Baseline from the value-chain walk: 7,200 new corporate and SME onboardings a year plus 9,500 periodic reviews; 140 FTE in the onboarding and KYC operations team; median end-to-end cycle time 23 business days for a standard corporate, 41 days for a complex structure; 31 percent of cases returned by the checker at least once; relationship managers spend an estimated 12 percent of their time on onboarding status chasing.

### 4.2 Decomposition and heatmap

Scores are 1 to 5 per dimension; heat is the sum (maximum 25). Scores marked with an asterisk are from system data; the others are operator consensus from the workshop. [ILLUSTRATIVE — composite]

| # | L2 stage | L3 task | Volume | Effort | Error | Delay | Friction | Heat |
|---|---|---|---|---|---|---|---|---|
| 1.1 | Intake and pre-screen | Capture request and basic entity details from RM | 4* | 2 | 2 | 1 | 2 | 11 |
| 1.2 | Intake and pre-screen | Pre-screen against risk appetite (sector, jurisdiction) | 4* | 1 | 1 | 1 | 1 | 8 |
| 1.3 | Intake and pre-screen | Determine required document set by entity type and jurisdiction | 4* | 2 | 4* | 3 | 3 | 16 |
| 2.1 | Document collection | Issue document request to client | 4* | 2 | 2 | 2 | 3 | 13 |
| 2.2 | Document collection | Chase outstanding documents | 5* | 4* | 1 | 5* | 5 | 20 |
| 2.3 | Document collection | Receive, classify and index documents | 5* | 3 | 3 | 2 | 1 | 14 |
| 2.4 | Document collection | Check completeness, validity and expiry | 5* | 3 | 4* | 3 | 3 | 18 |
| 3.1 | Entity verification | Verify legal entity against company | 4* | 2 | 2 | 1 | 1 | 10 |

| # | L2 stage | L3 task | Volume | Effort | Error | Delay | Friction | Heat |
|---|---|---|---|---|---|---|---|---|
| | | registry | | | | | | |
| 3.2 | Entity verification | Extract directors and shareholders from documents | 4* | 3 | 3 | 2 | 1 | 13 |
| 3.3 | Entity verification | Build ownership structure and identify UBOs across layers | 3* | 5* | 5* | 4 | 2 | 19 |
| 3.4 | Entity verification | Verify UBO identities (ID and V) | 3* | 2 | 2 | 3 | 3 | 13 |
| 3.5 | Entity verification | Reconcile discrepancies between client data and registry | 4* | 4 | 4* | 4 | 3 | 19 |
| 4.1 | Screening | Screen entity, directors and UBOs against sanctions and PEP lists | 5* | 1 | 1 | 1 | 1 | 9 |
| 4.2 | Screening | Adverse media search | 4* | 3 | 3 | 2 | 1 | 13 |
| 4.3 | Screening | Adjudicate screening hits and document disposition | 5* | 5* | 3 | 3 | 1 | 17 |
| 5.1 | Risk assessment | Determine customer risk rating | 4* | 2 | 3 | 1 | 1 | 11 |
| 5.2 | Risk assessment | Source of funds and wealth assessment (high risk) | 2* | 4 | 3 | 4 | 3 | 16 |
| 5.3 | Risk assessment | Draft enhanced due diligence narrative | 2* | 5* | 3 | 4 | 1 | 15 |
| 5.4 | Risk assessment | Capture expected activity profile | 4* | 2 | 4 | 1 | 2 | 13 |
| 6.1 | Decision | Checker quality review (four eyes) | 5* | 4* | 4* | 3 | 2 | 18 |
| 6.2 | Decision | Compliance and MLRO approval for high risk | 2* | 2 | 1 | 4 | 2 | 11 |
| 6.3 | Decision | Communicate decision and conditions to RM and client | 4* | 1 | 2 | 1 | 3 | 11 |
| 7.1 | Account setup | Set up accounts, mandates and signatories | 4* | 3 | 3 | 2 | 3 | 15 |
| 7.2 | Account setup | Tax classification (FATCA and CRS) | 4* | 2 | 3 | 1 | 1 | 11 |
| 8.1 | Handover | Set periodic review schedule and monitoring triggers | 4* | 1 | 2 | 1 | 1 | 9 |

Hot zones (heat 17 or above): document chasing and completeness (2.2, 2.4), ownership structure and discrepancy reconciliation (3.3, 3.5), screening hit adjudication (4.3), checker review (6.1). Together these six

tasks account for an estimated 58 percent of operations touch time and roughly 70 percent of the cycle-time variance between standard and complex cases [ESTIMATE — from the walk data; composite].

Non-AI findings handed to operations: document requirements matrix (1.3) is maintained in a spreadsheet with jurisdiction gaps; the registry lookup (3.1) is manual although an API is licensed; FATCA and CRS classification (7.2) could be rule-driven today.

## 4.3 Long list after the workshop and cleaning

The workshop produced 61 raw ideas; after normalising, de-duplicating and removing non-AI items, 12 candidate use cases remain. Pattern vocabulary follows Modules 1 and 3. [ILLUSTRATIVE — composite]

| UC | Use case | Stage | Pattern | Autonomy option | Baseline attached |
|---|---|---|---|---|---|
| UC1 | Document requirement determination and personalised request pack | 1.3, 2.1 | Rules plus LLM explanation | Auto-generate, RM approves | 7,200 packs a year; 31 percent incomplete first time |
| UC2 | Intelligent document intake: classify, extract, completeness and expiry check | 2.3, 2.4 | Document AI plus LLM validation | Auto with confidence threshold | 86,000 documents a year; 4.1 minutes handle each |
| UC3 | Client chase and clarification agent | 2.2 | Agent with tools (case system, email, portal) | Agent sends; escalates after two cycles | 2.2 chases per case; 6.4 days average wait |
| UC4 | Registry verification and data reconciliation agent | 3.1, 3.5 | Agent with registry APIs | Auto-reconcile; discrepancies to analyst | 4,100 discrepancy cases a year; 38 minutes each |
| UC5 | Ownership structure unwinding and UBO schedule drafting | 3.2, 3.3 | Plan-and-execute agent plus extraction | Draft for analyst confirmation | 2,900 complex structures a year; 2.6 hours each |
| UC6 | Screening hit adjudication copilot | 4.3 | RAG plus LLM narrative with rules pre-filter | Draft disposition; analyst decides | 48,000 hits a year; 94 percent false positive; 9 minutes each |
| UC7 | Adverse media search and summarisation | 4.2 | Retrieval plus summarisation | Auto-summary; analyst reads | 7,200 searches a year; 22 minutes each |
| UC8 | Customer risk rating pre-population and rationale | 5.1, 5.4 | Rules engine plus LLM rationale | Pre-populate; analyst confirms | 7,200 ratings a year; 14 percent corrected by checker |
| UC9 | EDD and source-of-wealth narrative drafting | 5.2, 5.3 | RAG over case file plus drafting | Draft; analyst edits; MLRO approves | 1,400 narratives a year; 3.8 hours each |
| UC10 | Checker quality-review copilot | 6.1 | LLM consistency checks across case file | Flags for checker; no auto-approval | 7,200 reviews a year; 31 percent return rate |
| UC11 | RM onboarding status and case-summary assistant | 6.3, cross-stage | Retrieval plus summarisation | Read-only assistant | 12 percent of RM time on status chasing |

| UC | Use case | Stage | Pattern | Autonomy option | Baseline attached |
|---|---|---|---|---|---|
| UC12 | Perpetual KYC: trigger-based review and pre-populated refresh | 8.1 and periodic review | Event-driven agent | Auto-refresh low risk; analyst for medium and high | 9,500 reviews a year; 2.1 hours each |

Clustering observation for the sponsor: UC2, UC3, UC4 and UC5 together cover the document and verification stages end to end. That is a "reshape" of stages 2 and 3 rather than four point solutions, and it should be assessed as a bundle in Module 7 as well as individually.

The so-what for the client: 58 percent of the team's touch time sits in six tasks, and ten of the twelve candidates act on those tasks. The domain is a credible "Transform" candidate because value is concentrated, not diffuse.

## 5. Real-world case

**HSBC and Google Cloud, anti-money-laundering transaction monitoring** [PUBLIC — Google Cloud announcement and HSBC statements, Jun 2023; figures VERIFY]. HSBC's work with Google Cloud's AML AI (built from HSBC's earlier "Dynamic Risk Assessment") is a discovery lesson as much as a model lesson. The team did not start from "apply AI to financial crime"; it targeted a specific, measured pain point in the value chain: rules-based transaction monitoring generating very high alert volumes with low true-positive rates, consuming investigator time. Google Cloud reported that the system identified two to four times more suspicious activity while cutting alert volumes by around 60 percent [PUBLIC — Google Cloud, Jun 2023, VERIFY]. The discovery lesson is that the use case was defined at the task level (alert generation and triage), with a baseline (alert volume, true-positive rate, investigator hours) that made the value measurable from day one.

Counter-example: **Commonwealth Bank of Australia** [PUBLIC — Australian press and Finance Sector Union statements, Aug 2025; VERIFY]. CBA announced 45 call-centre role reductions attributed to an AI voice bot, then reversed them after the union challenged the claim and the bank acknowledged that call volumes had not fallen as assumed. The discovery lesson is that the baseline (call volumes by type, deflection rates) was not established well enough to support the workforce claim, which is exactly the "attach the baseline" step in section 2.5.

## 6. Consultant language

> *"We assess AI fit at the task level and build business cases at the stage level. A department is not a unit of analysis."*

Use when a client asks for "AI for KYC" or "AI for operations" as a use case.

> *"The heatmap tells us where the pain is. The fit assessment tells us whether AI can reach it. They are different questions and we should not collapse them."*

Use when a stakeholder assumes the hottest task is the best AI candidate.

> *"Half of what came out of the workshop is not an AI problem; it is a rules, data or workflow problem. We should fix those first because they also make the AI cases cheaper."*

Use when presenting the parking lot of non-AI improvements.

> *"A use case without a baseline is a hypothesis. We cannot score it, cost it or defend it, so it goes back to discovery."*

Use when cleaning the long list in front of the sponsor.

> *"Four of these twelve act on the same two stages. That is a reshape of the stage, not four pilots, and we should assess it as a bundle."*

Use when clustering reveals concentration.

## 7. Common mistakes juniors make

- **Decomposing from documentation.** The documented process omits chasing, re-keying and discrepancy work, which is where most of the pain sits. Walk real cases.
- **Confusing heat with fit.** Ranking use cases by pain score alone sends the team at MLRO approval and checker sign-off, which are judgement tasks with regulatory accountability.
- **Letting the workshop produce technologies.** "Use GenAI for onboarding" is not a use case. Force the "pattern plus task plus outcome" format in the room.
- **Keeping non-AI fixes in the AI list.** It inflates the AI business case and hides the cheapest wins.
- **Skipping the baseline.** Discovery without volumes, handle times and error rates produces a long list that collapses at prioritisation and again at business case.
- **Over-decomposing.** Going to L4 everywhere produces 300 steps and workshop fatigue. Decompose to L4 only where an L3 task mixes automatable and judgement work.

## 8. Interview questions with model answers

**Q1. Walk me through how you would build the use-case long list for a bank's corporate onboarding function.**

I would run discovery in four steps over three to four weeks. First, decomposition: confirm the domain boundary with the sponsor, then build an L2 and L3 task map from five to ten real cases walked end to end with operators, capturing volume, handle time, cycle time, error and rework, systems touched, decision logic and controls for each of roughly 25 tasks. Second, heatmap: score each task on volume, effort, error, delay and friction, using system data where it exists, and identify hot zones; in onboarding these are typically document chasing, ownership unwinding, discrepancy reconciliation, screening adjudication and checker review. Third, a half-day ideation workshop with the process owner, operators, compliance, an architect and a product owner, using pattern prompt cards to force each idea into "pattern plus task plus outcome" form; I would expect 50 to 80 raw ideas. Fourth, cleaning: normalise, de-duplicate, cluster by stage, strip out rules and workflow fixes into an operations backlog, and attach a baseline to every surviving use case. The output is 10 to 15 use cases on one-page canvases, each with a sponsor and a baseline, ready for scoring. I would also hand the sponsor the non-AI fixes separately, because they are often the fastest value and they make the AI cases cheaper.

**Q2. Why should you separate pain-point heat from AI fit, and what happens if you do not?**

Heat measures where effort, error and delay concentrate; fit measures whether an AI pattern can take on the task given its variability, the judgement involved, the data available, the system access needed and the tolerance for error. The two are correlated only loosely. High-heat, low-fit tasks in onboarding include MLRO approval and checker sign-off: they are slow and effortful precisely because they carry regulatory accountability that cannot be delegated to a model under current supervisory expectations. Low-heat, high-fit tasks include registry lookups and tax classification. If you collapse the two, you prioritise the hottest tasks, propose automating judgement that regulators expect humans to exercise, and stall in model risk review. If you keep them separate, you can target high-fit tasks adjacent to the hot zones, for example drafting the EDD narrative so the MLRO reviews faster rather than trying to automate the approval itself. That is also the honest way to present the roadmap to a second-line function.

**Q3. A client's innovation team hands you a list of 40 GenAI use cases from vendor sessions. How do you treat it?**

I would treat it as raw input, not as a long list. Vendor-sourced ideas describe features, so the first pass is to rewrite each as pattern plus task plus outcome and discard those that do not map to a task in the client's process. Second, I would check for baselines: a vendor list never has the client's volumes, handle times or error rates, so most items will be unscoreable until we attach data from a value-chain walk. Third, I would test coverage against a heatmap: vendor lists cluster around generic productivity tools and under-represent the domain-specific hot zones where the measurable value sits. In my experience fewer than a third of such items survive into a clean long list, and the ones that do are usually the generic copilots [ILLUSTRATIVE — composite]. I would say this to the innovation team constructively: the list is useful as a catalogue of what is technically available, and we will test it against where the bank's pain actually is.

## 9. Self-check quiz

1. Explain why L3 is the right level for assessing AI fit and L2 the right level for building a business case. Give one onboarding example of each.
2. List the five standard heatmap dimensions and state which one you would weight most heavily for a client whose ambition is cycle-time reduction, and why.
3. From the Northgate heatmap, tasks 2.2 (chasing) and 6.1 (checker review) both score 18 or above. Which has the higher AI fit at a high autonomy level, and what is the single most important reason?
4. Rewrite "use GenAI to speed up KYC" as a properly normalised use case in "pattern plus task plus outcome" form, and name the baseline data you would attach.
5. Calculation: UC6 (screening hit adjudication) has 48,000 hits a year at 9 minutes each, with 94 percent false positives. If a copilot cuts handle time on false positives to 3 minutes and leaves true positives unchanged, how many analyst hours a year are saved, and how many FTE is that at 1,600 productive hours per FTE?

# Module 07 — Use-Case Assessment and Prioritisation

## 1. Why a client cares

Prioritisation is where the AI strategy becomes a capital allocation decision. The client has a long list of 10 to 40 use cases, a finite change capacity (typically two to four domain teams in a mid-sized bank), and a second-line function that will slow anything it does not understand. A defensible scoring method gives the executive committee a reason to say no to the loudest sponsor, gives the CFO a line of sight from score to business case, and gives the risk function an early view of what is coming. Done badly, prioritisation produces a 2x2 that everyone argues with and nobody funds.

## 2. Concept and mechanism

### 2.1 Three questions, three instruments

Assessment answers three distinct questions, and the common error is to answer them with one instrument.

| Question | Instrument | Unit of analysis | Output |
|---|---|---|---|
| How much is it worth? | Value scoring | Use case (L2 or L3) | Weighted value score, later a business case |
| Can we do it, and how hard is it? | Feasibility scoring | Use case | Weighted feasibility score |
| Should an agent do it, and how autonomously? | Agentic-fit assessment | Task (L3, sometimes L4) | Pattern and autonomy recommendation |
| What could go wrong and who must approve? | Risk overlay | Use case | Risk tier, governance route, autonomy cap |

Value and feasibility produce the 2x2. Agentic fit decides the solution pattern inside each use case. The risk overlay caps autonomy and routes governance. Sequencing then turns the result into waves.

### 2.2 Value scoring

Value is scored on the four levers that a bank CFO recognises: cost, revenue, risk and customer experience (Module 9 quantifies them). Each lever is scored 1 to 5 against anchors defined for the client, then weighted according to the ambition from Module 5.

| Lever | What it captures | Anchor for 5 | Anchor for 1 |
|---|---|---|---|
| Cost | Handle-time and FTE capacity released, vendor cost avoided | More than EUR 3 million a year run-rate | Under EUR 0.3 million |
| Revenue | Revenue pulled forward, abandonment reduced, cross-sell enabled | More than EUR 3 million a year | Negligible |
| Risk | Loss avoided, regulatory exposure reduced, control effectiveness | Material reduction in a Tier-1 risk (financial crime, credit, conduct) | No risk effect |
| Customer experience | Cycle time, effort, satisfaction, complaints | Cycle time cut by more than 50 percent on a customer-visible journey | Internal only |

Anchors are set in money or measurable outcome terms for the specific client, so that a score of 4 means the same thing across use cases. Without anchors, scoring degenerates into sponsor enthusiasm.

## 2.3 Feasibility scoring

Feasibility combines five dimensions. Scores run 1 to 5 where 5 is most feasible (so "integration complexity" is scored inversely: 5 means simple).

| Dimension | What it captures | Evidence to score it |
|---|---|---|
| Data readiness | Are inputs digital, accessible, of adequate quality, with a known owner? | Data readiness assessment (Module 4) |
| Integration complexity (inverse) | How many systems, with what access mode (API, RPA, manual)? | System inventory from the value-chain walk |
| Model and pattern maturity | Is the AI pattern proven for this task type at bank-grade accuracy? | Internal evals, peer deployments, vendor evidence |
| Regulatory exposure (inverse) | How heavily is the task regulated, and does automation change the control? | Regulatory mapping (Module 12) |
| Change effort (inverse) | How many roles, processes and controls change, and how willing is the owner? | Stakeholder assessment, workshop signals |

Weights again depend on the client. A bank with a strong platform and weak data weights data readiness heavily; a bank under regulatory remediation weights regulatory exposure heavily.

## 2.4 The agentic-fit assessment

Agentic AI is a solution pattern, not a value lever, so it is assessed at task level and feeds the solution design rather than the 2x2. The six-dimension framework below is a practitioner framework; variants are used across consulting firms and vendors and attribution is uncertain. Each dimension is scored 1 to 5 in the direction that favours agentic automation.

| Dimension | 1 (does not favour agent) | 5 (favours agent) | Why it matters |
|---|---|---|---|
| Task variability | Fully codifiable; a rules engine covers all cases | Inputs and paths vary too much for rules but stay within a bounded domain | Low variability means rules are cheaper and auditable; very high unbounded variability defeats agents too |
| Judgement bounded by policy | Expert discretion with personal regulatory accountability | Decisions follow documented policy that needs interpretation, not discretion | Agents can apply written policy; they cannot carry accountability |
| Tool and system access | Screen-only, manual, no APIs | APIs or well-defined tools for every action | Agents act through tools; no tools, no agent |
| Error tolerance and reversibility | Errors are severe and irreversible | Errors are low-impact and easily reversed | Determines autonomy level and HITL design |
| Data availability | Inputs on paper, in images, or inaccessible | Inputs digital, structured or reliably extractable | Agents reason over what they can read |
| Volume and frequency | Rare, ad hoc | High, continuous | Agent build and run cost needs volume to amortise |

Interpretation bands (total out of 30):

- **24 to 30:** candidate for autonomous agent execution with exception-based human review.
- **18 to 23:** agent with human-in-the-loop at defined gates (approve, or review above a risk threshold).
- **12 to 17:** copilot pattern; human performs the task with AI drafting or checking.
- **Below 12:** keep human, or use a rules engine if variability is low.

Two override rules apply regardless of total. If task variability scores 2 or below, default to a rules engine, because a deterministic solution is cheaper, auditable and does not need model risk validation of a stochastic component. If the risk overlay assigns Tier 1, cap the autonomy at human-in-the-loop regardless of score.

### 2.5 Risk overlay

The overlay has three parts, applied after scoring so that it modifies rather than contaminates the value and feasibility assessment:

1. **Model risk tier.** Using the client's model risk framework (aligned to SR 11-7 in the US and PRA SS1/23 in the UK [PUBLIC — Federal Reserve, 2011; PRA, May 2023, effective May 2024]), classify by materiality of decisions influenced and degree of autonomy. Tier 1 models require independent validation before use, which adds three to six months to the critical path [ESTIMATE — practice assumption].
2. **Regulatory classification.** Under the EU AI Act, creditworthiness assessment of natural persons is an Annex III high-risk use [PUBLIC — Regulation (EU) 2024/1689; VERIFY timelines, as a Nov 2025 Commission proposal would delay some high-risk obligations]. Sanctions and AML tasks carry conduct and criminal-law exposure even where the Act does not classify them as high risk. Consumer-facing decisions in the UK attract Consumer Duty scrutiny.
3. **Customer and reputational impact.** Does the use case make or materially influence a decision about a customer? Does it communicate directly with customers?

The overlay output is a tier (1 to 3), a governance route (which committees, which validation) and an autonomy cap.

### 2.6 Sequencing into waves

Prioritisation produces a ranking; sequencing produces a plan. Three considerations beyond the 2x2:

- **Dependencies.** Perpetual KYC depends on document intake and registry reconciliation being in place. A high-value use case that depends on three others belongs in a later wave even if it scores well.
- **Bundles.** Where several use cases act on adjacent stages, assess the bundle. A "reshape" of stages 2 and 3 of onboarding has a stronger business case than the four point solutions because benefits appear only when the stage's cycle time falls.
- **Capacity.** Each wave is limited by the number of product teams, the model risk validation throughput and the operations change capacity. Two to three use cases per team per wave is a typical limit [ESTIMATE — practice assumption].

The standard wave logic: Wave 1 quick wins (high value, high feasibility, low risk tier) and cheap fill-ins that build capability; Wave 2 foundational and bundled use cases that create the data and tooling for later waves; Wave 3 transformational use cases with high value and lower feasibility or higher risk tier.

## 3. Frameworks

### 3.1 Value versus feasibility 2x2

Attribution: generic portfolio tool, widely used; no single originator. Steps: define anchors; score; weight; plot; set thresholds (commonly the median or 3.0); label quadrants (quick wins, strategic bets, fill-ins, deprioritise); sanity-check the plot with the sponsor for surprises, and document why any score was changed.

### 3.2 Weighted scoring model

Attribution: generic multi-criteria decision analysis; no single originator. Steps: agree dimensions and weights with the sponsor before scoring; score with evidence; compute weighted sums; run a sensitivity check by flexing weights by plus or minus 10 points and confirming the top quartile is stable. If the ranking is unstable under small weight changes, the use cases are not meaningfully different and sequencing should be driven by dependencies instead.

### 3.3 Six-dimension agentic-fit assessment

Attribution: practitioner framework, widely used in variants, attribution uncertain. Steps in section 2.4. The output per task is a recommended pattern (rules, copilot, agent with HITL, autonomous agent) and an autonomy level on the 0 to 5 scale from Module 3.

### 3.4 Product-management scoring methods and when to use them

- **RICE** (Reach, Impact, Confidence, Effort), from Intercom, attributed to Sean McBride [PUBLIC — Intercom blog, 2016]. Useful inside a single product backlog once a use case is funded; less suited to cross-domain portfolio decisions because Reach is hard to normalise across functions.
- **ICE** (Impact, Confidence, Ease), attributed to Sean Ellis [PUBLIC — growth-hacking literature, 2010s]. Fast, coarse, useful for first triage of a raw idea list of 60 items.
- **MoSCoW** (Must, Should, Could, Won't), Dai Clegg, Oracle, 1994 [PUBLIC — 1994]. Useful for scoping a wave once the use cases are chosen.
- **Kano model**, Noriaki Kano, 1984 [PUBLIC — 1984]. Useful for customer-facing use cases to separate basic expectations from delighters; rarely used for back-office automation.

### 3.5 Risk tiering aligned to model risk management

Attribution: Federal Reserve SR 11-7 (2011) and PRA SS1/23 (2023) for the tiering logic; EU AI Act for regulatory classification [PUBLIC]. Steps in section 2.5.

## 4. Worked BFSI example

### 4.1 Portfolio and weights

[ILLUSTRATIVE — composite] Northgate Bank's prioritisation covers nine KYB use cases from Module 6 (UC2 to UC6, UC8 to UC10, UC12) plus three from other domains surfaced by the same method: agentic card dispute resolution, Level-1 disposition of transaction sanctions screening alerts, and SME credit underwriting memo drafting.

Weights agreed with the executive sponsor, reflecting a "Transform" ambition in onboarding with a cost and cycle-time emphasis:

| Value lever | Weight | Feasibility dimension | Weight |
|---|---|---|---|
| Cost | 35 | Data readiness | 25 |
| Revenue | 20 | Integration simplicity | 20 |
| Risk | 25 | Model and pattern maturity | 20 |
| Customer experience | 20 | Regulatory ease | 20 |
| | | Change ease | 15 |

### 4.2 Scored matrix

Raw scores 1 to 5; weighted totals out of 5. [ILLUSTRATIVE — composite; scores reflect the Module 6 baselines]

| Ref | Use case | Cost | Rev | Risk | CX | Value | Data | Integ | Matur | Reg | Change | Feas |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | UC2 Intelligent document intake | 5 | 2 | 3 | 4 | 3.70 | 4 | 3 | 5 | 4 | 4 | 4.00 |
| B | UC3 Client chase agent | 3 | 3 | 1 | 5 | 2.90 | 3 | 3 | 4 | 4 | 3 | 3.40 |

| Ref | Use case | Cost | Rev | Risk | CX | Value | Data | Integ | Matur | Reg | Change | Feas |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| C | UC4 Registry verification and reconciliation | 4 | 2 | 4 | 3 | 3.40 | 4 | 4 | 4 | 4 | 4 | 4.00 |
| D | UC5 Ownership and UBO unwinding | 4 | 3 | 5 | 3 | 3.85 | 2 | 2 | 3 | 3 | 3 | 2.55 |
| E | UC6 Screening hit adjudication copilot | 4 | 1 | 4 | 1 | 2.80 | 4 | 3 | 4 | 3 | 4 | 3.60 |
| F | UC8 Risk rating pre-population | 2 | 1 | 4 | 1 | 2.10 | 3 | 3 | 4 | 2 | 4 | 3.15 |
| G | UC9 EDD narrative drafting | 3 | 1 | 3 | 2 | 2.40 | 3 | 4 | 4 | 3 | 4 | 3.55 |
| H | UC10 Checker quality-review copilot | 3 | 2 | 5 | 2 | 3.10 | 3 | 3 | 3 | 3 | 3 | 3.00 |
| I | UC12 Perpetual KYC | 5 | 2 | 5 | 3 | 4.00 | 2 | 2 | 3 | 3 | 2 | 2.40 |
| J | Agentic card dispute resolution | 5 | 2 | 3 | 5 | 3.90 | 3 | 2 | 4 | 3 | 2 | 2.85 |
| K | Sanctions alert Level-1 disposition | 5 | 1 | 4 | 2 | 3.35 | 4 | 3 | 4 | 2 | 3 | 3.25 |
| L | SME underwriting memo drafting | 3 | 4 | 3 | 3 | 3.20 | 2 | 3 | 4 | 2 | 3 | 2.75 |

Worked arithmetic for row A: value = 0.35 x 5 + 0.20 x 2 + 0.25 x 3 + 0.20 x 4 = 1.75 + 0.40 + 0.75 + 0.80 = 3.70; feasibility = 0.25 x 4 + 0.20 x 3 + 0.20 x 5 + 0.20 x 4 + 0.15 x 4 = 1.00 + 0.60 + 1.00 + 0.80 + 0.60 = 4.00.

## 4.3 2x2 placement (threshold 3.0 on both axes)

| Quadrant | Use cases | Reading |
|---|---|---|
| Quick wins (value and feasibility at or above 3.0) | A document intake (3.70, 4.00); C registry reconciliation (3.40, 4.00); K sanctions alert disposition (3.35, 3.25); H checker copilot (3.10, 3.00) | Fund in Wave 1; H is borderline and should run as copilot only |
| Strategic bets (value at or above 3.0, feasibility below) | I perpetual KYC (4.00, 2.40); D UBO unwinding (3.85, 2.55); J disputes (3.90, 2.85); L underwriting memo (3.20, 2.75) | Highest value in the portfolio; feasibility gaps are data and integration, which are fixable enablers |
| Fill-ins (value below 3.0, feasibility at or above) | B chase agent (2.90, 3.40); E screening adjudication (2.80, 3.60); G EDD drafting (2.40, 3.55); F risk rating (2.10, 3.15) | Cheap; fund where they build capability for a strategic bet |
| Deprioritise | None below 3.0 on both | The long list was already cleaned in Module 6 |

Sensitivity check: flexing cost weight to 25 and CX weight to 30 moves J above I and B into quick wins; the top-quartile set (A, I, J, D) is otherwise stable, so the ranking is robust enough to sequence on.

## 4.4 Risk overlay

| Ref | Model risk tier | Regulatory note | Autonomy cap |
|---|---|---|---|
| A | 3 | Document handling; no customer decision | Autonomous above confidence threshold |
| C | 3 | Data reconciliation; discrepancies to analyst | Autonomous for matches; HITL for discrepancies |
| D | 2 | Influences UBO identification, an AML control | Draft for analyst confirmation |
| I | 2 | Changes periodic review control; MLRO sign-off on design | Autonomous for low-risk refresh only |
| J | 2 | Customer-facing decisions; consumer protection rules | Autonomous for low-value, clear-cut cases; HITL otherwise |
| K | 1 | Sanctions; false negatives carry criminal-law exposure | HITL always; agent proposes, analyst disposes |
| L | 1 to 2 | Credit decisions; if borrowers are sole traders the EU AI Act high-risk classification for creditworthiness may apply [VERIFY] | Draft only; underwriter decides |
| B, E, F, G, H | 3 (E: 2) | Copilots and communication; no autonomous decisions | Copilot or agent with HITL |

The overlay changes the plan: K scored as a quick win but its Tier 1 status means validation adds months and autonomy is capped at HITL, so its benefit is lower than the raw score implied. It stays in Wave 1 because the value is still strong at HITL (analyst time per alert falls even when the analyst decides).

## 4.5 Task-level agentic fit for eight KYB steps

Scores 1 to 5 per dimension in the direction that favours an agent; total out of 30. [ILLUSTRATIVE — composite]

| Task | Variability | Judgement bounded | Tool access | Error tolerance | Data | Volume | Total | Recommendation |
|---|---|---|---|---|---|---|---|---|
| 1.3 Determine document requirements | 2 | 2 | 4 | 4 | 4 | 4 | 20 | Rules engine (variability override); LLM only to explain the pack to the client |
| 2.2 Chase outstanding documents | 4 | 4 | 4 | 5 | 4 | 5 | 26 | Autonomous agent; escalate after two cycles |
| 2.4 Completeness and validity check | 3 | 3 | 4 | 4 | 4 | 5 | 23 | Agent with HITL below confidence threshold |
| 3.3 Ownership and UBO unwinding | 5 | 4 | 3 | 3 | 2 | 3 | 20 | Agent drafts UBO schedule; analyst confirms |

| Task | Variability | Judgement bounded | Tool access | Error tolerance | Data | Volume | Total | Recommendation |
|---|---|---|---|---|---|---|---|---|
| 3.5 Reconcile client versus registry data | 4 | 4 | 4 | 4 | 3 | 4 | 23 | Agent with HITL on unresolved discrepancies |
| 4.3 Adjudicate screening hits | 4 | 4 | 4 | 3 | 4 | 5 | 24 | Score says autonomous; Tier 1 overlay caps at HITL |
| 5.3 Draft EDD narrative | 5 | 3 | 3 | 3 | 3 | 2 | 19 | Copilot drafting with analyst edit and MLRO approval |
| 6.2 MLRO approval of high-risk client | 3 | 1 | 2 | 1 | 3 | 2 | 12 | Human; copilot case summary only |

### 4.6 Three-wave roadmap

| Wave | Timing | Use cases | Rationale | Enablers built |
|---|---|---|---|---|
| 1 Quick wins | Months 0 to 6 | A, C, K (HITL), H (copilot), E (fill-in) | High feasibility, Tier 2 or 3 except K; builds document and data foundations | Document AI pipeline, registry API integration, eval harness, LLM gateway hardening |
| 2 Foundational bundle | Months 6 to 15 | D plus B (with A and C, this reshapes stages 2 and 3 end to end), G, J pilot in one card portfolio | Stage-level cycle time benefit; disputes pilot proves agentic pattern with HITL | Entity and ownership data product, agent orchestration platform, model risk Tier 2 validation route |
| 3 Transformational | Months 15 to 30 | I perpetual KYC, J scale across portfolios, L underwriting memo | Highest value; depends on Wave 1 and 2 data products and on validated agent patterns | Event-driven architecture, Tier 1 validation, workforce transition plan |

Capacity check: two domain teams in Wave 1 carrying five use cases, of which two are copilots, is at the upper limit of the two to three per team guideline.

The so-what for the client: the four highest-value use cases are all strategic bets, and their feasibility gaps are the same two enablers, entity data and integration. Funding those enablers in Wave 1 converts roughly EUR 20 million of annual run-rate value from “bet” to “plan” [ESTIMATE — assumption, to be built out in the Module 9 business case].

## 5. Real-world case

**JPMorgan Chase, portfolio approach to AI** [PUBLIC — JPMorgan shareholder letters 2023 and 2024; press coverage 2024; figures VERIFY]. JPMorgan has described running several hundred AI use cases in production, governed through a central AI and data organisation, with an explicit focus on quantified value: Jamie Dimon’s 2023 letter stated that AI was already delivering value in fraud, marketing and risk, and the firm has publicly discussed internal value estimates in the USD 1 to 2 billion range [PUBLIC — JPMorgan, 2023–24, VERIFY]. The prioritisation lesson is institutional: a central function that scores and sequences use cases across lines of business, with the lines owning the benefits, is what allows a portfolio of hundreds rather than a scatter of pilots. The LLM Suite rollout to about 200,000 employees in 2024 [PUBLIC — Reuters and others, 2024, VERIFY] sits in BCG’s “Deploy” category and was sequenced alongside, not instead of, the domain-specific “Reshape” work.

**Klarna** [PUBLIC — Klarna, Feb 2024; CEO interviews, May 2025, VERIFY] is the cautionary case for the risk overlay. The customer-service assistant was a high-value, high-feasibility use case and was deployed at high autonomy. The 2025 walk-back on service quality shows what a risk overlay would have flagged: customer-facing autonomy needs a quality floor and an exception route, and the autonomy cap should be set by error tolerance, not by the cost case alone.

## 6. Consultant language

> *"We score value and feasibility separately, then overlay risk. If we blend them, the risk function will unblend them for us later, at a worse moment."*

Use when a sponsor asks for a single composite score.

> *"Agentic fit is a solution question, not a value question. It decides how we build a use case, not whether we fund it."*

Use when a stakeholder wants to rank use cases by how "agentic" they are.

> *"Your four most valuable use cases all fail feasibility for the same two reasons. That is not four problems; it is two enablers we should fund first."*

Use when presenting the strategic-bets quadrant.

> *"The score says this task could run autonomously. The risk tier says it must not. The tier wins, and the business case is built at the capped autonomy level."*

Use when the overlay reduces a quick win's benefit.

> *"The ranking is stable under a 10-point weight change, so we can stop arguing about weights and start arguing about sequencing."*

Use after the sensitivity check, to close the scoring debate.

## 7. Common mistakes juniors make

- **Scoring without anchors.** A 4 on value must mean the same money across use cases. Without anchors the matrix measures sponsor charisma.
- **Blending risk into feasibility.** Regulatory exposure belongs in feasibility as a difficulty signal, but the autonomy cap and governance route must be a separate overlay, or the plan will be wrong about what the use case can do.
- **Treating the 2x2 as the plan.** Dependencies and bundles move use cases between waves regardless of quadrant.
- **Ranking by agentic fit.** Agentic is a pattern. A task can be a perfect agent candidate and worth very little.
- **Ignoring the variability override.** Low-variability tasks belong to rules engines; proposing an agent for them adds cost, latency and model risk for no gain, and an experienced interviewer will catch it.
- **Skipping sensitivity.** If the ranking flips when a weight moves by 10 points, the differences are noise and you should say so.

## 8. Interview questions with model answers

**Q1. How would you prioritise 30 AI use cases for a bank, and how would you defend the result to the executive committee?**

I would separate three questions. First, value: score each use case 1 to 5 on cost, revenue, risk and customer experience against money-denominated anchors agreed with the CFO, weighted to the ambition; for a cost-

led ambition I would weight cost around 35 percent. Second, feasibility: score data readiness, integration simplicity, pattern maturity, regulatory ease and change ease, weighted to the bank's weaknesses, typically data. Plot the two, set the threshold at 3.0, and read the quadrants. Third, a risk overlay that assigns a model risk tier aligned to SR 11-7 or SS1/23 and a regulatory classification, including EU AI Act high-risk status for natural-person creditworthiness, and caps autonomy accordingly. I would then sequence into three waves using dependencies, bundles and team capacity rather than the quadrant alone. To defend it, I would show the anchors, the evidence behind each score, a sensitivity test proving the top quartile is stable under a 10-point weight change, and the reasons any score was changed after sponsor review. The committee can disagree with a weight; it cannot easily disagree with a process that is transparent and reproducible, and that is what makes the result fundable.

**Q2. Describe how you assess whether a task is suitable for an agent, with a banking example where the answer is no.**

I use a six-dimension task-level assessment: variability, whether judgement is bounded by written policy, tool and system access, error tolerance and reversibility, data availability, and volume. Each is scored 1 to 5 in the direction that favours an agent, totals of 24 or more suggest autonomous execution with exception review, 18 to 23 an agent with human gates, 12 to 17 a copilot, and below 12 a human or a rules engine. Two overrides matter. If variability is low, a rules engine is the right answer regardless, because it is cheaper and auditable. If the risk overlay puts the task in Tier 1, autonomy is capped at human-in-the-loop regardless of score. A banking "no" is MLRO approval of a high-risk client: judgement is discretionary and personally accountable under AML regulation, errors are severe and hard to reverse, and volume is low. It scores around 12. The right design is a copilot that produces a case summary and highlights inconsistencies so the MLRO decides faster. Sanctions hit adjudication is the subtler case: it scores about 24 on fit, but Tier 1 exposure caps it at agent-proposes, analyst-disposes.

**Q3. Your scoring puts a Tier 1 sanctions use case in the quick-wins quadrant. The sponsor wants it fully automated in six months. What do you say?**

I would agree it is a priority and disagree on the autonomy. The value is real because analyst handle time per alert drops even when the analyst makes the final decision, so I would keep it in Wave 1 at human-in-the-loop with the agent drafting the disposition and rationale. I would explain that Tier 1 means independent validation before go-live under the bank's model risk framework, which typically adds three to six months to the critical path, and that false negatives in sanctions carry criminal-law exposure, so no supervisor will accept autonomous disposition on day one. I would then offer a path: build the eval harness with a golden set of adjudicated alerts, run the agent in shadow mode to measure agreement with analysts, and propose a staged autonomy increase, for example auto-closing only the lowest-risk name-match categories once agreement exceeds an agreed threshold for a sustained period. That turns the disagreement into a roadmap with evidence gates, which is what the sponsor actually needs to take to the risk committee.

## 9. Self-check quiz

1. State the three questions that assessment must answer and the instrument for each. Why is agentic fit assessed at task level rather than use-case level?
2. Using the Northgate weights, compute the value score for a use case scoring Cost 4, Revenue 3, Risk 2, CX 5, and the feasibility score for Data 3, Integration 4, Maturity 4, Regulatory 2, Change 3. Which quadrant does it fall into at a 3.0 threshold?
3. Explain the two override rules in the agentic-fit framework and give a KYB example of each.
4. Perpetual KYC scored highest on value (4.00) but sits in Wave 3. Give two sequencing reasons.
5. A task scores 5 on variability, 2 on judgement bounded by policy, 4 on tool access, 2 on error tolerance, 4 on data and 4 on volume. Total it, state the band, and say what the judgement and error-tolerance scores imply for the human-in-the-loop design.

# Module 08 — Cost Factors and Total Cost of Ownership

## 1. Why a client cares

Every AI business case that fails a CFO review fails on cost, not on benefit. Benefits are assumptions the CFO expects to haircut; costs are commitments the CFO expects to be complete. The two most common omissions are run costs that scale with volume (inference, human review, evaluation) and lifecycle costs that recur (model refresh, re-validation, prompt migration when a vendor deprecates a model). A consultant who can lay out a three-year total cost of ownership with the unit economics per transaction, and show where it is sensitive, earns the right to talk about benefits.

## 2. Concept and mechanism

### 2.1 Total cost of ownership as a discipline

Total cost of ownership (TCO) is the full lifecycle cost of acquiring, operating, changing and retiring a capability, not the purchase or build price. The term was popularised by Gartner (Bill Kirwin, 1987) for IT assets [PUBLIC — Gartner, 1987]. Applied to AI systems it needs four extensions that classic IT TCO under-weights:

1. **Consumption-based run cost.** Inference is metered per token, so run cost scales with volume and with how the system is designed (prompt length, number of calls, model tier). It is an operating cost that engineering decisions can move by a factor of three or more.
2. **Human-in-the-loop cost.** Review, sampling, exception handling and appeals are part of the system's cost, not part of "business as usual".
3. **Assurance cost.** Evaluation sets, continuous evals, guardrails, monitoring and model risk validation are recurring, and in a bank they are non-negotiable.
4. **Lifecycle churn.** Foundation models are deprecated on 12 to 24 month cycles [ESTIMATE — observed vendor practice, VERIFY]. Each migration triggers re-evaluation, prompt adjustment and, for higher tiers, re-validation.

### 2.2 The cost taxonomy

| Phase | Cost category | What it includes | Typical share of 3-year TCO |
|---|---|---|---|
| Build | Discovery and design | Process redesign, solution architecture, control design, regulatory mapping | 4 to 8 percent |
| Build | Data preparation | Corpus curation, document digitisation, entity resolution, labelling, golden dataset construction, access provisioning | 6 to 12 percent |
| Build | Integration | Connectors to core systems, case management, scheme or network APIs, identity and access, event feeds | 12 to 20 percent |
| Build | Engineering | Orchestration, agent logic, tools, prompts, retrieval pipelines, UI for reviewers | 15 to 25 percent |
| Build | Evaluation and guardrails | Eval harness, test sets, safety and policy filters, red-teaming | 3 to 6 percent |

| Phase | Cost category | What it includes | Typical share of 3-year TCO |
|---|---|---|---|
| Build | Validation and assurance | Model risk validation, security testing, privacy assessment, legal review | 3 to 6 percent |
| Run | Inference | Token consumption across all calls, including guardrail and evaluation calls, retries | 2 to 15 percent |
| Run | Infrastructure and platform | Gateway, vector store, orchestration runtime, observability, compute for batch evals | 6 to 12 percent |
| Run | Licences | Agent frameworks, observability tools, document AI, vendor platforms | 3 to 10 percent |
| Run | Monitoring and evals | Drift monitoring, periodic eval runs, incident response, eval set maintenance | 4 to 8 percent |
| Run | Human review | Sampling QA of autonomous outputs, exception handling attributable to the system, appeals | 3 to 10 percent |
| Run | Product and support team | Product owner, engineers, support | 10 to 18 percent |
| Change | Training and adoption | Training, SOP rewrite, communications, floor-walking | 4 to 8 percent |
| Change | Workforce transition | Redeployment, retraining, where applicable severance | Reported separately |
| Lifecycle | Model refresh and re-validation | Migration to new model versions, prompt and eval re-baselining, re-validation | 3 to 6 percent |

[ESTIMATE — shares are indicative ranges from practice for bank-grade agentic systems at medium volume; the inference share is highly volume-dependent, VERIFY against the specific design]

The most important reading of this table: inference, the cost everyone argues about, is usually a minority of TCO for bank back-office systems at medium volume. People, integration and assurance dominate. The exception is very high-volume or token-heavy workloads (long documents, images, many agent steps), where inference can approach a third of run cost.

## 2.3 Token economics

Inference cost per transaction is the product of four things: number of model calls, tokens per call (input and output priced separately), model tier, and discount mechanisms.

- **Input versus output.** Output tokens are priced three to five times higher than input tokens at most vendors [ESTIMATE — order of magnitude, VERIFY]. Designs that generate long outputs (full narratives) cost more than designs that classify or extract.
- **Prompt caching.** Repeated prefixes (system prompts, policy documents, tool schemas) can be cached; cached input is typically discounted 75 to 90 percent [ESTIMATE — order of magnitude, VERIFY]. Agentic systems with long static instructions benefit heavily.
- **Batch processing.** Non-latency-sensitive work (overnight review preparation, back-book refresh) can use batch endpoints at roughly half price [ESTIMATE — VERIFY].

- **Model tiering and routing.** Route deterministic or low-judgement steps (classification, extraction, summarisation) to small or mid-tier models and reserve frontier models for reasoning steps. Frontier models cost roughly 10 to 30 times a small model per token [ESTIMATE — order of magnitude, VERIFY].
- **Overheads.** Guardrail classifiers, retries, evaluation calls and tracing add 20 to 40 percent to raw inference in production [ESTIMATE — practice assumption].

Order-of-magnitude pricing to use only as placeholders [ESTIMATE — 2025 orders of magnitude, VERIFY before any client use]: frontier models USD 2 to 5 per million input tokens and USD 8 to 20 per million output; mid-tier USD 0.5 to 1.5 and USD 2 to 6; small models USD 0.05 to 0.25 and USD 0.2 to 1.0.

### 2.4 Unit economics

The unit that a CFO understands is cost per transaction, or more precisely cost per successfully completed task. Three unit metrics matter:

- **AI cost per case:** inference plus the volume-driven share of platform and review cost, divided by cases handled.
- **Fully loaded cost per case after transformation:** AI cost per case plus residual human cost per case plus amortised build and change.
- **Cost per successful outcome:** the above divided by the success rate, where failures (escalations, reversals, re-work) are counted. A system that is cheap per attempt and wrong a quarter of the time is expensive per outcome.

### 2.5 Hidden costs

- **Vendor and model lock-in.** Prompts, evals and guardrails tuned to one model do not transfer cleanly. Budget a migration every 12 to 24 months.
- **Data egress and residency.** Cross-region inference can trigger egress charges and data-protection reviews; in-region or private endpoints cost more per token.
- **Model risk validation effort.** For Tier 1 and 2 systems, independent validation is a recurring cost with scarce internal capacity; it is also a schedule cost.
- **Eval set maintenance.** Golden datasets decay as policy and products change; budget for quarterly refresh.
- **Observability storage.** Full tracing of agent runs generates large volumes of logs that may have retention obligations.
- **Legal and privacy review of vendor terms.** Especially data-use, training-on-customer-data and liability clauses.

## 3. Frameworks

### 3.1 Three-year TCO template

Attribution: generic finance practice; structure below is practitioner convention. Steps:

1. Fix the volume forecast per year and the autonomy design (what share of cases the system handles without a human). Everything volume-driven flows from this.
2. Build the per-case inference model (calls, tokens, tier, caching, overhead).
3. Estimate build costs by category with named assumptions and a contingency (15 to 25 percent for first-of-kind agentic builds [ESTIMATE]).
4. Estimate run costs by category, with volume-driven items scaling with the forecast.
5. Add change and lifecycle costs by year.

6. Present per year and cumulative, with unit economics, then run sensitivities on volume, tokens per case, autonomy share and model price.

### 3.2 FinOps for AI

Attribution: FinOps Foundation (part of the Linux Foundation), cloud financial management framework, with AI cost management guidance added from 2024 [PUBLIC — FinOps Foundation, VERIFY]. The three phases (Inform, Optimise, Operate) apply directly: tag and attribute token spend to use cases and teams; optimise through caching, routing and prompt compression; operate with budgets and anomaly alerts. In a bank this sits naturally with the LLM gateway, which is the control point for metering.

### 3.3 Cost-per-outcome ladder

Attribution: practitioner convention. Compute cost per attempt, cost per completed case and cost per successful outcome; present all three, because the gap between them is where quality problems hide.

## 4. Worked BFSI example

### 4.1 Setting

[ILLUSTRATIVE — composite] A card issuer handles 600,000 cardholder disputes a year. Current operations: 320 FTE at a fully loaded USD 65,000 each (USD 20.8 million), plus dispute platform and vendor costs of USD 3.5 million, total USD 24.3 million, or USD 40.50 per dispute. The target design is a supervisor-led multi-agent system (Module 3) that resolves low-value, clear-cut disputes autonomously (target 55 percent of volume) and prepares the remaining 45 percent for human decision. All figures below are illustrative; pricing is order of magnitude and date-sensitive [ESTIMATE — VERIFY].

### 4.2 Per-dispute inference, frontier model only

Design assumptions: 12 model calls per dispute on average (intake classification, evidence retrieval and summarisation, scheme-rule reasoning, customer communication drafting, supervisor decisions). Each call averages 9,000 input tokens (3,000 system and tool schema, 4,000 retrieved context, 2,000 state) and 600 output tokens. Forty percent of input tokens are cache hits (system prompt, tool schemas, policy excerpts).

| Component | Tokens per dispute | Price per million | Cost per dispute |
|---|---|---|---|
| Uncached input (60 percent of 108,000) | 64,800 | USD 3.00 | USD 0.194 |
| Cached input (40 percent of 108,000) | 43,200 | USD 0.30 | USD 0.013 |
| Output | 7,200 | USD 15.00 | USD 0.108 |
| **Raw inference** | | | **USD 0.315** |
| Overheads (guardrails, retries, evals, tracing) at 30 percent | | | USD 0.095 |
| **Loaded inference per dispute** | | | **USD 0.410** |

Annual inference, frontier only: 600,000 x USD 0.410 = USD 246,000.

Without caching the raw figure would be USD 0.432 per dispute, so caching alone saves about 27 percent of raw inference.

## 4.3 Per-dispute inference, tiered routing

Route 4 of the 12 calls (scheme-rule reasoning, final supervisor decisions, complex communication) to the frontier model and 8 calls (classification, extraction, summarisation) to a mid or small model at a blended USD 0.50 input and USD 2.00 output, with cached input at USD 0.05.

| Component | Tokens per dispute | Price per million | Cost per dispute |
|---|---|---|---|
| Frontier uncached input (60 percent of 36,000) | 21,600 | USD 3.00 | USD 0.065 |
| Frontier cached input | 14,400 | USD 0.30 | USD 0.004 |
| Frontier output | 2,400 | USD 15.00 | USD 0.036 |
| Small-model uncached input (60 percent of 72,000) | 43,200 | USD 0.50 | USD 0.022 |
| Small-model cached input | 28,800 | USD 0.05 | USD 0.001 |
| Small-model output | 4,800 | USD 2.00 | USD 0.010 |
| **Raw inference** | | | **USD 0.138** |
| Overheads at 30 percent | | | USD 0.041 |
| **Loaded inference per dispute** | | | **USD 0.179** |

Annual inference, tiered: 600,000 x USD 0.179 = USD 107,000. Routing saves USD 139,000 a year, or 56 percent of inference, at the cost of extra engineering (routing logic, two eval sets) estimated at USD 150,000 of build.

## 4.4 Three-year TCO

| Cost line | Year 1 | Year 2 | Year 3 | 3-year total |
|---|---|---|---|---|
| Discovery, design, process and control redesign | 0.60 | | | 0.60 |
| Data preparation (dispute history, evidence stores, policy corpus, 3,000-case golden set) | 0.90 | | | 0.90 |
| Integration (card processor, scheme dispute platforms, case management, core banking, identity) | 2.80 | | | 2.80 |
| Engineering (orchestration, agents, tools, reviewer UI) | included above | | | |
| Evaluation harness and guardrails | 0.50 | | | 0.50 |
| Model risk validation (Tier 2), security, privacy, legal | 0.60 | | | 0.60 |
| **Build subtotal** | **5.40** | | | **5.40** |
| Inference (frontier only, half-year in Year | 0.12 | 0.25 | 0.25 | 0.62 |

| Cost line | Year 1 | Year 2 | Year 3 | 3-year total |
|---|---|---|---|---|
| 1) | | | | |
| Platform and infrastructure (gateway, vector store, runtime, observability) | 0.25 | 0.45 | 0.45 | 1.15 |
| Licences (agent framework, observability, document AI) | 0.15 | 0.30 | 0.30 | 0.75 |
| Monitoring, evals, drift, incident response (1.5 FTE plus compute) | 0.20 | 0.45 | 0.45 | 1.10 |
| Human review of autonomous outputs (5 percent sample, 10 minutes each, 3.5 FTE) | 0.12 | 0.23 | 0.23 | 0.58 |
| Product and support team (PO, 2 engineers, designer, L2 support) | 0.50 | 1.00 | 1.00 | 2.50 |
| **Run subtotal** | **1.34** | **2.68** | **2.68** | **6.70** |
| Training, SOP rewrite, communications, adoption | 0.70 | 0.20 | | 0.90 |
| Model refresh, prompt and eval re-baselining, re-validation | | 0.30 | 0.30 | 0.60 |
| **TCO (frontier only)** | **7.44** | **3.18** | **2.98** | **13.60** |
| Tiered routing variant: extra build 0.15, inference saving 0.07, 0.14, 0.14 | 7.52 | 3.04 | 2.84 | 13.40 |

All figures USD million [ILLUSTRATIVE — composite; pricing ESTIMATE, VERIFY]. Workforce transition costs (redeployment, retraining, any severance) are reported separately in the business case because they are policy decisions, not system costs; a placeholder of USD 1.5 million is used in Module 9.

## 4.5 Unit economics

| Metric | Value | Working |
|---|---|---|
| AI cost per dispute, steady state (Year 2) | USD 5.30 | 3.18 million / 600,000 |
| Of which inference | USD 0.41 | from 4.2 |
| Residual human cost per dispute | USD 16.25 | 150 FTE x USD 65,000 / 600,000 |
| Fully loaded cost per dispute, Year 2 | USD 21.55 | 5.30 + 16.25 |
| Current cost per dispute | USD 40.50 | 24.3 million / 600,000 |
| Cost per successful outcome at 92 percent first-time-right | USD 23.42 | 21.55 / 0.92 |

The residual FTE figure assumes 170 of 320 roles are released as 55 percent of disputes are resolved autonomously and the remaining cases take less handle time with agent preparation [ESTIMATE — assumption; Module 9 ramps this over time].

### 4.6 Sensitivities

| Scenario | Effect on 3-year TCO | Reading |
|---|---|---|
| Tokens per dispute doubles (richer evidence, more steps) | Frontier inference +0.62 million; tiered +0.27 million | Routing becomes a 0.35 million lever instead of 0.20 million |
| Volume rises to 1.2 million disputes | Inference and review double; platform +30 percent | AI cost per dispute falls to about USD 3.80; unit economics improve |
| Frontier price falls 50 percent (observed trend, VERIFY) | Frontier inference -0.31 million | Narrows the routing case further |
| Autonomy share falls from 55 to 35 percent | Residual human cost +USD 4.30 per dispute | Largest sensitivity in the whole model; it is a quality and policy variable, not a cost variable |
| Tier 1 classification instead of Tier 2 | Validation +0.4 million, go-live +4 months | Schedule cost often exceeds the money cost |

The so-what for the client: at this volume, inference is under 5 percent of TCO and model routing is a 1.5 percent lever on three-year cost. The decisions that move the TCO are integration scope, the autonomy share the risk function will accept, and the size of the product and assurance team. Negotiating token prices is the wrong meeting to spend time in.

## 5. Real-world case

**Klarna, customer-service AI assistant** [PUBLIC — Klarna press release, Feb 2024; CEO statements, 2025, VERIFY]. Klarna reported that its OpenAI-based assistant handled two-thirds of customer-service chats in its first month, equivalent to the work of about 700 full-time agents, with an estimated USD 40 million profit improvement in 2024. The TCO lesson is in the 2025 sequel: the CEO acknowledged that quality had suffered and that human agents would be re-hired for complex cases. In TCO terms, the "human review and exception handling" line was under-provisioned relative to the autonomy share assumed, and cost per successful outcome, which includes re-contacts and dissatisfaction, was higher than cost per handled chat suggested. The case is also a reminder that reported benefits of this kind are company statements, not audited figures.

**Goldman Sachs and coding agents** [PUBLIC — press coverage of Goldman Sachs piloting Cognition's Devin, Jul 2025, VERIFY]. Reported as a pilot of autonomous coding agents alongside roughly 12,000 developers, with the firm describing the agents as augmenting rather than replacing engineers. The TCO point: the run cost of an agent is not the licence; it is the review, integration and governance around it, and the firm's framing as augmentation reflects where the real costs and risks sit.

## 6. Consultant language

> *"Inference is the cost everyone argues about and the one that matters least at this volume. Integration, assurance and the team are the TCO."*

Use when a steering committee fixates on token prices.

> *"I would like the CFO to see cost per successful outcome, not cost per case. The gap between them is where quality problems hide."*

Use when presenting unit economics.

> *"The largest sensitivity in this model is the autonomy share the risk function will accept. That is a governance decision with a USD 4 per case price tag."*

Use to connect TCO to the risk conversation.

> *"Model routing is worth about 1.5 percent of three-year cost here. It becomes material if token use per case doubles, so we should design for it and not over-invest in it on day one."*

Use when an engineering team wants to build routing before volumes justify it.

> *"Budget a model migration every 12 to 24 months. The vendor will deprecate the model we build on, and re-validation is not optional in a bank."*

Use when the lifecycle line is missing from a draft budget.

## 7. Common mistakes juniors make

- **Pricing only the licences.** Platform and licence costs are rarely more than a fifth of TCO; omitting the team, review, assurance and lifecycle lines is the most common CFO rejection reason.
- **Assuming inference dominates.** It does at consumer scale with heavy token use; it rarely does in bank back-office systems. Do the per-case arithmetic before asserting either way.
- **Forgetting overheads.** Guardrail calls, retries, evaluation runs and tracing add 20 to 40 percent to raw inference.
- **Omitting human review.** Sampling QA of autonomous decisions is a permanent cost; it is also the control the regulator will ask about.
- **Treating workforce transition as free.** Redeployment, retraining and any severance belong in the business case even if reported separately from TCO.
- **Quoting token prices from memory.** Prices move quarterly. Label them as estimates, verify them before a client meeting, and show that the conclusion is robust to a 50 percent change.

## 8. Interview questions with model answers

**Q1. Walk me through the cost components of an agentic AI system in a bank and tell me which ones people get wrong.**

I group them into build, run, change and lifecycle. Build covers discovery and process redesign, data preparation including the golden dataset, integration with core and third-party systems, engineering of orchestration, agents, tools and reviewer tooling, evaluation and guardrails, and validation and assurance including model risk, security and privacy. Run covers inference, platform and infrastructure, licences, monitoring and evals, human review of autonomous outputs, and the ongoing product and support team. Change covers training, SOP rewrite and adoption, with workforce transition reported separately. Lifecycle covers model refresh, prompt and eval re-baselining and re-validation on roughly a 12 to 24 month cycle. The three things people get wrong are: first, over-estimating inference, which for a 600,000-case back-office process is typically under 5 percent of three-year TCO once you do the token arithmetic; second, omitting human review and assurance, which are permanent and which the regulator will ask about; third, omitting lifecycle churn, because vendors deprecate models and re-validation is mandatory for higher-tier systems. I would also insist on presenting cost per successful outcome, not just cost per case, because that is where quality failures show up as cost.

**Q2. How do you model token costs for an agentic workflow, and what levers reduce them?**

I start from the workflow design: number of model calls per case, average input and output tokens per call, and which calls need frontier reasoning versus classification or extraction. For a dispute workflow I might assume 12 calls, 9,000 input and 600 output tokens each, giving about 108,000 input and 7,200 output tokens per case. I price input and output separately because output is three to five times more expensive, apply a cache-hit share to the static prefix, which is typically discounted 75 to 90 percent, and add 20 to 40 percent for guardrails, retries and evaluation calls. At illustrative 2025 prices that gives roughly USD 0.40 per case loaded on a frontier model, which I would label as an order-of-magnitude estimate to verify. The levers, in order of impact, are routing low-judgement calls to small models, which can halve inference; caching

static prompts, worth 20 to 30 percent; shortening retrieved context through better chunking and re-ranking; using batch endpoints for non-real-time work; and reducing output length by structuring outputs as fields rather than prose. I would then show the CFO that even a 50 percent price change barely moves TCO at this volume, so the real decisions are elsewhere.

**Q3. A CFO says your TCO is too high because a vendor quoted a platform at a fifth of your figure. How do you respond?**

I would say the vendor quote is probably correct and that it is answering a different question. A platform quote covers licences and perhaps hosting; our TCO covers the integration into the bank's systems, the data preparation, the evaluation and guardrails, model risk validation, the human review of autonomous decisions, the product team that keeps it running, the change programme, and the model migrations over three years. In our worked case the platform and licence lines were about 14 percent of three-year TCO, which is consistent with a vendor quote being a fifth of the total. I would offer to reconcile line by line so the CFO can see exactly which costs the vendor excludes, and I would point out that the excluded costs are the ones that determine whether the system reaches production at all, which is where most AI investments have failed. Finally, I would show the sensitivity table so the CFO sees that the figure is robust to platform price and sensitive to autonomy share and integration scope, which refocuses the discussion on the decisions that matter.

## 9. Self-check quiz

1. List the four extensions that AI TCO needs beyond classic IT TCO, with one sentence each on why banks cannot omit them.
2. Calculation: a KYB document-processing agent makes 6 calls per case at 12,000 input and 400 output tokens, with 50 percent cache hits. At USD 3 per million input, USD 0.30 cached, USD 15 output, what is the raw inference cost per case? With 30 percent overheads and 86,000 cases a year, what is the annual loaded inference cost?
3. Explain why cost per successful outcome can diverge from cost per case, and give a dispute example.
4. In the worked example, routing saved 56 percent of inference but only 1.5 percent of three-year TCO. Explain why, and state the condition under which routing becomes material.
5. Name three hidden costs in AI TCO and say which budget line each should sit in.

# Module 09 — The AI Business Case

## 1. Why a client cares

Every AI programme in a bank competes for the same change budget as core-banking upgrades, regulatory remediation and product launches. The CFO does not fund "AI"; the CFO funds a cash-flow profile with a risk attached to it. A consultant who cannot build, defend and stress a business case will lose the room to the Finance business partner, however good the technology story is. The business case is also the control document for the whole programme: it fixes the baseline, names the benefit owners and defines what "success" means at every stage gate.

## 2. Concept and mechanism

### 2.1 The four value levers and how each is baselined

An AI business case is a claim that a system will change one or more measurable business quantities relative to a counterfactual. The four levers below are the standard decomposition; the discipline lies in the baseline, not the lever.

| Lever | What changes | Baseline you must capture | Typical evidence quality |
|---|---|---|---|
| Cost | Unit cost of a process falls (handling time, rework, vendor spend, error cost) | Volume, fully loaded cost per FTE, handle time per task, rework rate, vendor invoices | High if time-and-motion data exists; medium if estimated from interviews |
| Revenue | Conversion, cross-sell, pricing realisation, time-to-revenue improve | Funnel conversion by stage, abandonment, revenue per client, days to first transaction | Medium; attribution is contested |
| Risk | Expected loss, fines, provisions, false negatives fall; capital may improve | Loss history, regulatory findings, false-negative rate, SAR quality metrics | Low to medium; events are rare and lumpy |
| Customer experience | NPS, complaints, churn, cycle time | Survey data, complaint logs, churn cohort data, SLA breaches | Medium; monetisation requires a churn-to-revenue link |

Baselining means measuring the as-is on the same unit of account the benefit will be reported in. If the benefit is "analyst hours released", the baseline is analyst hours per case today, by case type, measured over a representative period (at least one quarter to capture seasonality). If the baseline is interview-estimated, say so in the case and apply a haircut (see 2.5).

### 2.2 Benefit types and why the CFO discounts some of them

| Benefit type | Definition | How a CFO treats it |
|---|---|---|
| Hard, cash-releasing | Reduces a cash cost line that will actually be cut: vendor contract, overtime, contractor spend, headcount that leaves the payroll | Counted at face value if an owner signs for it |
| Hard, cost-avoiding | Prevents a future cost that is already in the plan: planned hires to meet volume growth, planned vendor uplift | Counted if the avoided cost is in an approved plan; otherwise challenged |
| Capacity release | Hours freed in a team whose headcount does not change | Heavily discounted (often 0–30%) unless redeployment is specified and funded |
| Soft | Quality, employee satisfaction, brand | Not counted in NPV; noted qualitatively |

| Benefit type | Definition | How a CFO treats it |
|---|---|---|
| Risk-adjusted loss reduction | Expected-loss change, fines avoided | Counted as probability-weighted expected value, usually with a large haircut |

The capacity-release argument is the single most common source of inflated AI cases. Releasing 20% of 100 analysts' time does not reduce cost unless 20 roles are removed, redeployed into revenue-generating work with a measurable output, or used to absorb volume growth that would otherwise have required hiring. A manager-level case states explicitly which of these three the client is committing to, and books only that.

## 2.3 Benefit ramp curves

Benefits never arrive on day one. The ramp reflects three things: deployment coverage (what share of volume the system handles), adoption (what share of eligible users actually use it) and effectiveness (how much of the theoretical benefit is realised per unit handled). A simple multiplicative model is

Realised benefit in period t = Full-run-rate benefit × Coverage(t) × Adoption(t) × Effectiveness(t)

Typical shapes for a banking back-office agentic deployment, labelled [ESTIMATE — assumption]:

| Year | Coverage | Adoption | Effectiveness | Combined ramp |
|---|---|---|---|---|
| 1 | 40% | 70% | 70% | 20% |
| 2 | 90% | 90% | 85% | 69% |
| 3 | 100% | 95% | 95% | 90% |
| 4 | 100% | 95% | 100% | 95% |
| 5 | 100% | 95% | 100% | 95% |

Note the Year 1 compound effect: three individually reasonable assumptions multiply to a 20% ramp. Juniors often put 50% in Year 1 without decomposing it.

## 2.4 The financial metrics

**Net Present Value.** NPV = Σ_{t=0..N} CF_t / (1 + r)^t, where CF_t is net cash flow in period t (benefits minus all costs including run and change) and r is the discount rate. Banks typically use a hurdle rate between 8% and 12% for change investments [ESTIMATE — assumption; VERIFY with the client's Finance function]. NPV above zero at the hurdle rate means the project creates value after the cost of capital.

**Internal Rate of Return.** IRR is the r at which NPV = 0. It is solved iteratively. Use it for comparability with other projects but be aware it overstates attractiveness for projects with early, large benefits and can be multi-valued if cash flows change sign more than once.

**Payback period.** The first period in which cumulative undiscounted (or discounted, if the client prefers) cash flow turns positive. CFOs in banks often apply an informal ceiling of 24–36 months for technology change [ESTIMATE — assumption].

**Benefit-cost ratio.** PV of benefits / PV of costs. Useful for ranking under a budget constraint.

## 2.5 Sensitivity, scenarios and risk-adjustment

Three complementary techniques:

- **Sensitivity (tornado).** Flex one input at a time across its plausible range, hold the rest at base, record the NPV swing, and rank the swings. The chart shows the CFO which three assumptions matter and therefore where to focus validation.
- **Scenarios.** Internally consistent sets of assumptions (downside, base, upside). The downside is not "everything at the pessimistic end"; it is a coherent story such as "adoption stalls because the integration to the core system slips nine months".

- **Risk-adjustment.** Two methods. (a) Probability-weight the scenarios to get an expected NPV. (b) Apply evidence-based haircuts to each benefit line before discounting. A practitioner scale [ILLUSTRATIVE — composite; attribution uncertain, variants differ by firm]:

| Evidence level for a benefit line | Haircut |
|---|---|
| Measured in a production pilot on the client's data | 0–10% |
| Measured in a PoC on client data, non-production | 20–30% |
| Benchmarked from a comparable public deployment | 30–50% |
| Vendor claim or analyst estimate only | 50–70% |
| SME opinion only | 60–80% |

### 2.6 How a CFO challenges an AI business case and how to pre-empt it

| Challenge | What the CFO is really asking | Pre-emption |
|---|---|---|
| Double counting | Is this benefit already in another programme's case (digital onboarding, ops transformation, offshoring)? | Build a benefits register across programmes; show the net-of-overlap figure |
| Baseline inflation | Did you use the busiest quarter, the most expensive FTE rate, or the worst cycle time? | Use trailing twelve months; show fully loaded cost build-up with HR sign-off |
| Ramp optimism | Why is 60% of the benefit in Year 1 when the integration backlog is 18 months? | Decompose ramp into coverage, adoption, effectiveness; tie coverage to the IT roadmap |
| Run-cost omission | Where are inference costs, model re-validation, human review, vendor uplift, and the second-line team? | Carry a full TCO (see Module 8); show run cost as a percentage of gross benefit |
| Attribution | If conversion rises, how do I know it was the AI and not the pricing change or the market? | Design a control group or phased rollout; agree the attribution method before go-live |
| Capacity release | Which roles leave, and when, and who signs? | Headcount bridge by quarter signed by the business owner; book only the signed part |
| Terminal value | Why does the benefit persist in Year 5 when the model will be obsolete? | Include model refresh cost; no benefit growth beyond volume growth |
| Downside | What is the worst case I should budget for? | Present the probability-weighted NPV and the downside scenario with its trigger |

## 3. Frameworks

### 3.1 The benefits logic tree (generic; widely used in practice)

Decompose the headline benefit into measurable drivers and link each driver to a KPI, an owner and a data source. Structure: Financial outcome → Value lever → Operational driver → KPI → Data source → Owner. Applying it: start from the P&L line, not from the use case; work down until each leaf has a number you can measure today.

### 3.2 The benefits realisation approach (attribution: UK Office of Government Commerce, Managing Successful Programmes, 2000s; widely adapted)

Steps: identify benefits; map them to outcomes and changes; assign owners; set measures and baselines; plan realisation; track and review. The AI-specific addition is a "model performance to business KPI" link: the

business benefit only flows if the model KPI (for example, straight-through-processing rate) holds in production.

### 3.3 Five-step business case construction (practitioner method; attribution uncertain, variants differ by firm)

1. Baseline: volumes, unit costs, cycle times, error and loss rates, trailing twelve months, with source for each.
2. Benefit build: lever by lever, driver by driver, at full run-rate.
3. Ramp and haircut: apply coverage, adoption, effectiveness and evidence haircuts by year.
4. Cost build: build, run, change, refresh, from a TCO model.
5. Financials and stress: NPV, IRR, payback; tornado; three scenarios; probability-weighted NPV; headcount bridge.

### 3.4 McKinsey's observation on value capture (attribution: McKinsey Global Institute, June 2023, "The economic potential of generative AI")

MGI estimated USD 200–340 billion of annual value for banking [PUBLIC — MGI, 2023]. Useful as a top-down sanity check: value in that study is concentrated in a handful of functions (customer operations, software engineering, marketing and sales, risk and legal). If a client's bottom-up case attributes most value to a function outside that list, probe why.

## 4. Worked BFSI example

### 4.1 Context

[ILLUSTRATIVE — composite] A mid-sized corporate bank in the UK onboards 6,000 new corporate clients a year through a KYB (know-your-business) process: document collection, entity verification, ultimate beneficial owner (UBO) identification, screening, risk rating, and account set-up. The bank wants a five-year business case for a supervisor-led agentic system that automates document extraction, UBO graph construction, screening triage and case-file drafting, with human analysts approving every case. Discount rate 10% [ESTIMATE — assumption].

### 4.2 Baseline (trailing twelve months)

| Item | Value | Source |
|---|---|---|
| New onboardings per year | 6,000 | Case management system |
| KYB analysts (FTE) | 110 | HR |
| Fully loaded cost per analyst FTE | GBP 65,000 | Finance (salary, on-cost, premises, tooling) |
| Analyst cost base | GBP 7.15m | 110 × 65,000 |
| Analyst hours per case (average) | 24 | Time-and-motion sample, 400 cases |
| End-to-end cycle time (median) | 38 days | Case management system |
| Applicant abandonment rate | 12% | Funnel data |
| Average first-year revenue per onboarded client | GBP 9,500 | Finance |
| Rework rate (cases returned by QC) | 18% | QC logs |
| Periodic review backlog | 2,400 cases | Compliance |

Check: 110 FTE × ~1,600 productive hours = 176,000 hours; 6,000 cases × 24 hours = 144,000 hours; the remaining ~32,000 hours go to periodic reviews and backlog. The baseline reconciles, which is the first thing a CFO's analyst will test.

## 4.3 Benefit build at full run-rate

**Lever 1: Cost (analyst capacity).** Pilot measurement [ILLUSTRATIVE — composite, modelled as if measured in a non-production PoC] showed analyst hours per case falling from 24 to 11, as document extraction, UBO mapping and screening triage move to agents and the analyst reviews and approves.

- Hours released on new onboardings: 6,000 × (24 − 11) = 78,000 hours.
- FTE equivalent: 78,000 / 1,600 = 48.75 FTE.
- The business owner commits to: 30 FTE reduced through attrition and contractor release over 24 months; 10 FTE redeployed to clear the periodic review backlog (treated as cost avoidance of planned contractor spend, GBP 0.55m a year already in budget); the remaining 8.75 FTE treated as unbooked capacity.
- Booked cost benefit: 30 × 65,000 = GBP 1.95m plus GBP 0.55m avoided contractor spend = GBP 2.50m a year.
- Unbooked (reported, not in NPV): 8.75 × 65,000 = GBP 0.57m.

**Lever 2: Revenue (abandonment and time-to-revenue).** Cycle time median falls from 38 to 15 days [ESTIMATE — assumption based on pilot]. Abandonment is assumed to fall from 12% to 8% [ESTIMATE — benchmarked from internal funnel analysis showing abandonment concentrated after day 20].

- Applicants starting = 6,000 / (1 − 0.12) = 6,818.
- Additional completed onboardings = 6,818 × (0.12 − 0.08) = 273.
- Revenue uplift = 273 × 9,500 = GBP 2.59m a year. Apply a contribution margin of 45% [ESTIMATE — assumption] to convert revenue to profit: GBP 1.17m.
- Time-to-revenue: 23 days earlier on 6,000 clients at GBP 9,500 a year = 6,000 × 9,500 × 23/365 = GBP 3.59m of revenue pulled forward once, not recurring. Treated as a one-off Year 2 benefit at 45% margin: GBP 1.62m. Many CFOs exclude pull-forward entirely; we show it separately so it can be removed.

**Lever 3: Risk (rework and quality).** Rework falls from 18% to 6% [ESTIMATE — pilot QC sample]. Each rework costs 6 analyst hours. 6,000 × (0.18 − 0.06) × 6 = 4,320 hours = 2.7 FTE = GBP 0.18m. This is already partly inside the 24-to-11 hours measurement, so to avoid double counting we book zero here and note it. Avoided regulatory remediation is not booked; it is noted qualitatively.

**Lever 4: Customer experience.** NPS for onboarding is expected to rise; not monetised separately because the abandonment benefit already captures the cash effect.

**Full run-rate booked benefit (recurring):** GBP 2.50m (cost) + GBP 1.17m (revenue margin) = GBP 3.67m a year. One-off Year 2: GBP 1.62m.

## 4.4 Haircuts for evidence

| Benefit line | Evidence level | Haircut | Run-rate after haircut |
|---|---|---|---|
| Analyst capacity (cost) | PoC on client data | 25% | GBP 1.875m |
| Abandonment (revenue margin) | Internal funnel analysis plus assumption | 40% | GBP 0.70m |
| Time-to-revenue pull-forward (one-off) | Assumption | 50% | GBP 0.81m |

Haircut run-rate recurring benefit: GBP 2.58m a year.

## 4.5 Ramp

Using the ramp from 2.3 (20%, 69%, 90%, 95%, 95%) on the recurring GBP 2.58m, and placing the one-off pull-forward in Year 2:

| Year | Ramp | Recurring benefit | One-off | Total benefit |
|---|---|---|---|---|
| 1 | 20% | 0.52 | 0 | 0.52 |
| 2 | 69% | 1.78 | 0.81 | 2.59 |
| 3 | 90% | 2.32 | 0 | 2.32 |
| 4 | 95% | 2.45 | 0 | 2.45 |
| 5 | 95% | 2.45 | 0 | 2.45 |

All figures GBP millions.

## 4.6 Costs (TCO view, see Module 8 for the build-up)

[ESTIMATE — order of magnitude, VERIFY all vendor and model pricing]

| Cost line | Year 0 | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 |
|---|---|---|---|---|---|---|
| Build: engineering, integration, data preparation | 1.60 | 0.60 | 0 | 0 | 0 | 0 |
| Build: evaluation set, model risk validation | 0.25 | 0.15 | 0 | 0 | 0 | 0 |
| Run: inference (tokens) | 0 | 0.08 | 0.18 | 0.22 | 0.23 | 0.23 |
| Run: platform, vector store, observability | 0 | 0.15 | 0.20 | 0.20 | 0.20 | 0.20 |
| Run: product and engineering team (4 FTE) | 0 | 0.40 | 0.40 | 0.40 | 0.40 | 0.40 |
| Run: model risk, evals, guardrail maintenance | 0 | 0.15 | 0.15 | 0.15 | 0.15 | 0.15 |
| Change: training, process redesign, comms | 0.20 | 0.30 | 0.10 | 0 | 0 | 0 |
| Model refresh and re-validation | 0 | 0 | 0.10 | 0.20 | 0.10 | 0.20 |
| Total cost | 2.05 | 1.83 | 1.13 | 1.17 | 1.08 | 1.18 |

All figures GBP millions. Inference cost build-up: 6,000 cases × ~350,000 tokens per case at full run-rate (documents, UBO reasoning, drafting, retries) ≈ 2.1bn tokens a year; at a blended GBP 0.11 per million tokens for a tiered model mix this is ≈ GBP 0.23m [ESTIMATE — order of magnitude, VERIFY].

## 4.7 Cash flows, NPV, IRR, payback

| Year | Benefit | Cost | Net cash flow | Discount factor (10%) | PV | Cumulative net (undiscounted) |
|---|---|---|---|---|---|---|
| 0 | 0 | 2.05 | −2.05 | 1.000 | −2.05 | −2.05 |
| 1 | 0.52 | 1.83 | −1.31 | 0.909 | −1.19 | −3.36 |
| 2 | 2.59 | 1.13 | 1.46 | 0.826 | 1.21 | −1.90 |
| 3 | 2.32 | 1.17 | 1.15 | 0.751 | 0.86 | −0.75 |
| 4 | 2.45 | 1.08 | 1.37 | 0.683 | 0.94 | 0.62 |
| 5 | 2.45 | 1.18 | 1.27 | 0.621 | 0.79 | 1.89 |

All figures GBP millions.

- NPV at 10% = −2.05 − 1.19 + 1.21 + 0.86 + 0.94 + 0.79 = GBP 0.56m.
- IRR: at 15% the PVs are −2.05, −1.14, 1.10, 0.76, 0.78, 0.63, summing to 0.08; at 16% they sum to approximately −0.01. IRR ≈ 16%.
- Payback (undiscounted): cumulative turns positive during Year 4; interpolating, −0.75 + 1.37 × f = 0 gives f ≈ 0.55, so payback ≈ 3.5 years from Year 0 (about 42 months).
- Total cost of ownership over the horizon: GBP 8.44m. Total haircut benefit: GBP 10.33m. Benefit-cost ratio on PV basis: PV benefits = 0.47 + 2.14 + 1.74 + 1.67 + 1.52 = 7.54; PV costs = 2.05 + 1.66 + 0.93 + 0.88 + 0.74 + 0.73 = 6.99; ratio 1.08.

**Reading.** On haircut, ramped benefits with full TCO, this is a marginal case: positive NPV, IRR a few points above hurdle, payback at the edge of the CFO's informal ceiling. It is honest, and it tells the client where the value is actually created.

## 4.8 What the un-haircut case would have said

Without haircuts and with a 50% Year 1 ramp (the typical junior build), recurring benefit is GBP 3.67m, Year 1 benefit GBP 1.84m, and NPV at 10% rises to roughly GBP 4.1m with payback in Year 2. The gap between GBP 4.1m and GBP 0.56m is the credibility gap the CFO will find in the first meeting. Present the conservative case and show the upside as scenarios.

## 4.9 Scenarios

| Scenario | Story | Key changes | NPV at 10% |
|---|---|---|---|
| Downside | Core system integration slips nine months; adoption stalls at 70%; hours per case only falls to 15 | Ramp 10%/45%/75%/80%/80%; cost benefit −30%; build cost +GBP 0.5m | −GBP 1.9m |
| Base | As built above | As above | GBP 0.56m |
| Upside | Pilot results hold in production (haircuts removed); periodic review agentic extension adds GBP 0.6m a year from Year 3 | No haircuts; extra benefit Y3–Y5 | GBP 4.0m |

Probability weights [ESTIMATE — assumption]: downside 30%, base 50%, upside 20%. Expected NPV = 0.3 × (−1.9) + 0.5 × 0.56 + 0.2 × 4.0 = −0.57 + 0.28 + 0.80 = GBP 0.51m. The expected value is positive but the downside is material, so the recommendation includes a Year 1 gate: proceed to scale only if measured hours per case in the first production cohort are at or below 13.

### 4.10 Tornado (one-at-a-time sensitivity on base NPV)

| Input | Low value | High value | NPV at low | NPV at high | Swing |
|---|---|---|---|---|---|
| Hours per case after automation | 15 | 9 | −0.9 | 1.3 | 2.2 |
| Booked FTE reduction | 20 | 40 | −0.9 | 2.0 | 2.9 |
| Abandonment after | 10% | 7% | −0.1 | 0.9 | 1.0 |
| Year 1 ramp | 10% | 40% | 0.3 | 1.1 | 0.8 |
| Build cost | +30% | −20% | 0.0 | 0.9 | 0.9 |
| Discount rate | 12% | 8% | 0.3 | 0.9 | 0.6 |
| Inference unit cost | ×2 | ×0.5 | 0.1 | 0.8 | 0.7 |

All NPVs GBP millions, [ILLUSTRATIVE — composite]. The two drivers that matter are the hours-per-case outcome and how much of the released capacity is actually booked. Inference cost, which engineering teams obsess over, is near the bottom. That is the "so what": validation effort should go into the production measurement of hours per case and into securing a signed headcount bridge, not into token-price negotiation.

## 5. Real-world case

**Klarna, 2024–2025** [PUBLIC — Klarna press release, Feb 2024; CEO interviews, May 2025; VERIFY current figures]. In February 2024 Klarna announced that its OpenAI-powered customer service assistant was handling two-thirds of customer service chats in its first month, doing the equivalent work of roughly 700 full-time agents, and projected a USD 40 million profit improvement for 2024. The case is instructive for business-case construction in three ways. First, the headline benefit was a capacity figure (700 FTE equivalents) rather than booked headcount, and most of those agents were outsourced, so the cash-releasing mechanism was a reduction in outsourcing spend, which is a clean, contract-based benefit. Second, in 2025 the CEO publicly acknowledged that the company had gone too far on cost and would re-hire humans to lift service quality, which illustrates the unbooked cost side of a case: customer experience degradation does not appear in Year 1 cost lines but surfaces later as churn, complaints and reversal cost. Third, the revised position was a hybrid model, which is the shape most bank business cases should assume from the start: an AI-handled share with a human tier, not full substitution.

**Commonwealth Bank of Australia, 2025** [PUBLIC — Australian media and Finance Sector Union statements, Aug 2025; VERIFY]. CBA reversed 45 call-centre redundancies after a union challenge showed that the AI voice bot had not reduced call volumes as the business case assumed, and that staff were working overtime. For the business case, this is the attribution and baseline failure in 2.6: the benefit (volume reduction) was assumed, not measured, before the headcount action was taken.

## 6. Consultant language

> *"We have booked only the capacity the business owner has signed for; the rest is reported as unbooked upside."*

Use when presenting cost benefits to Finance, to pre-empt the capacity-release challenge.

> *"Every benefit line carries an evidence haircut, so the case gets stronger as the pilot data comes in, not weaker."*

Use at the steering committee to explain why your NPV is lower than the vendor's.

> *"The tornado tells us where the case lives: two assumptions drive 70% of the variance, and both are measurable in the first production cohort."*

Use to redirect debate from token prices to the metrics that matter.

> *"This is a net-of-overlap figure; we have reconciled it against the digital onboarding and operations transformation benefit registers."*

Use when the CFO raises double counting.

> *"Our recommendation is conditional: scale is gated on a measured hours-per-case at or below 13 in the first cohort."*

Use when closing; it turns the case into a decision rule rather than a forecast.

## 7. Common mistakes juniors make

- Counting released hours as cost savings with no headcount bridge or redeployment plan.
- Using a single Year 1 ramp percentage instead of decomposing coverage, adoption and effectiveness.
- Baselining on the most expensive quarter or the gross FTE rate rather than trailing twelve months and fully loaded cost with HR sign-off.
- Omitting run costs: inference, human review, model risk validation, evaluation maintenance and model refresh.
- Double counting the same hours under "efficiency" and "quality" (rework) levers.
- Presenting a point estimate with no sensitivity, so the first challenge collapses the whole case.
- Monetising NPS or CX separately when the cash effect is already captured in abandonment or churn.
- Including revenue pull-forward as recurring benefit.
- Quoting vendor-claimed outcomes without a haircut and without naming them as vendor claims.
- Writing the case after the technology choice instead of before it, so the case justifies a decision already taken.

## 8. Interview questions with model answers

**Q1. A client's AI chatbot case claims GBP 5m a year from "freeing 25% of contact-centre agent time". How do you test it?**

Model answer. I would structure the test around baseline, mechanism and booking. On baseline, I would ask for the trailing-twelve-month handle time and contact volume by intent, and check the fully loaded agent cost; a 25% time release on 400 agents at GBP 50,000 is GBP 5m only if every freed hour becomes cash. On mechanism, I would decompose the 25% into containment rate (share of contacts fully handled by the bot), deflection quality (are deflected contacts coming back through another channel) and handle-time change on contacts the bot assists but does not close. On booking, I would ask which of three things the business is committing to: headcount reduction through attrition, redeployment into measurable revenue work, or absorbing planned volume growth. Only the committed part is booked; the rest is unbooked capacity. Then I would apply an evidence haircut (a vendor benchmark deserves 50% or more), decompose the ramp into coverage, adoption and effectiveness, and add run cost: inference, human escalation tier, model risk and content maintenance. Typically a GBP 5m claim becomes GBP 1.5–2.5m of defensible benefit, and I would present that number with the upside as a scenario. The CBA 2025 reversal is a useful reference for what happens when volume reduction is assumed rather than measured [PUBLIC — Aug 2025; VERIFY].

**Q2. The CFO says the discount rate should be 15% for AI because it is risky. How do you respond?**

Model answer. I would separate two kinds of risk. Systematic, cost-of-capital risk belongs in the discount rate and should be the same hurdle the bank uses for comparable technology change, typically 8–12% [ESTIMATE]. Project-specific risk, such as adoption stalling or the model under-performing, is better handled explicitly in the cash flows through evidence haircuts, a decomposed ramp and probability-weighted scenarios, because that makes the risk visible and manageable rather than hidden in a rate. Inflating the discount rate penalises later cash flows disproportionately, which happens to be where AI benefits sit after the ramp, so it systematically biases against exactly the projects with sound, slower ramps. I would offer the CFO both views: NPV at the standard hurdle with risk in the cash flows, and the same cash flows at 15% as a stress. If the project only works at the standard rate with un-haircut benefits, the CFO is right to be sceptical. If it survives haircuts at the standard rate, a higher rate adds little information. I would also propose gating: release scale funding only when the first production cohort hits the measured KPI, which reduces the real risk rather than just repricing it.

**Q3. Walk me through how you would build the business case for an agentic sanctions-screening triage system.**

Model answer. Start with the baseline: alert volumes by list and channel, false-positive rate (often above 95% in payment screening [ESTIMATE — common practitioner range, VERIFY for the client]), analyst minutes per alert at level 1 and level 2, fully loaded cost, SLA breaches and any regulatory findings. The primary lever is cost: the agent triages and closes clear false positives with a documented rationale, so level-1 minutes per alert fall and a share of alerts never reach a human. I would measure that in a back-test on six months of historical alerts, with the compliance function validating the agent's dispositions against the analysts' originals, and I would book only the share of alerts where the agent's agreement with the human disposition exceeds the agreed threshold. Secondary levers are risk (consistency of dispositions, fewer missed true matches through better narrative quality) and speed (payments released faster, fewer customer complaints), both treated qualitatively or lightly haircut. Costs include build and integration with the screening engine, the inference cost per alert, a human sampling tier, independent model validation under the bank's model risk framework, and ongoing evaluation as sanctions lists change. I would ramp conservatively, because compliance will insist on a shadow-mode period, and I would present NPV, payback and a tornado, with the headline that the case depends on the validated auto-close rate and on the compliance function accepting that rate.

## 9. Self-check quiz

1. A benefit line is based on a vendor's published case study from another bank. What haircut range would you apply, and why?
2. Explain the difference between capacity release and cash-releasing cost benefit, and name the three ways capacity release can become bookable.
3. A project has net cash flows of −3.0, −0.5, 1.8, 2.0, 2.0 (GBP millions) in Years 0–4. Compute NPV at 10% and the undiscounted payback period.
4. Your Year 1 assumptions are coverage 50%, adoption 80%, effectiveness 75%. What is the combined Year 1 ramp, and what does this imply about a case that assumes 50% Year 1 benefit?
5. Name four challenges a CFO will raise against an AI business case and the specific pre-emption for each.

# Module 10 — Build vs Buy vs Partner, Model Selection and RFP Evaluation

## 1. Why a client cares

The sourcing decision locks in cost structure, speed, control and regulatory exposure for three to five years. Buying a packaged AI product gets a bank to a pilot in weeks but can leave it unable to explain the model to its regulator or to switch vendors when pricing changes. Building gives control but needs scarce talent and takes longer. Partnering shares risk but blurs accountability. Boards increasingly ask a second question: how exposed are we to a single model provider, and what happens when that provider changes its model, its terms or its price.

## 2. Concept and mechanism

### 2.1 The layered stack and where the decision actually sits

"Build vs buy" is not one decision. An AI solution in a bank is a stack, and the sourcing answer differs at each layer.

| Layer | What it contains | Typical bank posture | Why |
|---|---|---|---|
| Foundation model | The LLM or ML model weights and inference | Buy (API) or buy (open-weight, self-hosted); almost never build | Training frontier models costs hundreds of millions of dollars; no bank has the data or the rationale |
| Platform | Model gateway, orchestration, vector store, evaluation harness, observability, guardrails, prompt and agent registry | Buy hyperscaler or ISV components, assemble in-house | Commodity components, but the assembly and the control plane are a differentiating capability |
| Application | The agent workflows, prompts, tools, business rules, integration to core systems | Build for differentiating processes; buy for commodity functions | This layer encodes the bank's process knowledge, controls and risk appetite |
| Services | Design, integration, change, run | Partner for surge capacity and specialist skills; retain product ownership | Capability transfer matters more than labour arbitrage |

The consultant's job is to force the client to make the decision layer by layer rather than as a single "we are an X shop" statement.

### 2.2 Decision logic

Five criteria decide the posture at the application layer, where most of the debate happens:

1. **Differentiation.** Does the process create competitive advantage or regulatory risk that is specific to this bank? Dispute resolution logic tied to a bank's own card portfolio and network rules is differentiating; invoice OCR is not.
2. **Data sensitivity and residency.** Can the data leave the bank's environment? If customer transaction data or KYC documents must stay in-region and on controlled infrastructure, a SaaS product that processes data in a vendor cloud may be excluded or require a private deployment.
3. **Speed to value.** How fast must the client show results, and what is the cost of delay? Buy wins where the use case is commodity and the client's engineering capacity is committed elsewhere.

4. **Talent.** Does the bank have, or can it hire and retain, the engineering, LLMOps and model risk skills to build and run? If not, building creates a key-person dependency that the regulator will treat as operational risk.
5. **Lock-in and exit.** How expensive is it to switch? Lock-in is highest when prompts, evaluation data, fine-tuned models and workflow logic live inside a vendor's proprietary format with no export.

A compact rule set [ILLUSTRATIVE — practitioner heuristic]:

| Differentiating? | Data sensitive? | Talent available? | Posture |
|---|---|---|---|
| Yes | Yes | Yes | Build on bought platform |
| Yes | Yes | No | Partner to build, with capability transfer and IP ownership |
| Yes | No | Any | Build or partner |
| No | Yes | Any | Buy with private deployment or hyperscaler-native service |
| No | No | Any | Buy SaaS |

## 2.3 Model selection criteria

Model choice is now a portfolio decision: most production systems route different tasks to different models. Criteria to evaluate per task type:

- **Capability on domain evaluations.** Public benchmarks (MMLU, GPQA, SWE-bench and similar) are weak predictors of performance on a bank's own documents and tasks. The decisive evidence is the model's score on the client's golden dataset for the specific task (see Module 4). A model that leads public leaderboards can underperform a smaller model on structured extraction from a particular document type.
- **Cost per task.** Input and output token prices, with caching and batch discounts, multiplied by the realistic token profile of the task, including retries and tool-call overhead. Compare cost per completed task, not price per million tokens.
- **Latency.** Time to first token and tokens per second matter for interactive use; throughput and rate limits matter for batch. Agentic workflows multiply latency because each step is a model call.
- **Context window and long-context quality.** A large window is not the same as reliable use of information in the middle of it; test retrieval accuracy at the real document lengths.
- **Tool-calling reliability.** Share of calls where the model emits a schema-valid tool call with correct arguments. This is the dominant quality driver for agents and varies materially between models.
- **Instruction following and refusal behaviour.** Over-refusal on legitimate financial-crime content (for example, describing a laundering typology) is a real operational problem.
- **Data residency, retention and training-use terms.** Where is inference served, is prompt data retained, is it used for training? Enterprise tiers normally offer zero retention; verify contractually [VERIFY — vendor terms change].
- **Deployment options.** API only, hyperscaler-hosted, virtual private cloud, or open-weight self-hosted. Self-hosting open-weight models (for example Llama, Mistral, Qwen families [VERIFY — model availability and licences change]) gives residency control and predictable cost at the price of operational burden.
- **Licensing for open-weight models.** Read the licence: some restrict commercial use above a user threshold or require attribution; some are genuinely permissive.
- **Model lifecycle.** Deprecation notice periods, versioning and the ability to pin a version. A silent model update can change behaviour and invalidate an evaluation and a model risk approval.

- **Safety and security posture.** Prompt-injection resistance, jailbreak behaviour, content filters, and the vendor's own red-teaming disclosures.

## 2.4 Vendor selection and RFP evaluation

A bank RFP for an AI product or platform should score six families of criteria. Weights below are a defensible starting point [ILLUSTRATIVE — composite; adjust to the client's priorities]:

| Family | Weight | Examples of criteria |
|---|---|---|
| Functional fit | 25% | Coverage of the use case; accuracy on the client's test set; workflow and HITL features; configurability without vendor change requests |
| Non-functional | 15% | Latency, throughput, availability SLA, scalability, integration options (APIs, event hooks), observability exposed to the client |
| Security, compliance, model risk | 20% | Data residency; retention and training-use terms; certifications (ISO 27001, SOC 2, ISO/IEC 42001 [VERIFY]); explainability artefacts; audit logs; support for the bank's model validation; EU AI Act provider documentation |
| Commercials | 15% | 3-year TCO on the client's volume profile; pricing model (per seat, per transaction, per token, platform fee); price-change protections; professional services rates |
| Roadmap and viability | 10% | Financial stability; customer base in regulated FS; roadmap alignment; model-provider dependency and multi-model support; reference calls |
| Exit and portability | 15% | Export of prompts, configurations, evaluation data, fine-tuned weights, logs; transition assistance; termination for convenience; escrow where relevant |

Process: issue the RFP with the client's anonymised test set and require vendors to run it in a supervised proof of value; score blind where possible; separate commercial scoring from functional scoring so price does not colour capability judgement; weight exit heavily because it is where buyers regret most.

## 2.5 Contract clauses that matter for AI

| Clause | Why it matters | What to ask for |
|---|---|---|
| Data use and training | Default terms may allow the vendor to use prompts and outputs to improve models | Explicit prohibition on training use; zero or minimal retention; sub-processor list |
| Model change notification | A model upgrade can alter behaviour and break the bank's validation | Minimum 90 days' notice of model version changes; ability to pin a version for a defined period; re-validation support [ESTIMATE — negotiable] |
| Accuracy and performance SLAs | Standard SLAs cover uptime, not quality | Agreed quality metrics on the client's test set, measured periodically, with service credits or exit rights on sustained breach |
| Indemnity and IP | Output may infringe third-party rights | IP indemnity for outputs; clarity on who |

| Clause | Why it matters | What to ask for |
|---|---|---|
| | or produce harmful content | owns prompts, fine-tuned weights, evaluation data (the bank should) |
| Liability | Regulatory fines and customer harm from model errors | Liability caps that reflect regulatory exposure, not just fees paid; carve-outs for data breach and confidentiality |
| Regulatory cooperation | Supervisors will ask for documentation and may inspect | Obligation to provide documentation (model cards, risk assessments, EU AI Act technical documentation where the vendor is the provider), audit rights, and support for regulator requests |
| Exit and portability | Lock-in | Export formats, transition services, data deletion certification |
| Price protection | Token prices move; seat prices creep | Caps on annual increases; most-favoured-customer or benchmark clauses where obtainable |
| Operational resilience | DORA in the EU, PRA outsourcing rules in the UK [VERIFY — regulatory scope] | Resilience testing, incident notification timelines, exit plans, concentration risk disclosure |

# 3. Frameworks

## 3.1 Layered sourcing decision (practitioner framework; attribution uncertain, widely used)

Apply the five criteria of 2.2 at each layer of the stack in 2.1 separately, document the posture per layer, and then test the overall architecture for coherence (for example, a bought application that cannot use the bank's chosen model gateway is an integration problem, not a product problem).

## 3.2 Weighted scoring for RFP evaluation (generic method)

Steps: define criteria and sub-criteria; assign weights that sum to 100% with stakeholder sign-off before proposals are opened; define a 1–5 scoring scale with written anchors; score independently then calibrate in a panel; multiply and sum; run a sensitivity on weights to check the ranking is stable; document rationale for audit. The sensitivity step is the one juniors skip and procurement auditors look for.

## 3.3 Concentration and exit risk lens (regulatory origin: EBA outsourcing guidelines 2019, UK PRA SS2/21 outsourcing and third-party risk management 2021, EU DORA applicable from Jan 2025 [VERIFY])

For any material AI vendor, the bank must assess criticality, concentration (how many processes depend on this vendor or on the model provider behind it), substitutability and exit plan. Build this into the RFP so the regulatory analysis is done before signature.

## 3.4 Total cost of ownership comparison (TCO concept: Gartner, Bill Kirwin, 1987)

Compare options on the same volume profile over the same horizon with the same cost categories (licence or usage, infrastructure, integration, internal run team, change, validation, exit). See Module 8 for the categories.

# 4. Worked BFSI example

## 4.1 Context

[ILLUSTRATIVE — composite] A European universal bank needs an AML alert-narrative tool: for each transaction-monitoring alert escalated to level 2, the tool assembles the customer, transaction and screening context, drafts the investigation narrative and the suspicious activity report (SAR) rationale, and cites the evidence. Volume: 180,000 level-2 alerts a year. Analysts currently spend 55 minutes per narrative. Data must remain in the EU. The bank has a small platform team and a hyperscaler contract already in place.

Options evaluated:

- **Vendor A:** specialist RegTech SaaS with a pre-built AML investigation copilot; EU-hosted tenant; proprietary model orchestration with a choice of two LLM providers.
- **Vendor B:** the bank's existing transaction-monitoring vendor's add-on module; deepest integration; single LLM provider; narrative templates fixed.
- **Vendor C:** a horizontal "enterprise agent platform" that would need the bank to configure the AML workflow itself; strong tooling; US-headquartered with EU hosting option.
- **Build:** in-house application on the bank's hyperscaler platform using the model gateway, with a systems integrator partner for the first nine months.

## 4.2 Functional test on the bank's data

All four options ran a supervised proof of value on 400 historical alerts with analyst-written narratives as reference. Metrics: narrative completeness (share of required elements present), factual faithfulness to source records (checked by analysts), and analyst edit time.

| Option | Completeness | Faithfulness | Analyst edit time (min) | Functional score (1–5) |
|---|---|---|---|---|
| Vendor A | 91% | 96% | 14 | 4.5 |
| Vendor B | 84% | 97% | 22 | 3.5 |
| Vendor C (configured in PoV) | 88% | 93% | 18 | 3.5 |
| Build (prototype in PoV) | 89% | 95% | 16 | 4.0 |

[ILLUSTRATIVE — composite]

## 4.3 Scored evaluation

Scores 1–5; weights from 2.4.

| Criterion (weight) | Vendor A | Vendor B | Vendor C | Build |
|---|---|---|---|---|
| Functional fit (25%) | 4.5 | 3.5 | 3.5 | 4.0 |
| Non-functional (15%) | 4.0 | 4.5 | 4.0 | 3.5 |
| Security, compliance, model risk (20%) | 4.0 | 4.5 | 3.0 | 4.5 |
| Commercials (15%) | 3.0 | 3.5 | 3.5 | 3.5 |
| Roadmap and viability (10%) | 3.5 | 4.0 | 4.0 | 3.0 |
| Exit and portability (15%) | 2.5 | 2.0 | 3.5 | 5.0 |

Weighted totals:

- Vendor A: 4.5×0.25 + 4.0×0.15 + 4.0×0.20 + 3.0×0.15 + 3.5×0.10 + 2.5×0.15 = 1.125 + 0.60 + 0.80 + 0.45 + 0.35 + 0.375 = **3.70**
- Vendor B: 0.875 + 0.675 + 0.90 + 0.525 + 0.40 + 0.30 = **3.68**
- Vendor C: 0.875 + 0.60 + 0.60 + 0.525 + 0.40 + 0.525 = **3.53**
- Build: 1.00 + 0.525 + 0.90 + 0.525 + 0.30 + 0.75 = **4.00**

Weight sensitivity: if exit weight is cut to 5% and functional raised to 35%, Vendor A moves to 3.90 and Build to 3.90, a tie. If security weight rises to 30% and commercials fall to 5%, Build leads at 4.10 against Vendor B at 3.78. The Build ranking is robust except under a strongly functional-only weighting, which the panel documented.

## 4.4 Three-year cost comparison

[ESTIMATE — order of magnitude, VERIFY all pricing] EUR millions; 180,000 alerts a year; horizon three years from contract.

| Cost line | Vendor A | Vendor B | Vendor C | Build |
|---|---|---|---|---|
| Licence / platform fees | 2.70 (EUR 5 per alert) | 1.95 (module fee + per-alert) | 1.50 (platform seats + usage) | 0.45 (gateway, vector store, observability) |
| Model inference (pass-through or direct) | Included | Included | 0.60 | 0.55 |
| Integration and configuration | 0.60 | 0.30 | 1.10 | 1.40 (incl. SI partner) |
| Internal run team | 0.45 (1 FTE × 3 years) | 0.30 | 0.90 (2 FTE) | 1.35 (3 FTE) |
| Model risk validation and evals | 0.30 | 0.25 | 0.35 | 0.40 |
| Change and training | 0.25 | 0.20 | 0.25 | 0.25 |
| Exit provision (Year 3) | 0.30 | 0.40 | 0.15 | 0.05 |
| Total 3-year TCO | 4.60 | 3.40 | 4.85 | 4.45 |

Inference build-up for Build: 180,000 alerts × ~60,000 tokens per narrative (context assembly, drafting, citation checks, retries) = 10.8bn tokens a year; at a blended EUR 0.017 per thousand tokens using a mid-tier model for assembly and a frontier model for the final narrative, ≈ EUR 0.18m a year, EUR 0.55m over three years [ESTIMATE — order of magnitude, VERIFY].

## 4.5 Recommendation and reasoning

Vendor B is cheapest and the deepest-integrated but scores lowest on exit (narratives and configuration locked into the monitoring vendor, and the bank is already concentrated on that vendor for monitoring itself), and its narrative quality required the most analyst editing. Vendor A has the best out-of-box quality but a per-alert pricing model that scales with volume and weak portability. Build scores highest overall, is mid-table on cost, and gives the bank ownership of prompts, evaluation data and workflow, which the compliance function valued because the SAR narrative is a regulator-facing artefact the bank must be able to explain in its own terms.

Recommendation: Build on the existing hyperscaler platform with the SI partner for nine months and explicit capability transfer, and retain Vendor A as a benchmark through a 12-month option. Benefit side for context: 180,000 alerts × (55 − 16) minutes = 117,000 analyst hours a year ≈ 73 FTE equivalents; booked at 40% after haircut and headcount bridge ≈ EUR 2.0m a year [ILLUSTRATIVE — composite]. All four options pay back inside three years; the decision therefore rests on control, explainability and exit rather than cost. That is the “so what” for the client: when the benefit dwarfs the cost difference, choose on risk and control.

## 5. Real-world case

**Morgan Stanley and OpenAI, 2023–2024** [PUBLIC — Morgan Stanley press releases Sept 2023 and June 2024; VERIFY adoption figures]. Morgan Stanley took a buy-the-model, build-the-application posture: it licensed GPT-4 through a strategic relationship with OpenAI and built its own "AI @ Morgan Stanley Assistant" on top of its proprietary research and wealth-management content for roughly 16,000 advisers, followed by "Debrief" for meeting summaries in June 2024. Reported adoption among adviser teams was around 98%. The lesson for sourcing is that the differentiating asset was the firm's curated content library and its evaluation discipline (the firm publicly described building evaluation frameworks with advisers and compliance), not the model. The model was bought; the retrieval corpus, prompts, evaluation and workflow were built.

**JPMorgan Chase LLM Suite, 2024** [PUBLIC — press reporting 2024; VERIFY]. JPMorgan rolled out LLM Suite to around 200,000 employees as an internal platform that fronts external models, which illustrates the platform-layer decision: build the gateway and control plane, buy the models behind it, so that models can be swapped and usage governed centrally.

**NatWest, 2024–2025** [PUBLIC — NatWest and IBM announcements 2024; OpenAI collaboration Mar 2025; VERIFY]. NatWest's Cora+ built on IBM watsonx and its later OpenAI collaboration show a multi-provider posture at the model layer for a customer-facing assistant, which reduces single-provider concentration risk.

## 6. Consultant language

> *"This is not one build-or-buy decision; it is four decisions, one per layer of the stack."*

Use at the start of any sourcing discussion to reframe the debate.

> *"We score on the client's own test set, not on public leaderboards."*

Use when a vendor or an internal sponsor leads with benchmark claims.

> *"We weight exit at 15% because that is where buyers of AI products have regretted most."*

Use when procurement proposes a cost-heavy weighting.

> *"Model change notification and version pinning are model-risk controls, not commercial nice-to-haves."*

Use with the CRO and model risk team when reviewing contracts.

> *"When the benefit dwarfs the cost difference between options, we choose on control, explainability and exit."*

Use when presenting a recommendation that is not the cheapest.

## 7. Common mistakes juniors make

- Treating build vs buy as a single enterprise-level choice rather than a per-layer decision.
- Ranking models on public benchmarks instead of the client's own golden dataset for the specific task.
- Comparing price per million tokens instead of cost per completed task including retries and tool-call overhead.
- Letting commercial scores leak into functional scoring by running them in the same panel.
- Ignoring exit and portability, then discovering prompts, evaluation data and configurations cannot be exported.
- Accepting default data-use terms that permit training on the bank's prompts.

- Not asking for model change notification and version pinning, so a silent upgrade invalidates validation.
- Choosing the incumbent vendor's module without assessing concentration risk under outsourcing and resilience rules.
- Scoping the SI partner for delivery without capability transfer, creating permanent dependency.
- Running a proof of value on vendor-selected data rather than the bank's representative, anonymised sample.

## 8. Interview questions with model answers

**Q1. A bank CIO says "we are a buy-not-build shop" and wants to purchase an end-to-end agentic KYB product. How do you respond?**

Model answer. I would accept the principle at the foundation-model and platform layers, where buying is right for almost every bank, and challenge it at the application layer, where the KYB workflow encodes the bank's risk appetite, its policies and its regulator-facing decisions. I would apply five criteria: differentiation (KYB decisions are supervised and bank-specific, so high), data sensitivity (customer documents and UBO data, high), speed (buying gets a pilot in weeks, which is real value), talent (if the bank lacks LLMOps and model risk skills, building alone is risky), and exit (an end-to-end product tends to lock prompts, policies and case data into a proprietary format). My likely recommendation is a hybrid: buy the platform components and possibly a document-extraction product, build or co-build the KYB agent workflow on the bank's own gateway with a partner under a capability-transfer arrangement, and keep the policy logic in a rules layer the bank owns. I would also run a supervised proof of value on the bank's own anonymised cases before any decision, because packaged accuracy claims rarely survive contact with a bank's document mix. Finally, I would remind the CIO that under outsourcing and operational resilience rules the bank remains accountable for the decision regardless of who built the tool.

**Q2. How would you structure the evaluation criteria and weights for an RFP for a GenAI customer-service platform, and how would you defend the weights?**

Model answer. Six families: functional fit around 25%, non-functional 15%, security and compliance including model risk 20%, commercials 15%, roadmap and viability 10%, exit and portability 15%. I would defend them on three grounds. First, functional fit is capped at a quarter because every shortlisted vendor will pass a basic functional bar; the differentiation among them is smaller than the differentiation on risk and exit. Second, security, compliance and model risk at a fifth reflects that a customer-facing GenAI system is a conduct-risk surface: data residency, retention, training-use terms, audit logs and support for the bank's validation are gating. Third, exit at 15% reflects observed buyer regret: prompts, conversation data and configurations are the bank's intellectual property and must be exportable. I would get the weights signed off by the sponsor, risk and procurement before proposals open, use written scoring anchors, score functional criteria on the bank's own test set in a supervised proof of value, keep commercial scoring in a separate panel, and run a weight-sensitivity analysis to show whether the ranking is stable. If it flips under plausible alternative weights, that is a decision for the steering committee, not for the scoring model.

**Q3. What contract terms would you insist on with a foundation-model provider for a bank, and why?**

Model answer. Five groups. Data terms: zero or minimal retention, explicit prohibition on using prompts or outputs for training, named sub-processors and regional inference so data stays in-jurisdiction. Model lifecycle: minimum notice of model version changes, the ability to pin a version for a defined period, and deprecation timelines long enough to re-validate under the bank's model risk framework; a silent model change can invalidate an approved validation. Quality and resilience: standard uptime SLAs plus agreed quality metrics on the bank's test set measured periodically, incident notification timelines consistent with DORA and PRA outsourcing expectations [VERIFY scope], and concentration-risk disclosure. Liability and IP: output IP indemnity, clarity that prompts, fine-tunes and evaluation data belong to the bank, and liability caps that reflect regulatory exposure. Exit: export of all configurations and logs, transition assistance,

deletion certification. I would also ask for regulatory cooperation obligations, because supervisors increasingly expect banks to produce vendor documentation on request, and for price protection such as caps on annual increases given how quickly token prices and packaging change [VERIFY].

## 9. Self-check quiz

1. Name the four layers of the AI stack and state the typical bank sourcing posture at each.
2. Why is scoring models on the client's golden dataset preferred over public benchmarks? Give two reasons.
3. In the worked example, recompute Vendor A's weighted score if exit weight falls to 5% and functional fit rises to 35%. Does Vendor A overtake Build?
4. List five contract clauses specific to AI vendors and the risk each addresses.
5. A vendor offers per-alert pricing of EUR 5 at 180,000 alerts a year. Volume is expected to grow 15% a year. What is the three-year licence cost, and what pricing protection would you seek?

# Module 11 — From Proof of Concept to Scale: Stage Gates and Escaping Pilot Purgatory

## 1. Why a client cares

Most banks now have dozens of GenAI proofs of concept and very few systems in production carrying real volume. The board sees spend without P&L impact, the CRO sees ungoverned experiments, and the business sees tools that never reach the front line. The consultant who can define what each stage must prove, who must decide, and what has to be true before the first line of code is written is the one who turns a portfolio of demos into a portfolio of running systems.

## 2. Concept and mechanism

### 2.1 Stage definitions

The four stages differ in what question they answer, not just in size.

| Stage | Question it answers | Typical duration (banking) | Data | Users | Integration | Governance |
|---|---|---|---|---|---|---|
| Proof of concept (PoC) | Can the technique do the task at acceptable quality on our data? | 4–8 weeks | Historical, anonymised or synthetic sample | Project team, a few SMEs | None or mocked | Sandbox approval, data use sign-off |
| Pilot | Does it work in the real workflow with real users, and what is the measured benefit? | 8–16 weeks | Live or shadow production data, bounded scope | One team or site, 10–50 users | Read-only or shadow integration | Model risk tier assigned; HITL mandatory; pilot risk assessment |
| Minimum viable product (MVP) | Can it run in production with controls, at a subset of volume, and hold its KPIs? | 3–6 months | Production | Defined user population, partial volume | Production integration with core systems | Full model validation; change approval; monitoring live |
| Scale | Does the benefit hold across the full population and can the organisation run it? | 6–18 months | Production, full volume | All intended users, possibly multiple geographies | Hardened, resilient, observable | Ongoing monitoring, periodic re-validation, benefits tracking |

[ESTIMATE — typical ranges; durations vary with integration complexity and regulatory tier]

The commonest confusion is between PoC and pilot. A PoC proves technical feasibility on a sample; it never proves business value, because it runs outside the workflow. A pilot runs in the workflow and measures behaviour change and benefit. A programme that calls a PoC a pilot will book benefits it has not measured.

### 2.2 Exit criteria by stage

Exit criteria must be defined before the stage starts and should cover four dimensions.

| Dimension | PoC exit | Pilot exit | MVP exit | Scale checkpoint |
|---|---|---|---|---|
| Technical | Task accuracy on golden set above threshold; hallucination/faithfulness within tolerance; latency acceptable | Accuracy holds on live data; drift between golden set and live within tolerance | Accuracy, latency and availability hold under production load; monitoring alerts tested | Performance stable across segments and geographies; re-validation cadence met |
| Operational | Workflow design agreed; integration path identified and costed | Users adopt (usage rate); handle time or throughput improves; HITL override rate within band | Straight-through-processing rate and exception volumes within plan; run team staffed | Benefits realised against plan; operating model embedded |
| Business | Benefit hypothesis quantified with baseline | Benefit measured on pilot population and extrapolated with haircut | Benefit tracked monthly; business owner signs off | NPV tracking against case; next-wave decisions |
| Risk | Data use approved; initial model risk classification; no prohibited uses | Pilot risk assessment closed; incidents logged and reviewed; fairness checks on pilot outputs | Independent validation complete; controls tested; regulatory notifications made if required | Audit findings closed; periodic review complete |

## 2.3 Designing success criteria

Good success criteria are specific, measured by an agreed method, owned by a named person, and set before the work starts. Four rules:

1. **Pair every model metric with a business metric.** Extraction accuracy of 95% means nothing unless it is paired with "analyst review time per document falls from 12 to 5 minutes". The model metric is the leading indicator; the business metric is what gets funded.
2. **Set thresholds with the second line.** The accuracy threshold for a sanctions triage agent is a risk appetite statement, not an engineering choice. Compliance and model risk set it; engineering meets it.
3. **Define the measurement method.** Who samples, how many, how often, compared against what reference. A 2% difference in accuracy is noise on 100 samples and signal on 2,000.
4. **Include a stopping rule.** State what result ends the project. If nothing could stop it, it is not an experiment.

## 2.4 Pilot purgatory: causes and remedies

Pilot purgatory is the state in which a PoC or pilot shows promising results but never reaches production. External evidence that it is widespread: Gartner predicted in July 2024 that at least 30% of GenAI projects would be abandoned after proof of concept by the end of 2025, citing poor data quality, inadequate risk controls, escalating costs and unclear business value [PUBLIC — Gartner, Jul 2024; VERIFY]; Gartner predicted in June 2025 that over 40% of agentic AI projects would be cancelled by the end of 2027 [PUBLIC — Gartner, Jun 2025; VERIFY]; MIT NANDA's "GenAI Divide" report in August 2025 reported that around 95% of enterprise GenAI pilots delivered no measurable P&L impact, with methodological caveats including a small, self-reported sample [PUBLIC — MIT NANDA, Aug 2025; VERIFY]; and RAND found in 2024 that over 80% of AI projects fail, about twice the rate of non-AI IT projects [PUBLIC — RAND, 2024].

| Cause | How it shows up | Remedy |
|---|---|---|
| No production path defined | PoC built on a laptop or sandbox with no integration design; "productionising" becomes a new project nobody funded | Before PoC starts, document the target architecture, integration points, data access route and the budget for |

| Cause | How it shows up | Remedy |
|---|---|---|
| | | integration; fund the pilot and MVP in principle at PoC approval |
| No business owner | Innovation team or IT runs the PoC; the business sees it as “their” tool | Named business owner with P&L accountability signs the benefit hypothesis and the success criteria at PoC start |
| Data access not secured | PoC on synthetic or extracted data; production data access requires a six-month approval | Start data access and data protection approvals in parallel with the PoC; use the PoC to specify exactly what data is needed |
| Model risk engaged too late | Validation starts at MVP, finds the model untiered and the evals inadequate, and stops the launch | Classify the model and agree the evidence pack with model risk at PoC start; design the golden dataset with validation in mind |
| Success criteria vague or moving | “Promising results”; the demo impresses but nobody can say whether to proceed | Quantified exit criteria with a stopping rule, signed before the stage |
| Benefit not measured | Benefit remains a hypothesis; the business case cannot be updated | Pilot in the workflow with a control group or before/after measurement; haircut and extrapolate |
| Technology chosen before problem | Tool looking for a use case; weak fit; low adoption | Use-case assessment first (Module 7); the technology is an output |
| Run costs and run team ignored | No one owns the system after the consultants leave; costs appear as a surprise | TCO and operating model defined at pilot exit (Module 8, Module 13) |
| Change capacity absent | Users not trained; process not redesigned; the tool is bolted onto the old workflow | Change plan and process redesign funded as part of the MVP, not after |
| Point solutions without platform | Each PoC builds its own gateway, vector store and evals; the tenth PoC costs as much as the first | Platform and pattern library investment after the second or third PoC |

## 2.5 What to decide before a PoC starts

A PoC should not begin until the following are written down and signed:

- The business problem, the baseline metric and the benefit hypothesis in numbers.
- The business owner, the product owner and the model risk contact.
- The target production architecture and the integration path, with an order-of-magnitude cost.
- The data needed, where it lives, who approves access, and the expected approval timeline.
- The initial model risk classification and the evidence model risk will need.
- The success criteria, measurement method and stopping rule.
- The funding path: what is funded now, what is funded in principle if criteria are met, and who decides.
- The intended HITL design and autonomy level for the pilot (Module 3).

## 2.6 Scaling playbook

Scaling is an organisational problem more than a technical one. BCG’s 10-20-70 heuristic attributes roughly 10% of the effort to algorithms, 20% to technology and data and 70% to people, process and change [PUBLIC — BCG, 2023–24]. Elements of a scaling playbook:

- **Platform reuse.** A shared gateway, vector store, evaluation harness, observability and guardrail service, so that the marginal use case costs a fraction of the first. Measure the “time to second use case”.
- **Pattern library.** Reference architectures for the recurring patterns in banking: document extraction and summarisation, knowledge assistant with RAG, alert triage agent, case-file drafting, code modernisation. Each pattern carries its control set and evaluation template.
- **Reusable controls.** Pre-approved guardrails, logging standards and HITL patterns so each new use case inherits an approved baseline and model risk reviews the delta.
- **Change capacity.** Dedicated change and training resources per wave; adoption measured and managed as a KPI.
- **Benefits tracking.** A benefits register with owners, baselines and monthly actuals, reviewed by Finance.
- **Portfolio governance.** A single prioritised backlog across the bank, with a gate process that kills projects that miss criteria and reallocates funding.

# 3. Frameworks

## 3.1 Stage-gate process (attribution: Robert G. Cooper, Stage-Gate, 1980s–1990s; adapted to AI by practitioners)

Stages of work separated by gates at which a defined group makes a go, kill, hold or recycle decision against pre-set criteria. Application to AI: four stages as in 2.1; gate owners are the business owner, model risk and the AI portfolio lead; criteria as in 2.2; every gate has a kill option that is actually used.

## 3.2 Hypothesis-driven experimentation (Lean Startup, Eric Ries, 2011; adapted)

Frame each stage as a hypothesis with a measurable prediction, run the smallest experiment that can falsify it, and decide. In banking, “smallest” is constrained by control requirements, so the experiment design includes the controls.

## 3.3 Four-dimension success criteria grid (practitioner framework; attribution uncertain, widely used)

Technical, operational, business and risk criteria at each gate, as in 2.2. Apply by filling the grid with the second line in the room, then signing it before the stage begins.

## 3.4 Pilot-to-production readiness checklist (practitioner; composite)

Architecture and integration; data access and lineage; model risk validation evidence; controls and HITL; monitoring and incident process; run team and operating model; change and training; benefits tracking; regulatory notifications; exit and rollback plan. Apply at MVP gate; every item needs an owner and a status.

# 4. Worked BFSI example

## 4.1 Context

[ILLUSTRATIVE — composite] A regional bank’s payments-screening operation generates 420,000 sanctions alerts a year across SWIFT, SEPA and domestic rails. Over 97% are false positives [ESTIMATE — within common practitioner ranges for payment screening; VERIFY for any client]. Level-1 analysts spend an average of 9 minutes per alert to disposition; 38 FTE at a fully loaded GBP 55,000. The bank wants an agentic triage system that gathers context (name-match details, counterparty, payment purpose, prior dispositions, list entry metadata), proposes a disposition with a rationale, and auto-closes clear false positives within a defined confidence band, escalating the rest with a pre-filled narrative.

## 4.2 Stage-gate plan with quantified gates

**Gate 0 (before PoC): decisions recorded**

- Business owner: Head of Payments Screening. Model risk tier: provisional Tier 2 (material decision support with human control) [ILLUSTRATIVE].
- Production path: integrate with the screening engine's case API; read access to payment data store; write access only to case notes in pilot; auto-close permission deferred to MVP.
- Data: 12 months of historical alerts with analyst dispositions (≈420,000) for back-testing; data protection approval requested in Week 1.
- Benefit hypothesis: 55% of alerts auto-closable at agreed precision; level-1 minutes per remaining alert fall from 9 to 6.
- Funding: PoC GBP 180,000 funded; pilot GBP 350,000 and MVP GBP 900,000 approved in principle subject to gates [ESTIMATE].

**Stage 1: PoC (6 weeks), back-test on historical alerts**

| Criterion | Threshold | Measured | Pass? |
|---|---|---|---|
| Agreement with analyst disposition on proposed auto-closes (precision for "false positive") | ≥ 99.5% on a 5,000-alert stratified sample, validated by compliance QA | 99.6% | Yes |
| Share of alerts the agent proposes to auto-close (coverage) | ≥ 45% | 52% | Yes |
| True-match recall (no true matches among proposed auto-closes) | 100% on the 310 historical true matches in the 12 months | 100% (all 310 escalated) | Yes |
| Rationale faithfulness (rationale cites only real record fields) | ≥ 98% | 97.1% | No (recycle) |
| Latency per alert | ≤ 20 seconds | 11 seconds | Yes |

Decision: recycle on faithfulness. Two weeks of prompt and retrieval changes lifted faithfulness to 98.6% on a fresh sample; gate passed. The stopping rule had been: fail true-match recall at any point and the project stops; it did not trigger.

**Stage 2: Pilot (12 weeks), shadow mode then assisted mode on one rail (SEPA, ~35% of volume)**

Weeks 1–6 shadow mode: agent dispositions recorded but not acted on; analysts work normally.

| Criterion | Threshold | Measured | Pass? |
|---|---|---|---|
| Precision on proposed auto-closes, live data | ≥ 99.5% | 99.4% | Marginal |
| Coverage, live data | ≥ 45% | 49% | Yes |
| True-match recall, live | 100% (14 true matches in period) | 100% | Yes |
| Drift: golden-set accuracy vs live accuracy gap | ≤ 1.0 percentage point | 0.8 | Yes |

Weeks 7–12 assisted mode: analysts see the proposal and rationale; no auto-close yet.

| Criterion | Threshold | Measured | Pass? |
|---|---|---|---|
| Analyst minutes per alert (assisted) | ≤ 6.5 | 5.9 | Yes |

| Criterion | Threshold | Measured | Pass? |
|---|---|---|---|
| HITL override rate (analyst disagrees with proposed false-positive) | 0.3%–2.0% (too low suggests automation bias; too high suggests poor quality) | 0.9% | Yes |
| Override agreement: of overrides, share where QA sided with the analyst | ≥ 70% | 76% | Yes |
| User adoption (share of eligible alerts where analyst viewed rationale) | ≥ 85% | 91% | Yes |
| Pilot risk assessment: incidents | No severity-1 incidents | 0 | Yes |
| Fairness check: no disproportionate escalation by counterparty country beyond list-driven expectation | Reviewed by compliance | Passed | Yes |

Decision: proceed to MVP with a condition. Precision in shadow mode was 99.4% against a 99.5% threshold; the panel accepted because the shortfall was within sampling error on the sample size (confidence interval ±0.2 points [ESTIMATE]) and set a tighter confidence band for auto-close in the MVP (auto-close only above a higher model confidence, expected to reduce coverage to ~44% but lift precision above 99.7%).

**Stage 3: MVP (5 months), production on SEPA with auto-close enabled under controls**

| Criterion | Threshold | Measured at month 5 | Pass? |
|---|---|---|---|
| Precision on auto-closes (QA sample 2% of auto-closes, ≈ 1,500 a month) | ≥ 99.7% | 99.8% | Yes |
| Coverage (auto-closed share) | ≥ 40% | 44% | Yes |
| True-match recall | 100% | 100% (21 true matches, all escalated) | Yes |
| Level-1 minutes per escalated alert | ≤ 6.5 | 5.7 | Yes |
| Availability | ≥ 99.5% | 99.7% | Yes |
| Independent model validation | Complete, no high findings open | Complete; 2 medium findings closed | Yes |
| Measured benefit vs hypothesis | ≥ 70% of hypothesised FTE equivalent on SEPA volume | 93% | Yes |
| Run team in place | Product owner, 2 engineers, 1 model risk liaison | Staffed | Yes |

Decision: scale to SWIFT and domestic rails over the following nine months with a re-validation at each rail because name-match patterns and list behaviour differ.

## 4.3 Benefit arithmetic at MVP

SEPA volume ≈ 147,000 alerts a year. Auto-closed 44% = 64,700 alerts × 9 minutes = 9,700 hours saved. Remaining 82,300 alerts × (9 − 5.7) = 3.3 minutes = 4,530 hours saved. Total ≈ 14,230 hours ≈ 8.9 FTE equivalents on one rail. Hypothesis for SEPA had been 9.6 FTE (55% auto-close at 9 minutes plus 9-to-6 minutes on the rest); 93% achieved. Extrapolated to full volume at the same rates: ≈ 25 FTE equivalents ≈ GBP 1.4m a year before haircut and headcount bridge [ILLUSTRATIVE — composite]. Note what the business case books: only the signed bridge (Module 9).

### 4.4 Go/no-go decision log (extract)

| Gate | Date (relative) | Decision | Conditions | Decided by |
|---|---|---|---|---|
| G0 | Week 0 | Go to PoC | Data approval in parallel; stopping rule on true-match recall | Business owner, model risk, portfolio lead |
| G1 | Week 6 | Recycle | Faithfulness below 98% | Same |
| G1 | Week 8 | Go to pilot | None | Same |
| G2a | Week 14 | Go to assisted mode | Precision marginal; monitor | Same plus compliance |
| G2b | Week 20 | Go to MVP | Higher confidence band for auto-close; QA sample 2% | Same plus CRO delegate |
| G3 | Month 5 of MVP | Go to scale | Re-validate per rail; benefits tracked monthly by Finance | Steering committee |

The “so what” for the client: every decision in this log was made against a number set before the stage began, which is what allowed the CRO to approve auto-close and what allows Finance to trust the benefit.

## 5. Real-world case

**HSBC and Google Cloud AML AI, 2021–2023** [PUBLIC — Google Cloud announcement, Jun 2023; VERIFY]. HSBC worked with Google Cloud over several years to replace rules-based transaction monitoring with a machine-learning approach, publicly reported in 2023 as identifying two to four times more suspicious activity while cutting alert volumes by around 60%. The relevant lesson for staging is the length and shape of the journey: the system was developed and validated over a multi-year period with the bank’s financial crime and model risk functions and regulators engaged, and it was deployed market by market rather than in one global release. It is a counter-example to the idea that pilots stall only because of technology; the time went into validation, control design and jurisdictional rollout.

**Klarna, 2024–2025** [PUBLIC — Klarna, Feb 2024; CEO statements, May 2025; VERIFY]. Klarna’s move from announcement to full-scale customer-service deployment within months shows the opposite failure mode from purgatory: scaling before the quality criteria were stable. The 2025 adjustment to a hybrid human-plus-AI model is what a well-designed scale checkpoint with a customer-quality criterion would have produced earlier.

**Commonwealth Bank of Australia, 2025** [PUBLIC — Aug 2025; VERIFY]. The reversal of call-centre redundancies illustrates a missing pilot-exit criterion: measured contact-volume reduction on the pilot population before headcount actions are taken.

## 6. Consultant language

> *“A PoC proves the technique; a pilot proves the benefit. We will not book a benefit we have not measured in the workflow.”*

Use when a sponsor wants to put PoC results in the business case.

> *“The gate criteria are written before the stage begins, and every gate has a kill option we are prepared to use.”*

Use when setting up programme governance.

> *"The precision threshold is a risk-appetite statement owned by Compliance; engineering's job is to meet it."*

Use when engineers and compliance argue about thresholds.

> *"We fund the pilot and MVP in principle at PoC approval, released against the gates, so a successful PoC does not have to re-enter the budget cycle."*

Use with the CFO and portfolio office.

> *"Our scaling metric is time to the second use case on the platform."*

Use when making the case for platform investment.

## 7. Common mistakes juniors make

- Calling a PoC a pilot and extrapolating benefits from it.
- Starting a PoC without a production path, so productionising becomes an unfunded new project.
- Writing success criteria after seeing results, or using "promising" as a criterion.
- Treating accuracy thresholds as engineering choices rather than risk appetite statements owned by the second line.
- Measuring on samples too small to distinguish signal from noise, then declaring pass or fail.
- Ignoring the HITL override rate, or reading a very low override rate as success rather than possible automation bias.
- Engaging model risk at MVP instead of at PoC start.
- Scaling to new geographies or rails without re-validation.
- Letting each PoC build its own infrastructure instead of investing in a platform after the second one.
- Quoting the Gartner, MIT or RAND statistics without their caveats, or using them to argue against all pilots rather than for better gates.

## 8. Interview questions with model answers

**Q1. A bank has 40 GenAI PoCs and two in production. The COO asks you why and what to do. How do you answer?**

Model answer. I would say the pattern is common and diagnosable, then diagnose along five causes. First, production path: most PoCs were probably built in a sandbox with no integration design or integration budget, so "productionising" is a new project nobody funded. Second, ownership: PoCs run by innovation or IT without a business owner with P&L accountability rarely get pulled into the line. Third, controls: model risk and data protection engaged late, so each PoC hits a six-month approval wall. Fourth, criteria: success defined as "promising" rather than quantified gates with a kill option, so nothing is stopped and nothing is prioritised. Fifth, platform: 40 PoCs likely built 40 stacks, so each one carries full infrastructure cost. The remedy is a portfolio reset: score the 40 against value and feasibility, kill or merge most, select five to eight with named business owners, define four-dimension gate criteria with the second line, fund pilot and MVP in principle against gates, stand up a shared platform and pattern library, and measure time to second use case. I would cite Gartner's 2024 prediction that at least 30% of GenAI projects would be abandoned after PoC [PUBLIC; VERIFY] to normalise the kill decisions, and position the reset as the mark of a maturing portfolio rather than a failure.

**Q2. How would you set the exit criteria for a pilot of an agent that drafts customer complaint responses?**

Model answer. Four dimensions, set with the second line before the pilot. Technical: draft quality on a golden set of resolved complaints scored by a rubric agreed with the complaints team, with a threshold such as 85% of drafts rated "usable with minor edits", and a faithfulness criterion that drafts cite only facts in the

case record. Operational: handler minutes per response falls from the baseline, say 25 to 12, adoption above 80% of eligible cases, and an edit-distance or override rate within a band that indicates genuine use without automation bias. Business: measured handle-time benefit on the pilot team, extrapolated with a haircut, and a customer-outcome check such as no deterioration in complaint-upheld rate or Financial Ombudsman referrals in the UK context [VERIFY]. Risk: a pilot risk assessment closed, a fairness check that tone and outcomes do not vary by customer characteristics, a check against Consumer Duty expectations [VERIFY], zero severity-1 incidents, and an incident process tested. I would specify the measurement method, sample sizes and who scores, and a stopping rule: any draft sent to a customer containing a fabricated fact ends the pilot pending review. Then I would make the MVP funding conditional on the gate.

**Q3. What does “pilot purgatory” mean and what are the three interventions you would make first?**

Model answer. Pilot purgatory is the state where pilots show promising results but never reach production, so spend accumulates without P&L impact. RAND, Gartner and MIT NANDA have each published figures suggesting most AI projects fail to deliver measurable value, with caveats on method [PUBLIC — 2024–25; VERIFY]. The first three interventions, in order: one, insist on a defined production path and business owner before any PoC starts, including the integration design, data access route and funding in principle for pilot and MVP; two, replace vague success criteria with quantified, four-dimension gate criteria agreed with model risk and the business, each with a kill option, so decisions are made against numbers; three, invest in a shared platform and pattern library after the second or third PoC so that the marginal use case is cheap and inherits pre-approved controls. Behind those sit change capacity and an operating model for the run phase. I would avoid the two wrong responses: stopping all pilots, which freezes learning, and declaring everything production-ready, which creates risk incidents.

## 9. Self-check quiz

1. State the question each of the four stages answers, and explain why a PoC cannot prove business value.
2. List the four dimensions of gate criteria and give one example criterion for a pilot in each.
3. In the worked example, the MVP auto-closes 44% of 147,000 SEPA alerts at 9 minutes each and reduces the remaining alerts from 9 to 5.7 minutes. Recompute the hours saved and the FTE equivalent at 1,600 productive hours.
4. Why is a very low HITL override rate not automatically good news?
5. Name six causes of pilot purgatory and the remedy for each.

# Module 12 — Responsible AI, Regulation and Model Risk for AI in Banking

## 1. Why a client cares

A bank cannot deploy an AI system its regulator will not accept, its board cannot explain, or its customers can challenge successfully. In the EU, credit scoring of individuals is a high-risk AI use with hard obligations from 2026; in the UK, the PRA has brought qualitative and judgement-based models explicitly into model risk management; in India, the RBI has published a framework that will shape supervisory expectations. Responsible AI is therefore not a values exercise. It is the set of controls that determines whether an AI programme can go live, how long validation takes, and what the bank is liable for when the system is wrong.

Every regulatory statement in this module is date-sensitive. Treat each one as [VERIFY — date-sensitive] and confirm status before using it with a client.

## 2. Concept and mechanism

### 2.1 The regulatory landscape in one view

| Instrument | Nature | Who it binds | Status | Why it matters for AI in banking |
|---|---|---|---|---|
| EU AI Act, Regulation (EU) 2024/1689 | Binding law, risk-based | Providers and deployers of AI systems placed on or used in the EU market, wherever established | In force 1 Aug 2024; phased application 2025–2027 [VERIFY] | Credit scoring of natural persons and life/health insurance pricing are high-risk |
| NIST AI Risk Management Framework 1.0 | Voluntary framework | Any organisation | Published Jan 2023; GenAI Profile Jul 2024 | De facto reference vocabulary for AI risk programmes, including in non-US banks |
| ISO/IEC 42001:2023 | Certifiable management-system standard | Any organisation | Published Dec 2023 | The auditable "AI management system"; increasingly requested in vendor due diligence |
| US SR 11-7 / OCC 2011-12 | Supervisory guidance on model risk | US banks | 2011 | The parent of modern model risk management; defines model, validation, effective challenge |
| UK PRA SS1/23 | Supervisory statement, principles | UK banks, initially those with internal model permissions | Published May 2023; effective 17 May 2024 [VERIFY scope] | Explicitly brings qualitative and expert-judgement models into scope; SMF accountability |
| RBI FREE-AI | Committee framework and recommendations | Indian regulated entities (direction via RBI follow-up) | Report Aug 2025 [VERIFY implementation status] | Seven sutras and 26 recommendations shaping Indian supervisory expectations |
| MAS FEAT principles and Veritas | Principles and toolkit | Singapore financial institutions | FEAT 2018; Veritas toolkits 2021–2023; MAS AI model risk paper Dec 2024 [VERIFY] | Fairness, ethics, accountability, transparency assessments for AI in FS |

| Instrument | Nature | Who it binds | Status | Why it matters for AI in banking |
|---|---|---|---|---|
| UK FCA position | Principles-based, no AI-specific rulebook | UK regulated firms | AI Update Apr 2024; AI Lab 2024–25 [VERIFY] | AI governed through Consumer Duty, SM&CR, SYSC and existing conduct rules |

## 2.2 EU AI Act

**Risk tiers.** The Act classifies AI systems into: prohibited practices (Article 5, for example social scoring and certain manipulative or exploitative systems); high-risk systems (Article 6 with Annex I for AI embedded in regulated products and Annex III for standalone uses); systems with transparency obligations (Article 50, for example chatbots must disclose that the user is interacting with AI, and synthetic content must be marked); and minimal-risk systems with no specific obligations. General-purpose AI (GPAI) models are regulated separately (Chapter V), with heavier duties for models deemed to have systemic risk.

**Annex III relevance to banking.** Point 5(b) covers AI systems intended to evaluate the creditworthiness of natural persons or establish their credit score, except systems used to detect financial fraud. Point 5(c) covers risk assessment and pricing for life and health insurance for natural persons. Point 4 covers employment decisions, which touches HR analytics in any bank. Note what is not automatically high-risk: corporate lending to legal entities, AML transaction monitoring and sanctions screening, dispute resolution and most operational automation. These remain governed by model risk and conduct rules but not by the Annex III regime, unless they profile natural persons in a way that falls under another Annex III point.

**Timeline.** In force 1 August 2024. Prohibitions and AI literacy obligations (Article 4) from 2 February 2025. GPAI obligations, governance and penalties from 2 August 2025. Most high-risk (Annex III) and Article 50 transparency obligations from 2 August 2026. Annex I product-embedded high-risk from 2 August 2027. In November 2025 the European Commission proposed a “digital omnibus” that would, among other things, delay some high-risk obligations, with a proposed linkage to the availability of harmonised standards and a backstop in 2027; this is a proposal, not law [VERIFY — the status of the omnibus and the operative dates].

**Provider vs deployer.** A provider develops an AI system and places it on the market or puts it into service under its own name. A deployer uses an AI system under its authority. A bank buying a credit-scoring system is a deployer; a bank building its own is a provider and a deployer; a bank that substantially modifies a bought system, or puts its name on it, can become a provider (Article 25). Provider obligations for high-risk systems include a risk management system (Article 9), data governance (Article 10), technical documentation (Article 11), record-keeping (Article 12), transparency and instructions for use (Article 13), human oversight by design (Article 14), accuracy, robustness and cybersecurity (Article 15), a quality management system (Article 17), conformity assessment, registration in the EU database, and post-market monitoring. Deployer obligations (Article 26) include using the system in accordance with instructions, assigning human oversight to competent, trained persons, ensuring input data is relevant, monitoring and reporting serious incidents, keeping logs for at least six months, and informing affected persons. Deployers of Annex III 5(b) and 5(c) systems, which includes banks using credit scoring of natural persons, must also complete a fundamental rights impact assessment (Article 27) before first use [VERIFY exact scope].

**Financial-sector alignment.** The Act contains provisions intended to avoid duplication with financial services law: for regulated financial institutions, some quality-management and documentation obligations can be met through internal governance under existing prudential rules, and the financial supervisor is designated as the market surveillance authority for AI in financial services [VERIFY — Articles 17 and 74 and national implementation].

**Penalties.** Up to EUR 35m or 7% of global annual turnover for prohibited practices; up to EUR 15m or 3% for most other obligations; up to EUR 7.5m or 1% for supplying incorrect information, with lower caps for SMEs [VERIFY].

## 2.3 NIST AI Risk Management Framework

NIST AI RMF 1.0 (January 2023) organises AI risk management into four functions. **Govern** is cross-cutting: policies, accountability, culture, workforce, third-party risk. **Map** establishes context: intended purpose, users, affected groups, benefits and risks, including for components the organisation did not build. **Measure** uses quantitative and qualitative methods to analyse and track risks: performance, fairness, robustness, privacy, explainability, with test and evaluation. **Manage** allocates resources to prioritised risks, decides on treatment, and plans response and recovery. The RMF defines trustworthy AI characteristics: valid and reliable; safe; secure and resilient; accountable and transparent; explainable and interpretable; privacy-enhanced; fair with harmful bias managed. The Generative AI Profile (NIST AI 600-1, July 2024) adds twelve GenAI-specific risk categories, including confabulation, information integrity, information security, intellectual property, harmful bias and homogenisation, data privacy, and value chain and component integration, with suggested actions mapped to the four functions. Banks outside the US use the RMF as a vocabulary that regulators recognise and as a crosswalk to ISO/IEC 42001 and the EU AI Act.

## 2.4 ISO/IEC 42001:2023

ISO/IEC 42001 specifies requirements for an AI management system (AIMS) following the harmonised ISO management-system structure (context, leadership, planning, support, operation, performance evaluation, improvement), with an Annex A of controls organised under control objectives covering policies, internal organisation, resources, impact assessment, system lifecycle, data, information for interested parties, use of AI systems, and third-party relationships, and guidance in Annexes B–D [VERIFY control counts]. Like ISO 27001 it is certifiable, which is why it appears in vendor RFPs. Related standards: ISO/IEC 23894 (AI risk management guidance, 2023), ISO/IEC 22989 (concepts and terminology), ISO/IEC 42005 (AI system impact assessment, 2025) [VERIFY]. Certification demonstrates a management system exists; it does not certify that any individual model is accurate or fair.

## 2.5 US SR 11-7, the parent of model risk management

SR 11-7 (Federal Reserve) and OCC 2011-12 (April 2011) define a model as a quantitative method, system or approach that applies statistical, economic, financial or mathematical theories, techniques and assumptions to process input data into quantitative estimates. Model risk arises from fundamental errors and from incorrect or inappropriate use. The guidance requires three things: sound model development, implementation and use; a validation function with conceptual soundness review, ongoing monitoring and outcomes analysis; and governance, policies and controls including a model inventory and "effective challenge" by independent, competent and influential reviewers. Almost every bank's model risk framework worldwide descends from this structure. Its limit for GenAI is the definition: a foundation model producing text is not obviously "a quantitative estimate", which is one reason the UK PRA broadened the definition.

## 2.6 UK PRA SS1/23

SS1/23 "Model risk management principles for banks" was published in May 2023 and took effect on 17 May 2024, initially for firms with internal model permissions, with the PRA signalling broader relevance [VERIFY scope and any later extension]. Five principles:

1. **Model identification and model risk classification.** A model inventory and a tiering approach based on materiality and complexity.
2. **Governance.** Board oversight, a named Senior Management Function (SMF) accountable for the MRM framework, policies and reporting.
3. **Model development, implementation and use.** Standards for design, data, testing, documentation and use, including adjustments and overlays.
4. **Independent model validation.** Independent review, ongoing monitoring, periodic revalidation, and challenge.

5. **Model risk mitigants.** Post-model adjustments, restrictions on use, exception processes, and compensating controls where model risk is high.

The definition of a model in SS1/23 explicitly includes inputs that are quantitative, qualitative or expert-judgement based, and outputs that are quantitative or qualitative. That wording brings LLM-based systems inside the model definition and is the clearest regulatory statement that GenAI systems are models for MRM purposes [VERIFY wording]. SS1/23 also asks firms to self-assess against the principles and remediate gaps, and the PRA has stated in its AI-related communications that it regards SS1/23 as the primary vehicle for AI model risk [VERIFY].

**What "model" means for an LLM-based system.** The unit of validation is the system in use, not the foundation model alone. That system comprises the foundation model (often third-party and opaque), the prompts and system instructions, the retrieval corpus and retrieval logic, the tools and their permissions, the guardrails, the orchestration logic, and the human oversight design. A change to any component is a model change. Because the foundation model cannot be validated in the conventional sense (no access to weights or training data), validation shifts to the observable behaviour of the whole system: outcome testing on golden datasets, robustness testing, monitoring, and compensating controls. Third-party model risk expectations also apply.

## 2.7 RBI FREE-AI

The Reserve Bank of India constituted a committee in December 2024, chaired by Professor Pushpak Bhattacharyya of IIT Bombay, to develop a Framework for Responsible and Ethical Enablement of AI. The report, published in August 2025, set out seven guiding "sutras": trust is the foundation; people first; innovation over restraint; fairness and equity; accountability; understandable by design; and safety, resilience and sustainability. It made 26 recommendations across six pillars, grouped into innovation enablement (infrastructure, policy, capacity) and risk mitigation (governance, protection, assurance), including board-approved AI policies at regulated entities, AI incident reporting, a graded liability approach to encourage responsible innovation, shared digital infrastructure for AI, and sector-specific models [VERIFY — exact wording and which recommendations have become RBI directions]. The framework sits alongside existing RBI expectations on outsourcing, IT governance, digital lending and the fair practices code, and alongside India's Digital Personal Data Protection Act 2023 [VERIFY commencement].

## 2.8 MAS, HKMA and the UK FCA

The Monetary Authority of Singapore's FEAT principles (2018) set out Fairness, Ethics, Accountability and Transparency expectations for AI and data analytics in financial services; the Veritas consortium produced assessment methodologies and open-source toolkits (2021–2023); MAS published an information paper on AI model risk management in December 2024 following a thematic review, setting out good practices in governance, inventory, validation and monitoring, including for GenAI [VERIFY]. The HKMA launched a GenAI sandbox in 2024 [VERIFY]. The UK FCA has not issued an AI rulebook; its April 2024 AI Update explained how existing frameworks (Consumer Duty, the Senior Managers and Certification Regime, SYSC systems and controls, the Equality Act and data protection law) apply, and it has run an AI Lab and live-testing initiatives since late 2024 [VERIFY]. The joint Bank of England and FCA survey published in November 2024 reported around 75% of firms already using AI [PUBLIC — BoE/FCA, Nov 2024; VERIFY].

## 2.9 Explainability

**For conventional ML.** Global explanations describe the model overall: feature importance, partial dependence plots. Local explanations describe a single decision: LIME (Ribeiro, Singh and Guestrin, 2016) fits a simple surrogate around one prediction; SHAP (Lundberg and Lee, 2017) attributes the prediction to features using Shapley values from cooperative game theory, with consistency guarantees; counterfactual explanations (Wachter, Mittelstadt and Russell, 2017) state the smallest input change that would flip the decision, which maps naturally to adverse-action reasons in credit. Limits: SHAP values on correlated features can mislead; surrogates are approximations; and feature attributions are not causal.

**For LLM-based systems.** Three mechanisms are used, with different evidential weight. Citation grounding, where the system must cite the retrieved passages or record fields it relied on, is checkable: a reviewer can verify the citation supports the claim. Rationale generation, where the model writes out its reasoning, is useful for review but is not a faithful account of the computation; research has shown model-stated reasoning can diverge from the factors that actually drove the output (Turpin et al., 2023, "Language models don't always say what they think") [PUBLIC — 2023]. Mechanistic interpretability of model internals is an active research field and not yet an operational control. Consequently, for high-stakes decisions the defensible architecture is to keep the decision in an explainable model or rule set and use the LLM for information assembly, drafting and triage, with the explanation coming from the decision component.

## 2.10 Bias and fairness testing

Define protected attributes and proxies (postcode, name, device) in scope. Common metrics: disparate impact ratio (selection rate for the protected group divided by that for the reference group), with the four-fifths rule from the US EEOC Uniform Guidelines (1978) treating a ratio below 0.8 as evidence of adverse impact; demographic parity; equalised odds (equal true- and false-positive rates across groups); calibration within groups. Several of these cannot be satisfied simultaneously when base rates differ (Kleinberg, Mullainathan and Raghavan, 2016; Chouldechova, 2017), so the bank must choose and justify a fairness definition. In lending, the UK Equality Act 2010, EU non-discrimination law and the US Equal Credit Opportunity Act and Regulation B (which requires specific adverse-action reasons) set the legal frame [VERIFY jurisdictional detail]. For LLM systems add tests for differential tone, differential refusal rates and differential error rates across customer groups, because text outputs can discriminate in ways a scorecard cannot.

## 2.11 Auditability

Auditability means a competent third party can reconstruct what the system did and why. Requirements: immutable logging of inputs, retrieved context, prompts, model version, tool calls and outputs; lineage from training or retrieval data to output; versioning of every component of the system (Section 2.6); reproducibility, recognising that LLM outputs are not strictly deterministic even at temperature zero, so reproducibility is defined at the level of logged evidence rather than regenerated output; retention aligned to regulation (the EU AI Act requires deployers to keep logs for at least six months; banking record-keeping rules often require longer [VERIFY]); and access controls and tamper evidence for the logs themselves.

## 2.12 Classifying a GenAI or agentic system under a bank's model risk framework

Tiering drivers: materiality of the decision (financial exposure, customer impact); autonomy level (Module 3), from suggestion to unsupervised action; reversibility of errors; regulatory exposure (Annex III, conduct, financial crime); population affected; and opacity (third-party model, no access to weights). A practitioner tiering [ILLUSTRATIVE — composite; firms differ]:

| Tier | Characteristics | Example | Validation intensity |
|---|---|---|---|
| 1 | Material customer or financial decisions; high autonomy or regulatory high-risk | Credit decisioning support for individuals; auto-closing sanctions alerts | Full independent validation, annual revalidation, board-level reporting, FRIA where applicable |
| 2 | Decision support with mandatory human approval; moderate materiality | Case-file drafting for KYB; dispute recommendation with analyst approval | Independent validation at launch; monitoring; revalidation on material change or every two years |
| 3 | Internal productivity; low materiality; reversible | Knowledge assistant for policy questions; meeting summaries | Lighter review; usage policy; monitoring of misuse and leakage |

### 2.13 A Responsible AI control framework across the lifecycle

| Lifecycle stage | Controls | Evidence produced |
|---|---|---|
| Intake and classification | Use-case intake form; prohibited-use screen (Article 5); Annex III screen; model risk tier; data protection impact assessment trigger; FRIA trigger | Classification record; tier decision; DPIA and FRIA plans |
| Design | Autonomy level and HITL design; explainability approach (decision in explainable component); fairness definition chosen; data governance plan; third-party model due diligence | Design authority approval; architecture decision record |
| Data | Data lineage; representativeness assessment; bias assessment of training or retrieval data; consent and purpose limitation | Data governance sign-off |
| Build | Prompt and tool registry; guardrails (input, output, action); least-privilege tool permissions; secure development; adversarial testing including prompt injection | Build evidence pack |
| Evaluation and validation | Golden dataset; accuracy, faithfulness, robustness tests; fairness metrics by group; red-team report; independent validation report; conformity assessment where provider | Validation report; conformity documentation |
| Deployment | Change approval; user training and competence (Article 26, Article 4 AI literacy); customer disclosure (Article 50); regulatory notification or registration where required | Deployment approval; training records |
| Operation | Monitoring of performance, drift, override rates, incidents; logging and retention; complaint and challenge handling; serious-incident reporting | Monitoring dashboards; incident register |
| Change and retirement | Model change notification from vendors; re-validation triggers; periodic review; decommissioning and data deletion | Change log; revalidation reports |

## 3. Frameworks

### 3.1 NIST AI RMF Govern, Map, Measure, Manage (NIST, Jan 2023)

Apply by: establishing the governance structure first (policies, roles, inventory); mapping each use case's context, affected groups and risks; selecting metrics and tests for each identified risk; and deciding treatment with documented residual risk. Use the GenAI Profile's risk categories as a checklist for LLM systems.

### 3.2 PRA SS1/23 five principles (PRA, May 2023)

Apply by running a gap assessment of the bank's existing MRM framework against the five principles specifically for GenAI: does the inventory capture LLM systems, including shadow usage; does tiering account for autonomy and opacity; are development standards extended to prompts, retrieval and tools; can

validation operate without access to weights; are mitigants such as HITL and output filters formally recognised as model risk mitigants.

### 3.3 EU AI Act compliance pathway (Regulation (EU) 2024/1689)

Steps: inventory all AI systems; screen for prohibited practices; classify against Annex I and Annex III; determine role (provider, deployer, both); map obligations by role and date; conduct FRIA where required; align with existing financial-sector governance to avoid duplication; register, document, monitor. Build this as a crosswalk to the bank's model risk and third-party risk frameworks rather than a separate programme.

### 3.4 ISO/IEC 42001 AIMS (ISO/IEC, Dec 2023)

Apply by using the Annex A control objectives as the structure for the bank's AI policy suite, and by requiring ISO/IEC 42001 certification or equivalent evidence from material AI vendors.

### 3.5 FEAT principles (MAS, 2018) and the Veritas methodology

Apply by running the fairness, ethics, accountability and transparency assessment questions at design and at validation, using Veritas's fairness assessment methodology as a template where the bank has no equivalent.

## 4. Worked BFSI example

### 4.1 Context

[ILLUSTRATIVE — composite] A bank with operations in Germany, the UK and India wants an agentic credit-underwriting assistant for small-business lending up to EUR 250,000 equivalent. The applicant population includes sole traders (natural persons) and incorporated small companies. The assistant: collects and extracts financial documents; retrieves bureau and bank-statement data via tools; computes affordability ratios; drafts a credit memo with cited evidence; proposes a decision and pricing band; and routes to an underwriter who approves, modifies or declines. The bank builds the assistant on a hyperscaler platform with a third-party foundation model, and the final decision logic for individuals is a traditional scorecard the assistant calls as a tool.

### 4.2 Classification by jurisdiction

| Question | EU (Germany) | UK | India |
|---|---|---|---|
| Is this an Annex III high-risk system? | Yes, for sole traders and any natural-person assessment (5(b)); not for incorporated companies, but the system is one system, so treat it as high-risk in scope [VERIFY interpretation] | Not applicable (no AI Act); governed by PRA SS1/23 and FCA conduct rules | Not applicable; governed by RBI expectations and FREE-AI direction [VERIFY] |
| Bank's role | Provider and deployer (built in-house) | Model owner under SS1/23 | Regulated entity under RBI |
| Model risk tier | Tier 1 | Tier 1 | Tier 1 |
| Fundamental rights or impact assessment | FRIA required before first use (Article 27) plus GDPR DPIA | DPIA under UK GDPR; Consumer Duty fair-value and outcomes assessment | DPDP Act 2023 obligations; RBI fair practices; board-approved AI policy per FREE-AI [VERIFY] |
| Automated decision-making | GDPR Article 22 constraints where decisions are solely automated; human approval designed in | UK GDPR Article 22 as amended by the Data (Use and Access) Act 2025 [VERIFY]; human approval designed in | Human approval designed in; RBI digital lending guidelines on transparency [VERIFY] |

| Question | EU (Germany) | UK | India |
|---|---|---|---|
| Explainability expectation | Article 13 transparency and Article 86 right to explanation for high-risk decisions [VERIFY]; adverse decision reasons under consumer credit law | Consumer Duty; adverse decision reasons; Equality Act | Fair practices code: reasons for rejection on request; FREE-AI "understandable by design" |
| Disclosure to applicant | Article 50 where the applicant interacts with the system directly; Article 26 information to affected persons | FCA Principle 7 clear communications; Consumer Duty | RBI customer-protection expectations |
| Supervisor | BaFin as market surveillance authority for AI in financial services [VERIFY national designation] | PRA and FCA | RBI |

## 4.3 Obligations mapped to the system

| Obligation | Attaches in | How the design meets it |
|---|---|---|
| Risk management system across lifecycle (Article 9) | EU | Integrated with the bank's MRM framework; risk register per component |
| Data governance and bias examination (Article 10) | EU; SS1/23 Principle 3; FEAT-style fairness | Representativeness analysis of training data for the scorecard; retrieval corpus review; disparate-impact testing on gender, age and nationality proxies with four-fifths screen and equalised-odds review |
| Technical documentation (Article 11) | EU | Model card for the system; vendor documentation for the foundation model obtained contractually |
| Record-keeping and logs (Article 12, Article 26) | EU; UK record-keeping; RBI audit | Immutable logs of inputs, retrieved context, prompts, model version, tool calls, outputs; retention seven years per credit record rules [ESTIMATE — align to local rules] |
| Human oversight (Article 14, Article 26) | EU; SS1/23 Principle 5; RBI accountability | Underwriter approval mandatory; seeded test cases to detect automation bias; override rate band 3–15% monitored |
| Accuracy, robustness, cybersecurity (Article 15) | EU; SS1/23 Principles 3 and 4 | Golden dataset of 3,000 historical applications; extraction accuracy ≥ 98%; memo faithfulness ≥ 99%; prompt-injection testing on uploaded documents |
| Quality management system (Article 17) | EU, with alignment to CRD internal governance | Mapped to existing governance; delta documented |
| Conformity assessment and registration | EU | Internal control conformity assessment where permitted for Annex III; registration in the EU database before use [VERIFY] |
| FRIA (Article 27) | EU | Completed before first use; covers affected groups, risks to non-discrimination and access to credit, mitigation |

| Obligation | Attaches in | How the design meets it |
|---|---|---|
| Independent validation and SMF accountability | UK | Validation by model risk; CRO delegate as accountable SMF; annual revalidation |
| Explainability of decision | All | Decision and pricing come from the scorecard; adverse reasons generated from scorecard counterfactuals; the LLM memo cites evidence and never states a decision reason that is not in the scorecard output |
| Board-approved AI policy and incident reporting | India (FREE-AI) | AI policy approved by the India board; incident reporting process aligned to RBI expectations [VERIFY] |
| Consumer protection and fair value | UK (Consumer Duty); India (fair practices) | Outcome monitoring by segment; complaint analysis |

### 4.4 Control plan excerpt with numbers

[ILLUSTRATIVE — composite]

| Control | Threshold | Monitoring frequency | Owner |
|---|---|---|---|
| Document extraction accuracy | ≥ 98% field-level on monthly QA sample of 300 | Monthly | Product owner |
| Memo faithfulness (every stated fact traceable to a cited record) | ≥ 99% | Monthly | Model risk |
| Disparate impact ratio, approval rate by gender and age band | ≥ 0.80 and investigated below 0.90 | Quarterly | Fair lending officer |
| Underwriter override rate | 3–15% | Weekly | Credit operations |
| Override quality (QA sides with underwriter) | ≥ 70% | Monthly | Model risk |
| Prompt-injection detection on uploaded documents | 100% of seeded attacks blocked in quarterly red team | Quarterly | Security |
| Serious-incident reporting | Within statutory period (EU AI Act Article 73 timelines [VERIFY]) | Event-driven | Compliance |
| Foundation model version change | 90 days' notice; re-validation before adoption | Event-driven | Vendor manager |

The "so what" for the client: the architecture decision to keep the credit decision in an explainable scorecard and use the agent for assembly and drafting is what makes this system approvable in all three jurisdictions. Letting the LLM decide would make Article 13 explanation, UK adverse-reason and Indian fair-practice obligations very hard to meet.

## 5. Real-world case

**Apple Card and Goldman Sachs, 2019–2021** [PUBLIC — NYDFS report, Mar 2021]. In November 2019 public allegations that the Apple Card algorithm gave women lower credit limits than their spouses prompted a New York Department of Financial Services investigation. The March 2021 report found no unlawful discrimination under fair lending law, but criticised the issuer's customer service and lack of transparency, noting that customers could not get meaningful explanations of their limits. The lesson is that an

explainability and communications failure can create a regulatory event even when the model passes the legal fairness test, which is why adverse-reason generation and front-line scripts belong in the control plan.

**Air Canada chatbot, 2024** [PUBLIC — British Columbia Civil Resolution Tribunal, Moffatt v. Air Canada, Feb 2024]. The tribunal held the airline liable for a chatbot's incorrect statement about bereavement fares, rejecting the argument that the chatbot was a separate entity responsible for its own words. For banks this fixes the accountability principle: statements made by a deployed system are the firm's statements, which is also the position under UK Consumer Duty and FCA Principle 7 [VERIFY].

**HSBC and Google Cloud AML AI, 2023** [PUBLIC — Google Cloud, Jun 2023; VERIFY]. The publicly reported outcome (two to four times more suspicious activity identified, around 60% fewer alerts) was achieved with explainability features designed so investigators and regulators could understand risk scores, illustrating that explainability was a precondition for regulatory acceptance of replacing rules with ML in financial crime.

**Amazon recruiting tool, 2018** [PUBLIC — Reuters, Oct 2018]. An ML hiring tool trained on historical CVs learned to penalise indicators associated with women and was scrapped. Relevant to banks because HR analytics is Annex III high-risk and because the failure was in data representativeness, which Article 10 and SS1/23 Principle 3 both address.

## 6. Consultant language

> *"We validate the system in use, not the foundation model: prompts, retrieval, tools and oversight are all in the model boundary."*

Use with model risk when scoping validation of an LLM system.

> *"The decision stays in an explainable component; the agent assembles and drafts. That is what makes the explanation obligation meetable."*

Use with the CRO and product teams during architecture decisions.

> *"Under the AI Act we are both provider and deployer of this system, so the obligations stack."*

Use when the client assumes building in-house avoids provider duties.

> *"SS1/23 brings qualitative and judgement-based models into scope, so the question is not whether this is a model but which tier it sits in."*

Use with UK clients who argue GenAI is outside MRM.

> *"Fairness metrics conflict when base rates differ, so we will choose one definition, justify it, and monitor the others."*

Use when fairness testing is being designed.

## 7. Common mistakes juniors make

- Treating the EU AI Act as applying to all AI in banking; most operational use cases are not Annex III high-risk.
- Forgetting that a bank building its own high-risk system is a provider as well as a deployer.
- Quoting EU AI Act dates without checking the status of the digital omnibus and national implementation.
- Describing SR 11-7 as a US-only concern rather than the parent of every bank's model risk framework.
- Validating only the foundation model, or asserting it cannot be validated, instead of validating the system in use.
- Presenting LLM-generated rationales as explanations of the model's decision.

- Choosing a fairness metric without acknowledging the impossibility results and documenting the choice.
- Scoring ISO/IEC 42001 certification as evidence that a vendor's model is accurate or fair.
- Designing HITL as a rubber stamp without controls for automation bias.
- Treating Responsible AI as a policy document rather than a set of controls with thresholds, owners and evidence.

## 8. Interview questions with model answers

**Q1. A UK bank's CRO says GenAI tools are "not models" and so sit outside model risk management. How do you respond?**

Model answer. I would point to the PRA's SS1/23, effective from May 2024, whose model definition explicitly includes qualitative and expert-judgement inputs and qualitative outputs, which brings LLM-based systems inside scope; the PRA has also indicated that it sees SS1/23 as the primary vehicle for AI model risk [VERIFY]. So the question is not whether a GenAI tool is a model but which tier it belongs to. I would propose a tiering based on materiality, autonomy, reversibility, regulatory exposure and opacity: a knowledge assistant answering policy questions is Tier 3 with light review and a usage policy; a case-drafting agent with mandatory human approval is Tier 2; anything influencing customer credit or financial crime decisions is Tier 1 with full independent validation. I would then address the practical objection behind the CRO's statement, which is that a third-party foundation model cannot be validated conventionally. The answer is to define the model boundary as the system in use, including prompts, retrieval, tools, guardrails and oversight, and to validate behaviour through golden datasets, robustness and fairness testing, monitoring and compensating controls, with third-party model risk applied to the vendor. Finally, I would note the benefit: a clear tiering lets the bank move faster on Tier 3 tools rather than treating all GenAI as equally risky.

**Q2. Which banking AI use cases are high-risk under the EU AI Act, and what does that mean in practice for a bank deploying one?**

Model answer. Annex III makes two financial use cases high-risk: creditworthiness assessment or credit scoring of natural persons, excluding fraud detection, and risk assessment and pricing in life and health insurance; employment-related AI is also high-risk and affects HR. Most other bank uses, such as AML monitoring, sanctions screening, disputes and operational automation, are not Annex III high-risk, though they remain under model risk and conduct rules, and chatbots carry Article 50 disclosure duties. In practice, for a credit-scoring system a bank that buys it is a deployer and must use it per the instructions, assign trained human oversight, ensure input data relevance, keep logs for at least six months, monitor and report serious incidents, inform affected persons, and complete a fundamental rights impact assessment before first use. A bank that builds it is also a provider and must meet the full provider set: risk management system, data governance, technical documentation, record-keeping, transparency, human oversight by design, accuracy and robustness, a quality management system aligned to existing prudential governance, conformity assessment and registration. Timing: most Annex III obligations apply from 2 August 2026, subject to the Commission's November 2025 omnibus proposal that may delay them [VERIFY]. My advice to a bank is to treat the Act as a crosswalk onto its existing MRM, third-party and conduct frameworks rather than as a new standalone programme, and to run the inventory and classification now.

**Q3. How would you make an agentic underwriting assistant explainable and fair enough to satisfy regulators?**

Model answer. Architecturally, I would separate assembly from decision. The agent gathers documents, extracts data, retrieves bureau and bank-statement information, computes ratios and drafts the memo with citations, but the credit decision and pricing come from an explainable scorecard or constrained model that the agent calls as a tool. That lets the bank generate adverse-action reasons from the scorecard, using counterfactual or SHAP-style attributions, and lets the memo cite evidence for every stated fact without pretending that the LLM's prose is the reason for the decision; research shows model-stated reasoning is not

a faithful account of the computation. On fairness, I would define protected attributes and proxies, choose a primary metric such as disparate impact with the four-fifths screen plus equalised-odds review, document why that choice was made given the impossibility of satisfying all metrics when base rates differ, test at validation and quarterly in operation, and add LLM-specific checks for differential tone, refusal and error rates across groups. On oversight, mandatory underwriter approval with automation-bias controls such as seeded test cases and an override-rate band. On evidence, immutable logs of inputs, retrieved context, prompts, model versions, tool calls and outputs, with retention aligned to credit record rules. And I would frame the package against the regulator's own vocabulary: EU AI Act Articles 10, 13, 14 and 27, PRA SS1/23 principles, and RBI FREE-AI's "understandable by design" and "fairness and equity" sutras [VERIFY].

## 9. Self-check quiz

1. Name the two Annex III points of the EU AI Act most relevant to financial services and state one bank use case that is not high-risk under Annex III.
2. State the five principles of PRA SS1/23 and explain the significance of its model definition for GenAI.
3. A model approves 62% of male applicants and 47% of female applicants. Compute the disparate impact ratio and state whether it fails the four-fifths rule.
4. Why is an LLM's written rationale not an explanation in the model risk sense, and what architectural choice addresses this?
5. List the four NIST AI RMF functions and give one GenAI-specific risk category from the NIST Generative AI Profile.

# Module 13 — AI Operating Model and Talent

## 1. Why a client cares

Most banks that stall on AI do not stall on technology. They stall because nobody owns the portfolio, the business cannot get engineering capacity, risk has no way to approve an agent, and the data team is a bottleneck for everyone. The operating model is the mechanism that converts a use-case list into shipped, governed, adopted systems. A CEO who asks “why have we spent GBP 40m on AI and have three things in production” is asking an operating-model question, not a model question.

## 2. Concept and mechanism

### 2.1 What an AI operating model is

An operating model is the set of structures, roles, decision rights, processes, funding flows and platforms through which an organisation does a category of work. For AI, the category of work is unusual in four ways, and each creates a design requirement:

- **The work is cross-functional by construction.** A single use case needs a business owner, a data engineer, an ML or agent engineer, a model-risk validator, a control owner, a change lead and a platform. In a conventional bank these sit in five reporting lines. The operating model has to create a team that spans them without a six-month resourcing negotiation.
- **The work is probabilistic.** A traditional IT change either works or does not. An AI system works 94% of the time and the operating model must decide who owns the 6%, who monitors drift, who re-validates after a model refresh, and who can switch it off.
- **The platform is reusable; the use case is not.** Retrieval infrastructure, evaluation harnesses, prompt registries, guardrail services, agent orchestration and observability are shared. Building them inside each use case is the single most common source of duplicated spend. The operating model has to separate platform (shared, funded centrally, product-managed) from use cases (business-funded, outcome-owned).
- **The risk surface is new.** Model risk, conduct risk, data-protection risk and operational resilience all attach to the same artefact. Three-lines-of-defence has to be extended rather than assumed.

### 2.2 The four archetypes

The archetypes below are described consistently across McKinsey/QuantumBlack and BCG practitioner writing on AI and analytics operating models since roughly 2018; the labels are generic and the exact definitions vary by firm (attribution: widely used in practice; no single originator).

| Archetype | Where capability sits | Who sets priorities | Typical fit |
|---|---|---|---|
| Centralised CoE | One central team does discovery, build and run | Centre, with business input | Low maturity, few use cases, scarce talent, high regulatory caution |
| Hub-and-spoke | Hub owns platform, standards, scarce specialists; spokes in business units own use cases and have embedded delivery capacity | Jointly: hub sets guardrails and allocates platform; spokes set business priorities | Medium maturity, several domains with real demand, 20 to 100 use cases |
| Federated | Each business unit has its own AI team and budget; a thin central function sets policy | Business units | High maturity, strong divisional P&Ls, very different data domains |
| Embedded (fully | AI engineers sit in product | Product owners | Digital-native firms; rarely |

| Archetype | Where capability sits | Who sets priorities | Typical fit |
|---|---|---|---|
| decentralised) | squads with no central function | | appropriate in a regulated bank |

The mechanism that distinguishes them is where three things sit: **talent**, **money** and **decision rights**. A centralised CoE holds all three. Hub-and-spoke splits them: the hub holds platform money, standards and the scarcest talent (evaluation, model risk engineering, agent platform); the spokes hold use-case money, business decision rights and domain-embedded delivery. Federated gives nearly everything to the spokes and keeps only policy centrally.

## 2.3 Typical evolution path and its failure points

Most banks move CoE, then hub-and-spoke, then partially federated. Each transition has a characteristic failure:

- **CoE to hub-and-spoke** fails when the hub releases people to spokes before it has built a platform. The spokes rebuild retrieval, evals and guardrails five times, and the central team becomes a help desk.
- **Hub-and-spoke to federated** fails when model-risk and data governance are federated with everything else. Standards fragment, and the regulator sees three different definitions of "model inventory" in one bank.
- **Staying centralised too long** fails through queueing. A central team of 40 serving 15 business units becomes a prioritisation committee, and the business starts shadow-AI procurement.

The rule of thumb used by practitioners is: centralise what is scarce, risky or reusable; decentralise what needs domain knowledge and speed. For AI in a bank that resolves to a hub holding platform, standards, model risk engineering and evaluation, and spokes holding use-case ownership and domain-embedded delivery.

## 2.4 Roles

The roles below are the set a hub-and-spoke model in a bank typically needs. Titles vary; the responsibilities are what matter.

| Role | Core accountability | Hub or spoke | Indicative ratio |
|---|---|---|---|
| AI product owner | Owns the use case outcome, backlog, KPIs and adoption; the business's accountable person | Spoke | 1 per 1 to 2 live use cases |
| Domain SME (process expert) | Supplies ground truth, labels golden data, defines exceptions, signs off process change | Spoke (part-time) | 1 to 2 per use case during build |
| Agent / prompt engineer (AI engineer) | Designs agent scoping, prompts, tool contracts, retrieval, evaluation loops | Spoke, hub for platform patterns | 2 to 3 per use case in build; 0.5 in run |
| ML engineer | Classical models, fine-tuning, feature pipelines | Hub pool, assigned | 1 per 2 to 3 use cases |
| Data engineer | Pipelines, data quality, access, lineage | Spoke-aligned, hub standards | 1 to 2 per use case in build |
| Platform engineer (LLMOps) | Shared gateway, vector stores, eval harness, observability, guardrail services | Hub | 6 to 12 total for a mid-sized bank |
| AI risk and validation | Independent validation, | Hub-adjacent, reports to CRO | 1 validator per 4 to 6 |

| Role | Core accountability | Hub or spoke | Indicative ratio |
|---|---|---|---|
| | tiering, ongoing monitoring sign-off (second line) | | models/agents in inventory |
| Responsible AI / AI ethics lead | Policy, fairness testing standards, regulatory mapping (EU AI Act, SS1/23) | Hub, reports into Risk or Legal | 2 to 4 for the bank |
| Change and adoption lead | Role redesign, training, incentive alignment, adoption KPIs | Spoke, hub playbook | 1 per domain |
| AI architect | Reference architectures, build vs buy, integration patterns | Hub | 2 to 4 for the bank |
| Portfolio and value lead (AI PMO) | Pipeline, stage gates, benefits tracking, funding allocation | Hub | 3 to 5 for the bank |

Ratios are `[ESTIMATE — assumption, based on practitioner experience]`. The two most under-provisioned roles in banks are independent validation (regulators require it, and LLM validation skills are rare) and the domain SME (business leaders volunteer the SME's time and then recall them when the quarter gets busy).

## 2.5 The AI target operating model: six layers

A target operating model (TOM) for AI can be described in six layers. Each layer has a design question and an artefact that answers it.

| Layer | Design question | Artefact |
|---|---|---|
| Strategy and governance | Who decides ambition, domains and risk appetite; how is the AI council composed | AI strategy; AI council charter; risk appetite statement for AI |
| Demand and portfolio | How do use cases enter, get scored, funded, staged and retired | Intake process; prioritisation scorecard; stage-gate criteria; benefits register |
| Delivery | How do teams form, build, test, release and hand to run | Squad model; delivery playbook; definition of done including evals and validation |
| Platform | What is shared; how is it funded and productised | Platform roadmap; service catalogue (gateway, RAG, evals, guardrails, observability); chargeback rules |
| Risk and compliance | How is an AI system tiered, validated, monitored, and who can stop it | AI model risk policy; tiering matrix; validation standards; kill-switch authority |
| Talent | What roles, how many, sourced how, developed how | Role catalogue; capacity plan; hiring and reskilling plan; partner model |

The error to avoid is designing the layers independently. Demand and portfolio must reference the risk tiering (a Tier 1 agent needs validation capacity booked before funding is released). Platform must reference delivery (the definition of done cannot require an eval harness the platform has not shipped).

## 2.6 Funding models and incentive effects

| Funding model | Mechanism | Incentive effect | Risk |
|---|---|---|---|
| Central budget | Group funds all AI centrally | Fast start, easy to standardise | Business treats it as free; weak ownership of benefits; central team becomes order- |

| Funding model | Mechanism | Incentive effect | Risk |
|---|---|---|---|
| | | | taker |
| Chargeback | Business units pay the hub for platform and services | Disciplines demand; platform must prove value | Under-use of platform; shadow procurement of cheaper vendor tools |
| Business-funded | Each use case funded from the sponsoring unit's budget | Strong ownership; benefits land in the P&L that paid | Starves cross-cutting and foundational work; no one funds the platform |
| Venture-style stage gates | Central fund releases tranches at PoC, pilot, MVP, scale on evidence | Kills weak cases early; forces evidence | Governance overhead; can slow genuinely strategic bets |
| Hybrid (most common target) | Central funding for platform and foundations; business funding for use cases; a stage-gated innovation pool for new domains | Balances ownership with reuse | Requires a competent AI PMO to run |

The incentive point to make with a CFO is that the funding model decides where benefits are counted. If the hub is funded centrally and benefits land in the business units, the hub will always look like cost. The remedy is a benefits register with business-unit sign-off, tracked by the AI PMO and reported alongside platform spend.

### 2.7 Building the central platform team

A platform team is a product team, not a shared-services pool. It needs a product manager, a roadmap, consumers it treats as customers, and service levels. The services a bank's AI platform typically productises, in the order they tend to be needed:

1. Model gateway (single authenticated entry point to approved models, with logging, cost attribution, PII redaction and rate limits).
2. Retrieval service (document ingestion, chunking, embedding, vector store, access-control-aware retrieval).
3. Evaluation harness (golden dataset management, LLM-as-judge configuration, regression runs on every prompt or model change).
4. Guardrail service (input classification, output checking, policy enforcement, tool-call allow-lists).
5. Agent orchestration runtime (tool registry, memory, HITL queue, audit trail).
6. Observability (traces, cost per task, latency, override rates, drift alerts).
7. Prompt and agent registry (versioning, approval status, link to model inventory).

A team of 8 to 12 can deliver and run items 1 to 4 within 9 to 12 months for a mid-sized bank `[ESTIMATE — assumption]`; items 5 to 7 follow as agentic use cases enter pilot.

## 3. Frameworks

### 3.1 McKinsey "Rewired" capabilities applied to operating model (Lamarre, Smaje, Zemmel, 2023) [PUBLIC — Wiley, 2023]

"Rewired" names six capabilities: business-led digital roadmap; talent; operating model; technology; data; adoption and scaling. Two of the six (talent, operating model) are the subject of this module. The book's operating-model argument is that the unit of delivery should be a persistent, cross-functional, product-oriented team ("pod" or "squad") with a business owner, and that enterprises should move from project funding to persistent-team funding. Apply it by: (1) listing current AI work and classifying it as project or persistent team; (2) identifying the domains with enough demand to justify a persistent team; (3) designing the platform team as its own product; (4) moving funding to the persistent-team model domain by domain.

### 3.2 BCG 10-20-70 (BCG, 2023–24) [PUBLIC — BCG publications, 2023–24]

BCG's rule of thumb is that roughly 10% of the effort and value in AI is algorithms, 20% is technology and data, and 70% is people, process and business change. As an operating-model heuristic it says the talent plan and the change capacity, not the model team, should dominate the resourcing plan. Apply it as a check: if 70% of your AI headcount is engineers and data scientists, your model is inverted.

### 3.3 Hub-and-spoke design method (attribution: widely used in practice; variants by firm)

A step-by-step method for designing a hub-and-spoke model:

1. **Inventory demand** by business unit: number of live and pipeline use cases, their risk tier, their shared pattern (RAG assistant, document extraction, agentic workflow, classical scoring).
2. **Classify capabilities** as scarce, risky, reusable, or domain-specific. Scarce, risky and reusable go to the hub.
3. **Size the hub** from the platform roadmap and the validation workload (inventory size divided by validator ratio).
4. **Size the spokes** from the use-case pipeline (product owners, embedded engineers, SMEs) and set a minimum viable spoke (a unit with fewer than three pipeline use cases is served from the hub instead).
5. **Define decision rights** in a RACI across intake, prioritisation, funding release, tiering, validation sign-off, go-live, kill switch, and benefits sign-off.
6. **Set the funding model** per Section 2.6 and agree benefits-register rules with Finance.
7. **Sequence the transition** so the platform reaches a minimum service level before people move to spokes.

### 3.4 Three lines of defence extended to AI (Institute of Internal Auditors model; extension to AI by practitioners)

First line: use-case owners and delivery teams own controls, monitoring and evals. Second line: AI risk and model validation set standards, tier systems, validate independently and can veto. Third line: internal audit tests both. The AI-specific extension is that the second line must own the AI inventory and tiering, and the first line must produce evidence (eval results, monitoring dashboards, override logs) in a form the second line can consume. UK PRA SS1/23 `[PUBLIC – PRA, 2023; VERIFY scope]` makes the SMF (senior management function) accountability for model risk explicit, which in practice pushes the second-line role up to the CRO's direct reports.

## 4. Worked BFSI example

`[ILLUSTRATIVE – composite]` Northgate Bank: a mid-sized UK bank, GBP 45bn assets, 7,000 staff, three business units (Retail, Commercial, Wealth), a 25-model classical inventory, four GenAI pilots run by a 12-person central "Data Science CoE", and a stated ambition of 30 AI use cases in production within 3 years with GBP 60m annual run-rate benefit.

### 4.1 Demand inventory

| Business unit | Live | Pipeline (scored) | Dominant patterns | Highest tier |
|---|---|---|---|---|
| Retail | 2 | 11 | Servicing assistant, disputes agent, collections scoring | Tier 1 (collections) |
| Commercial | 1 | 9 | KYB onboarding agent, credit memo drafting, payments investigations | Tier 1 (credit) |

| Business unit | Live | Pipeline (scored) | Dominant patterns | Highest tier |
|---|---|---|---|---|
| Wealth | 0 | 4 | Adviser assistant, suitability report drafting | Tier 2 |
| Group functions (Risk, Finance, Ops) | 1 | 8 | AML alert narratives, reconciliation, policy Q&A, code modernisation | Tier 1 (AML) |

Conclusion: Retail, Commercial and Group functions justify spokes. Wealth, with four pipeline cases, is served from the hub until demand grows.

## 4.2 Target headcount and cost

Loaded cost assumptions `[ESTIMATE — UK, 2025, fully loaded including employer costs and overhead]`: engineer or data scientist GBP 130k; product owner GBP 120k; validator GBP 140k; platform engineer GBP 145k; change lead GBP 110k; architect GBP 160k; PMO GBP 100k; Responsible AI lead GBP 150k; SME time charged at GBP 90k equivalent.

| Role | Hub | Retail spoke | Commercial spoke | Group spoke | Total | Annual cost (GBP m) |
|---|---|---|---|---|---|---|
| Platform engineers | 10 | 0 | 0 | 0 | 10 | 1.45 |
| AI architects | 3 | 0 | 0 | 0 | 3 | 0.48 |
| AI risk and validation | 6 | 0 | 0 | 0 | 6 | 0.84 |
| Responsible AI | 2 | 0 | 0 | 0 | 2 | 0.30 |
| AI PMO and value | 4 | 0 | 0 | 0 | 4 | 0.40 |
| Shared ML engineers | 5 | 0 | 0 | 0 | 5 | 0.65 |
| AI product owners | 1 | 4 | 3 | 3 | 11 | 1.32 |
| Agent / AI engineers | 4 | 8 | 7 | 6 | 25 | 3.25 |
| Data engineers | 2 | 4 | 4 | 4 | 14 | 1.82 |
| Change and adoption | 1 | 2 | 1 | 1 | 5 | 0.55 |
| Domain SMEs (FTE-equivalent) | 0 | 3 | 3 | 3 | 9 | 0.81 |
| Total | 38 | 21 | 18 | 17 | 94 | 11.87 |

Arithmetic check on the largest line: 25 agent engineers x GBP 130k = GBP 3.25m. Total people cost GBP 11.9m a year at steady state, against a GBP 60m benefit ambition. Platform run costs (cloud, model consumption, tooling licences) are modelled separately in Module 8; here we assume GBP 4m a year at steady state `[ESTIMATE]`, giving a total operating-model run cost of around GBP 16m, or 27% of targeted benefit.

## 4.3 Validation capacity check

Target inventory at year 3: 25 classical models plus 30 AI use cases, of which an estimated 12 are Tier 1 or Tier 2 requiring full independent validation and annual re-validation. At a ratio of one validator per 5 systems

in scope for full validation, plus a lighter touch on the rest, 6 validators is adequate [ESTIMATE]. The current CoE has one part-time validator, which means validation, not engineering, is the binding constraint on the roadmap in year 1.

### 4.4 Funding model

Hybrid: hub platform and risk functions (GBP 4.1m people plus GBP 4m run) funded centrally from the Group technology budget; spoke teams funded from business-unit budgets with benefits booked in those units; a GBP 3m a year stage-gated innovation pool held by the AI council for new domains, released at PoC/pilot/MVP gates. Benefits register owned by the AI PMO, signed quarterly by business-unit CFOs.

### 4.5 24-month transition plan

| Months | Hub actions | Spoke actions | Exit criteria |
|---|---|---|---|
| 0 to 3 | Appoint Head of AI; charter AI council; adopt tiering matrix; hire 2 validators; define platform roadmap | Name product owners for top 6 use cases; agree spoke budgets | Council live; tiering approved by CRO; roadmap signed |
| 4 to 9 | Ship gateway, retrieval service, eval harness v1; validate first Tier 1 agent; set chargeback rules | Form Retail and Commercial spokes with 4 engineers each seconded from CoE and 6 external hires; 3 use cases to pilot using hub services | Two platform services in production use; first Tier 1 validation complete |
| 10 to 15 | Guardrail service and orchestration runtime; validation capacity to 5; Responsible AI policy adopted | Group functions spoke formed; 8 use cases in production; change leads in each spoke | Eight use cases live; no use case using unapproved model routes |
| 16 to 24 | Observability and registry; chargeback live; third-party assurance of AI control framework | 18 to 20 use cases live; spokes self-sufficient on standard patterns; Wealth demand reviewed for spoke | Benefits register shows GBP 25m run-rate [ILLUSTRATIVE]; audit finds no high-severity gaps |

So what for the client: the bank's constraint is not engineers, it is validation capacity and the absence of a platform; fund those two first, and the use-case roadmap becomes achievable with a 94-person model costing roughly a quarter of the targeted benefit.

## 5. Real-world case

**DBS Bank** [PUBLIC — DBS annual reports and press, 2023–2025; VERIFY figures]. DBS has reported a portfolio of around 800 AI/ML models supporting around 350 use cases, with projected economic value above SGD 1bn in 2025. The operating-model features DBS has described publicly include a central data and AI platform, industrialised MLOps, business-embedded data teams, and a governance framework for responsible AI (its PURE principles: Purposeful, Unsurprising, Respectful, Explainable). The lesson for operating-model design is that scale came from treating the platform and governance as products while use-case ownership sat in the business. DBS also announced in February 2025 that it expected around 4,000 temporary and contract roles to reduce over three years as AI absorbed work, which is a reminder that the talent layer of the TOM includes workforce transition, not only hiring.

**JPMorgan Chase** [PUBLIC — company statements and shareholder letters, 2023–2024; VERIFY figures]. JPMorgan's rollout of its LLM Suite to around 200,000 employees in 2024 was possible because of a central platform and a firm-wide AI and data organisation under a Chief Data and Analytics Officer, with business-aligned teams. The public narrative emphasises a central platform with enterprise-wide access controls and business-led use cases, a hub-and-spoke pattern in substance.

## 6. Consultant language

> *"Centralise what is scarce, risky or reusable; decentralise what needs domain knowledge and speed."*

Use when the client asks whether to build a CoE or push AI into the business. It gives a rule rather than an opinion.

> *"Your binding constraint is validation capacity, not engineering capacity."*

Use when a bank has a long roadmap and a tiny second line. It redirects a hiring conversation to the real bottleneck.

> *"The platform is a product with customers, a roadmap and service levels, not a shared-services pool."*

Use when the central team is being treated as an order-taker and the spokes are rebuilding the same components.

> *"The funding model decides where benefits are counted, so we design the benefits register before we design the budget."*

Use with the CFO when the hub looks like pure cost.

> *"We are moving from project funding to persistent-team funding, domain by domain."*

Use when describing the delivery layer; it signals familiarity with the Rewired argument without name-dropping.

## 7. Common mistakes juniors make

- **Drawing an org chart instead of designing decision rights.** The RACI across intake, tiering, validation, go-live and kill switch is the operating model; the boxes are secondary.
- **Sizing the team from the use-case count only.** Validation workload, platform roadmap and change capacity drive headcount at least as much.
- **Recommending federated to a low-maturity bank** because the business asked for autonomy. Without a platform and standards it produces five vector databases and no model inventory.
- **Ignoring the second line.** Proposing an operating model with no independent validation role is a regulatory finding waiting to happen under SS1/23 or SR 11-7.
- **Treating the SME as free.** Domain SME time for labelling and exception design is the most frequently missing line in the resourcing plan.
- **Confusing the AI CoE with the data office.** They overlap on data governance, but the AI function needs evaluation, model risk engineering and agent platform skills the data office does not have.
- **Presenting a target state with no transition.** The sequencing (platform before people move) is what makes it credible.

## 8. Interview questions with model answers

### Q1. A UK bank with 6,000 staff has a 15-person central data-science team and 40 GenAI ideas from the business. What operating model do you recommend and why?

Model answer: I would start by diagnosing three things: demand concentration by business unit, the maturity of the shared platform, and second-line capacity. With 40 ideas across presumably three or four units, a pure CoE of 15 will become a queue, and a federated model would fragment standards before any exist. I would recommend a hub-and-spoke model: the hub keeps platform engineering, architecture, evaluation, independent validation and Responsible AI; each business unit with at least three credible

pipeline cases gets a spoke with a product owner, two to four AI engineers and embedded data engineering. I would sequence it so the hub ships a model gateway, a retrieval service and an eval harness in the first six to nine months before engineers move into spokes, because otherwise the spokes rebuild the plumbing. Funding would be hybrid: central for platform and risk, business-funded for use cases, with a stage-gated pool for new domains. The number I would put on the table is a target of roughly 60 to 90 people at steady state including spokes, with validation capacity sized from the tiered inventory rather than from headcount ratios. The caveat is that this depends on the bank having at least one business leader willing to own outcomes; without that, I would keep the CoE and spend the first year on platform and governance.

### Q2. How would you size the independent validation function for an AI portfolio?

Model answer: From the inventory and tiering, not from a ratio of engineers. First, I would agree a tiering matrix with the CRO: materiality, autonomy, customer impact and regulatory exposure drive a Tier 1 to Tier 3 classification. Second, I would estimate the steady-state inventory by tier from the pipeline. Third, I would apply effort norms: a Tier 1 agentic system needs initial validation of perhaps 8 to 12 weeks of a validator's time, plus annual re-validation and event-driven re-validation on each material model or prompt change; Tier 3 might be a desk review. Dividing total annual validation effort by productive capacity per validator gives the headcount; in practice that lands around one validator per four to six Tier 1 and Tier 2 systems. I would present that as an estimate and stress-test it against the roadmap: if the plan puts six Tier 1 agents into pilot in the same quarter, validation is the critical path, and the roadmap needs to flex or validation needs external support. I would also say that LLM validation skills are scarce, so the sourcing plan matters as much as the number.

### Q3. The CFO says the central AI team is a cost centre with no measurable return. How do you respond?

Model answer: I would agree with the observation and reframe its cause. A centrally funded hub whose benefits land in business-unit P&Ls will always look like cost, because the accounting is asymmetric. The fix is structural: a benefits register that attributes run-rate benefit to each use case, signed quarterly by the business-unit CFO, and reported alongside hub spend. Then I would separate the hub's cost into platform, which should be judged on reuse (number of use cases served, cost per use case served, time to first pilot), and risk functions, which should be judged on control outcomes, not return. Finally I would propose chargeback for platform services once they are mature, which disciplines demand and gives the hub a revenue line. With illustrative numbers: if the hub costs GBP 8m and serves 20 use cases with GBP 40m of signed run-rate benefit, that is a reasonable 5x; the point is to make the number visible rather than to argue about whether it exists.

## 9. Self-check quiz

1. Name the three things whose location distinguishes a centralised CoE from a hub-and-spoke model from a federated model.
2. A bank plans 40 AI systems in inventory at year 3, of which 14 are Tier 1 or Tier 2 requiring full independent validation. Using a ratio of one validator per five such systems, plus one validator for the remaining lighter-touch systems, how many validators are needed? What is the binding assumption?
3. Which funding model most often leaves the shared platform unfunded, and what hybrid fixes it?
4. State BCG's 10-20-70 rule and explain what it implies for the composition of an AI operating model's headcount.
5. Why should the platform reach a minimum service level before engineers move from a central team into business-unit spokes?

# Module 14 — Change Management, Adoption and the KPI Framework

## 1. Why a client cares

An AI system that is accurate but unused delivers nothing, and an AI system that is used but unmeasured cannot be defended at the next budget round. Clients have been told by their own boards that "95% of GenAI pilots show no P&L impact" `[PUBLIC — MIT NANDA, Aug 2025; VERIFY, small self-reported sample]`, and the two most common causes are adoption failure and measurement failure. This module covers the mechanics of both: how people actually come to rely on an AI system, and how to build a KPI chain from model behaviour to the P&L that a CFO will accept.

## 2. Concept and mechanism

### 2.1 Adoption is a trust-calibration problem

Human reliance on an automated system is governed by calibration: the degree to which the user's trust matches the system's actual reliability. The research base is in human factors (Lee and See, "Trust in Automation", 2004; Parasuraman and Riley, "Humans and Automation: Use, Misuse, Disuse, Abuse", 1997) `[PUBLIC — Human Factors journal]`. Four states matter:

- **Calibrated trust:** reliance matches reliability. The target state.
- **Over-trust (misuse):** users accept outputs without checking. In an agentic dispute-resolution system this shows up as human reviewers approving 99% of agent recommendations in under ten seconds, which means the control has become theatre.
- **Under-trust (disuse):** users re-do the work or route around the system. Shows up as low usage, parallel spreadsheets, and "I just check it myself anyway".
- **Abuse:** the system is deployed for tasks it was not designed or validated for.

Two failure mechanisms are specific to AI in operations:

- **Automation bias and complacency.** When a system is right most of the time, humans stop scrutinising. Documented in aviation and clinical decision support; directly transferable to AML alert triage and KYC document review. The design counter-measures are: forced disagreement sampling (show the reviewer cases where the agent is uncertain), confidence display calibrated to actual accuracy, random blind audits, and rotating reviewers between assisted and unassisted work.
- **Skill atrophy.** If junior analysts never do first-pass work, the bank loses the pipeline that produces senior analysts who can judge the agent's output. Mitigations include deliberate unassisted rotations, using the agent as a training tool (explain-the-reasoning mode), and redefining the junior role around exception handling and quality assurance.

### 2.2 Role redesign is the real change

AI in a process changes what a role does rather than removing the role in one step. For a disputes analyst, the shift is from case assembly (gather evidence, read merchant response, apply reason-code rules) to adjudication of exceptions and supervision of the agent's queue. That changes required skills (judgement over procedure), performance measures (quality and override accuracy over cases per hour), career path, and span of control. If performance measures stay on cases per hour, the analyst is incentivised to rubber-stamp, which destroys the control. Role redesign therefore has to be done jointly by Operations, HR and Risk, and before go-live, not after.

## 2.3 Workforce transition

The honest frame for a client is capacity: an agentic system releases hours. What the bank does with the hours is a management decision with four options: redeploy to backlog or growth work, redeploy to quality and control work, reduce attrition backfill, or reduce headcount. The business case should state which. The public record shows the cost of getting this wrong in both directions: Klarna announced in 2024 that its assistant did the work of around 700 agents and in 2025 said it had over-rotated on cost and would re-hire humans for quality `[PUBLIC — Klarna, Feb 2024; CEO interviews, May 2025; VERIFY]`; Commonwealth Bank of Australia reversed 45 call-centre redundancies in 2025 after a union challenge showed call volumes had not fallen as assumed `[PUBLIC — Aug 2025; VERIFY]`.

Reskilling has a cost line. For an operations workforce the typical plan has three tracks: AI-supervisor skills for the retained analysts (prompting, interpreting confidence, escalation), quality and controls skills for a smaller group that becomes the human assurance layer, and transition support for roles that end.

## 2.4 Incentive alignment

Adoption follows incentives more than training. Design checks:

- Are team leaders measured on throughput in a way that penalises taking time to review agent output properly?
- Does a reviewer who overrides the agent face more work (writing justification) than one who approves? If so, approvals will be inflated.
- Is the business unit credited with benefits only if headcount falls, which makes managers resist adoption to protect teams?
- Do the people whose work changes have any say in the exception taxonomy and the HITL thresholds? Co-design is the cheapest adoption lever available.

## 2.5 The KPI framework: three layers and a causal chain

A defensible KPI framework for an AI system has three layers, and the value of the framework is in the arrows between them.

| Layer | What it measures | Examples | Owner | Cadence |
|---|---|---|---|---|
| Model and system KPIs | Does the system behave as designed | Accuracy against golden set; groundedness or faithfulness; hallucination rate; precision/recall per decision class; latency p50/p95; cost per task; tool-call success rate; override rate; escalation rate; drift indicators | AI product owner, platform | Daily to weekly |
| Operational KPIs | Did the process change | Cycle time; straight-through processing rate; touches per case; rework rate; backlog age; analyst hours per case; adoption rate (share of eligible cases routed via the system); active-user rate | Process owner | Weekly to monthly |
| Business KPIs | Did the P&L or risk position change | Cost per unit (per dispute, per | Business-unit leader, Finance | Monthly to quarterly |

| Layer | What it measures | Examples | Owner | Cadence |
|---|---|---|---|---|
| | | onboarding); FTE capacity released; revenue retained or won; loss rate (chargeback write-offs, fraud losses); regulatory breaches; complaints; NPS or CSAT; time-to-revenue | | |

The causal chain for a dispute-resolution agent reads: higher accuracy and groundedness (model) allow a higher auto-adjudication threshold, which raises straight-through rate and lowers touches per case (operational), which lowers cost per dispute and cycle time, which improves recovery rate on merchant chargebacks and customer satisfaction (business). A KPI framework that lists metrics without these arrows cannot explain why a business KPI moved, and a CFO will attribute the movement to something else.

## 2.6 Leading versus lagging indicators

Leading indicators move first and predict the business result: adoption rate, override rate, straight-through rate, eval accuracy on new golden cases, user confidence survey. Lagging indicators confirm it: cost per unit, loss rate, NPS, FTE released. In the first six months of a deployment, the steering committee should be managing on leading indicators, because the lagging ones are dominated by noise and ramp. A common governance failure is to review only lagging indicators quarterly, discover a problem two quarters late, and then blame the model.

## 2.7 KPI tree construction

Build the tree top-down from the business case lever and bottom-up from the system's measurable behaviour, and make them meet:

1. Start with the business-case benefit lines (for example, cost per dispute down 35%).
2. Decompose each into operational drivers (touches per case, handle time per touch, straight-through rate).
3. For each operational driver identify the system behaviours that move it (auto-adjudication threshold, accuracy at that threshold, evidence-retrieval completeness).
4. For each system behaviour name the eval or monitoring metric and its target.
5. Mark each node leading or lagging and assign an owner and a data source.
6. Define the baseline measurement method and period for every node before go-live.

## 2.8 Measurement pitfalls

- **Baseline inflation and drift.** The pre-AI baseline is measured in a quarter with unusual volume, or the process changes for other reasons during the pilot. Use a 6 to 12 month baseline and a control group where feasible (a region, a product line, or a randomised share of cases).
- **Hawthorne effect.** Pilot teams perform better because they are observed. Control groups and a sustainment check at month 9 to 12 address this.
- **Attribution.** Other initiatives (a new case-management tool, a policy change) move the same KPI. Agree attribution rules with Finance up front; where a control group is impossible, use a pre-agreed attribution share.
- **Survivorship in eval sets.** Golden datasets built from historical closed cases over-represent the easy cases; sample from live traffic quarterly.
- **Goodhart's law.** When straight-through rate becomes the target, thresholds get lowered and quality falls. Pair every efficiency KPI with a quality KPI (override accuracy, audit error rate).

- **Measuring hours released, not hours redeployed.** Finance will only count what is redeployed or removed; the KPI framework should track both and show the gap.

# 3. Frameworks

## 3.1 Kotter's eight steps (John Kotter, "Leading Change", 1996) [PUBLIC — Harvard Business School Press]

Create urgency; build a guiding coalition; form a strategic vision; enlist a volunteer army; enable action by removing barriers; generate short-term wins; sustain acceleration; institute change. Applied to AI programmes: the "short-term wins" step maps directly to the wave-one quick wins in the use-case roadmap, and "remove barriers" is usually about data access, risk approval paths and performance measures rather than attitudes.

## 3.2 ADKAR (Prosci, Jeff Hiatt, 2003) [PUBLIC — Prosci]

Awareness, Desire, Knowledge, Ability, Reinforcement, assessed at the individual level. ADKAR is more useful than Kotter for operations teams because it diagnoses where a specific population is stuck. For an AI deployment, the common gap is Desire (fear about roles) and Reinforcement (performance measures unchanged). Apply by surveying each affected role against the five elements before go-live and at month three, and targeting interventions at the lowest-scoring element.

## 3.3 Trust-in-automation calibration (Lee and See, 2004) [PUBLIC — Human Factors]

Use as the design frame for HITL: the aim is calibrated reliance, achieved through transparency (show reasoning and evidence), displayed confidence that matches real accuracy, and feedback (show the reviewer when they were right to override). Pair with Parasuraman and Riley's misuse/disuse taxonomy when explaining override-rate anomalies to a steering committee.

## 3.4 Three-layer KPI framework with causal chain (attribution: generic practitioner framework; the model/operational/business layering is common across consultancies)

Steps are in Section 2.7. The framework's test is whether every business KPI has at least one leading operational KPI and one system KPI beneath it with named owners and baselines.

## 3.5 Balanced scorecard (Kaplan and Norton, 1992) [PUBLIC — Harvard Business Review]

Useful when the client wants AI KPIs integrated into existing performance management: financial, customer, internal process, learning and growth. Map business KPIs to the first two perspectives, operational KPIs to internal process, and adoption and skills metrics to learning and growth.

# 4. Worked BFSI example

`[ILLUSTRATIVE — composite]` Meridian Cards, a UK card issuer with 4.2 million cards, processes 420,000 cardholder disputes a year in an operations centre of 210 disputes analysts and 18 team leaders. A supervisor-led multi-agent dispute system (evidence gathering, reason-code classification, merchant-response analysis, draft adjudication, with HITL approval) goes live in January.

## 4.1 Baseline (12-month average, prior year)

| Metric | Baseline |
|---|---|
| Disputes per year | 420,000 |
| Average touches per case | 3.4 |
| Average handle time per touch | 11 minutes |

| Metric | Baseline |
|---|---|
| Analyst hours per case | 0.62 |
| Cycle time to provisional decision | 6.8 days |
| Straight-through (no analyst touch) | 0% |
| Cost per dispute (fully loaded) | GBP 31 |
| Chargeback recovery rate (merchant liable cases) | 61% |
| Dispute-related complaints per 1,000 disputes | 14 |
| Dispute CSAT | 62 |

Cost per dispute check: 210 analysts at GBP 52k loaded plus 18 team leaders at GBP 68k = GBP 12.1m; plus systems and overhead GBP 0.9m; total GBP 13.0m / 420,000 = GBP 31.

## 4.2 KPI tree with targets at 3, 6 and 12 months

| Layer | KPI | Type | Baseline | Month 3 | Month 6 | Month 12 | Owner |
|---|---|---|---|---|---|---|---|
| System | Adjudication accuracy vs golden set (n=1,500) | Leading | n/a | 91% | 94% | 96% | AI product owner |
| System | Groundedness of draft rationale (LLM-judge, sampled) | Leading | n/a | 95% | 97% | 98% | AI product owner |
| System | Override rate by reviewers | Leading | n/a | 22% | 14% | 9% | AI product owner |
| System | Override accuracy (share of overrides that were correct on audit) | Leading | n/a | 85% | 88% | 90% | Quality lead |
| System | Cost per case (tokens and infra) | Leading | n/a | GBP 1.10 | GBP 0.85 | GBP 0.70 | Platform |
| System | Latency p95 for draft decision | Leading | n/a | 90 s | 60 s | 45 s | Platform |
| Operational | Share of eligible disputes routed via agent (adoption) | Leading | 0% | 40% | 75% | 95% | Ops director |
| Operational | Straight-through rate (auto-adjudicated low-risk classes) | Leading | 0% | 10% | 25% | 40% | Ops director |
| Operational | Touches per case | Leading | 3.4 | 2.6 | 1.9 | 1.5 | Ops director |
| Operational | Cycle time to provisional decision (days) | Lagging | 6.8 | 5.0 | 3.2 | 2.0 | Ops director |
| Operational | Rework rate (cases reopened) | Lagging | 7% | 7% | 5% | 4% | Quality lead |
| Business | Cost per dispute | Lagging | GBP 31 | GBP 28 | GBP 23 | GBP 19 | CFO BU |
| Business | Analyst capacity released (FTE-equivalent) | Lagging | 0 | 20 | 60 | 85 | CFO BU |
| Business | Chargeback recovery rate | Lagging | 61% | 62% | 65% | 68% | Head of disputes |
| Business | Complaints per 1,000 disputes | Lagging | 14 | 14 | 11 | 9 | Conduct risk |
| Business | Dispute CSAT | Lagging | 62 | 63 | 68 | 72 | CX lead |

Arithmetic for the month-12 cost per dispute: analyst hours per case fall from 0.62 to roughly 0.28 (1.5 touches x 11 minutes, plus review time on agent drafts). Capacity released of 85 FTE out of 210 is 40%; GBP 13.0m x (1 - 0.40 x 12.1/13.0) = GBP 8.2m of people cost; add agent run cost 420,000 x GBP 0.70 = GBP 0.29m and platform allocation GBP 0.5m; total GBP 8.0m / 420,000 = GBP 19 `[ESTIMATE]`. Recovery-rate improvement is worth more than the cost saving: 420,000 disputes, 55% merchant-liable, average GBP 140, 7 percentage points of recovery = GBP 2.3m a year `[ESTIMATE]`.

## 4.3 Adoption plan

| Phase | Months | Actions | Adoption KPI gate |
|---|---|---|---|
| Co-design | -3 to 0 | 12 analysts and 3 team leaders design exception taxonomy, HITL thresholds and the review screen; HR redesigns role profile (adjudicator and quality analyst); performance measures changed from cases per hour to quality-weighted throughput | Role profiles and measures signed by HR, Ops and Risk |
| Pilot cohort | 1 to 3 | Two teams (40 analysts), assisted mode only (no straight-through); daily calibration huddle reviewing override cases; forced disagreement sampling at 10% | 40% routing; override rate below 25%; override accuracy above 85% |
| Expand | 4 to 6 | All teams; straight-through enabled for two lowest-risk reason-code classes; reviewers rotate one day a week unassisted to preserve skill; monthly ADKAR pulse | 75% routing; 25% straight-through; complaints flat or down |
| Sustain | 7 to 12 | Straight-through extended to further classes on evidence; quality analyst track formed (25 roles); redeployment of released capacity to fraud investigations backlog and collections; quarterly blind audit | 95% routing; override accuracy above 90%; audit error rate below 2% |

## 4.4 Dashboard specification

- **Audience and cadence:** AI product owner daily (system layer); operations steering weekly (system and operational); executive and CFO monthly (all layers with the causal chain shown left to right).
- **Structure:** one page per layer plus a summary page showing the chain: accuracy and override rate, then straight-through and touches per case, then cost per dispute and recovery rate, each with baseline, target and actual.
- **Data sources:** eval harness (accuracy, groundedness), orchestration audit log (override, escalation, latency, cost), case-management system (touches, cycle time, rework), general ledger and workforce system (cost, FTE), complaints system, CSAT survey.
- **Controls shown alongside efficiency:** override accuracy, blind-audit error rate, complaint rate, and a drift indicator (accuracy on the latest 200 live-sampled cases vs golden set).
- **Attribution note on the summary page:** control group of 10% of cases held in manual processing for the first six months to isolate the agent's effect from seasonality and policy changes.

So what for the client: the business case rests on two leading indicators, override rate and straight-through rate, and the adoption plan exists to move those; the CFO sees cost per dispute and recovery rate, but the steering committee should manage the leading pair weekly.

## 5. Real-world case

**Morgan Stanley Wealth Management** `[PUBLIC — Morgan Stanley and OpenAI statements, 2023–2024; VERIFY figures]`. The AI @ Morgan Stanley Assistant, built on GPT-4 for around 16,000 financial advisers, launched in September 2023, with the Debrief meeting-summary tool following in June 2024. Reported adoption of around 98% of adviser teams is unusually high for an enterprise GenAI tool. The publicly described drivers were: a tightly scoped first use case (retrieval over the firm's own research and procedures) where the answer quality was testable; an evaluation programme run with the vendor before rollout; adviser co-design; and a second use case (Debrief) that removed a disliked task rather than adding a tool. The adoption lesson is that the KPI to manage early was adviser usage and answer acceptance, and the firm invested in evaluation and content curation to move it.

**Klarna** `[PUBLIC — Klarna, Feb 2024; CEO interviews, May 2025; VERIFY]`. The February 2024 announcement quantified the assistant's impact in operational KPIs (two-thirds of chats handled, resolution time from 11 minutes to under 2, repeat inquiries down 25%) and a business KPI (estimated USD 40m profit improvement). By 2025 the CEO acknowledged the firm had over-weighted cost and under-weighted quality and would re-introduce human agents for customers who wanted them. The KPI lesson is that the framework lacked a paired quality KPI with equal weight to the efficiency KPI, and the workforce-transition decision was made before the lagging customer indicators had matured.

## 6. Consultant language

> *"The control is only as good as the override rate is honest."*

Use when the client reports a very low override rate as a success; it reframes it as a possible automation-bias signal.

> *"We will manage the programme on leading indicators and report it on lagging ones."*

Use when designing steering-committee governance; it sets expectations about what the CFO will see and when.

> *"Every efficiency KPI gets a paired quality KPI with equal weight."*

Use to pre-empt Goodhart's law in target setting.

> *"Capacity released is a management option, not a benefit, until you decide what to do with it."*

Use with the CFO and the COO together when the business case counts hours.

> *"Role redesign happens before go-live, with HR and Risk in the room, or adoption will be whatever the old performance measures reward."*

Use when the client plans to "roll it out and see".

## 7. Common mistakes juniors make

- **Treating change management as training and communications.** The levers that move adoption are performance measures, role design, incentive alignment and co-design; training is necessary and insufficient.
- **Reporting adoption as logins.** Adoption is the share of eligible work routed through the system and the share of recommendations acted on; logins measure curiosity.

- **Listing KPIs without a causal chain.** If the business KPI moves, nobody can say why; if it does not, nobody can say which upstream metric to fix.
- **Celebrating a near-zero override rate.** It usually signals automation bias or a review step that has become a rubber stamp.
- **Using a one-month baseline.** Seasonality in disputes, onboarding and collections is large; use 6 to 12 months and a control group.
- **Counting hours released as savings.** Finance will haircut them to zero unless redeployment or headcount decisions are explicit.
- **Ignoring skill atrophy.** Removing all first-pass work from juniors destroys the future supply of people who can judge the system.

## 8. Interview questions with model answers

### Q1. How would you design the KPI framework for an agentic AI deployment so that a CFO accepts the benefits?

Model answer: I would build three layers and insist on the links between them. The system layer measures whether the agent behaves as designed: accuracy against a golden dataset, groundedness, override rate, override accuracy, latency and cost per task. The operational layer measures whether the process changed: routing or adoption rate, straight-through rate, touches per case, cycle time, rework. The business layer measures the P&L and risk effect: cost per unit, capacity released and what was done with it, loss or recovery rates, complaints, satisfaction. The CFO conversation is won on three things. First, baselines agreed before go-live over a 6 to 12 month window, with a control group where possible. Second, attribution rules agreed in advance for KPIs that other initiatives also move. Third, the distinction between capacity released and capacity redeployed or removed, with the business case stating which. I would manage the programme weekly on the leading indicators, override rate and straight-through rate in particular, and report monthly on the lagging ones, and I would pair every efficiency KPI with a quality KPI so that the targets cannot be hit by lowering thresholds. As an illustration, in a disputes deployment I would expect cost per dispute to be the headline, but the steering committee should be watching whether override accuracy stays above 90% as the straight-through rate rises.

### Q2. Six months into a deployment, reviewer override rate has fallen from 20% to 2%. The operations head calls it a success. What is your reaction?

Model answer: I would treat it as a signal to investigate rather than a result to celebrate, because a 2% override rate is consistent with two very different stories. Either the agent's accuracy has genuinely risen to around 98% on the routed population, or reviewers have stopped reviewing, which is automation bias. The way to distinguish them is to look at the paired metrics: blind audit error rate on a random sample of approved cases, median review time per case, override accuracy, and whether accuracy on the latest live-sampled golden cases has actually improved. If audit error rate is up or review time has collapsed to a few seconds, the human-in-the-loop control has become theatre and the bank is exposed, particularly if the system is Tier 1 under its model-risk framework. The remedies are design changes, not exhortation: forced disagreement sampling where the reviewer must decide a case before seeing the agent's view, calibrated confidence display, rotation into unassisted work, and performance measures that reward correct overrides. I would also raise it with the second line, because the validation of the system assumed a functioning review step.

### Q3. What is the difference between leading and lagging indicators in an AI programme, and why does it matter for governance?

Model answer: Leading indicators move early and predict the outcome: adoption rate, override rate, straight-through rate, eval accuracy on new cases, user confidence. Lagging indicators confirm the outcome:

cost per unit, loss rates, satisfaction, capacity released. It matters because lagging indicators in the first two quarters are dominated by ramp, seasonality and noise, so a steering committee that reviews only lagging indicators quarterly will discover problems two quarters late and will have no way to diagnose them. Governance should therefore run at two cadences: weekly on leading indicators owned by the product and operations leads, with pre-agreed thresholds that trigger action (for instance, override rate above 25% pauses threshold increases), and monthly or quarterly on lagging indicators for the executive and the CFO. The causal chain between the layers is what lets you explain a lagging movement using a leading cause, which is also what makes the benefit attributable rather than coincidental.

## 9. Self-check quiz

1. Name the four states in the trust-in-automation taxonomy and give one operational symptom of each in an AML alert-triage deployment.
2. A disputes operation handles 300,000 cases a year at 3.0 touches per case and 12 minutes per touch, with analysts costing GBP 50 per productive hour. If an agent reduces touches to 1.6 and adds 2 minutes of review time per case, what is the annual change in analyst cost? Show the arithmetic.
3. Which element of ADKAR is most often the gap in operations teams facing AI deployment, and which single design choice addresses it most directly?
4. Why should every efficiency KPI be paired with a quality KPI? Name the principle and give an example pair.
5. What are two methods to protect the benefit measurement from baseline drift and the Hawthorne effect?

# Module 15 — AI in Managed Services: AI-led Delivery, the Labour-model Shift and Outcome-based Pricing

## 1. Why a client cares

Banks outsource large volumes of KYC refresh, AML alert review, disputes processing, payments investigations and reconciliations to managed-service providers on contracts priced per FTE or per transaction. Agentic AI removes a large share of the labour inside those contracts. The client's questions are direct: should my provider's price fall, by how much, who owns the AI, who carries the risk when it is wrong, and should I insource it instead. For the consulting firm, the same shift changes its own business model, from selling people-hours to selling outcomes delivered partly by software.

## 2. Concept and mechanism

### 2.1 From labour arbitrage to automation arbitrage

The traditional managed-services model earns a margin on labour arbitrage: the provider delivers the same work from a lower-cost location with a productivity uplift from process standardisation. The unit of sale is an FTE or a transaction, and the provider's margin is the difference between the price per unit and the fully loaded labour cost per unit. Scale economics are weak because every additional unit of work needs additional labour.

Agentic automation changes the cost function. A large share of each transaction is performed by software whose marginal cost (tokens, inference infrastructure, monitoring) is a fraction of labour cost, and the remaining human work concentrates in exceptions, quality assurance and supervision. The provider's margin now comes from automation arbitrage: the gap between a price still anchored to the labour-based unit and a cost base that has partly become software. The commercial tension in every contract renegotiation in 2025–26 is who captures that gap.

### 2.2 The cost stack before and after

A KYC-refresh managed service in 2024 has a cost stack dominated by analyst labour (roughly 70 to 80% of cost), with the remainder in team leadership, quality, technology licences, facilities and management `[ESTIMATE — assumption, typical BPO structure]`. After agentic automation, four things happen to the stack:

- Analyst labour falls sharply but does not disappear; what remains is more senior and more expensive per head (exception handling, QA, supervising the agent queue).
- A new cost line appears: AI run cost (model consumption, platform, evaluation, guardrails, observability) and AI engineering to maintain the agents, prompts and integrations.
- Transition cost appears once: building and validating the agents, integrating with the client's systems, re-training the retained workforce.
- Risk cost rises: the provider now owns an accuracy liability that used to be diffused across hundreds of individual analysts' judgement.

### 2.3 AI-led delivery models

| Model | Description | Who owns the AI | Typical fit |
|---|---|---|---|
| Provider-embedded AI | Provider deploys its own agents inside its delivery; client sees a lower price or faster SLA | Provider | Commodity processes, low client data sensitivity, client has no AI platform |
| Client-platform, provider- | Agents run on the client's | Client | Regulated processes where |

| Model | Description | Who owns the AI | Typical fit |
|---|---|---|---|
| operated | platform and models; provider supplies the human layer and process operation | | the client wants model-risk ownership and audit trail |
| Joint build, shared IP | Co-developed agents with agreed IP and gain-share | Shared | Strategic partnerships; client wants to retain option to insource |
| AI product plus thin service layer | Vendor sells software with a small managed exception-handling team | Vendor | Narrow, well-defined tasks (document extraction, sanctions name screening) |

The ownership question drives model-risk accountability. Under UK PRA SS1/23 `[PUBLIC — PRA, 2023; VERIFY scope]` and the broader outsourcing and third-party risk rules (PRA SS2/21 on outsourcing and third-party risk management, EBA outsourcing guidelines 2019, and the EU Digital Operational Resilience Act in force from January 2025 `[PUBLIC; VERIFY]`), the regulated bank cannot outsource its accountability for a model's outcomes. If the provider's agent makes KYC risk-rating decisions, the bank needs model inventory entries, validation evidence and audit access as if it had built the agent itself.

## 2.4 The labour-model shift: pyramid flattening and agent-to-human ratios

A 400-FTE KYC-refresh operation in 2024 has a pyramid: perhaps 320 analysts, 50 senior analysts and QC, 24 team leaders, 6 managers `[ILLUSTRATIVE]`. Agentic automation removes most of the base tier's work and shifts the retained roles upward:

- The base of the pyramid shrinks; the agent does first-pass document collection, data extraction, screening-hit analysis and case assembly.
- The middle grows relatively: exception adjudicators, QA reviewers who audit agent output, and "agent supervisors" who manage the HITL queue and the escalation taxonomy.
- New roles appear: agent engineers and prompt maintainers, evaluation analysts who curate golden datasets from live cases, and a model-risk liaison to the client's second line.

A useful planning parameter is the agent-to-human ratio: the number of agent-processed cases per retained human hour. In a mature agentic KYC refresh, 55 to 70% of effort removed implies the retained human handles 2.5 to 3.5 times the case volume of the 2024 analyst `[ESTIMATE]`. This is also the parameter that makes outcome-based pricing viable, because the provider's cost per case becomes predictable once the ratio stabilises.

## 2.5 Pricing models and risk allocation

| Pricing model | Mechanism | Risk sits with | Incentive effect on provider | Fit with AI-led delivery |
|---|---|---|---|---|
| FTE-based (time and materials) | Price per named or equivalent FTE per month | Client (volume and productivity) | None to automate; automation reduces revenue | Poor; provider resists automation |
| Transaction-based | Price per case completed, with quality SLAs | Shared (provider carries productivity, client carries volume) | Strong to automate within quality floor | Good; natural first step |
| Outcome-based | Price tied to business outcomes (cycle time, backlog cleared, risk-rating accuracy, regulatory findings avoided) | Provider carries delivery and partly business risk | Strongest; provider optimises the whole process | Good when outcomes are measurable and attributable |
| Gain-share | Base fee plus share of verified savings or | Shared | Strong, but disputes over baseline and | Useful for transition periods |

| Pricing model | Mechanism | Risk sits with | Incentive effect on provider | Fit with AI-led delivery |
|---|---|---|---|---|
| | benefits | | attribution | |
| Subscription (platform plus service) | Fixed fee for agent platform and exception-handling capacity, tiered by volume band | Provider carries cost, client carries demand within band | Automate aggressively; manage exceptions tightly | Good for steady-state, commoditised processes |

Outcome-based pricing fails when the outcome is not under the provider's control (a backlog driven by client policy changes), when measurement is contested (no agreed baseline), or when the provider lacks the balance sheet to carry downside. Transaction-based pricing with quality SLAs is the most common landing point in 2025–26 because it rewards automation while keeping measurement simple `[ESTIMATE — practitioner observation]`.

## 2.6 "Service-as-software"

The phrase inverts software-as-a-service: instead of selling software that people use to do work, the vendor sells the work itself, performed by software with human oversight, and prices it against the labour budget it replaces rather than against a software licence budget. The framing was popularised in venture-capital essays in 2024 (Foundation Capital, Sequoia) `[PUBLIC — 2024; VERIFY exact attribution]` arguing that AI expands the addressable market for software from IT budgets to services budgets. For a bank, the practical consequence is that the procurement counterparty for a "KYC refresh agent" might be a software company pricing per case, not a BPO pricing per FTE, and the evaluation criteria have to cover both software (security, model risk, integration) and service (SLAs, exceptions, accountability).

## 2.7 Contracting for AI-delivered services

Clauses that matter and that standard BPO contracts do not contain:

- **Accuracy and quality SLAs defined at the decision level**, not only turnaround: for example, risk-rating accuracy against an agreed golden sample, false-negative rate on screening hits, and audit error rate, each with measurement method and sample size.
- **Model and prompt change notification**: material changes to models, prompts, retrieval sources or thresholds notified in advance, with re-validation evidence; client right to veto for Tier 1 processes.
- **Audit and validation access**: client second line and regulators can inspect evaluation results, logs and the HITL audit trail.
- **Data use restrictions**: client data not used to train or fine-tune provider models without consent; data residency; retention.
- **Liability and indemnity** for decision errors, with caps that reflect regulatory exposure (a missed sanctions hit is not a service credit matter).
- **Exit and portability**: ownership of golden datasets, prompts, exception taxonomies and process documentation; transition assistance; escrow of agent configurations.
- **Benefit-sharing mechanics**: how productivity gains flow to price over the term (automatic step-downs, open-book costing, or gain-share), and how a baseline is set and frozen.
- **Human oversight commitments**: minimum HITL coverage by risk tier, reviewer qualification, override logging.

## 2.8 How a client should evaluate an AI-led managed-services proposal

1. Ask for the cost stack, not just the price: what share is labour, AI run, engineering, margin. A provider unwilling to show it is pricing on automation arbitrage.
2. Ask where the agents run and who owns the model risk; map the answer to SS1/23, SR 11-7 and outsourcing rules.

3. Ask for the evaluation evidence: golden dataset size, accuracy by decision class, override rates in live operation at other clients.
4. Test the price trajectory: does the per-unit price step down as automation matures, or is the gain captured entirely by the provider?
5. Test the exit: what does the client own at the end?
6. Compare with the insource option on a 3 to 5 year TCO, including the client's own platform cost and validation effort.

### 2.9 Implications for the consulting firm's own model

Consulting firms face the same shift. Leverage models built on pyramids of analysts doing research, modelling and slide production are being compressed by GenAI; firms are responding with AI-enabled delivery (internal assistants, code generation), asset-based consulting (reusable agents, accelerators, evaluation harnesses sold alongside advice), and outcome-linked fees. Public examples include McKinsey's internal "Lilli" assistant (2023) and the large consultancies' announced AI investments and workforce restructuring `[PUBLIC – various, 2023–2025; VERIFY specific figures]`. For an AI strategy consultant the implication is personal: the defensible value is in problem structuring, judgement about risk and sequencing, and the client relationship, not in producing the first draft.

## 3. Frameworks

### 3.1 Cost-stack decomposition and automation arbitrage analysis (attribution: standard sourcing-advisory practice; no single originator)

Steps: (1) obtain or estimate the provider's current cost stack by line; (2) classify each labour line by automatability using the task-level agentic-fit dimensions from Module 7; (3) estimate the effort removed by line and the new cost lines added; (4) compute the new cost per unit and the implied margin at the current price; (5) decide the target split of the gap between client price reduction and provider margin; (6) select the pricing model that enforces the split over the term.

### 3.2 Pricing-model selection matrix (attribution: generic sourcing practice)

Choose by two axes: measurability and attributability of the outcome (low to high), and provider control over the drivers (low to high). High on both: outcome-based. High measurability, mixed control: transaction-based with SLAs. Low measurability: subscription or FTE with gain-share during transition.

### 3.3 Regulatory outsourcing overlay (PRA SS2/21, EBA Guidelines on outsourcing 2019, DORA 2025) [PUBLIC; VERIFY current status]

Apply as a checklist on top of any AI-led service: materiality assessment, due diligence on the provider's AI controls, written agreement with audit and access rights, exit plan, register of outsourcing arrangements, and for DORA, ICT third-party risk requirements including concentration risk. For AI specifically, add the model-risk requirements of SS1/23 where the provider's system meets the bank's definition of a model.

### 3.4 Service-as-software framing (Foundation Capital, Sequoia, 2024) [PUBLIC; VERIFY]

Use as a lens on the vendor landscape: for each outsourced process, ask whether a software-first entrant could sell the outcome per unit. Where yes, the incumbent BPO's price is under structural pressure and the client has negotiating leverage.

## 4. Worked BFSI example

`[ILLUSTRATIVE – composite]` A European bank outsources periodic KYC refresh of 180,000 corporate and SME customer files a year to a provider operating a 400-FTE team. The current contract is FTE-based. The

provider proposes an agentic refresh platform that removes 55% of human effort and asks to move to a new pricing model. All figures are EUR, annual, `[ESTIMATE — assumption]`.

## 4.1 Current state cost stack and price

| Line | Headcount | Unit cost (EUR k) | Annual cost (EUR m) |
|---|---|---|---|
| Analysts | 320 | 38 | 12.16 |
| Senior analysts and QC | 50 | 52 | 2.60 |
| Team leaders | 24 | 65 | 1.56 |
| Managers | 6 | 110 | 0.66 |
| Technology licences, workflow, screening tools | | | 1.40 |
| Facilities, training, overhead | | | 1.90 |
| Total cost | 400 | | 20.28 |
| Price to client (FTE-based, 18% margin on cost) | | | 23.93 |
| Price per file (180,000 files) | | | 133 |

Check: 20.28 x 1.18 = 23.93; 23.93m / 180,000 = EUR 133 per file.

## 4.2 Future-state cost stack after 55% effort removal

Effort removal is not uniform. Analyst work falls 70% (document collection, extraction, screening-hit first review, case assembly are automated); senior and QC work falls 20% but shifts to auditing agent output; team leader numbers fall with span; new AI lines appear.

| Line | Headcount | Unit cost (EUR k) | Annual cost (EUR m) |
|---|---|---|---|
| Analysts (exception handling) | 96 | 42 | 4.03 |
| Senior analysts and QC (agent-output audit) | 40 | 54 | 2.16 |
| Agent supervisors (HITL queue, escalation taxonomy) | 12 | 60 | 0.72 |
| Team leaders | 10 | 65 | 0.65 |
| Managers | 4 | 110 | 0.44 |
| Agent engineers and eval analysts | 8 | 95 | 0.76 |
| AI run cost (model consumption, platform, observability) | | | 1.10 |
| Technology licences (reduced workflow seats, screening tools) | | | 1.10 |
| Facilities, training, overhead | | | 0.95 |
| Total run cost | 170 | | 11.91 |
| One-off transition (build, integrate, validate, retrain), amortised over 3 years | | | 1.20 |
| Total cost including amortised transition | | | 13.11 |

AI run cost check: 180,000 files, average 9 agent steps per file, roughly 60,000 tokens per file across steps at a blended EUR 4 per million tokens = EUR 0.24 per file in model cost, EUR 43k a year; platform, vector store, observability, evaluation runs and guardrail services add roughly EUR 1.05m, giving EUR 1.10m `[ESTIMATE — order of magnitude, VERIFY pricing]`. Model consumption is a small share; platform and people dominate, which is typical.

Headcount falls from 400 to 170 (57.5%), cost falls from EUR 20.28m to EUR 13.11m (35%). The percentage cost reduction is smaller than the effort reduction because the retained roles are more expensive and new AI lines appear. This gap is the first thing to show a client who expects a 55% price cut.

## 4.3 Three pricing-model outcomes

Assume the client requires a minimum 20% price reduction and the provider requires at least its current absolute margin (EUR 3.65m) to invest in the transition.

| Pricing model | Mechanism | Price (EUR m) | Price per file (EUR) | Client saving vs today | Provider margin (EUR m) | Provider margin % |
|---|---|---|---|---|---|---|
| A. FTE-based, re-based to 170 FTE | 170 FTE at uplifted blended rate with 18% margin on new cost | 15.47 | 86 | 35% | 2.36 | 18% |
| B. Transaction-based | EUR 100 per file with quality SLAs (risk-rating accuracy 97%, audit error below 2%) | 18.00 | 100 | 25% | 4.89 | 37% |
| C. Outcome-based with gain-share | Base EUR 16.5m plus 30% share of verified savings above a EUR 5m baseline reduction; backlog and cycle-time outcomes with penalties | 17.24 (expected) | 96 | 28% | 4.13 | 31% |

Model C arithmetic: verified savings vs today = 23.93 - 16.50 = 7.43m; above the 5m baseline = 2.43m; 30% share = 0.73m; expected price 16.50 + 0.73 = 17.23m; margin 17.23 - 13.11 = 4.12m `[ESTIMATE]`.

Interpretation:

- **Model A** gives the client the full automation gain and leaves the provider with less absolute margin than today despite investing in the platform. The provider will not accept it, and if forced, has no incentive to automate further.
- **Model B** splits the gain: the client saves 25%, the provider's margin doubles in percentage terms, and the provider is paid per file so it automates further on its own account. The client should demand a step-down schedule (for example, EUR 100 falling to EUR 88 in year 3) so the provider does not capture all subsequent gains.
- **Model C** aligns on outcomes the bank cares about (backlog, cycle time, accuracy) and shares upside, but needs an agreed, frozen baseline and a dispute mechanism. It fits a client with a strong benefits-tracking function.

### 4.4 Insource comparison

If the bank insources on its own AI platform: 170 FTE at bank cost (EUR 14.8m, higher unit costs), platform allocation EUR 1.5m, validation effort EUR 0.6m, transition EUR 4.5m one-off amortised EUR 1.5m, total EUR 18.4m a year for three years `[ESTIMATE]`, compared with EUR 18.0m under model B. Insourcing is roughly cost-neutral and gives model-risk ownership and data control, at the price of execution risk and management attention. The recommendation would depend on the bank's AI operating-model maturity (Module 13): a bank without a platform and validation function should take model B with strong audit and exit rights and revisit at year 3.

So what for the client: the headline 55% effort reduction becomes a 35% cost reduction and a 25 to 28% price reduction under the models a provider will accept; the negotiation is about the step-down schedule, audit rights and what the bank owns at exit, not about the first-year price.

## 5. Real-world case

**DBS Bank workforce announcement** `[PUBLIC — Feb 2025; VERIFY]`. DBS said it expected around 4,000 temporary and contract roles to reduce over three years as AI took over work, while permanent staff were not expected to be affected. The relevance to managed services is that contract and temporary labour, including outsourced operations, is where the first labour-model shift lands, because it carries the least transition friction.

**Klarna** `[PUBLIC — Feb 2024 and May 2025; VERIFY]`. Klarna's customer-service assistant replaced work previously done partly through outsourced agents; its 2024 announcement quantified the equivalent of around 700 FTE and USD 40m of profit improvement. Its 2025 recalibration towards human service for quality shows the limit of pure cost-based outcome measures and the importance of paired quality SLAs in any AI-led service contract.

**Industry context** `[PUBLIC — various analyst and press reports, 2024–2025; VERIFY]`. Large IT and BPO providers have publicly described shifts away from headcount-linked revenue, including statements that revenue growth is decoupling from headcount growth and announcements of outcome-based and "AI-first" service offerings. Specific figures vary by firm and period and should be checked before citing.

## 6. Consultant language

> *"The price should follow the cost stack, not the headcount."*

Use when the provider proposes an FTE-based re-basing or the client expects a price cut equal to the effort reduction.

> *"You cannot outsource the accountability for a model's decisions; you can only outsource its operation."*

Use when discussing who owns model risk in a provider-embedded AI arrangement.

> *"We want a step-down schedule, not a one-time discount."*

Use in pricing negotiations to capture future automation gains.

> *"Quality SLAs at the decision level, with sample sizes, or the efficiency gain is not real."*

Use when the provider offers only turnaround-time SLAs.

> *"What do we own at exit: the golden data, the prompts, the exception taxonomy?"*

Use when testing portability and lock-in.

## 7. Common mistakes juniors make

- **Equating effort reduction with cost reduction with price reduction.** Retained roles cost more per head, new AI lines appear, and the provider keeps a share; the three numbers differ materially.
- **Ignoring model-risk ownership.** A provider's agent making risk-rating decisions is a model in the bank's inventory under SS1/23 or SR 11-7, with validation and audit consequences.
- **Recommending outcome-based pricing without a measurable, attributable outcome and a frozen baseline.** It becomes a permanent dispute.
- **Forgetting the exit.** Prompts, golden datasets and exception taxonomies are the real assets; if the provider owns them the client is locked in.
- **Modelling AI run cost as mostly tokens.** Platform, evaluation, guardrails and engineering dominate; tokens are usually a small share.
- **Treating the consulting firm's own leverage model as unaffected.** Interviewers expect you to have a view on how AI changes your own firm's economics.

## 8. Interview questions with model answers

### Q1. A bank's KYC-refresh provider proposes replacing a 400-FTE contract with an "AI-led" service at a 20% lower price. How do you advise the bank?

Model answer: I would start by separating three numbers the provider has blended: the effort the AI removes, the cost reduction that implies after retained roles get more senior and AI costs are added, and the price. With 55% effort removal the cost reduction is typically closer to 35%, so a 20% price cut leaves a large share of the gain with the provider; that may be acceptable as the price of them carrying the transition, but it should be visible. I would ask for the cost stack and propose a transaction-based price per file with a step-down schedule over the term, so the client captures later gains automatically. Second, I would ask where the agents run and who owns model risk; if the provider's agent sets risk ratings, the bank's second line must have validation evidence and audit access under SS1/23 and outsourcing rules, and that needs contract terms the standard BPO template does not have. Third, I would insist on quality SLAs at the decision level with sample sizes, paired with the efficiency SLAs. Fourth, I would test exit: the bank should own the golden datasets, exception taxonomy and prompts. Finally I would run an insource comparison on a 3 to 5 year TCO; it is often close to neutral, and the decision then rests on the bank's operating-model maturity rather than price.

### Q2. When is outcome-based pricing appropriate for an AI-delivered service, and when is it not?

Model answer: It is appropriate when three conditions hold. The outcome is measurable with an agreed method and sample size, for instance risk-rating accuracy against a golden sample or backlog cleared within a cycle-time standard. The outcome is attributable to the provider, meaning the provider controls most of the drivers; a backlog driven by the client's policy changes fails this test. And the provider has the balance sheet and governance to carry downside. When those hold, outcome-based pricing aligns incentives better than any alternative because the provider optimises the whole process rather than its own unit cost. It is not appropriate when the baseline cannot be frozen, when multiple initiatives move the same outcome, or when the outcome is a risk event that is rare and lumpy, such as a regulatory finding, where penalties become uninsurable for the provider. In those cases I would use transaction-based pricing with quality SLAs and, during transition, a gain-share with a pre-agreed baseline. I would also note that in practice transaction-based with SLAs is the most common landing point because it rewards automation while keeping measurement simple.

### Q3. How does agentic AI change the economics of a consulting firm, and what does that mean for how you would position yourself?

Model answer: The traditional model earns leverage on a pyramid of junior staff producing research, analysis and documents at hourly rates. GenAI compresses that pyramid because first drafts, data gathering, and much of the modelling can be produced faster by fewer people with AI assistance. Firms are responding in three ways: AI-enabled delivery to protect margin, asset-based offerings where reusable agents, accelerators and evaluation harnesses are sold alongside advice, and outcome-linked fees where the firm shares in the value it helps create. Several large firms have described internal assistants and restructuring of their delivery models publicly. For me the implication is that my value has to sit in what the model cannot do for the client: structuring an ambiguous problem, making judgement calls about sequencing and risk appetite, defending numbers to a CFO, navigating the regulatory position, and owning the client relationship. The experience I bring from payments and from working on agentic use-case assessment and prototypes is useful precisely because it lets me judge what is feasible and what is not, rather than because I can build the thing myself.

## 9. Self-check quiz

1. Explain the difference between labour arbitrage and automation arbitrage, and state who captures automation arbitrage under an unchanged FTE-based contract.
2. A provider's cost falls from EUR 20m to EUR 13m after automation. The client wants a 25% price cut from the current EUR 24m price. What is the provider's new margin in EUR and percent, and how does it compare with the current margin?
3. Name four contract clauses an AI-led managed service needs that a standard BPO contract usually lacks.
4. Why does a 55% reduction in effort typically produce a materially smaller reduction in cost? Give two reasons.
5. Under UK rules, can a bank transfer accountability for a KYC risk-rating model's decisions to its managed-service provider? Name the relevant supervisory statements.

# Module 16 — BFSI Use-case Library

## 1. Why a client cares

A bank executive does not want a lecture on agents. They want to know which of their processes AI can change, by how much, with what risk, and what others are doing. This library gives you a consistent answer for the eleven processes that dominate AI conversations in banking and payments. Each entry is structured so that you can speak to the process, the right AI pattern, the value lever, the KPIs and the risks in two minutes, and then go deeper if asked.

## 2. Concept and mechanism

### 2.1 How to read each entry

Every use case below has the same structure: a process summary; the AI pattern (rules, classical ML, GenAI, agentic, and which agent pattern from Module 3); the dominant value lever (cost, revenue, risk, customer experience); three or four KPIs with illustrative baseline-to-target ranges; key risks and the controls that address them; data needs; and a market maturity note for 2025–26. All baseline-to-target figures are `[ILLUSTRATIVE — composite ranges from practitioner experience, not from any single client]` unless labelled otherwise, and maturity notes are `[VERIFY — date-sensitive]`.

### 2.2 The recurring pattern across the library

Across all eleven processes the same decomposition appears. Each process contains four kinds of work:

- **Deterministic rule application** (eligibility, thresholds, reason-code logic): rules engine; do not use an LLM.
- **Pattern recognition on structured data** (fraud scores, risk ratings, propensity): classical ML, with explainability requirements.
- **Unstructured information handling** (reading documents, emails, chat transcripts, drafting narratives): GenAI, usually RAG-grounded.
- **Multi-step orchestration across systems with judgement at branches** (gather, check, decide, act, escalate): agentic, almost always with human-in-the-loop gates at the decision step in a regulated process.

The consulting skill is to decompose a process to the task level (Module 6), assign each task to the right kind of work, and only then decide whether an agent is the right container for the whole thing. Most “agentic” use cases in banking in 2025–26 are in fact agentic orchestration around a core of rules and classical models, with GenAI handling the unstructured steps `[ESTIMATE — practitioner observation]`.

### 2.3 Library

#### KYC and KYB (onboarding and periodic refresh)

| Dimension | Summary |
|---|---|
| Process | Collect identity and ownership documents, verify, screen parties, assess risk, rate, approve; refresh periodically by risk tier |
| AI pattern | Agentic orchestration (supervisor-led): document extraction (GenAI), screening-hit analysis (GenAI + rules), beneficial-ownership graph resolution (ML + rules), risk rating (rules + ML), human approval at rating and exceptions |
| Value lever | Cost (analyst effort), revenue (time-to-revenue for corporate clients), risk (consistency, audit trail) |

| Dimension | Summary |
|---|---|
| KPIs | Cycle time to onboard corporate: 25 to 40 days to 8 to 15; analyst hours per file: 6 to 10 to 2 to 4; first-time-right documentation: 45% to 80%; outreach touches per customer: 4 to 1.5 |
| Risks and controls | Mis-extraction of ownership (dual extraction and reconciliation); false clearing of screening hits (human review of all hits above threshold); regulatory expectation of analyst judgement (HITL at rating; full audit log); data-protection (residency, redaction) |
| Data needs | Document store, registry APIs, screening lists, CRM, case history for golden data |
| Maturity 2025–26 | Document extraction and outreach automation widely deployed; end-to-end agentic refresh in pilot to early production at large banks; regulators accepting with HITL at decision `[VERIFY]` |

KYB is the strongest agentic candidate in the library because it is high volume, document heavy, system fragmented and judgement is concentrated at a few identifiable points. The design decision that matters most is the autonomy level by risk tier: low-risk refresh can be near-straight-through with sampling review, high-risk and complex ownership structures need full human adjudication. The economics are driven as much by time-to-revenue for corporate clients as by analyst cost, which is why the business sponsor should be the corporate bank, not operations.

## AML transaction monitoring

| Dimension | Summary |
|---|---|
| Process | Rules and models generate alerts on transactions; analysts triage (level 1), investigate (level 2), file suspicious activity reports (SARs) |
| AI pattern | Classical ML for alert scoring and false-positive reduction; GenAI for investigation narrative drafting and evidence summarisation; agentic for evidence gathering across systems; rules remain the regulatory anchor |
| Value lever | Cost (alert review), risk (detection quality, consistency), regulatory standing |
| KPIs | False-positive rate: 90 to 95% to 60 to 75%; level-1 handle time: 25 to 40 minutes to 10 to 15; SAR narrative drafting time: 2 to 3 hours to 30 to 45 minutes; SAR quality score (regulator or QA): up 10 to 20 points |
| Risks and controls | Missed suspicious activity when alerts are suppressed (back-testing, below-the-line sampling, champion-challenger); narrative hallucination (grounding to evidence with citations, mandatory analyst sign-off); model governance (AML models are Tier 1 almost everywhere) |
| Data needs | Transactions, customer profiles, alert and case history with dispositions, SAR library, external data |
| Maturity 2025–26 | Alert scoring is mature at large banks; narrative drafting in production at several; agentic evidence gathering in pilot `[VERIFY]` |

The HSBC and Google Cloud AML AI work `[PUBLIC — Google Cloud, June 2023; VERIFY]` reported 2 to 4 times more suspicious activity identified with around 60% fewer alerts, which is the clearest public benchmark for ML-based alert scoring. The consulting caution is that regulators scrutinise alert suppression

closely; any reduction in alert volume needs below-the-line testing evidence. GenAI's safest high-value role is narrative drafting from an evidence pack the analyst has approved.

## Sanctions screening

| Dimension | Summary |
|---|---|
| Process | Screen customers and payments against sanctions lists; resolve name-match hits (mostly false positives); escalate true matches; block or release payments |
| AI pattern | Rules and fuzzy matching remain the screening core; ML for hit scoring; GenAI for hit-resolution reasoning (comparing identifiers, context in payment messages, adverse media); agentic for gathering evidence; human decision on any non-trivial hit |
| Value lever | Cost (hit review), customer experience (payment delay), risk (consistency of resolution) |
| KPIs | False-positive hits per 1,000 screened: 20 to 50 to 8 to 20; average resolution time per hit: 8 to 15 minutes to 3 to 6; payments delayed beyond SLA: down 50 to 70%; level-1 to level-2 escalation accuracy: up 10 to 15 points |
| Risks and controls | False negative on a true match (never auto-release above a conservative threshold; full logging; periodic blind testing); regulator expectation that the screening engine itself is deterministic and tuned, so AI sits in resolution not in screening |
| Data needs | Screening hits with dispositions, payment messages (ISO 20022 structured fields help materially), customer data, list data |
| Maturity 2025–26 | Hit-scoring ML is common; GenAI-assisted resolution in production at several large banks and in vendor products; fully autonomous release of hits is rare and conservative `[VERIFY]` |

ISO 20022 is relevant here: structured name and address fields reduce false positives and make GenAI resolution more reliable than parsing free-text MT messages, which is a point a payments specialist can make credibly. The risk asymmetry is extreme: a missed true match is a regulatory breach with fines; a slow false positive is a cost. That asymmetry dictates the autonomy level.

## Disputes and chargebacks

| Dimension | Summary |
|---|---|
| Process | Cardholder raises dispute; issuer classifies reason code, gathers evidence, decides liability, raises chargeback or provisional credit, handles merchant representment, arbitration |
| AI pattern | Agentic orchestration: intake classification (GenAI), evidence gathering from transaction and merchant systems (tools), scheme-rule application (rules engine), draft adjudication with rationale (GenAI), human approval by risk class; straight-through for low-value, low-risk classes |
| Value lever | Cost (analyst effort), revenue (recovery rate), customer experience (cycle time), risk (scheme compliance deadlines) |
| KPIs | Cost per dispute: GBP 25 to 40 to GBP 15 to 22; cycle time to provisional decision: 5 to 10 days to 1 to 3; straight-through rate: 0 to 10% to 30 to 50%; recovery rate on merchant-liable |

| Dimension | Summary |
|---|---|
| | cases: up 4 to 8 points; scheme deadline breaches: down 60 to 80% |
| Risks and controls | Wrong liability decision (HITL by class, override accuracy monitoring); scheme-rule changes (rules engine owned by scheme experts, not embedded in prompts); fairness to vulnerable customers (conduct review of auto-decisions) |
| Data needs | Dispute case history with outcomes, transaction data, merchant data, scheme rulebooks, customer communications |
| Maturity 2025–26 | Among the most advanced agentic use cases in issuers and processors; several vendors offer agentic dispute products; straight-through adjudication of low-risk classes in production `[VERIFY]` |

Disputes are an ideal teaching case because they contain all four kinds of work and have a natural straight-through boundary defined by value and reason code. The scheme rules must live in a rules engine with a scheme-expert owner; an LLM reasoning over a rulebook PDF is a design error that will fail on the next rule update. Recovery-rate improvement is often worth more than cost reduction.

## Fraud detection and prevention

| Dimension | Summary |
|---|---|
| Process | Real-time scoring of transactions and account events; rules and models decide to approve, step-up, decline or refer; investigation of referred cases; customer contact |
| AI pattern | Classical and deep-learning models for real-time scoring (latency-bound, milliseconds); GenAI for investigation summaries, customer contact drafting and analyst assistance; agentic for investigation case assembly; rules for hard controls and regulatory limits |
| Value lever | Risk (fraud losses), customer experience (false declines), cost (investigation) |
| KPIs | Fraud loss basis points: down 10 to 25%; false-decline rate: down 15 to 30%; investigation handle time: down 30 to 50%; model refresh cycle: quarterly to monthly |
| Risks and controls | Adversarial adaptation (continuous monitoring, rapid retraining); bias in declines (fairness testing); explainability for customer disputes of declines; latency constraints exclude LLMs from the real-time path |
| Data needs | Transaction streams, device and behavioural data, confirmed fraud labels, customer contact outcomes |
| Maturity 2025–26 | Real-time ML scoring is mature; GenAI in investigations is emerging; Mastercard reported generative AI improving fraud detection rates by an average of around 20% and up to 300% in some cases in its Decision Intelligence Pro `[PUBLIC — Mastercard, Feb 2024; VERIFY]` |

The important consulting point is that GenAI is not the scoring engine. Real-time authorisation decisions happen in tens of milliseconds and need deterministic, explainable, monitored models. GenAI's role is in the investigation and customer-contact layers and in feature discovery. Mastercard's use of generative techniques is at the network level on transaction patterns, which is a different mechanism from LLM reasoning.

## Credit underwriting (SME and corporate)

| Dimension | Summary |
|---|---|
| Process | Collect financials and documents, spread financial statements, analyse, apply policy, draft credit memo, committee decision, covenant setting |
| AI pattern | GenAI for document extraction and financial spreading, memo drafting from structured analysis; classical ML for probability of default and early-warning signals; rules for policy; agentic orchestration of the pack assembly; human credit officer decides |
| Value lever | Cost (analyst time), revenue (time-to-decision, win rate), risk (consistency, early warning) |
| KPIs | Time to decision for SME: 10 to 20 days to 3 to 7; analyst hours per credit memo: 12 to 20 to 5 to 8; spreading accuracy vs manual: 95%+ with exception flagging; early-warning lead time: up 30 to 90 days |
| Risks and controls | Regulatory classification (EU AI Act Annex III treats creditworthiness assessment of natural persons as high-risk; corporate credit is outside that annex but still model-risk governed `[PUBLIC – Regulation (EU) 2024/1689; VERIFY]`); explainability of decisions; bias; memo hallucination (grounding, officer sign-off); model risk tiering under SS1/23 or SR 11-7 |
| Data needs | Financial statements, bank statements, bureau data, internal ratings history with outcomes, policy documents, memo library |
| Maturity 2025–26 | Spreading and memo drafting in production at several banks; agentic pack assembly in pilot; autonomous credit decisions for natural persons are constrained by regulation `[VERIFY]` |

Underwriting is where the regulatory overlay is heaviest, and it is where a consultant should distinguish between the decision (which stays with a credit officer or a validated scoring model) and the preparation (which is where GenAI and agents deliver most of the time saving). Your own experience in as-is mapping of underwriting via SME interviews is directly relevant: the pain points are typically in document chasing, spreading and memo assembly, not in the decision itself.

## Collections and recoveries

| Dimension | Summary |
|---|---|
| Process | Identify delinquent accounts, segment, choose contact strategy and channel, negotiate arrangements, manage promises to pay, escalate to recoveries |
| AI pattern | Classical ML for propensity to pay, roll rates and treatment optimisation; GenAI for conversational agents and agent-assist in negotiation; agentic for arrangement setup within policy; rules for regulatory treatment of vulnerable customers |
| Value lever | Risk (loss rate, roll rate), cost (contact cost), customer experience and conduct (fair treatment) |
| KPIs | Roll rate from early to late delinquency: down 5 to 15% relative; cost to collect: down 20 to 35%; promise-to-pay kept rate: up 5 to 10 points; vulnerable-customer identification rate: up materially with conduct review |

| Dimension | Summary |
|---|---|
| Risks and controls | Conduct risk and fairness (UK FCA Consumer Duty `[PUBLIC — FCA, 2023; VERIFY]`, vulnerable-customer rules); over-contact; bias in treatment; conversational agent making commitments outside policy (hard policy limits in tools, HITL for arrangements above thresholds) |
| Data needs | Account and payment history, contact history and outcomes, customer circumstances, complaint data |
| Maturity 2025–26 | ML segmentation mature; conversational GenAI in early production for digital collections; agentic arrangement setup in pilot with tight policy bounds `[VERIFY]` |

Collections is a Tier 1 conduct area. The value is real but the controls must be designed first: a collections agent that offers an arrangement outside policy or fails to identify vulnerability is a conduct breach, not a bug. The strongest positioning with a client is “better outcomes for customers in difficulty and lower losses”, not “fewer collectors”.

## Customer servicing (contact centre and digital)

| Dimension | Summary |
|---|---|
| Process | Handle customer queries and requests across voice, chat, email and app; authenticate, resolve, escalate; after-call work |
| AI pattern | GenAI virtual assistants for digital self-service (RAG over procedures and account data via tools); agent-assist for human agents (real-time knowledge, summarisation, next-best-action); agentic for executing service requests (address change, card replacement) with authentication gates; classical ML for intent routing and churn |
| Value lever | Cost (contact deflection, handle time, after-call work), customer experience (resolution, availability), revenue (next-best-offer) |
| KPIs | Containment or deflection rate: 20 to 30% to 45 to 60%; average handle time: down 15 to 25%; after-call work: down 50 to 70%; first-contact resolution: up 5 to 10 points; CSAT: flat to up |
| Risks and controls | Incorrect information with liability (Air Canada tribunal, Feb 2024 `[PUBLIC]`); grounding and refusal design; authentication before any account action; conduct and vulnerability; brand risk from tone |
| Data needs | Knowledge base and procedures (curated), transcripts with outcomes, account data via secure APIs |
| Maturity 2025–26 | Most mature GenAI domain in banking; Bank of America’s Erica (since 2018) reported more than 2bn interactions `[PUBLIC — BofA, 2024; VERIFY]`; Wells Fargo’s Fargo reported around 245m interactions in 2024 `[PUBLIC; VERIFY]`; agentic execution of service requests spreading `[VERIFY]` |

Servicing is where most banks started, and where the public cautionary tales sit: Air Canada was held liable for its chatbot’s incorrect statement, and Klarna’s 2025 recalibration showed the cost of over-weighting deflection against quality. The consulting advice is to measure containment together with resolution quality and complaint rates, and to keep account actions behind authentication and policy gates.

## Payments operations (exceptions, investigations, ISO 20022)

| Dimension | Summary |
|---|---|
| Process | Handle payment exceptions and repairs, investigations (recalls, beneficiary claims non-receipt, returns), ISO 20022 message handling, nostro breaks, SLA management across correspondents |
| AI pattern | Rules for STP repairs where deterministic; ML for repair prediction and routing; GenAI for reading unstructured investigation messages (camt.027/camt.029, gpi, email), drafting responses, and summarising case history; agentic for end-to-end investigation case handling with tool access to payment systems and correspondent channels |
| Value lever | Cost (operations effort), customer experience (resolution time), risk (regulatory deadlines, liability), liquidity (faster release of funds) |
| KPIs | Investigation cycle time: 3 to 7 days to under 1 to 2; manual repairs per 10,000 payments: down 30 to 60%; cases handled per operations FTE: up 2 to 3 times; correspondent SLA breaches: down 50%+ |
| Risks and controls | Wrong-beneficiary release (never auto-execute fund movements without a second control); message standard changes (rules engine for format logic); correspondent data quality |
| Data needs | Payment messages (structured ISO 20022 fields are a major enabler), investigation case history, correspondent SLAs, nostro data |
| Maturity 2025–26 | Rule-based STP mature; GenAI-assisted investigations in production at some banks and in Swift and vendor offerings; agentic end-to-end investigations in pilot `[VERIFY]` |

This is the domain where your payments depth is a differentiator. Points to make: ISO 20022 structured data (structured addresses, LEIs, purpose codes, UETR) makes investigation automation materially easier than MT; investigations are a natural agentic use case because they are multi-step, cross-system and message-driven; fund movements must stay behind deterministic controls. Swift's own AI pilots on anomaly detection and investigations `[PUBLIC — Swift announcements, 2024–2025; VERIFY]` are worth knowing.

## Reconciliation (cash, nostro, ledger, securities)

| Dimension | Summary |
|---|---|
| Process | Match transactions between systems or counterparties; investigate and resolve breaks; post adjustments; report ageing |
| AI pattern | Rules for deterministic matching; ML for probabilistic matching of residual breaks and break classification; GenAI for interpreting unstructured break commentary and drafting resolutions; agentic for investigating breaks across systems and proposing (not posting) adjustments |
| Value lever | Cost (break investigation), risk (unreconciled exposure, financial reporting), control quality |
| KPIs | Auto-match rate: 85 to 92% to 96 to 98%; breaks older than 30 days: down 50 to 70%; investigator hours per break: down 40 to 60%; adjustment error rate: flat or down |
| Risks and controls | Incorrect auto-match (confidence thresholds, sampling); posting of adjustments must remain human-approved with |

| Dimension | Summary |
|---|---|
| | maker-checker; audit trail for financial controls |
| Data needs | Transaction feeds from both sides, historical breaks with resolutions, reference data |
| Maturity 2025–26 | ML matching is mature in vendor reconciliation platforms; GenAI break interpretation emerging; agentic break investigation in pilot `[VERIFY]` |

Reconciliation is a control process, so the standard for change is high and the sponsor is usually the CFO or Head of Operations Control. The value case is often strongest on risk (unreconciled exposure, audit findings) rather than cost. Keep adjustment posting human-approved with maker-checker regardless of agent confidence.

### Code modernisation and software engineering

| Dimension | Summary |
|---|---|
| Process | Understand legacy systems (COBOL, PL/I, older Java), document, refactor or migrate, test, maintain; general software delivery |
| AI pattern | GenAI coding assistants (completion, explanation, test generation); agentic coding agents for multi-file changes, migration tasks and test suites; RAG over code and documentation; classical ML rarely |
| Value lever | Cost (engineering productivity), risk (legacy knowledge loss, resilience), speed (time to market) |
| KPIs | Developer productivity on measured tasks: up 15 to 40% (highly variable by task and seniority); legacy documentation coverage: up materially; test coverage: up 10 to 30 points; migration effort: down 20 to 40% |
| Risks and controls | Security vulnerabilities in generated code (SAST, review); licence contamination; over-reliance reducing understanding; correctness of migrated business logic (equivalence testing); IP and data leakage to model providers |
| Data needs | Source code, documentation, test suites, defect history |
| Maturity 2025–26 | Coding assistants widely deployed in banks (Goldman Sachs, JPMorgan, Citi have described large rollouts `[PUBLIC — 2024–2025; VERIFY]`); agentic coding in pilot at several, including Goldman Sachs' pilot of Cognition's Devin `[PUBLIC — July 2025; VERIFY]`; mainframe modernisation with GenAI offered by hyperscalers `[VERIFY]` |

The productivity numbers quoted in the market vary widely and are often vendor-sourced; present them as ranges and insist on measured internal baselines. For a bank, the strategic value is as much in legacy knowledge recovery and resilience as in productivity. Governance points: security scanning, code review, and clarity on whether code goes to an external model.

## 3. Frameworks

### 3.1 The four-kinds-of-work decomposition (attribution: practitioner synthesis; consistent with the rules, ML, GenAI and agentic taxonomy in Module 1)

Steps: (1) decompose the process to task level (Module 6); (2) classify each task as deterministic rule application, structured-data pattern recognition, unstructured information handling, or multi-step

orchestration with judgement; (3) assign the AI pattern per class; (4) decide whether an agent container adds value over a workflow engine (it does when branches require judgement over unstructured inputs and tool selection varies by case); (5) place HITL gates at decisions with regulatory or financial consequence.

### 3.2 Use-case card standard (attribution: generic; the card structure used here)

Process; AI pattern; value lever; KPIs with baseline and target; risks and controls; data needs; maturity. Use the card as the unit of the use-case library and as the input to the Module 7 scoring matrix. A card without baseline-to-target KPIs and without named controls is not ready for prioritisation.

### 3.3 Risk-asymmetry test for autonomy (attribution: practitioner heuristic)

For each decision point ask: what is the cost of a false positive versus a false negative, and is the action reversible? Sanctions release (irreversible, catastrophic false negative) and fund movements (irreversible) get no autonomy; low-value dispute credits (reversible, bounded) get high autonomy; everything else gets sampled or threshold-based review. This test produces the autonomy level by decision class in a form a regulator will recognise.

### 3.4 McKinsey Global Institute value estimate (June 2023) [PUBLIC — MGI, 2023]

MGI estimated generative AI could add USD 200 to 340bn a year to banking, largely through productivity in customer operations, marketing and sales, software engineering and risk and legal. Use it as context for ambition setting, with the caveat that it is a top-down potential estimate, not a benchmark for a single bank's business case.

## 4. Worked BFSI example

`[ILLUSTRATIVE — composite]` Harbourline Bank, a mid-sized universal bank, asks which of three use cases to fund first: KYB onboarding and refresh, sanctions hit resolution, and card disputes. Figures are annual, GBP, `[ESTIMATE — assumption]`.

### 4.1 Value estimate

| Driver | KYB | Sanctions resolution | Card disputes |
|---|---|---|---|
| Volume per year | 60,000 files | 1.2m hits | 300,000 disputes |
| Current unit cost (fully loaded) | GBP 180 per file | GBP 9 per hit | GBP 31 per dispute |
| Current annual cost | GBP 10.8m | GBP 10.8m | GBP 9.3m |
| Effort reduction (steady state) | 50% | 45% | 40% |
| Cost benefit (capacity released) | GBP 5.4m | GBP 4.9m | GBP 3.7m |
| Non-cost benefit | Time-to-revenue: 2,000 corporate onboardings a year, 15 days faster, GBP 400 revenue per client per day of earlier activation = GBP 12.0m revenue pull-forward, of which GBP 2.4m is incremental annual revenue `[ESTIMATE]` | SLA breach reduction and customer retention: GBP 0.6m | Recovery rate up 6 points on 55% merchant-liable at GBP 140 average = 300,000 x 0.55 x 0.06 x 140 = GBP 1.4m |
| Risk benefit (qualitative) | Audit consistency, fewer regulatory findings | Consistency of resolution; faster true-match escalation | Scheme deadline compliance |
| Total quantified annual | GBP 7.8m | GBP 5.5m | GBP 5.1m |

| Driver | KYB | Sanctions resolution | Card disputes |
|---|---|---|---|
| benefit at steady state | | | |

Check on KYB cost: 60,000 x 180 = GBP 10.8m; 50% = GBP 5.4m. Check on disputes: 300,000 x 31 = GBP 9.3m; 40% = GBP 3.72m.

## 4.2 Feasibility and risk scores (1 low to 5 high, feasibility higher is better)

| Dimension (weight) | KYB | Sanctions resolution | Card disputes |
|---|---|---|---|
| Data readiness (25%) | 3 (documents unstructured, registry APIs available) | 4 (hit data structured, dispositions logged) | 4 (case history complete, scheme data structured) |
| Integration complexity (20%) | 2 (6 systems, legacy CRM) | 4 (single screening platform) | 3 (card platform plus case management) |
| Model maturity for the pattern (20%) | 4 (extraction and orchestration proven) | 4 (resolution reasoning proven in vendor tools) | 5 (agentic disputes products in market) |
| Regulatory exposure, inverted (20%) | 3 (HITL at rating accepted) | 2 (false-negative asymmetry; conservative autonomy) | 4 (bounded financial exposure, conduct review needed) |
| Change effort, inverted (15%) | 2 (corporate bank, ops, compliance all affected) | 4 (one team) | 3 (one operations centre) |
| Weighted feasibility | 2.85 | 3.60 | 3.85 |

Arithmetic for KYB: 3 x 0.25 + 2 x 0.20 + 4 x 0.20 + 3 x 0.20 + 2 x 0.15 = 0.75 + 0.40 + 0.80 + 0.60 + 0.30 = 2.85. Sanctions: 1.00 + 0.80 + 0.80 + 0.40 + 0.60 = 3.60. Disputes: 1.00 + 0.60 + 1.00 + 0.80 + 0.45 = 3.85.

## 4.3 Placement and recommendation

| Use case | Annual benefit (GBP m) | Feasibility | Indicative build and 3-year run cost (GBP m) | 3-year net benefit at 40/80/100% ramp (GBP m) | Placement |
|---|---|---|---|---|---|
| KYB | 7.8 | 2.85 | 9.5 | 17.2 - 9.5 = 7.7 | High value, lower feasibility: foundational wave 2 |
| Sanctions resolution | 5.5 | 3.60 | 4.0 | 12.1 - 4.0 = 8.1 | Medium value, high feasibility: wave 1 |
| Card disputes | 5.1 | 3.85 | 4.5 | 11.2 - 4.5 = 6.7 | Medium value, highest feasibility: wave 1 |

Ramp arithmetic for KYB: 7.8 x (0.4 + 0.8 + 1.0) = 7.8 x 2.2 = GBP 17.2m. Sanctions: 5.5 x 2.2 = 12.1. Disputes: 5.1 x 2.2 = 11.2.

Recommendation: fund disputes and sanctions resolution in wave 1 as the two quick-to-prove agentic deployments that also build the platform components (orchestration, HITL queue, eval harness) KYB will need; start KYB data and integration foundations in parallel and move it to build in wave 2. KYB has the largest prize but its feasibility score says the first year would be spent on integration; sequencing it second is what makes its benefit achievable.

So what for the client: the biggest number is not the first thing to build; the first wave is chosen to prove the pattern and build the platform, so that the biggest number lands in year 2 with lower risk.

## 5. Real-world case

**HSBC AML with Google Cloud** `[PUBLIC — Google Cloud announcement, June 2023; VERIFY]`. Google Cloud reported that its AML AI product, developed with HSBC, identified 2 to 4 times more suspicious activity while reducing alert volumes by around 60% compared with the rules-based approach. This is the clearest public benchmark for the AML entry above and shows the shape of the value case: better detection and fewer alerts at the same time, which is only credible with rigorous back-testing evidence for the regulator.

**Bank of America Erica** `[PUBLIC — Bank of America, 2024; VERIFY]`. Launched in 2018, Erica surpassed 2bn client interactions and is often cited as the most-used virtual assistant in banking. The relevant lessons are that it started with narrow, well-understood servicing intents on structured account data, scaled through years of intent tuning rather than a single model, and that the bank publicly reports usage metrics, which has supported continued investment.

**JPMorgan COiN and LLM Suite** `[PUBLIC — 2017 and 2024; VERIFY figures]`. COiN (Contract Intelligence, 2017) used ML to review commercial loan agreements and was reported to save large volumes of lawyer and loan-officer hours, an early example of the document-handling pattern. LLM Suite's 2024 rollout to around 200,000 employees shows the servicing and knowledge-work patterns scaled through a central platform.

## 6. Consultant language

> *"Let us decompose this to the task level and assign each task to rules, models, GenAI or an agent; most of the value is in getting that split right."*

Use at the start of any use-case conversation to signal method over enthusiasm.

> *"The autonomy level follows the risk asymmetry at each decision: irreversible and asymmetric means human; bounded and reversible means straight-through."*

Use when the client asks how much the agent will decide on its own.

> *"Scheme rules and policy live in a rules engine with a named owner, not in a prompt."*

Use in disputes, sanctions, collections and payments operations conversations.

> *"Recovery rate and time-to-revenue are usually worth more than the analyst saving; let us make sure the business, not operations, sponsors this."*

Use when the business case is being built as a pure cost case.

> *"ISO 20022 structured data is the enabler here; investigation automation on MT free text would have been a research project."*

Use in payments operations, where your expertise is the differentiator.

## 7. Common mistakes juniors make

- **Pitching GenAI as the fraud or screening engine.** Real-time scoring and list screening are deterministic or classical ML by regulatory expectation and latency; GenAI sits in resolution and investigation.
- **Embedding scheme rules or credit policy in prompts.** They change, they need an owner, and an LLM is not an auditable rules engine.
- **Quoting vendor productivity figures as benchmarks.** Present ranges, label them, and insist on measured baselines.

- **Building the KYB business case on analyst cost only.** Time-to-revenue and first-time-right documentation are often the larger levers.
- **Ignoring EU AI Act Annex III** for anything touching creditworthiness of natural persons.
- **Treating all eleven use cases as "agentic".** Most are agentic orchestration around rules and models with GenAI on the unstructured steps; say so.
- **Forgetting conduct risk in collections and servicing.** Consumer Duty and vulnerability rules make these Tier 1 conduct areas regardless of model tier.

## 8. Interview questions with model answers

### Q1. Walk me through how you would apply AI to sanctions screening, and where you would and would not use an LLM.

Model answer: I would separate screening from resolution. Screening itself, matching names and identifiers against lists, stays a deterministic, tuned fuzzy-matching engine because regulators expect it to be explainable and testable, and because recall on true matches is non-negotiable. The problem is the hit volume: typically 95% or more of hits are false positives. Classical ML can score hits to prioritise and route them, which is well established. GenAI's role is in resolution: reading the payment message and customer context, comparing identifiers, pulling adverse media, and drafting a reasoned disposition for an analyst, and an agent can orchestrate that evidence gathering across systems. ISO 20022 structured fields make this much more reliable than parsing MT free text. The autonomy decision follows the risk asymmetry: a missed true match is a regulatory breach, a slow false positive is a cost, so I would allow automated release only for hits below a conservative score threshold with blind sampling, and route everything else to a human with the agent's draft. The KPIs I would set are false positives per thousand screened, resolution time per hit, payments delayed beyond SLA, and escalation accuracy, with a quality KPI from periodic blind testing. The illustrative benefit for a bank with over a million hits a year is a 40 to 50% reduction in resolution effort, plus customer and liquidity benefits from faster release.

### Q2. A COO wants to start with KYB because it has the biggest prize. What do you say?

Model answer: I would agree that KYB is usually the largest prize in operations, and then show why it should probably not be the first build. KYB scores high on value because it combines analyst effort, time-to-revenue for corporate clients and audit consistency, but it scores low on feasibility: it typically spans six or more systems, documents are unstructured, registry data quality varies, and it affects the corporate bank, operations and compliance simultaneously. In a scored comparison it often lands as high value with the lowest feasibility of the candidate set. My recommendation would be to start wave one with one or two use cases that prove the agentic pattern quickly and build the platform components KYB needs, for example disputes and sanctions hit resolution, while running KYB's data and integration foundations in parallel so that it moves to build in wave two. That way the largest benefit lands in year two with lower delivery risk, rather than the first year being consumed by integration. I would also reframe the sponsor: KYB's biggest lever is time-to-revenue, so the corporate bank should own it with operations, which changes the business-case conversation from cost to growth.

### Q3. Where is agentic AI genuinely the right pattern in banking operations, and where is it over-sold?

Model answer: Agentic AI is the right container when a process has multiple steps across systems, the path through the steps varies by case, the inputs include unstructured information, and there are identifiable decision points where a human can be placed. Disputes, KYB refresh, payments investigations, AML evidence gathering and credit-pack assembly all fit that description. In each, the agent orchestrates tools, handles the unstructured steps with GenAI, applies rules through a rules engine, and hands decisions with regulatory or financial consequence to a human or a validated model. It is over-sold when it is proposed as the decision

engine itself in real-time fraud or sanctions screening, where latency and determinism rule it out; when it replaces a rules engine for scheme rules or policy; and when a single well-grounded LLM call or a workflow engine would do the job with less risk and cost. In practice most agentic use cases in banking today are agentic orchestration around a core of rules and classical models. The consulting value is in making that decomposition explicit and in setting autonomy by risk asymmetry at each decision point, which is also what regulators are asking for.

## 9. Self-check quiz

1. For each of the following, name the right AI pattern and one reason: real-time card authorisation fraud scoring; drafting an AML SAR narrative; applying card-scheme chargeback time limits; investigating a cross-border payment non-receipt claim.
2. A bank has 1.5m sanctions hits a year at GBP 8 per hit. If GenAI-assisted resolution reduces effort by 45% and run cost is GBP 0.60 per hit, what is the net annual benefit? Show the arithmetic.
3. Which use case in the library is most directly affected by EU AI Act Annex III, and what distinction limits that effect?
4. State the risk-asymmetry test for autonomy and apply it to two decision points: releasing a payment after a sanctions hit, and issuing a provisional credit on a GBP 20 dispute.
5. Why does ISO 20022 adoption increase the feasibility of AI in sanctions resolution and payments investigations?

# Module 17 — Public case-study teardowns

## 1. Why a client cares

Every bank executive has read the same press releases you have. What they cannot get from a press release is the pattern behind the headline: which AI pattern was used, what was actually measured, what was later walked back, and what that means for their own programme. A consultant who can cite seven public cases with the measured outcome and the caveat in one breath earns the right to challenge the client's ambition. A consultant who repeats vendor claims loses that right in the first meeting.

## 2. Concept and mechanism

### 2.1 How to read a public AI case

Public AI announcements follow a predictable structure, and each element needs a different level of scepticism.

| Element of the announcement | What it usually is | How to treat it |
|---|---|---|
| Scale claim ("rolled out to 200,000 employees") | Licence or access count | Input, not outcome. Access is not adoption, adoption is not value |
| Productivity claim ("saves 2 hours a week") | Self-reported survey or vendor telemetry | Directionally useful; discount 30 to 50% for selection and novelty effects |
| FTE-equivalent claim ("work of 700 agents") | Volume handled divided by historical productivity | Only a saving if the FTE cost actually leaves the cost base |
| Financial claim ("USD 40m profit improvement") | Forecast, rarely audited | Check whether a later annual report confirms it |
| Detection uplift ("2 to 4x more suspicious activity") | Model-level metric from a pilot period | Ask what happened to downstream investigation capacity and regulator acceptance |
| Walk-back ("re-hiring humans") | Correction of over-automation | The most informative part of any case; this is where the operating lesson sits |

### 2.2 The four outcome archetypes

Across the public record, bank AI programmes land in one of four archetypes. Naming the archetype quickly is the mechanism that makes a teardown useful in a client conversation.

1. **Scaled enabler.** Broad employee-facing assistant, measured in adoption and time saved, value diffuse and hard to book. JPMorgan LLM Suite, Citi, BBVA, Goldman Sachs sit here.
2. **Vertical workflow win.** A single high-volume process with a hard baseline, measured in cost, cycle time or detection rate. HSBC AML AI, Mastercard Decision Intelligence Pro, Morgan Stanley adviser assistant.
3. **Customer-facing automation with correction.** Customer service automation that delivered the volume claim and then had to be rebalanced for quality, liability or labour relations. Klarna, Commonwealth Bank of Australia, Air Canada.
4. **Model failure in a core economic decision.** The model drove a pricing or credit decision and the error compounded into losses or regulatory scrutiny. Zillow Offers, Apple Card and Goldman Sachs.

### 2.3 The teardown template

Use the same template for every case, in this order: what they did; AI pattern (rules, ML, GenAI, agentic, and the integration pattern); measured outcomes as publicly reported; what went wrong or was walked back;

lesson for a strategy consultant; one-line interview citation. The discipline of filling the "what went wrong" row for every case, including the successes, is what distinguishes a strategy view from a vendor view.

# 3. Frameworks

## 3.1 Claimed versus realised value ladder (practitioner framework; attribution: widely used in benefits realisation practice, no single originator)

| Rung | Evidence type | Typical haircut applied by a CFO |
|---|---|---|
| 1. Vendor or press-release claim | Announcement | 60 to 80% |
| 2. Pilot metric | Controlled sample, short period | 40 to 60% |
| 3. Production telemetry | Live system, 6+ months | 20 to 30% |
| 4. Booked in P&L or annual report | Audited financials | 0 to 10% |

Place every public case on this ladder before quoting it. Most 2023 to 2025 GenAI cases sit at rung 1 or 2. HSBC AML AI and Klarna moved to rung 3. Very few have reached rung 4 with an explicit line item.

## 3.2 The teardown lens: five failure drivers

When a case is walked back, the cause is almost always one of five drivers. This is a composite practitioner lens, not an attributed framework.

1. **Baseline error.** The automation was assumed to remove demand that it did not remove (Commonwealth Bank of Australia).
2. **Quality-cost trade-off mispriced.** Cost saving was booked without pricing the customer-experience cost (Klarna).
3. **Liability not designed in.** The system spoke with authority the organisation had not agreed to stand behind (Air Canada).
4. **Model used outside its competence envelope.** A predictive model drove irreversible economic commitments under regime change (Zillow).
5. **Explainability and fairness not evidenced.** The outcome may have been defensible, but the firm could not show it (Apple Card and Goldman Sachs).

## 3.3 How to use cases in a client room

- Lead with the case closest to the client's own archetype, not the most famous one.
- Quote the measured outcome, then the caveat, in the same sentence.
- Translate to the client's scale: "At your volumes, the HSBC-type alert reduction would release roughly X analysts."
- Always label what is public versus what you are estimating.

# 4. Worked BFSI example: cross-case synthesis

The table below is the synthesis a partner would carry into a steering committee. Public facts are labelled; the "realised vs claimed" column is the consultant's estimate.

| Case | Archetype | AI pattern | Headline claim [PUBLIC] | Realised vs claimed [ESTIMATE — assumption] | Dominant driver |
|---|---|---|---|---|---|
| Klarna (2024 to 25) | Customer-facing with correction | GenAI chat assistant, RAG over policies | Two-thirds of chats, ~700 FTE, USD 40m profit | Volume claim held; cost claim partly offset by | Quality-cost trade-off |

| Case | Archetype | AI pattern | Headline claim [PUBLIC] | Realised vs claimed [ESTIMATE — assumption] | Dominant driver |
|---|---|---|---|---|---|
| | | | uplift | quality rework and re-hiring; estimate 50 to 70% of claimed net benefit | |
| HSBC AML AI (2023) | Vertical workflow win | Supervised ML risk scoring replacing rules thresholds | 2 to 4x more suspicious activity found, ~60% fewer alerts | High: model metric is directly operational; estimate 70 to 90% realised, subject to regulator acceptance in each jurisdiction | Hard baseline, regulator-aligned |
| Morgan Stanley (2023 to 24) | Vertical workflow win | GenAI assistant, RAG over research corpus, meeting summarisation | ~98% adviser-team adoption, Debrief tool | Adoption is real; financial value diffuse; estimate 40 to 60% of implied productivity converted to revenue or capacity | Curated corpus, narrow user group |
| JPMorgan LLM Suite (2024) | Scaled enabler | GenAI portal over multiple models, no customer exposure | ~200,000 employees with access | Low measurability by design; estimate 20 to 40% of licence base in weekly active use | Scale first, value tracking later |
| DBS (2024 to 25) | Scaled enabler plus vertical wins | ~800 ML models, ~350 use cases, GenAI assistants | SGD 1bn+ economic value projected 2025; ~4,000 contract roles reduced over three years | Medium to high; economic value is internally defined, not audited; estimate 60 to 80% | Decade-long data foundation |
| Bank of America Erica (2018 to 24) | Vertical workflow win | Rules plus NLU intent classification, later GenAI elements | 2bn+ interactions cumulative | High on deflection; value is cost-to-serve and engagement rather than booked savings; estimate 60 to 80% | Narrow scope, long iteration |
| Commonwealth Bank of Australia (2025) | Customer-facing with correction | Voice bot for call deflection | 45 roles made redundant, then reinstated | Negative in the period; baseline assumption failed | Baseline error |
| Air Canada (2024) | Customer-facing with correction | Chatbot giving policy answers | None claimed; tribunal liability | Negative; reputational and legal cost exceeded any deflection saving | Liability not designed in |
| Apple Card and Goldman Sachs (2019 to 21) | Model failure in core decision | ML credit-limit model | None claimed; regulator cleared on discrimination | Neutral financially; high reputational cost; weak | Explainability not evidenced |

| Case | Archetype | AI pattern | Headline claim [PUBLIC] | Realised vs claimed [ESTIMATE — assumption] | Dominant driver |
|---|---|---|---|---|---|
| | | | | explainability process | |
| Zillow Offers (2021) | Model failure in core decision | Predictive pricing model driving purchases | None; USD 500m+ write-downs, ~25% workforce reduction | Severely negative | Competence envelope exceeded |

So what for the client: the cases with hard baselines and narrow scope (HSBC, Erica, Morgan Stanley) realised most of what they claimed; the cases that booked labour savings before validating demand or quality (Klarna, Commonwealth Bank) had to reverse; the cases where a model made irreversible economic commitments (Zillow) destroyed value. Design the portfolio accordingly.

## 5. Real-world case teardowns

Each teardown follows the template in 2.3. All figures are [PUBLIC] with the source and year given; those marked VERIFY are known to move or to be reported inconsistently across sources.

### 5.1 Klarna: customer-service assistant and the walk-back

- **What they did.** In February 2024 Klarna announced an OpenAI-powered customer service assistant live in 23 markets. In its first month it handled two-thirds of customer service chats, equivalent to the work of about 700 full-time agents, with resolution time down from 11 minutes to under 2 minutes and repeat enquiries down 25%, with an estimated USD 40m profit improvement for 2024. [PUBLIC — Klarna press release, Feb 2024] [VERIFY — date-sensitive]
- **AI pattern.** GenAI conversational assistant with retrieval over policy and account context, escalation to humans, multilingual. Customer-facing, Level 2 to 3 autonomy (acts within policy, escalates exceptions).
- **Measured outcomes.** Volume share, handling time, repeat-contact rate and an FTE-equivalent claim. The USD 40m figure was a forecast at announcement. Klarna also stated a headcount reduction trajectory through attrition and a hiring freeze in 2024. [PUBLIC — Klarna statements, 2024] [VERIFY]
- **What was walked back.** In 2025 CEO Sebastian Siemiatkowski said publicly that Klarna had over-indexed on cost, that quality had suffered, and that the company would again hire human agents so that customers could always reach a person, piloting a flexible, remote agent workforce. [PUBLIC — CEO interviews reported by Bloomberg and others, May 2025] [VERIFY]
- **Lesson for a strategy consultant.** The volume claim was true and the operating lesson is about the quality-cost frontier. The business case booked the cost side of the trade-off and did not price the customer-experience side. A strategy consultant should have insisted on a quality floor (CSAT, first-contact resolution, complaint rate) as a hard constraint in the case, and on a target human-agent ratio rather than a target of zero.
- **Interview citation.** “Klarna proved the volume claim in 2024 and corrected the quality trade-off in 2025; it is the best public evidence that you design the human-agent ratio, you do not let the cost case set it.”

### 5.2 HSBC and Google Cloud AML AI: a vertical workflow win

- **What they did.** HSBC replaced rules-based transaction-monitoring thresholds with a machine-learning risk-scoring approach, developed with Google Cloud and productised as “AML AI” (HSBC’s internal name was Dynamic Risk Assessment). Google Cloud reported that the system identified 2 to

4 times more suspicious activity while reducing alert volumes by around 60%. [PUBLIC — Google Cloud announcement, Jun 2023] [VERIFY]

- **AI pattern.** Supervised ML on customer-level behavioural features, scoring risk across the customer rather than per transaction rule. Not GenAI. Decision-support, with human investigators retaining the suspicious activity reporting decision.
- **Measured outcomes.** Detection uplift and alert reduction, both model-level metrics tied directly to analyst capacity and regulatory effectiveness. HSBC has stated it uses the approach across multiple markets. Exact investigator-capacity savings have not been disclosed. [PUBLIC — HSBC and Google Cloud statements, 2023 to 24] [VERIFY]
- **What went wrong or is still open.** Nothing public was walked back. The open issues typical of this archetype are regulator acceptance jurisdiction by jurisdiction, model-risk validation of a system that replaces auditable thresholds with scores, and the investigator-capacity question: fewer but richer alerts shift effort rather than remove it in the first year.
- **Lesson for a strategy consultant.** The win came from a hard baseline (alerts per month, suspicious-activity yield), a narrow scope, and years of data foundation. It is also a reminder that "AI in AML" in a Tier-1 bank is still predominantly supervised ML; GenAI sits on top for narrative drafting, not for the scoring decision.
- **Interview citation.** "HSBC's AML AI is the cleanest public example of ML beating rules on a regulator-visible metric: more suspicious activity found with 60% fewer alerts, and it is ML, not GenAI."

## 5.3 Morgan Stanley: adviser assistant and Debrief

- **What they did.** Morgan Stanley Wealth Management launched the AI @ Morgan Stanley Assistant in September 2023, built with OpenAI's GPT-4, giving around 16,000 financial advisers and support staff retrieval over roughly 100,000 internal research and procedure documents. In June 2024 it added Debrief, which summarises client meetings with consent, drafts follow-up emails and notes to the CRM. The firm reported adoption around 98% of adviser teams. [PUBLIC — Morgan Stanley and OpenAI statements, 2023 to 24] [VERIFY]
- **AI pattern.** GenAI with RAG over a curated, access-controlled corpus; meeting transcription and summarisation; human adviser remains the decision-maker and the client interface.
- **Measured outcomes.** Adoption and usage, qualitative time saved, Debrief usage counts. No disclosed revenue or cost figure attributable to the tool. [VERIFY]
- **What went wrong or is open.** No public walk-back. Open questions are the standard ones for scaled enablers: how much saved time converts to revenue-generating activity, and the governance of client-meeting recordings. The firm invested heavily in evaluation, reportedly building an expert-graded test set that gated release, which is a significant part of why it is cited as a quality benchmark.
- **Lesson for a strategy consultant.** The corpus was curated and the user group was narrow and high-value, so each unit of adviser time saved has a clear revenue path. Evaluation discipline before release is a design choice, not an afterthought.
- **Interview citation.** "Morgan Stanley is the reference for curated-corpus RAG with expert-graded evals: 98% adoption because the corpus and the user were narrow, and because evals gated the release."

## 5.4 JPMorgan Chase: from COiN to LLM Suite

- **What they did.** JPMorgan's COiN (Contract Intelligence) applied ML to commercial loan agreement review from around 2017, with the bank stating it removed a large volume of manual review hours. In 2024 the bank rolled out LLM Suite, an internal GenAI portal giving access to multiple large language models for writing, summarisation and ideation, to around 200,000 employees. Its annual

technology spend is around USD 17bn. [PUBLIC — JPMorgan statements and shareholder letters, 2017 to 2024] [VERIFY]

- **AI pattern.** COiN: supervised ML and NLP on structured document types. LLM Suite: GenAI enabler with model-agnostic access layer, no direct customer exposure, with a stated intent to track value by use case.
- **Measured outcomes.** Access counts and management commentary on productivity; the CEO's letters position AI as transformational across the firm. Specific booked financial outcomes have not been disclosed at the tool level. [VERIFY]
- **What went wrong or is open.** No public walk-back. The structural risk of the scaled-enabler archetype is diffuse value: broad access generates usage statistics, not P&L. JPMorgan has mitigated this by pairing the portal with targeted vertical use cases and a central AI organisation.
- **Lesson for a strategy consultant.** Scaled enablers are a platform decision, not a business case. Make the case on option value and talent, and attach value tracking to the vertical use cases that ride on the platform.
- **Interview citation.** "JPMorgan's LLM Suite shows the scaled-enabler play: 200,000 seats is an input metric, and the value sits in the vertical use cases the platform enables."

## 5.5 DBS Bank: a decade of data foundation and the workforce signal

- **What they did.** DBS has reported deploying around 800 AI and ML models across roughly 350 use cases, with an internally projected economic value above SGD 1bn in 2025, and a long-running programme of hyper-personalised nudges and operational models. In February 2025 the CEO said the bank expected to reduce around 4,000 temporary and contract roles over three years as AI absorbed work, while permanent staff would not be affected. [PUBLIC — DBS annual reporting and CEO statements, 2024 to 25] [VERIFY]
- **AI pattern.** Predominantly predictive ML at scale on a modern data platform, with GenAI assistants added for employees and customer service; a central AI and data function with embedded business ownership.
- **Measured outcomes.** Internal economic value, number of models and use cases, and a workforce trajectory. The economic value definition is internal and not audited. [VERIFY]
- **What went wrong or is open.** No public walk-back. The workforce statement attracted scrutiny on how value is shared with contract staff; the bank framed it as attrition of temporary roles rather than redundancies.
- **Lesson for a strategy consultant.** The SGD 1bn claim rests on a decade of data platform investment; it is not replicable in a two-year programme without that foundation. It is also the clearest public statement of the labour-model shift, which makes it useful when a client asks about workforce implications.
- **Interview citation.** "DBS is the proof that AI value at the billion level follows a decade of data foundation, and the first major bank to quantify the contract-labour impact publicly."

## 5.6 Bank of America Erica: narrow scope, long iteration

- **What they did.** Bank of America launched the Erica virtual assistant in its mobile app in 2018. By 2024 the bank reported more than 2 billion cumulative client interactions and tens of millions of users, with Erica handling balance queries, transaction search, bill reminders and routing to humans. The bank has stated it will spend around USD 4bn on new technology initiatives in 2025, with AI a significant component. [PUBLIC — Bank of America statements, 2018 to 2025] [VERIFY]
- **AI pattern.** Intent classification and natural-language understanding over a bounded set of banking tasks, rules-driven fulfilment, escalation to humans; GenAI elements introduced later for employees and selected functions.

- **Measured outcomes.** Interactions, users, containment and engagement metrics, and management commentary on efficiency. No disclosed line-item saving. [VERIFY]
- **What went wrong or is open.** No public walk-back. The design choice to keep Erica bounded and largely deterministic is why it has avoided the liability problems seen in generative customer-facing bots.
- **Lesson for a strategy consultant.** Customer-facing automation in banking has worked best when the scope is bounded and the fulfilment is deterministic. Erica is the counter-example to "put a GenAI chatbot in front of customers" and a strong reference for the rules-plus-NLU pattern.
- **Interview citation.** "Erica is seven years of bounded, deterministic customer automation at 2 billion interactions; it shows scope discipline beats model sophistication for customer-facing use."

## 5.7 Commonwealth Bank of Australia: the baseline that was not there

- **What they did.** In 2025 Commonwealth Bank announced the redundancy of 45 customer-service roles, citing an AI voice bot that was expected to reduce call volumes. The Finance Sector Union challenged the decision at the Fair Work Commission. In August 2025 the bank reversed the redundancies and apologised, acknowledging that call volumes had not fallen as assumed and that the roles were still required. [PUBLIC — reported by Australian media and the Finance Sector Union, Aug 2025] [VERIFY]
- **AI pattern.** Voice assistant for call deflection; customer-facing automation.
- **Measured outcomes.** The intended outcome (call reduction) did not materialise in the measurement window; the realised outcome was reinstatement and reputational cost.
- **What went wrong.** The labour saving was booked against a forecast deflection, not an observed one. The sequencing error was to remove capacity before the demand reduction was proven in production.
- **Lesson for a strategy consultant.** Sequence benefits after evidence: prove deflection in production telemetry for at least one full seasonal cycle before releasing capacity, and design the business case with capacity redeployment first and headcount release second. It is also a labour-relations lesson: in unionised markets, the sequencing is a legal and reputational matter, not only a financial one.
- **Interview citation.** "CBA's 2025 reversal is the textbook baseline error: capacity was removed on a forecast deflection, and the call volumes did not fall."

## 5.8 Air Canada: liability by chatbot

- **What they did.** Air Canada's website chatbot told a customer that bereavement fares could be claimed retroactively, contradicting the airline's actual policy. When the airline refused the refund, the customer took the case to the British Columbia Civil Resolution Tribunal, which in February 2024 held Air Canada responsible for the chatbot's statement and rejected the argument that the chatbot was a separate entity. [PUBLIC — Moffatt v. Air Canada, BC Civil Resolution Tribunal, Feb 2024]
- **AI pattern.** Customer-facing conversational assistant giving policy information without grounding to the authoritative source or a disclaimer that was accepted by the tribunal.
- **Measured outcomes.** Small financial award; large precedent: an organisation is liable for what its chatbot says.
- **What went wrong.** The system spoke with authority on policy without citation grounding or a human check, and the organisation had no agreed position on standing behind its output.
- **Lesson for a strategy consultant.** For any customer-facing GenAI in a bank, grounding to the policy source, citation, and a liability position agreed with Legal and Compliance are design requirements, not nice-to-haves. This case is why "the model said so" is not a defence under consumer protection law.
- **Interview citation.** "Air Canada 2024 established that the firm owns what its chatbot says; in banking that means grounding, citation and a Legal-approved liability position before go-live."

## 5.9 Apple Card and Goldman Sachs: explainability not evidenced

- **What they did.** In November 2019 public complaints alleged that the Apple Card, issued by Goldman Sachs, gave women materially lower credit limits than their spouses with similar or better financial profiles. The New York Department of Financial Services investigated and in March 2021 found no evidence of unlawful discrimination under fair lending law, while criticising the customer service response and the lack of transparency in explaining decisions. [PUBLIC — NYDFS report, Mar 2021]
- **AI pattern.** ML credit-limit model within an underwriting process; the model did not use gender as an input.
- **Measured outcomes.** No regulatory finding of discrimination; significant reputational cost and a regulatory critique of explainability and appeal processes.
- **What went wrong.** The firm could not explain individual outcomes to customers or the press at the moment it mattered. The legal position was defensible; the process for evidencing it was not ready.
- **Lesson for a strategy consultant.** Fairness testing and explainability are operational capabilities that must exist before the first complaint. The appeal process and the customer-facing explanation are part of the model deployment, not of customer service.
- **Interview citation.** "Apple Card shows that being compliant and being able to show you are compliant are different capabilities; NYDFS cleared the model and still criticised the explanation process."

## 5.10 Zillow Offers: the competence envelope

- **What they did.** Zillow used its Zestimate-based pricing models to buy homes directly through Zillow Offers. In late 2021, after the models over-priced purchases in a shifting market, the company wound down the business, took write-downs in excess of USD 500m and cut around 25% of its workforce. [PUBLIC — Zillow Q3 2021 results and statements, Nov 2021] [VERIFY exact write-down figure]
- **AI pattern.** Predictive ML model driving irreversible capital commitments at scale with limited human override.
- **Measured outcomes.** Direct losses, business exit, workforce reduction.
- **What went wrong.** The model was used outside the regime it was trained on; the business committed capital on each prediction without a sufficient margin of safety or a kill-switch keyed to drift.
- **Lesson for a strategy consultant.** Not banking, but the closest public analogue to letting a model drive balance-sheet decisions. In BFSI terms: any model that commits capital or credit needs drift monitoring, exposure limits and human override designed as controls, and the reversibility of the action should set the autonomy level.
- **Interview citation.** "Zillow is the reference for competence-envelope failure: a pricing model drove irreversible purchases through a regime change, and USD 500m of write-downs followed."

## 5.11 Goldman Sachs and Citi: scaled enablers in 2025 (brief)

- Goldman Sachs rolled out its GS AI Assistant firm-wide in 2025 and announced a pilot of Cognition's Devin coding agent in July 2025, positioning it as a "hybrid workforce". [PUBLIC — Goldman Sachs statements, 2025] [VERIFY]
- Citi deployed Citi Assist and Citi Stylus GenAI tools to around 140,000 employees across 2024 to 2025. [PUBLIC — Citi statements, 2024 to 25] [VERIFY]
- Both are scaled enablers; the interesting signal is the explicit move from assistants to agents in engineering, where output is testable and reversibility is high. That is where agentic AI is reaching production first in large banks.

## 6. Consultant language

> *"Let me separate what was claimed, what was measured, and what has since been corrected."*

Use when a client cites a press release as a target for their own programme.

> *"That figure is an input metric; the outcome metric we would track is the one that reaches your P&L."*

Use when a client quotes seat counts, interaction counts or model counts as evidence of value.

> *"The cases that held up had a hard baseline and a narrow scope; the ones that reversed booked the saving before the demand had moved."*

Use to frame sequencing and benefit-realisation discipline.

> *"At your volumes, an HSBC-type alert reduction translates to roughly this much analyst capacity; let me show you the arithmetic and the assumptions."*

Use to move from anecdote to the client's numbers.

> *"The Klarna correction is the design lesson: we set the human-agent ratio as a quality constraint, not as a residual of the cost case."*

Use when the client's ambition is "fully automate".

## 7. Common mistakes juniors make

- **Quoting the headline as the outcome.** "Klarna saved 700 FTE" is not what Klarna said and not what happened; the chats were handled, the cost base moved less, and quality was later re-funded.
- **Treating all AI as GenAI.** HSBC AML AI and Erica are predominantly supervised ML and rules plus NLU; calling them GenAI in an interview signals shallow understanding.
- **Ignoring the walk-backs.** Juniors present the successes; managers present the corrections, because that is where the design lessons are.
- **Not translating to the client's scale.** A case is only useful when you can say what it would mean in the client's volumes and cost base.
- **Mislabelling estimates as facts.** Any "realised vs claimed" judgement is the consultant's estimate and must be labelled as such.
- **Using non-banking cases without a bridge.** Zillow and Air Canada are valuable, but only when you state the banking analogue explicitly.

## 8. Interview questions with model answers

### Q1. "Give me a public example where an AI deployment in financial services was walked back, and what you take from it."

Model answer. I would use Klarna because it is the most complete public arc. In February 2024 Klarna announced that its OpenAI-based assistant handled two-thirds of customer service chats in its first month, equivalent to about 700 agents, with a forecast USD 40m profit improvement [PUBLIC — Klarna, 2024]. In 2025 the CEO said publicly that the company had over-weighted cost, that service quality had suffered, and that it would hire human agents again [PUBLIC — 2025, VERIFY]. My read is that the volume claim was true and the business case was incomplete: it booked the cost side of the quality-cost trade-off and did not price the customer-experience side. The design lesson I apply is to set a quality floor, typically CSAT, first-contact resolution and complaint rate, as a hard constraint in the case, and to define the human-agent ratio as a design parameter rather than letting the cost target drive it to zero. A second, more operational example is Commonwealth Bank of Australia in August 2025, which reinstated 45 roles because the voice bot had not

reduced call volumes as forecast [PUBLIC, VERIFY]; the lesson there is sequencing: remove capacity only after deflection is observed in production.

### Q2. “Which public bank AI case do you consider the strongest evidence of measurable value, and why?”

Model answer. HSBC’s AML transaction monitoring with Google Cloud, reported in June 2023, where the ML risk-scoring approach identified 2 to 4 times more suspicious activity while cutting alert volumes by about 60% [PUBLIC — Google Cloud, 2023, VERIFY]. I rate it highest for three reasons. First, the metric is operational and regulator-visible: alerts per period and suspicious-activity yield are the AML function’s own KPIs, so there is no translation gap to value. Second, the scope is narrow and the baseline is hard, which is the pattern that holds up across the public record. Third, it is supervised ML replacing rules, which means the model-risk pathway is well understood by validators and regulators. The caveats I would add are that the realised analyst-capacity saving has not been disclosed, that fewer alerts are richer alerts so first-year effort shifts rather than disappears, and that acceptance has to be secured regulator by regulator. For a client in payments or AML, I would translate it as: at your alert volumes a 60% reduction releases roughly this many investigator hours, and here is how we would evidence it to your regulator.

### Q3. “A client wants to put a GenAI chatbot in front of retail customers next quarter. What public evidence would you put on the table?”

Model answer. Three cases, in this order. Bank of America’s Erica, which has handled more than 2 billion interactions since 2018 by staying bounded and largely deterministic in fulfilment [PUBLIC, VERIFY]; that is the pattern that has worked in banking. Air Canada, where a tribunal in February 2024 held the airline liable for its chatbot’s incorrect policy statement [PUBLIC]; in banking that means grounding to the policy source, citation and a Legal-approved liability position are launch gates. And Klarna, which proved the volume case and then re-hired humans for quality [PUBLIC, VERIFY]. My recommendation would be a bounded-intent design: deterministic fulfilment for account and payment actions, GenAI for understanding and for drafting grounded responses, a human-escalation ratio set as a design target, and a quality floor in the business case. I would also show the client the autonomy level we are proposing and the controls that go with it, so that Compliance can sign it off before the quarter ends rather than after.

## 9. Self-check quiz

1. Place Klarna’s 2024 claims and HSBC’s 2023 AML results on the claimed-versus-realised value ladder, and justify the rung for each.
2. Which of the ten cases is an example of the “baseline error” failure driver, and what sequencing rule would have prevented it?
3. A client says “Erica proves GenAI chatbots work for banks.” Correct the statement in two sentences.
4. JPMorgan’s LLM Suite reached around 200,000 employees. Explain why that figure is an input metric and name two outcome metrics you would track instead.
5. Translate HSBC’s reported 60% alert reduction into analyst capacity for a bank that generates 240,000 AML alerts a year, with 45 minutes average handling time and 1,600 productive hours per analyst. Show the arithmetic and state one reason the first-year saving will be lower.

# Module 18 — Three end-to-end engagement simulations

## Purpose and how to use this module

This module walks three complete engagements from client ask to benefits, in the shape a partner would actually run them. Every client, number and finding here is [ILLUSTRATIVE — composite]: assembled from patterns typical of such programmes, with the arithmetic shown so you can defend or rebuild every figure. Public frameworks and regulations are attributed and labelled [PUBLIC]; the orders of magnitude for AI pricing are [ESTIMATE — order of magnitude, VERIFY].

Use the simulations in three ways. First, as a template: the H3 structure (client ask, diagnostic, hypotheses, workstreams, week-by-week plan, key analyses, deliverables, recommendation, benefits and risks) is the structure you should be able to reproduce for any AI engagement in an interview. Second, as a numbers drill: rebuild the business case tables with different assumptions until the arithmetic is automatic. Third, as a language model: the action titles in the storyline tables show how a strategy consultant states a finding.

### Common assumptions used across all three simulations

To keep the simulations numerically consistent, the following assumptions are shared and reused. All are [ESTIMATE — assumption, VERIFY against the client's actuals].

| Assumption | Value | Note |
|---|---|---|
| Loaded FTE cost, UK operations analyst | GBP 62,000 a year | Salary, employer costs, workplace, tooling |
| Loaded FTE cost, UK KYB specialist | GBP 70,000 a year | Senior analyst or approver |
| Loaded FTE cost, EU mid-sized bank operations | EUR 80,000 a year | Northern European composite |
| Loaded FTE cost, US card operations (blended onshore and nearshore) | USD 52,000 a year | 45% onshore, 55% nearshore |
| Loaded FTE cost, India offshore operations | USD 22,000 a year (about GBP 17,500) | Captive or vendor |
| Productive hours per FTE | 1,600 a year | After leave, training, shrinkage |
| Frontier model API price | USD 3 per million input tokens, USD 15 per million output tokens | Order of magnitude only |
| Mid-tier model API price | USD 0.80 input, USD 4 output per million tokens | Order of magnitude only |
| Small model API price | USD 0.15 input, USD 0.60 output per million tokens | Order of magnitude only |
| Prompt caching discount on cached input | 75% | Applies to repeated system prompts and policy context |
| Discount rate for NPV | 10% | Typical bank hurdle for operational investments |
| FX for cross-checks | 1 EUR = 1.10 USD; 1 GBP = 1.17 EUR | Illustrative |

# Simulation A: AI strategy and use-case assessment for a mid-sized bank

## Client ask

[ILLUSTRATIVE — composite] “Nordbank” is a universal bank with EUR 52bn of assets, 1.6 million retail customers, 45,000 SME and corporate clients, 6,400 FTE, operating income of EUR 2.1bn and a cost base of EUR 1.3bn (cost-income ratio 62%). The core banking platform is a 1990s mainframe with COBOL batch processing; a cloud migration is about 30% complete; seven machine-learning models are in production (retail credit scoring, card fraud, two churn models, three marketing propensity models). Two GenAI pilots ran in 2025 (a contact-centre knowledge assistant and a coding copilot trial) and neither reached production.

The CEO’s ask to the partner: “The board wants an AI strategy by the next strategy cycle. Tell us where the value is, what it would take, and what we should do first. We do not want a 200-page report; we want a decision.”

Engagement shape: 12 weeks, team of one partner (about 20% time), one engagement manager, four consultants, two AI engineers, one data scientist for the two mid-engagement deep dives. Illustrative fee range EUR 1.4m to 1.9m [ESTIMATE].

## Diagnostic

The first three weeks established the fact base. Findings below carry the numbers the team actually used.

**Maturity assessment.** Scored on a 1 to 5 scale across six dimensions adapted from the McKinsey “Rewired” capability set [PUBLIC — Lamarre, Smaje, Zemmel, 2023], with a peer benchmark from the team’s own experience [ESTIMATE].

| Dimension | Nordbank score | Peer median (mid-sized European banks) | Gap |
|---|---|---|---|
| Business-led AI roadmap and ambition | 1.5 | 2.5 | 1.0 |
| Talent (AI product, engineering, risk) | 2.0 | 2.5 | 0.5 |
| Operating model and governance | 1.5 | 2.5 | 1.0 |
| Technology (platform, APIs, cloud) | 1.5 | 3.0 | 1.5 |
| Data (quality, access, lineage) | 2.0 | 3.0 | 1.0 |
| Adoption and scaling (change, value tracking) | 2.0 | 2.5 | 0.5 |
| Average | 1.75 | 2.67 | 0.92 |

**Cost base decomposition.** Of the EUR 1.3bn cost base, EUR 610m is personnel in operations, servicing, risk and compliance functions where work is document- and language-heavy; EUR 290m is technology; the balance is premises, marketing and other. The team sized the “AI-addressable” pool as the personnel cost in high-volume, rule- and judgement-mixed processes: EUR 385m (about 4,800 FTE at EUR 80,000) [ESTIMATE — from HR headcount by cost centre].

**Benchmark of value potential.** The McKinsey Global Institute estimate for generative AI in banking is USD 200bn to 340bn a year globally, equivalent to roughly 2.8% to 4.7% of industry revenue [PUBLIC — McKinsey Global Institute, June 2023]. Applied to Nordbank’s EUR 2.1bn operating income, that implies EUR 60m to 100m a year of potential at full maturity [ESTIMATE — top-down cross-check only]. The bottom-up use-case assessment below landed at EUR 75m gross run-rate, inside that range, which is the cross-check a CFO expects to see.

**Pilot post-mortem.** Both 2025 pilots stalled for the same reasons: no named business owner, no production path (the knowledge assistant ran on an unsupported sandbox), and no integration budget. Classic pilot purgatory, consistent with Gartner's forecast that at least 30% of GenAI projects would be abandoned after proof of concept by end 2025 [PUBLIC — Gartner, July 2024, VERIFY].

## Hypotheses

Falsifiable hypotheses set in week 1 and tested through the diagnostic.

1. **H1.** At least 60% of the three-year value lies in six to eight high-volume operations processes (KYC, AML, servicing, payments investigations, credit operations, engineering), not in customer-facing GenAI. Tested through the scored portfolio. Result: confirmed; 68% of risk-adjusted value in those domains.
2. **H2.** The binding constraint is technology and data access, not model capability; a shared AI platform with retrieval, evaluation and gateway services is a prerequisite for any use case to scale. Tested through architecture review and the pilot post-mortem. Result: confirmed.
3. **H3.** The bank can fund the programme from first-wave savings within 24 months if wave 1 is restricted to use cases with hard baselines. Tested through the business case. Result: confirmed with a 10-month gap bridged by central investment.
4. **H4.** A hub-and-spoke operating model is appropriate at Nordbank's scale; a fully federated model would fragment scarce talent. Tested through interviews and talent inventory (14 people with relevant skills across the bank). Result: confirmed.
5. **H5.** Regulatory exposure under the EU AI Act is manageable because the first two waves avoid Annex III high-risk use (creditworthiness assessment of natural persons) except where existing model-risk governance already applies. Tested through a legal and compliance review. Result: confirmed, with the next-best-offer and SME underwriting use cases flagged for deployer obligations [PUBLIC — Regulation (EU) 2024/1689, VERIFY timeline].

## Workstreams

| Workstream | Lead | Scope | Key outputs |
|---|---|---|---|
| 1. Ambition and maturity | Partner and EM | Executive interviews (18), maturity scoring, ambition options | Ambition statement, maturity baseline |
| 2. Use-case discovery and assessment | Two consultants | Value-chain walk, long list (64), scoring (18), prioritisation (12), waves | Scored matrix, roadmap |
| 3. Platform and data | Two AI engineers | Architecture review, platform target state, data readiness | Target platform blueprint, data remediation plan |
| 4. Business case and funding | One consultant and data scientist | Baselines, benefit build, TCO, funding model | Five-year business case, funding ask |
| 5. Operating model, governance and risk | One consultant | Operating-model options, roles, AI Act and model-risk mapping | Operating model design, governance charter |

## Week-by-week plan

| Week | Focus | Key activities | Steering committee |
|---|---|---|---|
| 1 | Mobilise | Hypotheses, interview schedule, data requests, workshop design | Kick-off |
| 2 | Diagnose | 18 executive interviews, cost- | |

| Week | Focus | Key activities | Steering committee |
|---|---|---|---|
| | | base decomposition, architecture review | |
| 3 | Diagnose | Maturity scoring, pilot post-mortems, data readiness assessment | SteerCo 1: diagnostic read-out |
| 4 | Discover | Three ideation workshops (operations, risk and compliance, retail and digital); long list of 64 | |
| 5 | Discover | Screening 64 to 38 to 18; scoring criteria agreed with CFO office | |
| 6 | Assess | Scoring of 18 use cases; value baselining with Finance | SteerCo 2: long list and scoring approach |
| 7 | Assess | Deep dive 1: AML alert triage (data sample, feasibility); platform target state | |
| 8 | Assess | Deep dive 2: contact-centre agent assist; TCO modelling | |
| 9 | Synthesise | Prioritised portfolio of 12, three waves, business case v1 | SteerCo 3: portfolio and case |
| 10 | Design | Operating model and governance; funding model; AI Act mapping | |
| 11 | Decide | Business case v2 after CFO challenge; roadmap; executive storyline | |
| 12 | Commit | Board paper, final presentation, mobilisation plan for wave 1 | Board presentation |

## Key analyses

**Analysis 1: the scored portfolio.** Value and feasibility were each composite scores on a 1 to 5 scale. Value weights: cost 40%, revenue 20%, risk and loss 20%, customer experience 20%. Feasibility weights: data readiness 25%, integration complexity 25%, model and pattern maturity 20%, regulatory exposure 15%, change effort 15%. Weights were agreed with the CFO office in week 5; this matters because agreeing the weights before scoring removes the later argument about the ranking.

| Use case | Value (1–5) | Feasibility (1–5) | Gross run-rate value, EUR m [ESTIMATE] | Wave |
|---|---|---|---|---|
| Engineering copilots and COBOL modernisation assist | 4.1 | 4.2 | 10 | 1 |
| Contact-centre agent assist (retail) | 4.0 | 4.0 | 9 | 1 |
| AML alert triage and narrative drafting | 4.2 | 3.2 | 7 | 1 |
| Internal knowledge assistant (policies, | 3.0 | 4.6 | 5 | 1 |

| Use case | Value (1–5) | Feasibility (1–5) | Gross run-rate value, EUR m [ESTIMATE] | Wave |
|---|---|---|---|---|
| procedures) | | | | |
| Payments investigations (ISO 20022 exceptions, enquiries) | 3.2 | 4.0 | 4 | 1 |
| KYC periodic review automation (retail and SME) | 4.0 | 3.0 | 8 | 2 |
| SME credit memo drafting and underwriting assist | 4.3 | 3.0 | 11 | 2 |
| Mortgage document processing | 3.1 | 3.2 | 4 | 2 |
| Dispute and chargeback handling | 3.0 | 3.8 | 3 | 2 |
| Collections next-best-action and conversational agent | 3.2 | 2.9 | 4 | 3 |
| Next-best-offer personalisation | 3.9 | 2.2 | 8 | 3 |
| Regulatory reporting and compliance QA | 2.2 | 2.1 | 2 | 3 |
| Total | | | 75 | |

How a value score was built, for one row [ILLUSTRATIVE]. AML alert triage: 180,000 alerts a year, 48 minutes average handling, so 144,000 analyst hours, or 90 FTE at 1,600 hours, costing EUR 7.2m. A 55% reduction in handling effort (narrative drafting, data gathering, false-positive pre-disposition) releases EUR 4.0m. Risk value: better-evidenced suspicious activity reports and fewer missed cases, valued at EUR 2.0m of avoided remediation cost [ESTIMATE]. CX value: nil. Revenue: nil. Total EUR 7m gross, which scored 4 on cost (top quartile of the portfolio), 1 on revenue, 4 on risk, 1 on CX; weighted 0.4×4 + 0.2×1 + 0.2×4 + 0.2×1 = 2.8 before normalisation across the portfolio, which rescaled to 4.2.

**Analysis 2: the 2x2 and wave logic.** Use cases with feasibility 3.8 or above and value 3.0 or above went to wave 1 (quick wins with hard baselines). Wave 2 held high-value use cases needing data remediation or integration (KYC, SME underwriting). Wave 3 held use cases with regulatory exposure or weak data (next-best-offer, collections). The rule the team applied: wave 1 must contain at least one use case per major operations domain so that the platform is proven on real variety, and no wave 1 use case may be customer-facing GenAI.

**Analysis 3: platform TCO.** The shared platform (model gateway, retrieval service, evaluation harness, observability, guardrails, agent runtime) was costed at EUR 14m over three years: EUR 6m build, EUR 2.7m a year run [ESTIMATE]. Use-case build costs averaged EUR 2.5m for wave 1 and EUR 3.5m for wave 2, dominated by integration and change, not model costs. Token costs across the full portfolio at scale were estimated at EUR 1.1m a year [ESTIMATE — order of magnitude], below 2% of total programme cost, which is a finding worth stating to a board because it reframes the cost conversation away from model pricing and towards integration and change.

**Analysis 4: five-year business case, risk-adjusted.** Gross run-rate of EUR 75m was haircut by evidence level: wave 1 at 75% (hard baselines, proven patterns), wave 2 at 60%, wave 3 at 40%, giving a risk-adjusted run-rate of EUR 48m by year 3 [ESTIMATE].

| Year | Investment, EUR m | Run cost, EUR m | Risk-adjusted benefit, EUR m | Net cash flow, EUR m | Cumulative, EUR m |
|---|---|---|---|---|---|
| 1 | 18 | 3 | 6 | -15 | -15 |
| 2 | 22 | 7 | 24 | -5 | -20 |
| 3 | 12 | 9 | 42 | 21 | 1 |
| 4 | 4 | 10 | 48 | 34 | 35 |
| 5 | 4 | 10 | 48 | 34 | 69 |

NPV at 10%: -15/1.10 + -5/1.21 + 21/1.331 + 34/1.4641 + 34/1.6105 = -13.6 - 4.1 + 15.8 + 23.2 + 21.1 = EUR 42.3m. Payback during year 3, roughly month 35 from programme start. Investment of EUR 60m over five years against a run-rate benefit of EUR 48m is a payback profile a bank board will accept; the sensitivity that matters most is wave 2 slipping by a year, which moves payback to month 43 and cuts NPV to about EUR 25m.

**Analysis 5: operating model.** Hub-and-spoke: a central AI hub of 28 FTE by end of year 2 (platform engineering 12, AI product and delivery 8, AI risk and evaluation 5, change and adoption 3) with embedded product owners in each of the five domains. Funding: central budget for platform and wave 1, business-funded from wave 2 with a gain-share rule that 50% of validated savings are reinvested in the business unit's next use case. Governance: an AI steering committee chaired by the COO, a model-risk gate run by the existing model validation function extended with LLM evaluation competence, and an AI Act register maintained by Compliance.

## Deliverables and slide-by-slide storyline

The final board document was 24 slides plus an appendix. Each title is an action title stating the finding.

| Slide | Action title | Key content | Source analysis |
|---|---|---|---|
| 1 | Nordbank can release EUR 48m a year by year 3 from 12 AI use cases, funded by wave-1 savings after month 35 | Executive summary, three decisions requested | All |
| 2 | The board asked three questions: where is the value, what does it take, what first | Restating the ask | Ask |
| 3 | Nordbank's AI maturity is 1.75 of 5 against a peer median of 2.7, with technology the largest gap | Maturity table | Analysis: maturity |
| 4 | Two pilots stalled for the same three reasons, all fixable | Pilot post-mortem | Diagnostic |
| 5 | EUR 385m of the cost base sits in language-heavy processes that AI can address | Cost decomposition | Diagnostic |
| 6 | Top-down, GenAI potential for a bank of this size is EUR 60m to 100m a year | MGI cross-check [PUBLIC] | Benchmark |
| 7 | We screened 64 ideas to 12 use cases using criteria agreed with Finance | Funnel, criteria, weights | Analysis 1 |
| 8 | Twelve use cases carry EUR | Scored matrix | Analysis 1 |

| Slide | Action title | Key content | Source analysis |
|---|---|---|---|
| | 75m of gross run-rate value, two-thirds in operations and risk | | |
| 9 | Five use cases are ready now; they have hard baselines and proven patterns | 2x2 with wave 1 highlighted | Analysis 2 |
| 10 | AML alert triage: EUR 7m from 90 analysts' effort, with a regulator-visible metric | Deep dive 1 | Deep dive |
| 11 | Contact-centre agent assist: EUR 9m from handling time and first-contact resolution, not from headcount alone | Deep dive 2 | Deep dive |
| 12 | Wave 2 unlocks EUR 26m but needs KYC data remediation and core integration first | Wave 2 dependencies | Analysis 2 |
| 13 | A shared AI platform is a prerequisite: EUR 14m over three years, reused by every use case | Platform blueprint and TCO | Analysis 3 |
| 14 | Model costs are below 2% of programme cost; integration and change are 70% | Cost structure | Analysis 3 |
| 15 | Risk-adjusted, the programme delivers EUR 48m a year by year 3 and NPV of EUR 42m | Business case table | Analysis 4 |
| 16 | The case is most sensitive to wave 2 timing, not to model pricing | Tornado | Analysis 4 |
| 17 | A hub-and-spoke model with 28 central FTE fits Nordbank's talent base | Operating model | Analysis 5 |
| 18 | Business units fund from wave 2 under a 50% reinvestment rule | Funding model | Analysis 5 |
| 19 | Governance extends existing model-risk and compliance functions rather than creating new ones | Governance charter | Analysis 5 |
| 20 | Two use cases attract EU AI Act deployer obligations; neither is in wave 1 | AI Act register [PUBLIC, VERIFY] | Workstream 5 |
| 21 | The 36-month roadmap sequences platform, wave 1, wave 2, wave 3 | Roadmap | Synthesis |
| 22 | Wave 1 mobilises in 90 days with five named business owners | Mobilisation plan | Synthesis |
| 23 | Three decisions for the board today | Decision slide | Synthesis |

| Slide | Action title | Key content | Source analysis |
|---|---|---|---|
| 24 | Appendix index | Baselines, scoring detail, interview list | |

### Recommendation

Commit to a EUR 60m five-year programme in three waves, starting with a shared AI platform and five operations use cases with hard baselines; stand up a 28-FTE hub-and-spoke AI function under the COO; fund wave 1 centrally and waves 2 and 3 from business-unit budgets under a reinvestment rule; and treat the two pilots as closed, re-starting the knowledge assistant as a wave 1 use case with a named owner and a production path.

### Benefits and risks

Benefits: EUR 48m risk-adjusted run-rate by year 3, NPV EUR 42m, payback month 35; a reusable platform that halves the build cost of each subsequent use case; a governance structure that satisfies the EU AI Act deployer obligations in advance of the high-risk deadlines [PUBLIC, VERIFY].

Risks and mitigations: wave 2 slippage from KYC data remediation (mitigate by starting remediation in month 1); talent attrition in the hub (mitigate with a retention plan and partner model); value leakage if capacity released is not converted (mitigate with a quarterly benefits review co-signed by Finance); regulatory change (mitigate with an AI Act register and a horizon-scanning role in Compliance).

## Simulation B: Agentic AI transformation of corporate onboarding (KYB) at a UK Tier-1 bank

### Client ask

[ILLUSTRATIVE — composite] “Albion Bank” is a UK Tier-1 bank. Its Corporate and Commercial Banking division onboards 36,000 new business clients a year (24,000 SME, 9,000 mid-corporate, 3,000 large or complex structures) and performs 110,000 periodic KYB reviews (80,000 SME, 22,000 mid-corporate, 8,000 complex). KYB operations employ 920 FTE: 540 in the UK and 380 in an Indian captive centre. Median cycle times are 9 days (SME), 27 days (mid-corporate) and 54 days (complex). Mid-corporate abandonment during onboarding is 18%.

The Head of Corporate Banking’s ask: “Our competitors are onboarding mid-corporates in under two weeks. I want a credible plan to halve cycle time and take a third out of the cost of KYB using agentic AI, and I want to know how our model-risk and financial-crime people will sign it off. Design it and get a pilot live in 16 weeks.”

Engagement shape: 16 weeks design-to-pilot; partner (15%), engagement manager, three consultants (one with financial-crime depth), three AI engineers, one model-risk specialist. Illustrative fee GBP 1.6m to 2.2m plus pilot build [ESTIMATE].

### Diagnostic

**Cost baseline.** Personnel: 540 UK FTE at GBP 62,000 = GBP 33.5m; 380 India FTE at GBP 17,500 = GBP 6.6m; total GBP 40.1m, average GBP 43,600 per FTE. Technology, registry data and screening vendors: GBP 6.2m. Total KYB cost GBP 46.3m a year [ESTIMATE — from cost-centre data].

**Effort baseline.** Analyst hours per case from a time-and-motion sample of 240 cases [ILLUSTRATIVE]:

| Case type | Volume a year | Hours per case | Total hours |
|---|---|---|---|
| SME new onboarding | 24,000 | 6.0 | 144,000 |
| Mid-corporate new | 9,000 | 22.0 | 198,000 |

| Case type | Volume a year | Hours per case | Total hours |
|---|---|---|---|
| onboarding | | | |
| Complex new onboarding | 3,000 | 55.0 | 165,000 |
| SME periodic review | 80,000 | 2.5 | 200,000 |
| Mid-corporate periodic review | 22,000 | 9.0 | 198,000 |
| Complex periodic review | 8,000 | 24.0 | 192,000 |
| Direct case hours | 146,000 cases | | 1,097,000 |

Capacity is 920 × 1,600 = 1,472,000 hours, so 375,000 hours (25%) are indirect: quality control, rework, document chasing, management and training. That indirect pool is a target in its own right.

**Where the time goes.** Share of direct hours by L3 task, from the sample: document collection and chasing 12%, document classification and extraction 14%, UBO identification and ownership unwrapping 15%, screening alert disposition 12%, EDD narrative drafting 12%, four-eyes QC 10%, intake 6%, registry verification 5%, approval 4%, account setup and communications 4%, risk rating 3%, screening execution 3%.

**Revenue at risk.** Mid-corporate first-year revenue averages GBP 38,000; complex clients GBP 190,000 [ESTIMATE]. Abandonment of 18% on 9,000 mid-corporate cases is 1,620 lost clients, GBP 61.6m of first-year revenue; 11% on 3,000 complex cases is 330 clients, GBP 62.7m. Interviews with relationship managers attributed roughly half of abandonment to cycle time. The team assumed 20% of revenue at risk is recoverable by halving cycle time, GBP 24.9m, at a 40% contribution margin, giving GBP 10m a year of contribution [ESTIMATE — deliberately conservative, agreed with Finance].

**Technology.** Workflow tool with APIs, document store, a screening vendor with an API, Companies House and commercial registry feeds via API, a rules-based customer risk rating (CRR) engine already in the model inventory, and no vector store or model gateway. Data for evaluation is abundant: four years of closed cases with analyst decisions and QC outcomes.

## Hypotheses

1. **H1.** More than half of direct effort sits in tasks with high input variability and bounded judgement (extraction, UBO unwrapping, alert disposition, narrative drafting), which is the profile where agents outperform rules. Result: confirmed; 53% of direct hours in those four tasks.
2. **H2.** Blended effort can fall by about 45% at steady state without moving any regulatory decision away from a human. Result: confirmed at 47% before human-in-the-loop overhead, 45% after.
3. **H3.** The multi-agent system can be classified under the bank's SS1/23 framework as a Tier 2 model system with component-level validation, which allows a pilot within 16 weeks. Result: confirmed with the model-risk function, subject to the validation plan [PUBLIC — PRA SS1/23, 2023; VERIFY].
4. **H4.** SME onboarding and SME periodic review are the right pilot scope because volume is high, structures are simple, and errors are detectable before approval. Result: confirmed.
5. **H5.** Token costs are immaterial relative to people and integration costs, so model choice should be driven by accuracy and controllability rather than price. Result: confirmed; about GBP 0.36 per case.

## Workstreams

| Workstream | Scope | Outputs |
|---|---|---|
| 1. Process and task decomposition | L1 to L4 decomposition, time and motion, pain points | Task inventory with effort shares |
| 2. Agentic-fit assessment and target design | Six-dimension scoring, automation bands, supervisor-led architecture, HITL design | Scored task matrix, target architecture |

| Workstream | Scope | Outputs |
|---|---|---|
| 3. Model risk, financial crime and legal | SS1/23 classification, validation plan, MLRO sign-off, DPIA | Model-risk pack, control framework |
| 4. Pilot build and evaluation | Golden dataset, eval harness, agents, integration, shadow run | Pilot system, gate results |
| 5. Business case and change | Baseline, benefits, TCO, workforce plan, role redesign | Business case, change plan |

## Week-by-week plan

| Week | Focus | Key activities | Gate |
|---|---|---|---|
| 1 | Mobilise | Hypotheses, data access, team onboarding, model-risk engagement | |
| 2 | Diagnose | Time-and-motion sample (240 cases), interviews with 22 analysts and approvers | |
| 3 | Diagnose | Task decomposition to L4, effort shares, pain-point heatmap | SteerCo 1 |
| 4 | Assess | Six-dimension agentic-fit scoring workshop with operations and financial crime | |
| 5 | Design | Target architecture v1: supervisor and worker agents, tools, HITL gates | |
| 6 | Design | Control framework, SS1/23 classification proposal, DPIA scoping | SteerCo 2: design approval |
| 7 | Build | Golden dataset (1,200 closed SME cases), eval harness, platform set-up | |
| 8 | Build | Intake, extraction and registry agents; integration with workflow tool | |
| 9 | Build | UBO, screening-disposition and narrative agents; supervisor orchestration | |
| 10 | Build | QC agent, HITL console, audit logging, guardrails | Internal gate: eval thresholds on golden set |
| 11 | Validate | Independent model validation of components; MLRO review; security testing | |
| 12 | Shadow | Shadow run on 2,000 live SME cases, no production effect | SteerCo 3: shadow results |
| 13 | Shadow | Error analysis, prompt and retrieval tuning, threshold calibration | |

| Week | Focus | Key activities | Gate |
|---|---|---|---|
| 14 | Live assist | 1,000 live SME cases with analyst acceptance of agent outputs | |
| 15 | Live assist | Gate measurement, override analysis, analyst feedback | |
| 16 | Decide | Pilot gate review, scale business case, board paper | Board gate: scale decision |

## Key analyses

**Analysis 1: task-level agentic-fit scoring.** Six dimensions, each scored 1 to 5 where 5 means "strong fit for agentic automation", with weights agreed in week 4. This is a practitioner framework; variants differ by firm, and attribution to a single source would be wrong.

| Dimension | Weight | 5 means | 1 means |
|---|---|---|---|
| D1 Input variability | 20% | Varied, unstructured inputs where rules fail | Fully structured, rules suffice |
| D2 Bounded judgement | 20% | Judgement needed but codifiable in policy with examples | Either none, or open-ended expert judgement |
| D3 Tool and system access | 15% | All systems reachable by API | Manual, swivel-chair |
| D4 Error tolerance and reversibility | 20% | Errors cheap, visible and reversible | Irreversible or regulatory consequence |
| D5 Data for evaluation | 15% | Abundant labelled history | None |
| D6 Volume and frequency | 10% | High volume, daily | Rare |

Bands: 3.5 and above, agentic automation with sampling; 2.5 to 3.5, agent-assist with mandatory human decision; below 2.5, human or deterministic rules.

| L3 task | D1 D2 D3 D4 D5 D6 | Weighted score | Band and design decision |
|---|---|---|---|
| Intake and data capture | 3 2 4 5 4 5 | 3.70 | Automate; agent with deterministic validation |
| Document collection and chasing | 3 2 3 5 3 5 | 3.40 | Assist; communications agent drafts, analyst releases |
| Document classification and extraction | 5 3 4 4 5 5 | 4.25 | Automate; confidence-gated, low confidence to human |
| Registry verification | 2 2 5 4 5 5 | 3.60 | Automate; API-first, agent reconciles discrepancies |
| UBO identification and unwrapping | 5 4 3 2 3 4 | 3.50 | Assist; agent proposes structure, human confirms all layered structures |
| Screening execution | 2 1 5 3 5 5 | 3.20 | Keep deterministic vendor tool; not agentic |
| Screening alert disposition | 4 4 4 2 5 5 | 3.85 | Automate false-positive discounting with 100% QC in pilot; true-match confirmation human |
| Customer risk rating | 1 2 5 2 5 5 | 3.00 | Keep rules engine; agent assembles inputs only |
| EDD narrative drafting | 5 4 3 4 4 3 | 3.95 | Automate drafting; approver |

| L3 task | D1 D2 D3 D4 D5 D6 | Weighted score | Band and design decision |
|---|---|---|---|
| | | | edits and signs |
| Four-eyes QC | 4 4 4 3 4 5 | 3.90 | Agent QC first pass; human QC on risk-based sample |
| Approval decision | 3 2 4 1 4 4 | 2.80 | Human; agent prepares decision pack |
| Account setup and communications | 2 1 4 4 4 5 | 3.10 | Deterministic automation; agent drafts communications |

Worked arithmetic for one row: UBO identification = 0.20×5 + 0.20×4 + 0.15×3 + 0.20×2 + 0.15×3 + 0.10×4 = 1.00 + 0.80 + 0.45 + 0.40 + 0.45 + 0.40 = 3.50. It sits exactly on the band boundary; the design decision went to assist because D4 (reversibility) is low and the regulatory consequence of a missed beneficial owner is severe. This is the point to make in an interview: the score informs, the risk dimension decides.

**Analysis 2: effort reduction build.** Reduction assumptions per task were set from the design decision, the pilot evidence base from comparable programmes [ESTIMATE], and analyst challenge.

| Task | Share of direct hours | Assumed reduction | Contribution to direct reduction |
|---|---|---|---|
| Intake | 6% | 60% | 3.6 points |
| Document collection | 12% | 40% | 4.8 points |
| Extraction | 14% | 80% | 11.2 points |
| Registry verification | 5% | 70% | 3.5 points |
| UBO unwrapping | 15% | 35% | 5.3 points |
| Screening execution | 3% | 10% | 0.3 points |
| Alert disposition | 12% | 55% | 6.6 points |
| Risk rating | 3% | 20% | 0.6 points |
| EDD narrative | 12% | 65% | 7.8 points |
| QC | 10% | 50% | 5.0 points |
| Approval | 4% | 15% | 0.6 points |
| Setup and communications | 4% | 60% | 2.4 points |
| Total | 100% | | 51.7 points |

Direct hours fall 51.7%: 1,097,000 × 0.517 = 567,000 hours. Indirect hours fall 35% (less rework and chasing): 375,000 × 0.35 = 131,000 hours. Total 698,000 of 1,472,000 hours, or 47.4%. Human-in-the-loop overhead (QC sampling of agent output, exception handling, agent supervision roles) adds back about 2.5 points, leaving 45% at steady state. Capacity released: 920 × 0.45 = 414 FTE-equivalent; at the GBP 43,600 blended cost that is GBP 18.0m a year [ESTIMATE]. The team was explicit that this is capacity, and that the workforce plan converts it to cost through attrition (UK KYB attrition 11% a year, about 60 FTE), captive flex-down in India, and redeployment of 90 FTE to the periodic-review backlog, which was 14,000 cases overdue.

**Analysis 3: target architecture.** A supervisor-led (hierarchical) multi-agent system.

| Component | Role | Tools | Human gate |
|---|---|---|---|
| Case supervisor agent | Plans the case, sequences workers, tracks state, escalates | Workflow API, case store, policy retrieval | Escalation rules |

| Component | Role | Tools | Human gate |
|---|---|---|---|
| Intake and document agent | Validates application, classifies and extracts from documents | Document store, OCR and extraction, schema validator | Low-confidence fields |
| Registry and UBO agent | Pulls registry data, builds ownership graph, identifies UBOs | Companies House and commercial registry APIs, graph builder | All structures with three or more layers, any trust or nominee |
| Screening disposition agent | Reviews vendor alerts, discounts false positives with evidence | Screening vendor API, adverse media retrieval | All potential true matches; 100% QC during pilot |
| Risk-rating adapter | Assembles inputs and calls the existing CRR rules engine | CRR engine API | None (deterministic) |
| Narrative agent | Drafts EDD and case rationale with citations to evidence | Policy retrieval, case evidence store | Approver edits and signs |
| QC agent | Checks completeness, consistency, policy adherence | Policy checklist, case store | Human QC on risk-based sample |
| HITL console | Presents agent proposals, evidence and confidence to analysts and approvers | | All approvals |

Guardrails: input validation and prompt-injection screening on uploaded documents; output schema enforcement; citation requirement on every narrative claim; action allow-lists (no agent can approve, close an alert as a true match, or change a risk rating); full audit log of prompts, retrieved evidence, outputs and human actions; fallback to manual workflow on any component failure.

**Analysis 4: model risk classification under SS1/23.** The PRA's SS1/23 sets five principles: model identification and risk classification; governance; development, implementation and use; independent validation; and model risk mitigants [PUBLIC — PRA SS1/23, May 2023, effective May 2024; VERIFY scope and any later amendments]. The team's proposal, agreed with the model-risk function:

- Identification: each LLM-based component (extraction, UBO reasoning, alert disposition, narrative, QC) is recorded in the model inventory as a component of one "KYB agentic model system"; the CRR engine keeps its existing inventory entry and tier.
- Classification: Tier 2 (material process, but every regulatory decision retains a human approver and errors are detectable before decision). Alert disposition is flagged for review towards Tier 1 if auto-closure of false positives is expanded beyond the pilot sampling regime.
- Development and use: documented intended use, known limitations, prohibited uses (no approvals, no true-match closure), and vendor model change notification requirements.
- Independent validation: pre-deployment validation against the golden dataset; conceptual soundness review of prompts and retrieval; outcome testing on the shadow run; annual revalidation and trigger-based revalidation on any model version change.
- Mitigants: HITL gates, QC sampling rates by risk, override monitoring, manual fallback, and a kill switch owned by the MLRO.

Also recorded: a Data Protection Impact Assessment for beneficial-owner personal data under UK GDPR; MLRO sign-off on the screening disposition design; and a note that the EU AI Act does not apply to the UK pilot but would for the bank's Irish subsidiary [PUBLIC, VERIFY].

**Analysis 5: token cost.** Blended tokens per case about 220,000 input and 14,000 output (documents, registry extracts, policy context). Routing: mid-tier model for extraction and classification (about 70% of tokens), frontier model for UBO reasoning and narrative (about 30%). Mid-tier: 154,000 × USD 0.80 per million = USD 0.123 plus 9,800 × USD 4 per million = USD 0.039, total USD 0.162. Frontier: 66,000 × USD 3 per million =

USD 0.198 plus 4,200 × USD 15 per million = USD 0.063, total USD 0.261. Sum USD 0.42; caching on policy context saves about 20% and retries plus evaluation runs add about 30%, so USD 0.45 a case, about GBP 0.36. Across 146,000 cases: GBP 53,000 a year [ESTIMATE — order of magnitude, VERIFY pricing]. Against GBP 46m of KYB cost, model pricing is irrelevant to the decision; accuracy and controllability are not.

**Analysis 6: pilot gates.** Pilot scope: SME new onboarding and SME periodic review; 2,000 shadow cases (weeks 12 to 13) then 1,000 live-assist cases (weeks 14 to 15).

| Gate metric | Baseline | Threshold to pass | Result in simulation [ILLUSTRATIVE] |
|---|---|---|---|
| Field-level extraction accuracy vs gold set | 96% (human) | 97% or above | 98.1% |
| UBO resolution agreement with senior analyst | n/a | 92% or above; 100% of layered structures to human | 93.5% |
| Missed true matches on agent-discounted alerts (100% QC) | 0 | 0 | 0 |
| False-positive discount precision | n/a | 98% or above | 98.7% |
| EDD narrative accepted without material edit | n/a | 80% or above | 83% |
| Analyst hours per SME case | 6.0 | 3.6 or below | 3.4 |
| Human override rate on agent proposals | n/a | Below 15%, trending down | 11% |
| SME cycle time, median days | 9 | 4 or below | 4 |
| Reportable incidents and audit-trail gaps | 0 | 0 | 0 |

**Analysis 7: business case.** Build GBP 14m across 16 months (design and pilot GBP 3.2m; scale build GBP 10.8m dominated by integration, change and validation). Run GBP 4.5m a year: platform and infrastructure 1.2, licences and data 0.8, AI team of ten at GBP 110,000 = 1.1, evaluation, monitoring and guardrails 0.5, model validation 0.4, HITL quality uplift 0.5, tokens 0.1 with headroom. Benefit ramp: cost benefit 15%, 55%, 100% in years 1 to 3; revenue benefit 10%, 50%, 100%.

| Year | Build, GBP m | Run, GBP m | Cost benefit, GBP m | Revenue contribution, GBP m | Net, GBP m | Cumulative, GBP m |
|---|---|---|---|---|---|---|
| 1 | 9.0 | 1.5 | 2.7 | 1.0 | -6.8 | -6.8 |
| 2 | 5.0 | 4.5 | 9.9 | 5.0 | 5.4 | -1.4 |
| 3 | 0 | 5.0 | 18.0 | 10.0 | 23.0 | 21.6 |
| 4 | 0 | 4.5 | 18.0 | 10.0 | 23.5 | 45.1 |
| 5 | 0 | 4.5 | 18.0 | 10.0 | 23.5 | 68.6 |

NPV at 10%: -6.8/1.10 + 5.4/1.21 + 23.0/1.331 + 23.5/1.4641 + 23.5/1.6105 = -6.2 + 4.5 + 17.3 + 16.1 + 14.6 = GBP 46.3m. Payback in month 25. Cost-only view, which is what the CFO will ask for: year 1 net -7.8, year 2 +0.4, year 3 +13.0, years 4 and 5 +13.5; NPV GBP 20.6m; payback month 31. The recommendation was made on the cost-only view with revenue as upside.

## Deliverables and slide-by-slide storyline

| Slide | Action title | Key content | Source analysis |
|---|---|---|---|
| 1 | Agentic KYB can halve SME | Executive summary | All |

| Slide | Action title | Key content | Source analysis |
|---|---|---|---|
| | and mid-corporate cycle time and release 45% of KYB effort, with every regulatory decision still made by a human | | |
| 2 | KYB costs GBP 46m a year and loses an estimated GBP 124m of first-year revenue to abandonment | Baseline | Diagnostic |
| 3 | Four tasks consume 53% of analyst time and share the same profile: varied inputs, bounded judgement | Effort shares | Analysis 1 |
| 4 | We scored twelve tasks on six dimensions; eight are fit for agentic automation or assist | Scored matrix | Analysis 1 |
| 5 | Three tasks stay deterministic or human, by design, because reversibility is low | Band decisions | Analysis 1 |
| 6 | A supervisor-led multi-agent system orchestrates seven workers and four human gates | Target architecture | Analysis 3 |
| 7 | No agent can approve a client, close a true match or change a risk rating | Guardrails | Analysis 3 |
| 8 | The system classifies as a Tier 2 model system under SS1/23 with component-level validation | Model-risk pack [PUBLIC, VERIFY] | Analysis 4 |
| 9 | Effort falls 47% before human-in-the-loop overhead and 45% after, releasing 414 FTE of capacity | Effort build | Analysis 2 |
| 10 | Capacity converts to cost through attrition, captive flex and redeployment to the review backlog | Workforce plan | Analysis 2 |
| 11 | Token costs are GBP 53,000 a year; model choice is about accuracy, not price | Token maths | Analysis 5 |
| 12 | The pilot passed all nine gates on 3,000 SME cases | Gate results | Analysis 6 |
| 13 | Analysts overrode 11% of proposals, falling to 8% by week 15 | Override analysis | Analysis 6 |
| 14 | Scaling costs GBP 14m and GBP 4.5m a year to run | TCO | Analysis 7 |
| 15 | On cost alone the programme pays back in | Business case | Analysis 7 |

| Slide | Action title | Key content | Source analysis |
|---|---|---|---|
| | month 31 with NPV of GBP 21m; revenue takes NPV to GBP 46m | | |
| 16 | Mid-corporate is the next segment; complex structures stay assist-only for 18 months | Scale roadmap | Synthesis |
| 17 | Three decisions: scale to mid-corporate, fund the platform, approve the workforce plan | Decisions | Synthesis |
| 18 | Appendix: scoring detail, validation plan, DPIA, gate data | | |

### Recommendation

Scale the supervisor-led system from SME to mid-corporate over the next two quarters, keep complex structures in assist mode until 18 months of override data support expansion, fund the GBP 14m build and GBP 4.5m run from the Corporate Banking change budget, approve the workforce plan that converts 414 FTE of capacity through attrition, captive flex-down and redeployment, and formalise the Tier 2 classification and annual revalidation with the model-risk function.

### Benefits and risks

Benefits: GBP 18m of annual capacity release and an estimated GBP 10m of revenue contribution at steady state; SME cycle time from 9 to 4 days; a clearance of the 14,000-case review backlog within a year using redeployed analysts; a model-risk pattern reusable for AML and sanctions.

Risks and mitigations: a missed beneficial owner or true match (mitigated by hard gates, 100% QC in pilot, risk-based sampling at scale, kill switch); vendor model changes degrading accuracy (mitigated by version pinning, change notification clauses, regression evals); analyst de-skilling (mitigated by rotation through complex cases and QC roles); prompt injection via uploaded documents (mitigated by document sanitisation, isolation of untrusted content, action allow-lists); labour relations in the UK and India (mitigated by early consultation and an attrition-first plan).

## Simulation C: An AI business case for an agentic dispute-resolution platform, defended to a CFO

### Client ask

[ILLUSTRATIVE — composite] “Meridian Card Services” is a US card issuer with 14 million active cards. It receives 600,000 cardholder disputes a year with an average disputed amount of USD 142, so USD 85.2m of disputed value. Disputes operations employ 410 FTE (blended onshore and nearshore) at USD 52,000 loaded, costing USD 21.3m, plus USD 4.2m of technology and vendor cost: USD 25.5m a year, or USD 42.50 per dispute. Write-offs attributable to dispute handling (unrecovered provisional credits, chargebacks filed late or with weak evidence, representment losses) run at USD 9.4m a year, about 11% of disputed value. Only 12% of disputes are resolved without human touch today, mostly low-value automatic write-offs. The COO sponsors an agentic dispute platform; the CFO has refused two previous AI business cases as “capacity dressed up as savings”.

The ask from the COO: “Build the business case the CFO will sign. Eight weeks. Every number has to survive her.”

Engagement shape: 8 weeks; partner (15%), engagement manager, two consultants, one AI engineer for architecture and cost, one Finance-seconded analyst to tie out baselines. Illustrative fee USD 650,000 to 900,000 [ESTIMATE].

## Diagnostic

**Baseline tie-out.** The team reconciled the USD 25.5m to the general ledger cost centres with the seconded Finance analyst in week 2, producing a signed baseline. Cost per dispute USD 42.50; handle time 1.24 hours per human-handled dispute (656,000 productive hours over 528,000 human-handled disputes); 38% of disputes fraud-coded, 62% merchant or service disputes; 2.1% of disputes escalate to a formal complaint.

**Regulatory frame.** Regulation E requires provisional credit within 10 business days where an investigation runs longer, and completion of the investigation within 45 days (90 in defined cases); Regulation Z sets billing-error resolution within two billing cycles and no more than 90 days [PUBLIC — 12 CFR 1005.11 and 12 CFR 1026.13; VERIFY current text]. Late filing against network chargeback time limits is a direct loss driver. These timelines make cycle time a compliance metric, not only a cost metric.

**Process.** Twelve L3 steps from intake to closure; the heaviest are evidence gathering across transaction, merchant and authorisation data (29% of handle time), customer correspondence (18%), reason-code classification and network filing (17%), and investigation write-up (14%).

**Loss drivers.** Of the USD 9.4m: USD 3.9m from chargebacks not filed or filed late; USD 3.1m from representments lost for weak evidence; USD 2.4m from provisional credits on low-value disputes written off under policy.

## Hypotheses

1. **H1.** At least half of disputes can reach a resolution proposal without human handling by year 3, with human sign-off retained on fraud-coded and high-value cases. Result: confirmed at 50%, with 60% as upside.
2. **H2.** Faster, better-evidenced network filing reduces dispute write-offs by at least 20%. Result: confirmed at 25%, applying only to the late-filing and weak-evidence pools.
3. **H3.** The CFO will accept capacity release only where a workforce mechanism converts it to cost within 18 months. Result: confirmed; attrition of 14% a year (about 57 FTE) plus nearshore contract flexibility covers 100% of the planned release by year 3.
4. **H4.** Model costs are below 1% of run cost. Result: confirmed; about USD 0.17 per dispute.
5. **H5.** A hybrid of buying the network-filing connector and building the orchestration beats a pure SaaS purchase. Result: confirmed on scope coverage and three-year cost.

## Workstreams

| Workstream | Scope | Outputs |
|---|---|---|
| 1. Baseline and value levers | GL tie-out, process metrics, loss decomposition | Signed baseline |
| 2. Solution and TCO | Target architecture, build and run cost, buy comparison | TCO model |
| 3. Benefits and ramp | Straight-through model, effort build, loss model, ramp curve | Benefit model |
| 4. Financial case and CFO challenge | NPV, IRR, payback, scenarios, tornado, pre-empted challenges | Business case and challenge log |

## Week-by-week plan

| Week | Focus | Key activities | Milestone |
|---|---|---|---|
| 1 | Mobilise | Hypotheses, Finance secondment, data pull of 12 months of disputes | |
| 2 | Baseline | GL tie-out, handle-time analysis, loss decomposition | Signed baseline |
| 3 | Solution | Target architecture, straight-through eligibility rules, HITL design | |
| 4 | Cost | Build and run TCO, token modelling, buy option quotes | CFO pre-read 1 |
| 5 | Benefits | Effort build, loss model, ramp, workforce mechanism | |
| 6 | Case | NPV, IRR, payback, risk-adjusted case, tornado, scenarios | CFO pre-read 2 |
| 7 | Challenge | Mock CFO session with Finance, revisions, challenge log | |
| 8 | Decide | CFO and COO decision meeting, investment committee paper | Decision |

## Key analyses

**Analysis 1: target design in one table.**

| Agent | Function | Autonomy |
|---|---|---|
| Intake and triage agent | Captures dispute, classifies reason code, checks eligibility rules | Automated; fraud-coded and over USD 500 flagged |
| Evidence agent | Gathers transaction, authorisation, device and merchant data; queries merchant collaboration networks | Automated |
| Resolution agent | Proposes outcome (credit, deny, chargeback) with evidence summary | Automated below USD 500 non-fraud; human sign-off otherwise |
| Filing agent | Files chargebacks and representments in network format within time limits | Automated with deterministic validation |
| Communications agent | Drafts regulatory notices and customer updates | Template-bound, human release on denials |
| Supervisor | Orchestrates, tracks Reg E and Z clocks, escalates | Escalation rules |

**Analysis 2: effort and straight-through build.** Straight-through eligibility at year 3: non-fraud disputes under USD 500 with complete evidence and a confident classification, plus fraud disputes with confirmed compromised-card markers; modelled at 50% of volume. Human-handled disputes fall from 528,000 to 300,000, and handle time on those falls 30% to 0.87 hours because evidence and drafting arrive prepared. Hours: 300,000 × 0.87 = 260,000; QA sampling of 10% of straight-through cases at 0.25 hours = 7,500; total 267,500 hours, or 167 FTE, plus a 12-FTE supervision and exceptions team. Future state 179 FTE against 410 today: a reduction of 231 FTE, 56%, worth USD 12.0m a year at USD 52,000 [ESTIMATE].

**Analysis 3: loss reduction.** Applied only to the late-filing (USD 3.9m) and weak-evidence (USD 3.1m) pools, USD 7.0m. Filing within time limits and evidence completeness were modelled to recover 34% of that pool, USD 2.4m a year, about 25% of total write-offs. The low-value write-off pool was excluded because it is a policy choice, not a handling failure; the CFO was told this explicitly. [ESTIMATE]

**Analysis 4: other benefits.** Complaint escalations fall from 2.1% to 1.4% of disputes, avoiding USD 0.8m of complaint handling and remediation; retirement of a legacy case-tracking licence saves USD 0.3m. Total gross run-rate benefit: 12.0 + 2.4 + 0.8 + 0.3 = USD 15.5m a year [ESTIMATE].

**Analysis 5: TCO.** Build, over ten months: integration with card platform, network interfaces and case tool USD 2.5m; agent engineering and orchestration USD 2.0m; golden dataset, evaluation harness and guardrails USD 0.8m; change, training and role redesign USD 0.7m; model-risk validation USD 0.4m; programme management USD 0.4m. Total USD 6.8m. Run, a year at scale: platform and infrastructure 1.2; vendor connectors and merchant-network fees 0.9; AI engineering team of eight at USD 160,000 = 1.3; monitoring, evaluation and guardrail operations 0.6; model-risk revalidation 0.3; tokens 0.1. Total USD 4.4m, plus a USD 0.6m model refresh in year 3. [ESTIMATE]

Token arithmetic per dispute: about 60,000 input tokens and 6,000 output tokens across the agent chain. Mid-tier model for 70% of steps: 42,000 × USD 0.80 per million = USD 0.034 plus 4,200 × USD 4 per million = USD 0.017, total USD 0.051. Frontier model for 30%: 18,000 × USD 3 per million = USD 0.054 plus 1,800 × USD 15 per million = USD 0.027, total USD 0.081. Sum USD 0.13; add 30% for retries and evaluation runs: USD 0.17 a dispute; 600,000 disputes: USD 0.10m a year, about 2% of run cost [ESTIMATE — order of magnitude, VERIFY pricing]. Cross-check: Module 8 models a different composite issuer with the same 600,000 volume and arrives at USD 0.18 per dispute with tiered routing, a build of USD 5.4m and run of USD 2.7m a year; the two sit within the same order of magnitude, which is the test a CFO will apply.

**Analysis 6: the financial case.** Ramp: 20% of run-rate benefit in year 1, 60% in year 2, 100% from year 3. Year 1 run cost is half a year, USD 2.5m, as the platform goes live mid-year.

| Year | Build, USD m | Run, USD m | Benefit, USD m | Net, USD m | Cumulative, USD m |
|---|---|---|---|---|---|
| 0 | 6.8 | 0 | 0 | -6.8 | -6.8 |
| 1 | 0 | 2.5 | 3.1 | 0.6 | -6.2 |
| 2 | 0 | 4.4 | 9.3 | 4.9 | -1.3 |
| 3 | 0 | 5.0 | 15.5 | 10.5 | 9.2 |
| 4 | 0 | 4.4 | 15.5 | 11.1 | 20.3 |
| 5 | 0 | 4.4 | 15.5 | 11.1 | 31.4 |

NPV at 10%: -6.8 + 0.6/1.10 + 4.9/1.21 + 10.5/1.331 + 11.1/1.4641 + 11.1/1.6105 = -6.8 + 0.55 + 4.05 + 7.89 + 7.58 + 6.89 = USD 20.2m. IRR about 66% (the discount rate at which the stream sums to zero; at 60% it is +0.8, at 70% it is -0.5). Payback in month 26 from the start of the build year.

Risk-adjusted case, applying a 30% haircut to all benefits, which is the evidence-level haircut for a pilot-stage pattern: benefits 2.2, 6.5, 10.9, 10.9, 10.9; net -6.8, -0.3, 2.1, 5.9, 6.5, 6.5; NPV USD 7.4m; IRR about 34%; payback month 34. The team presented both and recommended on the risk-adjusted case.

**Analysis 7: sensitivity (tornado), NPV impact on the USD 20.2m base.**

| Driver | Downside assumption | NPV, USD m | Upside assumption | NPV, USD m |
|---|---|---|---|---|
| Capacity converted to cost | 60% converted | 7.1 | 100% (base) | 20.2 |
| Straight-through rate at year 3 | 35% | 13.1 | 60% | 24.8 |
| Ramp timing | Six-month delay | 13.1 | On plan | 20.2 |

| Driver | Downside assumption | NPV, USD m | Upside assumption | NPV, USD m |
|---|---|---|---|---|
| | | | | |
| Loss reduction | Nil | 13.6 | 35% of pool | 22.5 |
| Build cost | +40% overrun | 17.5 | On budget | 20.2 |
| Token prices | Three times base | 19.7 | Base | 20.2 |

The ranking is the message: whether capacity becomes cost matters three times more than whether the technology works as well as planned, and model pricing is irrelevant to the decision.

**Analysis 8: buy comparison.** A dispute-automation SaaS quoted at USD 1.90 per dispute (USD 1.14m a year) plus USD 1.5m implementation covered intake, rules and network filing and was modelled to reach 30% straight-through with limited evidence reasoning. Three-year cost USD 4.9m against USD 15.3m for the build, but three-year benefit of USD 13m against USD 28m. The recommendation was a hybrid: buy the network-filing connector (USD 0.4m a year, included in run cost), build orchestration, evidence and resolution agents where the differentiation and the straight-through rate are won. [ILLUSTRATIVE]

**Analysis 9: the CFO's eight challenges and the answers given.**

1. **"This is capacity, not cost. Show me where USD 12m leaves the P&L."** Answer: 231 FTE over three years. Attrition at 14% is about 57 FTE a year, 170 over three years, replaced only by exception; nearshore contract flex covers the remaining 61 in year 3. Headcount glide path by quarter attached, co-owned by the COO and HR. The 60% conversion downside is in the tornado at NPV USD 7.1m, still positive.
2. **"Your baseline is inflated."** Answer: USD 25.5m tied to GL cost centres and signed by your seconded analyst in week 2; USD 42.50 a dispute reconciles to the cost-per-dispute metric your team already reports.
3. **"Your ramp is optimistic."** Answer: 20/60/100 is below the volume ramp Klarna reported publicly in 2024, and we have priced the quality correction they later made by keeping a 12-FTE exceptions team and a quality floor [PUBLIC — Klarna, 2024 and 2025, VERIFY]. A six-month delay takes NPV to USD 13.1m.
4. **"Where are the run costs? The last case had none."** Answer: USD 4.4m a year, itemised; people and platform, not tokens. Tokens are USD 0.1m; tripling them moves NPV by USD 0.5m.
5. **"You are double counting with the fraud programme's loss savings."** Answer: the loss benefit applies only to the late-filing and weak-evidence pools, USD 7.0m of the USD 9.4m; fraud-model savings sit in a different pool and a different cost centre. Attribution map attached.
6. **"What happens when Reg E or a CFPB examiner looks at this?"** Answer: the supervisor tracks Reg E and Z clocks per case, every automated resolution carries an evidence record and a reason, denials require human release, and the system is classified under model risk with validation before go-live [PUBLIC — Reg E and Z, VERIFY]. Cycle-time compliance improves, which is a benefit we have not monetised.
7. **"You are locking us into a model vendor."** Answer: model-agnostic gateway, two models qualified, regression evaluation suite to requalify a replacement in under four weeks, and a USD 0.6m refresh in year 3 already in the case.
8. **"Why not buy the vendor's product for a quarter of the cost?"** Answer: it reaches 30% straight-through, not 50%, and three-year benefit of USD 13m against USD 28m; we buy the filing connector and build the parts that win the straight-through rate.

A ninth challenge came from the CRO rather than the CFO: "Easier disputes invite first-party fraud." The answer was a first-party-abuse model in the triage agent, velocity limits per cardholder, and a monitored dispute-rate KPI with a trigger to tighten eligibility.

## Deliverables and slide-by-slide storyline

| Slide | Action title | Key content | Source analysis |
|---|---|---|---|
| 1 | An agentic dispute platform returns USD 7m of risk-adjusted NPV and pays back in month 34, with cost release secured through attrition | Executive summary | All |
| 2 | Disputes cost USD 25.5m and write off USD 9.4m a year; the baseline is tied to the GL | Baseline | Diagnostic |
| 3 | Evidence gathering, correspondence and filing consume 64% of handle time | Process metrics | Diagnostic |
| 4 | Six agents under a supervisor resolve half of disputes without human handling by year 3 | Target design | Analysis 1 |
| 5 | Humans keep every fraud-coded, high-value and denial decision | Autonomy and gates | Analysis 1 |
| 6 | Effort falls 56%, from 410 to 179 FTE, worth USD 12m a year | Effort build | Analysis 2 |
| 7 | Capacity converts to cost through attrition and contract flex within three years | Workforce glide path | Analysis 2 and challenge 1 |
| 8 | Better-evidenced, on-time filing recovers USD 2.4m of write-offs a year | Loss model | Analysis 3 |
| 9 | The platform costs USD 6.8m to build and USD 4.4m a year to run; tokens are 2% of run cost | TCO | Analysis 5 |
| 10 | Base case NPV is USD 20m; risk-adjusted NPV is USD 7m | Financial case | Analysis 6 |
| 11 | The case turns on converting capacity, not on the technology | Tornado | Analysis 7 |
| 12 | A hybrid beats pure SaaS by USD 15m of three-year benefit | Buy comparison | Analysis 8 |
| 13 | Regulatory clocks and evidence records improve compliance, unmonetised | Controls | Analysis 9 |
| 14 | We answer the eight challenges raised on prior cases | Challenge log | Analysis 9 |
| 15 | Decision: approve USD 6.8m and the workforce plan; gate scale on month-12 metrics | Decision | Synthesis |

| Slide | Action title | Key content | Source analysis |
|---|---|---|---|
| 16 | Appendix: GL tie-out, model detail, vendor quotes, scenario tables | | |

### Recommendation

Approve USD 6.8m of build investment and USD 4.4m of annual run cost on the risk-adjusted case (NPV USD 7.4m, IRR about 34%, payback month 34), with scale beyond 35% straight-through gated on month-12 evidence; adopt the attrition-first workforce plan co-owned by the COO and HR; buy the network-filing connector and build the orchestration and evidence agents; and classify the system under model risk with validation before go-live.

### Benefits and risks

Benefits: USD 15.5m gross and USD 10.9m risk-adjusted run-rate; cost per dispute from USD 42.50 to about USD 24 at scale [ESTIMATE]; faster regulatory clock compliance; a reusable evidence and filing capability for fraud operations.

Risks and mitigations: capacity not converted (attrition-first plan, quarterly glide-path review); first-party fraud (abuse model, velocity limits, dispute-rate trigger); regulatory finding on automated denials (human release on all denials, evidence records); vendor model change (gateway, regression suite); customer harm from wrong automated credits or denials (quality floor, complaint-rate trigger, sampling).

## How to talk about these simulations in an interview

Interviewers want three things from an engagement walkthrough: structure, numbers you can defend, and judgement about what mattered. Use the following pattern for any of the three simulations.

- **Open with the ask and the shape.** “A mid-sized European bank asked for an AI strategy and a first-wave decision in twelve weeks.” One sentence, then the team and the deliverable.
- **State the diagnostic finding as a number.** “Maturity 1.75 against a peer median of 2.7, with technology the widest gap; EUR 385m of cost in language-heavy processes.”
- **Give the hypothesis that drove the work and whether it held.** “We hypothesised two-thirds of value sat in six operations domains, not customer-facing GenAI; it did, 68% on the risk-adjusted view.”
- **Describe one analysis in depth.** Pick the scored matrix, the agentic-fit scoring or the tornado, and walk the arithmetic. This is where you show you own the numbers.
- **State the recommendation and the decision the client took.** “EUR 60m over five years, three waves, hub-and-spoke; the board approved wave 1 and the platform.”
- **Close with what you would do differently.** For Simulation A, start data remediation in week 1. For B, bring the MLRO into week 1 rather than week 4. For C, secure the workforce mechanism before building the benefit model.

Make your own role explicit in every sentence that matters: “I ran the scoring workshop”, “I built the TCO model”, “I presented the tornado to the CFO”. Interviewers probing strategy versus operations are listening for who owned the judgement calls, not who wrote the code.

## Self-check quiz

1. In Simulation A, recompute the five-year NPV if wave 2 slips by a year, assuming year 3 benefit falls from EUR 42m to EUR 24m and year 4 from EUR 48m to EUR 42m, with all other figures unchanged. Show the arithmetic.

2. In Simulation B, the UBO identification task scored exactly 3.50, on the boundary between “automate” and “assist”. State which dimension decided the design and why, and name the SS1/23 principle most relevant to that decision.
3. In Simulation C, explain why the CFO’s challenge on capacity conversion matters more to NPV than the straight-through rate, using the tornado figures.
4. Across all three simulations, token costs were below 2% of run cost. Give the per-case token arithmetic for Simulation C and explain what this means for how you present model choice to a client.
5. Simulation C excluded the USD 2.4m low-value write-off pool from the loss benefit. Why, and what does that choice signal to a CFO?

# Module 19 — Positioning Your Own Experience as an AI Strategy Consultant

## 1. Why a client cares

A partner selling an AI engagement is selling judgement, not hours. The client wants to know that the person across the table has made the hard calls before: where to draw agent boundaries, when to stop a pilot, how to defend a number to a CFO. Interviewers are proxies for that client. When they probe "what was *your* role", they are testing whether you own decisions or only execute them. Your three projects already contain strategy-grade decisions. This module makes them visible, honestly, and in the language a manager-level AI strategy consultant uses.

## 2. Concept and mechanism

### How an MBB-style interviewer scores an experience story

Experience interviews at MBB and MBB-adjacent AI practices (QuantumBlack, BCG X, Bain Vector) are scored on four things, in roughly this priority order. Attribution: this is a practitioner synthesis of published MBB interview guidance and partner practice; the exact rubrics are proprietary and vary by firm.

| Dimension | What they listen for | Red flag |
|---|---|---|
| Ownership | "I decided", "I recommended", "I pushed back" | Passive voice; "the team did" throughout |
| Judgement | Trade-offs you weighed and why you chose one side | Listing activities without a decision |
| Impact | A quantified or clearly described client outcome you can defend | Vague "improved efficiency" |
| Altitude | Can you move between task detail and the client's P&L in one breath | Stuck at task level, or stuck at slogan level |

The "strategy vs ops" probe is really an altitude test. An ops answer stays at task level ("I mapped the process"). A strategy answer connects the same task to a decision the client had to make ("I mapped the process to establish which steps had the judgement profile that justified an agent rather than a rule, which set the scope and the business case").

### The three mechanisms that convert an implementer story into a strategy story

1. **Decision ledger.** For each project, list every decision made during the engagement, who made it, and which ones you influenced with analysis. Interviewers reward decisions you shaped even when a senior person signed them off. "I built the analysis that led the partner to recommend X" is a legitimate ownership claim.
2. **Altitude ladder.** Every task sits on a ladder: task → workstream output → engagement recommendation → client outcome. Practise stating your contribution at all four rungs in two sentences. Your as-is process mapping (task) produced a pain-point heatmap (workstream output) that determined which three steps went into the agentic scope (recommendation) and therefore the size of the benefit case (client outcome).
3. **De-risking framing.** Prototypes, process maps and scoring frameworks are strategy artefacts when they remove uncertainty from a decision. Your vibe-coded prototype was not "coding"; it was a cheap experiment that tested the agent-boundary hypothesis before the client committed engineering budget. That is exactly what a strategy consultant should do before writing a business case.

### Honesty boundary

You may claim: analysis you performed, recommendations you drafted, decisions you influenced, artefacts you built, client conversations you led or co-led. You may not claim: decisions you were not in the room for, production outcomes you did not measure, or work by engineers you did not direct. Where you do not know a number, say so and give the range the team used. Interviewers at this level check references and can smell inflation; an honest "my part was X, the team's result was Y" scores higher than an inflated "I delivered Y".

## 3. Frameworks

### Framework A: STAR to consulting-narrative conversion

STAR (Situation, Task, Action, Result) is the standard behavioural-interview structure; attribution uncertain, widely used in HR practice since the 1980s. It is fine for HR screens and weak for consulting interviews because it has no slot for the decision or the hypothesis. Convert it as follows.

| STAR element | Consulting narrative element | What changes |
|---|---|---|
| Situation | Client ask and why it mattered to the P&L or regulator | Add the business stake and the client sponsor's level |
| Task | The question we had to answer, stated as a hypothesis | Replace "I was asked to" with "the question was whether" |
| Action | Key decisions and why, with the analysis behind each | Replace activity list with two or three decisions and trade-offs |
| Result | Client benefit, your personal contribution, what you would do differently | Add the "so what" and a reflective point |

The reflective point is deliberate. Partners trust people who can name a weakness in their own work.

### Framework B: the four-rung altitude ladder

State every project at four rungs. Practise until each rung is one sentence.

1. **Task rung:** what you physically did.
2. **Workstream rung:** what output the workstream produced.
3. **Engagement rung:** what recommendation that output enabled.
4. **Client rung:** what the client gained, in money, risk or time.

### Framework C: the probe-defence table

For each project, anticipate the five most likely probes and prepare a weak answer (to recognise and avoid) and a strong answer. The tables below are your drafts; refine with real figures.

### Framework D: the experience fact sheet

Before interviews, fill one page per project with only facts you can defend. Template at the end of Section 4.

## 4. Worked BFSI example: your three projects, rewritten

All three narratives below are [ILLUSTRATIVE — composite] in their numbers. The square-bracket placeholders must be replaced with real figures from your engagement files, or deleted. Do not insert numbers you cannot defend.

### Project A: Agentic-fit assessment of corporate KYB at a UK Tier-1 bank

**30-second version.**

> *At a UK Tier-1 bank we were asked which parts of corporate onboarding, the KYB process, were genuinely suited to agentic AI rather than conventional automation. I decomposed the process to task level and scored each step on a six-dimension agentic-fit framework covering task variability, judgement required, system access, error tolerance and reversibility, data availability, and volume. That scoring was the basis for the scope the bank adopted: a supervisor-led multi-agent design covering [N] of [M] steps, with human approval retained on the risk-rated decisions. The business case rested on that scope, so the scoring was where the value was decided.*

**2-minute version.**

*Client ask.* The bank's corporate onboarding took [X] days end to end and consumed [Y] FTE across document collection, entity resolution, UBO (ultimate beneficial owner) identification, screening adjudication and risk rating. The COO sponsor wanted to know whether agentic AI, which several vendors were pitching, would actually change the economics, or whether it was conventional automation with a new label.

*What we delivered.* A task-level decomposition of KYB into [approximately 25] steps; a six-dimension agentic-fit score for each; a target design of a supervisor-led (hierarchical) multi-agent system with specialist agents for document intake, entity and UBO resolution, screening adjudication and case narrative, under an orchestrating supervisor; a human-in-the-loop design that kept risk rating and any adverse decision with an analyst; and the inputs to the business case.

*Key decisions and why.* First, we rejected the vendor's "agent for everything" scope. Roughly [a third] of steps were deterministic, for example document completeness checks and registry lookups, and belonged in rules or API integrations; putting them in an agent would have added cost and non-determinism for no gain. Second, we chose a supervisor-led pattern rather than peer-to-peer agents because the bank's model risk team needed one accountable control point for every decision path; a hierarchical design gives you one place to log, test and explain. Third, we kept the final risk rating human because the error tolerance dimension scored lowest there and the regulatory consequence of a wrong onboarding decision is asymmetric.

*My personal role.* I owned the decomposition and the scoring. I ran the SME sessions that produced the step-level inputs, calibrated the six dimensions with the risk function so that the scores would survive model-risk scrutiny, and wrote the scoring rationale that went into the steering committee paper. The target architecture was designed with the engineering lead; I owned the question of what went inside the agent boundary and what stayed outside it.

*Client benefit.* The adopted scope targeted [Y%] of handling effort and a cycle-time reduction from [X] to [X'] days. [State only the figures from the actual business case; if you only know the range, give the range and say so.] The structural benefit was that the bank had a defensible, reusable method for deciding where agents belong, which it later applied to [other processes if true].

*What I would do differently.* I would have brought model risk into the scoring calibration a week earlier. We lost time re-justifying dimension weights after the fact.

**Probe-defence table.**

| Likely probe | Weak answer | Strong answer |
|---|---|---|
| "Who designed the six dimensions?" | "It was the firm's framework" | "We adapted a practitioner framework; I led the calibration of weights with the client's risk function, which is where the real judgement sat, because the weights decided scope" |
| "Why supervisor-led and not a single large agent?" | "That is the standard pattern" | "Accountability and testability. One supervisor gives model risk one control point and lets us test specialist agents |

| Likely probe | Weak answer | Strong answer |
|---|---|---|
| | | in isolation against golden sets” |
| “What did you personally decide?” | “I supported the team” | “I decided which steps were scored as rules-not-agents and defended that to the vendor and the sponsor” |
| “What went wrong?” | “Nothing major” | “Our first weights overvalued volume; two high-volume deterministic steps scored as agent-fit until risk pushed back. We re-weighted error tolerance” |
| “Did it go live?” | Overclaim | State exactly what stage it reached and what was measured |

## Project B: Agentic AI points of view across disputes, sanctions, KYC and credit underwriting

**30-second version.**

> *I worked on a set of agentic AI points of view for core banking operations: disputes, sanctions screening, KYC and credit underwriting. On underwriting I owned the as-is diagnostic, running SME interviews to map the end-to-end process and surface pain points. That diagnostic is what decides whether an agentic proposition is credible, because it tells you which pain points are judgement problems an agent can address and which are data or policy problems that no model fixes. The output shaped the proposition’s scope and its value hypothesis.*

**2-minute version.**

*Client ask.* [Internal practice-building or client-funded; state which.] The firm needed defensible, bank-ready points of view on where agentic AI changes the economics of four operational processes, so that partners could open client conversations with a hypothesis rather than a generic deck.

*What we delivered.* For each process: an as-is process map, a pain-point analysis, a target agentic pattern, a value hypothesis and the risk and control implications. For underwriting I ran [N] SME interviews across credit analysts, relationship managers and credit risk, and produced the as-is map and pain-point heatmap.

*Key decisions and why.* The underwriting diagnostic surfaced that the largest pain point was not analysis time but information assembly: analysts spent [a share] of their time gathering and normalising financial statements, covenant data and bureau information from disconnected systems. That shifted the proposition from “AI writes the credit memo” to “an orchestrated set of agents assembles, reconciles and pre-populates, with the analyst owning the judgement”. The second decision was to treat credit decisioning itself as out of scope for autonomy because it is a regulated model under SR 11-7 style regimes and, for natural persons in the EU, a high-risk use under the EU AI Act Annex III [PUBLIC — Regulation (EU) 2024/1689, 2024; VERIFY status]. Positioning the agent as a pre-processor rather than a decider kept the proposition inside what a bank’s model risk function could approve.

*My personal role.* I owned the as-is diagnostic for underwriting end to end: interview guide, interviews, synthesis, pain-point heatmap, and the “what this means for the agentic scope” section. I contributed the payments and disputes process knowledge to the disputes point of view.

*Client benefit.* [If client-funded: state the outcome.] For the practice: a proposition with a specific, defensible value hypothesis; propositions built on a real as-is diagnostic are the ones that survive a client COO’s first question, which is always “do you understand how we actually do this today”.

*What I would do differently.* Quantify the pain points during the interviews, not afterwards. Asking “how many hours a week” in the room is worth more than estimating it later.

**Probe-defence table.**

| Likely probe | Weak answer | Strong answer |
|---|---|---|
| “So you did process mapping, that is ops work” | “Yes, I did the mapping” | “The map was the evidence base for a scoping decision: it moved the proposition from memo-writing to information assembly, which is where the value and the regulatory room were” |
| “What was the biggest pain point?” | Generic list | One specific finding with the share of analyst time, and the decision it drove |
| “Why not automate the credit decision?” | “Too risky” | “Regulatory classification: high-risk under the EU AI Act for natural persons and a Tier-1 model under most bank MRM frameworks. Autonomy there needs validation effort that would wipe out the case” |
| “How did you validate SME claims?” | “We trusted them” | “Triangulated with system timestamps and case samples where available, and labelled the rest as SME estimate” |
| “What was the value hypothesis?” | No number | A ranged estimate with the lever named, labelled as hypothesis |

## Project C: Working multi-agent dispute-resolution prototype

**30-second version.**

> *I built a working multi-agent prototype for card dispute resolution: an orchestrator, scoped specialist agents with their own system prompts, retrieval over scheme rules with deliberate chunking choices, a rules engine for the deterministic checks and human-in-the-loop gates on the decisions that mattered. I built it to settle design questions cheaply before anyone committed engineering budget: where the agent boundary sits, what must stay in rules, and where a human has to approve. Those answers are what went into the scoping and risk discussions.*

**2-minute version.**

*Client ask or purpose.* [State whether this was for a client, a pursuit, or practice development.] The open questions were design questions that decks cannot answer: does an orchestrator-plus-specialists design actually hold together on real dispute types; which steps break when given to a model rather than a rule; and how much human review the design needs to be safe.

*What we delivered.* A working prototype with an orchestrator agent, specialist agents for intake and classification, evidence gathering, scheme-rule application and customer communication, a retrieval layer over scheme dispute rules, a rules engine for reason-code validity and timeframe checks, and human-in-the-loop gates on write-off decisions and customer-facing outcomes.

*Key decisions and why.* First, reason-code validity and chargeback timeframes went into a rules engine, not an agent, because they are deterministic and a model’s error rate on them is pure downside. Second, chunking of the scheme rules was by rule clause rather than by fixed token window, because retrieval quality on clause-level questions was materially better and the agent’s citations became auditable. Third, the human gate sat on financial outcomes above a threshold and on any customer-facing adverse decision, mirroring how a bank’s operational risk function would want it. Fourth, each specialist agent had a narrow system prompt and a restricted tool list; a broad “dispute agent” with all tools hallucinated actions it had no business taking.

*My personal role.* I designed and built the prototype myself using AI-assisted coding. The design decisions were mine; I validated them against [N] sample dispute scenarios and documented the failure modes.

*Client benefit.* The prototype converted architecture debates into evidence. It let the team scope the proposition to the agent-suited steps, size the human review effort realistically, and show a sponsor a working flow rather than a diagram. [If any measured results exist, for example accuracy on sample cases, state them with the sample size; otherwise say "qualitative".]

*What I would do differently.* Build the evaluation set first. I retrofitted a golden set of scenarios after the prototype worked, which is the wrong order.

**Probe-defence table.**

| Likely probe | Weak answer | Strong answer |
|---|---|---|
| "Vibe coding is not strategy" | Defensive | "Agreed, and I did not build it to ship. I built it to de-risk three scoping decisions before the client spent engineering money. That is what a strategy team should do" |
| "Why a rules engine inside an agentic system?" | "Hybrid is best practice" | "Determinism where the answer is deterministic. Timeframes and reason codes have one right answer; an agent adds cost and error there" |
| "How did you measure it?" | "It worked" | Sample size, scenario coverage, failure modes documented; honest about what was not measured |
| "Would it survive production?" | "Yes" | "No. It would need an eval harness, observability, model-risk validation and integration with the case system. Its value was in settling the design, not in being production code" |
| "What broke?" | "Nothing" | A specific failure: the broad agent attempting out-of-scope actions, the fixed-window chunking giving weak citations |

## Section A: answering "were you strategy or implementation?"

The question is a trap only if you pick a side. The manager-level answer owns both and states the altitude at which you add most value.

> *"Both, deliberately, and in a specific order. My strategy work is where I spend most of my time: scoping which processes and steps justify agentic AI, framing the business case, and setting the human-in-the-loop and risk posture. I go into implementation detail for one reason: an AI strategy that has not been tested against how the system actually behaves is a slide, not a strategy. The prototype work and the process mapping are how I make sure the recommendations I give a COO will survive contact with their model risk team and their engineers. If you need someone to run a production engineering team, that is not my role. If you need someone to decide what gets built, defend why, and know when the engineers are telling you something the business case has missed, that is."*

Three follow-ups to prepare:

- "Give an example where implementation knowledge changed a strategy recommendation." Use the rules-vs-agent decision from Project C, or the information-assembly finding from Project B.
- "Give an example where you stayed out of the detail." Use the point where you handed the architecture to the engineering lead in Project A and owned only the agent boundary.
- "Where do you want to be in three years?" Leading AI engagements end to end: diagnostic, scoping, business case, delivery oversight, with engineering partners owning the build.

## Section B: payments depth as an AI differentiator

Most AI consultants come from data science or generic digital. Deep payments knowledge is rare in AI practices and is directly relevant. Position it in three ways.

| Payments asset | Why it matters for AI | How to say it |
|---|---|---|
| ISO 20022 expertise | Structured, richer remittance and party data is an AI enabler: better entity resolution, better sanctions screening context, fewer free-text investigations | "ISO 20022 migration gives banks the structured data that AI in payments operations has been missing. I know which fields matter and which banks have actually populated them" |
| Real-time payments | Fraud and authorised push payment decisions must happen in sub-second windows; this constrains model latency, favours small models and pre-computed features, and shapes where agents can and cannot sit | "In RTP the decision window is milliseconds. That rules out heavy LLM calls in the authorisation path and pushes agentic AI to pre-transaction education and post-transaction investigation" |
| Cross-border and investigations | Exceptions and investigations are text-heavy, multi-party and rule-bound: an ideal agentic domain, and one where you know the message flows and the SLAs | "Cross-border investigations are a textbook agentic use case: structured messages, documented rules, high volume, human sign-off at the end. I have mapped these flows for years" |
| Cards and disputes | Scheme rules are codified, dispute volumes are large, and the human judgement is narrow and describable | "Scheme dispute rules are effectively a rulebook. That is why a hybrid rules-plus-agents design works, and I have built one" |
| Open banking | Consent, data access and API maturity determine data readiness for AI use cases in lending and servicing | "Open banking data is the feasibility dimension most AI assessments in retail lending get wrong" |

## Section C: experience fact sheet template

Fill one per project. Only facts you can defend.

| Field | Entry |
|---|---|
| Client type and geography | [UK Tier-1 bank, corporate banking] |
| Sponsor level | [COO / Head of Onboarding] |
| Engagement type and length | [Design and scoping, N weeks] |
| Team size and my position | [N consultants, I owned workstream X] |
| The question we answered | [One sentence] |
| Baseline facts | [Cycle time, FTE, volumes, error rates, with source] |
| Key analyses I owned | [List] |
| Decisions I influenced | [List with the analysis that drove each] |
| Recommendation | [One sentence] |
| Stage reached | [Scoping / pilot approved / pilot live / scaled] |
| Measured results | [Only measured; label estimates] |
| What I would do differently | [One sentence] |
| Public cases I would pair it with | [e.g. HSBC AML AI for screening; Klarna for service automation lessons] |

### Section D: pairing your projects with public cases

Pairing an own-experience story with a public case signals that you read the market and can benchmark. Use the anchors from Module 17.

- KYB and screening: pair with HSBC's AML AI work with Google Cloud, where Google reported 2–4x more suspicious activity identified and ~60% fewer alerts [PUBLIC — Google Cloud, June 2023; VERIFY].
- Disputes and servicing automation: pair with Klarna's 2024 assistant claims and its 2025 partial reversal on service quality [PUBLIC — Klarna, Feb 2024; CEO interviews, May 2025; VERIFY]. The lesson you draw: automation scope must be set by error tolerance, not by cost ambition, which is exactly what your HITL gates did.
- Workforce and adoption: pair with Commonwealth Bank of Australia's 2025 reversal of call-centre redundancies when assumed volume reductions did not materialise [PUBLIC — Aug 2025; VERIFY]. The lesson: baseline the volume effect before taking out the cost.

## 5. Real-world case

[PUBLIC — Morgan Stanley, Sept 2023 and June 2024; VERIFY] Morgan Stanley's adviser-facing assistant built with OpenAI and its later "Debrief" meeting tool are the public case most analogous to a scoping-led approach. The firm started with a narrow, well-bounded use (retrieval over its own research library for advisers), measured adoption (reported at roughly 98% of adviser teams), and only then expanded to note-taking and drafting. Public commentary from the firm emphasised evaluation of answer quality by advisers before rollout. When you tell your KYB story, Morgan Stanley is the reference for "scope narrowly, evaluate, then expand", and it lets you make the point that the scoping decision is the strategy.

## 6. Consultant language

> *"The scoping decision is where the value is decided; everything after it is execution risk."*

Use when an interviewer suggests your assessment work was preparatory rather than strategic.

> *"I built the prototype to retire the three design uncertainties that would otherwise have been argued on slides."*

Use when challenged on the vibe-coded prototype.

> *"Deterministic steps belong in rules; judgement steps belong with the agent; consequential decisions stay with a human. The six dimensions are how we drew those lines."*

Use to summarise your agentic-fit philosophy in one breath.

> *"My role was to own the question, not every answer."*

Use to state ownership honestly when a larger team delivered the final product.

> *"Payments gives me the data and the process; AI gives me the lever. Few people in this market have both."*

Use in the fit conversation to position your differentiation.

## 7. Common mistakes juniors make

- **Activity lists instead of decisions.** "I ran interviews, built a map, created a heatmap" says nothing about judgement. Lead with the decision the work enabled.
- **Claiming the team's result as personal delivery.** It unravels in references. Say what you owned.

- **Apologising for the prototype.** It is a strength if framed as de-risking; it is a weakness only if framed as “I like coding”.
- **Numbers without a label.** If a figure was a hypothesis, say “our hypothesis was”. Partners discount unlabelled precision.
- **Letting the interviewer define you as ops.** Answer the altitude test by moving up the ladder in the same sentence.
- **No reflective point.** Every story needs one thing you would do differently; it is the fastest way to demonstrate seniority.
- **Over-long 2-minute versions.** Rehearse with a timer. Two minutes is roughly 280 words spoken.

## 8. Interview questions with model answers

**Q1. “Walk me through the KYB agentic-fit work. What did you actually do?”**

Structure the answer by the four rungs. “The question was which steps of corporate onboarding justified an agent rather than a rule or a human. I decomposed the process to about [25] steps with SMEs, scored each on six dimensions, task variability, judgement required, system access, error tolerance and reversibility, data availability, and volume, and calibrated the weights with the risk function. Roughly [a third] of steps scored as deterministic and went to rules or integrations; a cluster around document interpretation, entity and UBO resolution and screening adjudication scored as agent-fit; risk rating stayed human on error tolerance. That scoring set the scope of the supervisor-led multi-agent design the bank adopted and therefore the business case. I owned the decomposition, the scoring and the rationale paper; the architecture was co-designed with the engineering lead. What I would change: involve model risk in weight calibration a week earlier.” Numbers are placeholders; use real ones.

**Q2. “You say you did process mapping on underwriting. Why should I see that as strategy work?”**

“Because the map changed the proposition. Before the diagnostic the hypothesis was that AI would draft credit memos. The interviews showed the dominant pain point was information assembly across disconnected systems, which is where analyst time went and where no regulatory approval for autonomous decisioning was needed. We re-scoped to orchestrated assembly and pre-population with the analyst owning judgement. That is a strategy decision, scope and value hypothesis, and the map was its evidence. Without the diagnostic we would have pitched the wrong thing to a credit risk officer who would have rejected it on model-risk grounds in the first meeting.”

**Q3. “Tell me about the prototype. Isn’t that an engineer’s job?”**

“An engineer’s job is to build something that ships. Mine was to answer design questions a client would otherwise have paid engineers to find out. Three questions: where does the agent boundary sit, what must stay deterministic, and where does a human have to approve. The prototype answered all three: reason codes and timeframes to a rules engine, clause-level chunking for auditable citations, narrow tool lists per specialist agent after the broad agent attempted out-of-scope actions, and human gates on financial and adverse outcomes. I would not put that code in production and I say so. Its value was that the scope and risk conversations happened with evidence.”

**Q4. “What makes you different from the data scientists we already have?”**

“Two things. I know the processes and data of payments and financial crime operations at the level where AI scoping decisions are actually made: ISO 20022 fields, RTP decision windows, scheme dispute rules, KYB evidence chains. And I have made the scoping and risk decisions on agentic designs, not only modelled them. Your data scientists will build better models than I will. I will make sure they are building the right thing for a client who has to defend it to a regulator.”

**Q5. “What has your AI work taught you that changed how you consult?”**

“That the business case is downstream of the agent boundary. In conventional automation the scope follows the process; in agentic AI the scope follows error tolerance and reversibility, because a model’s mistakes are probabilistic. So I now start every AI engagement with the question ‘where can the system be wrong and what does it cost’, before I ask ‘where can it save money’. Klarna’s 2024 claims and 2025 reversal [PUBLIC — Feb 2024 and May 2025; VERIFY] are the public illustration of what happens when you do it the other way round.”

## 9. Self-check quiz

Attempt before checking the answer key in the appendix.

1. State the four rungs of the altitude ladder and give the four-rung version of your Project B in four sentences.
2. An interviewer says: “Scoring steps on a framework is analyst work.” Give a two-sentence response that reframes it without being defensive.
3. Which of your three projects best demonstrates the “deterministic steps belong in rules” principle, and what specific example would you cite?
4. Name two public cases you would pair with your disputes prototype and the one-line lesson you would draw from each.
5. Write the “strategy vs implementation” answer in your own words in under 80 words, including the phrase that states where you add most value.

# Module 20 — Interview Arsenal: 40 Questions, 5 Case Drills, 3 Guesstimates and a Full Mock

## How to use this module

Work it in four passes. First, read Part A once for coverage and mark the questions where your answer would have differed materially. Second, do the five case drills in Part B aloud, against a timer, before reading the model structure. Third, do the guesstimates in Part C on paper, then compare logic rather than the final number. Fourth, have someone read the mock in Part D to you and score you with the rubrics. All numbers in this module are [ILLUSTRATIVE — composite] or [ESTIMATE — assumption] unless marked [PUBLIC — source, year]. Pricing and regulatory items carry [VERIFY — date-sensitive].

A model answer in an AI strategy interview has a recognisable shape: structure first (two or three headings), then content under each, then one number or example, then the risk or caveat. Answers below follow that shape. Aim to deliver each in 60–90 seconds spoken.

## Part A: Top 40 AI-consulting interview questions with model answers

### Theme 1: Concepts

**Q1. Explain the difference between machine learning, generative AI and agentic AI to a COO in two minutes.**

Three layers, each adding a capability. Machine learning learns a mapping from historical data to a prediction or classification: fraud score, default probability, next best offer. Its output is a number or label, and it is well understood by model risk functions under SR 11-7 style guidance [PUBLIC — Federal Reserve, 2011]. Generative AI, built on large language models, produces content, text, code, structured extractions, from a prompt and context; it reasons over language rather than fitting a target variable, which makes it good at unstructured work and inherently probabilistic in its errors. Agentic AI wraps a model in a loop with tools, memory and planning so it can take actions towards a goal across several steps: query a system, decide, call another system, ask a human. The COO's takeaway: ML predicts, GenAI produces, agents act. Each step up adds value and adds a new class of risk, from wrong predictions to wrong content to wrong actions, so the control design must step up with it.

**Q2. What is a context window and why does it matter commercially?**

The context window is the maximum number of tokens the model can attend to in one call, covering system prompt, retrieved documents, conversation history and the output. Commercially it matters in three ways. Cost: you pay per token, so every document you stuff into context is a recurring charge; a 50,000-token KYB case file processed once per step across a ten-step workflow is 500,000 input tokens per case. Quality: attention degrades with very long inputs, the "lost in the middle" effect, so bigger is not automatically better. Architecture: the window size decides whether you need retrieval-augmented generation, summarisation or memory structures. The consultant point is that context is a design variable with a price tag. At frontier pricing of the order of USD 2–5 per million input tokens [ESTIMATE — order of magnitude, VERIFY], the KYB case above costs around USD 1–2.50 in input alone before caching, which is why prompt caching and model tiering appear in every serious TCO.

**Q3. What is RAG, and when is it the wrong answer?**

Retrieval-augmented generation (Lewis et al., 2020 [PUBLIC]) retrieves relevant passages from an indexed corpus, typically via embeddings and a vector store, and places them in the model's context so the answer is grounded in the organisation's own documents rather than the model's parametric memory. It is the default pattern for policy, procedure and knowledge questions because it is cheaper and more controllable than fine-tuning and gives you citations for audit. It is the wrong answer in three situations. When the task needs the model to behave differently rather than know more, for example a consistent output format or tone,

fine-tuning or structured prompting is the lever. When the corpus is small enough to fit the context, retrieval adds latency and failure points for nothing. And when the question requires computation or live system data, retrieval of static text is the wrong tool; you need tool calling. A junior mistake is to propose RAG for every use case; a manager asks what the failure mode is, retrieval miss or generation error, and designs for it.

**Q4. Why do LLMs hallucinate, and what actually reduces it?**

A language model generates the most probable continuation given its context; it has no internal notion of truth, only of plausibility. Hallucination arises when the probable continuation is not the correct one: the model has no relevant knowledge, the retrieved context is missing or contradictory, or the prompt invites confident completion. Four mitigations work, in descending order of effect. Grounding: give the model the facts via retrieval or tools and instruct it to answer only from them, with citations. Constraint: restrict the output to structured schemas and enumerated options, which turns open generation into selection. Verification: a second pass that checks claims against sources, or a deterministic rule that validates fields. Human review: sampled or exception-based, proportionate to error cost. What does not work on its own is "better prompting" as a control; it reduces frequency but is not auditable. In banking, the standard a model risk function will accept is measured groundedness on a golden set, with a defined threshold and a monitoring process.

**Q5. Frontier model or small open-weight model: how do you choose?**

Decide on five criteria. Capability on your evaluation set, not on public benchmarks: run the golden set through both. Cost per task at expected volume: small models can be ten to fifty times cheaper per token [ESTIMATE — order of magnitude, VERIFY], which dominates at high volume for narrow tasks. Latency: sub-second requirements, as in payment authorisation paths, generally exclude large models. Data residency and control: open-weight models can be hosted in the bank's own environment, which some risk functions and regulators prefer. Operational burden: self-hosting shifts patching, scaling and safety tuning onto the bank. The usual answer is a tiered design: a frontier model for the few judgement-heavy steps, small models for high-volume classification and extraction, with a router. The consultant framing is that model choice is a portfolio decision made per task, not a one-time vendor choice.

## Theme 2: Frameworks

**Q6. How do you assess an organisation's AI maturity?**

Use a multi-dimensional model, score each dimension on a defined scale and compare with the stated ambition. Dimensions should cover strategy and leadership ambition, data foundations, technology platform, talent and operating model, governance and risk, and value delivery and adoption. Gartner publishes an AI maturity model with five levels [PUBLIC — Gartner; VERIFY current version], and McKinsey's "Rewired" (Lamarre, Smaje and Zemmel, 2023 [PUBLIC]) frames six capabilities: business-led digital roadmap, talent, operating model, technology, data, and adoption and scaling. The method matters more than the brand. Evidence each score with artefacts, not interviews alone: a data catalogue, a model inventory, a funded roadmap. Then set the ambition explicitly, defend, extend or transform, because a bank aiming to defend its cost position needs a different maturity target from one aiming to reinvent servicing. The output is a gap-to-ambition view that becomes the enabler workstream in the roadmap. The mistake is to produce a maturity score without a decision attached to it.

**Q7. Explain BCG's 10-20-70 rule and how you use it.**

BCG's heuristic [PUBLIC — BCG, 2023–24] says that in AI programmes roughly 10% of the effort and value is in algorithms and models, 20% in technology and data, and 70% in people, process and business change. I use it in three places. In the business case, to sanity-check the cost split: a programme budget that is 70% model licences and 10% change is inverted and will under-deliver. In the operating model, to argue for change capacity, process redesign and adoption measurement as first-class workstreams. In vendor conversations, to push back on propositions that price the model and hand-wave the integration and

adoption. The caveat I add: the ratio is a heuristic about effort, not a formula, and in regulated banking the governance and validation share is larger than the rule implies. Pair it with McKinsey's "Rewired" capabilities for the fuller picture of what the 70% consists of.

**Q8. Walk me through how you build and prioritise a use-case portfolio.**

Five steps. Discovery: walk the value chain, decompose processes to task level, and build a long list with pain points from SMEs and data. Cleaning: de-duplicate, cluster into domains, and write each use case as a problem statement with a value hypothesis. Assessment: score value, across cost, revenue, risk and customer experience, and feasibility, across data readiness, integration complexity, model maturity, regulatory exposure and change effort, with weights agreed with the sponsor before scoring. Overlay: a risk tier and a dependency map, because the top-scoring use case is often blocked by a data foundation nobody scored. Sequencing: waves, with quick wins to fund credibility, foundational investments to unlock the next wave, and transformational bets that need the foundations. Value-vs-feasibility matrices are generic tools; RICE (Intercom) and ICE (Sean Ellis) are product-management variants [PUBLIC]. The manager-level point is that weights are a political decision, so agree them in the room with the sponsor and the risk function.

**Q9. How do you assess whether a process step is suited to an agent?**

Score the step, not the process. The framework I use has six dimensions: task variability (does the input vary in form and content, which rules cannot handle), judgement required (interpretation versus lookup), system and tool access (can the agent actually act), error tolerance and reversibility (what a wrong action costs and whether it can be undone), data availability (is the evidence machine-readable), and volume and frequency (does the economics justify the build). Attribution: a practitioner framework, variants differ by firm. Deterministic, low-variability steps go to rules or integrations; high-variability, high-judgement, reversible steps are agent candidates; irreversible, high-consequence decisions keep a human. In a KYB assessment [ILLUSTRATIVE — composite] about a third of steps typically score as rules, half as agent-fit with review, and a handful, such as final risk rating and adverse decisions, stay human. The scope that falls out is the business case.

**Q10. What is "Deploy, Reshape, Invent" and why does the distinction matter?**

It is BCG's framing of three levels of GenAI ambition [PUBLIC — BCG, 2023–24]. Deploy: adopt available tools to raise individual productivity, such as coding assistants and document drafting, with low integration and quick but diffuse value. Reshape: redesign end-to-end processes and functions around AI, such as agentic onboarding or claims, where the value is large and the change effort is where the risk sits. Invent: create new products, services or business models that AI makes possible. It matters because each needs a different funding model, governance posture and success metric. Deploy value is measured in adoption and time saved and is routinely overstated because saved minutes do not become saved cost. Reshape value is measured in unit cost and cycle time and requires process ownership. Invent value is measured in revenue and option value and needs venture-style gates. A portfolio should be explicit about which of the three each initiative is.

## Theme 3: Agentic AI

**Q11. Describe the main agent architectures and when you would use each.**

Single ReAct agent (Yao et al., 2022 [PUBLIC]): one model reasons and acts in a loop with tools; suitable for bounded tasks with few tools. Plan-and-execute: a planner decomposes the goal into steps, executors carry them out; better for long tasks where you want the plan reviewable before action. Supervisor or hierarchical: an orchestrator routes work to specialist agents with narrow scopes and tool lists; the pattern I recommend for regulated operations because it gives model risk one accountable control point, allows specialists to be evaluated in isolation, and keeps each agent's prompt and tools small enough to constrain behaviour. Orchestrator-worker with parallel fan-out: for independent sub-tasks such as evidence gathering from several sources. Multi-agent collaboration or debate: agents critique each other; useful for quality on high-

stakes drafting, costly in tokens and hard to explain. The choice follows accountability and testability, not novelty.

**Q12. What is the Model Context Protocol, and should a bank care?**

MCP, released by Anthropic in November 2024 [PUBLIC], is an open protocol that standardises how models connect to tools and data sources: a server exposes tools, resources and prompts, and any MCP-capable client can use them without bespoke integration code. Google's Agent2Agent protocol (April 2025, donated to the Linux Foundation in June 2025 [PUBLIC; VERIFY adoption]) addresses the adjacent problem of agents communicating with other agents. A bank should care for three reasons. Integration cost: tool integration is usually the largest build cost, and standard interfaces reduce it and the lock-in. Security surface: every exposed tool is a new access path and must be governed, scoped and logged like an API. Strategy: the protocol layer is where platform choices become reversible or not. My advice is to adopt the standards for internal tool exposure with the same access controls as APIs, and to treat agent-to-agent interoperability as emerging and worth piloting rather than betting on. Verify the current state before citing adoption figures.

**Q13. How do you decide the autonomy level of an agent?**

Use an explicit scale and tie it to error cost. A workable scale is: level 0, human does the task with AI information; level 1, AI recommends, human decides and acts; level 2, AI acts, human approves before effect; level 3, AI acts, human reviews a sample after the fact; level 4, AI acts with exception-only escalation; level 5, full autonomy. Attribution: practitioner scale, variants exist. Assign the level per action, not per system, using error tolerance and reversibility, regulatory consequence, and measured accuracy on a golden set. A dispute-resolution agent might sit at level 4 for evidence assembly, level 3 for low-value provisional credits, level 2 for write-offs above a threshold and level 1 for any customer-facing adverse decision [ILLUSTRATIVE — composite]. Autonomy then becomes a dial you move up as evidence accumulates, which is how you talk about it to a regulator: earned, measured and reversible.

**Q14. What guardrails does an agentic system in a bank need?**

Four layers. Input guardrails: prompt-injection detection, data classification checks and scope filters so the agent only receives what it should. Output guardrails: schema validation, groundedness checks against sources, toxicity and PII filters, and confidence thresholds that route low-confidence outputs to humans. Action guardrails: allow-lists of tools and operations, value and volume limits, idempotency and rollback, and mandatory approval gates on irreversible actions. Policy and governance guardrails: logging of every reasoning step and tool call, model and prompt versioning, kill switches, and periodic re-validation under the model risk framework. The key design principle is that guardrails must be enforced outside the model, in deterministic code, because a model-based check can be manipulated by the same inputs that compromise the model. In interview, say that explicitly; it separates people who have built one from people who have read about them.

**Q15. When is agentic AI the wrong tool?**

Four cases. When the task is deterministic: reason-code validity, timeframe checks, eligibility rules, where a rules engine is cheaper, faster and auditable, and an agent only adds probabilistic error. When a single model call suffices: summarisation, classification and extraction do not need a loop, tools or memory, and wrapping them in an agent adds cost and failure modes. When the error cost is high and the action is irreversible: there the correct design is decision support at autonomy level 1, not an agent. When the data is not machine-readable or the systems have no interfaces: an agent with nothing to call is a chatbot. The honest framing for a client is that agents are the right tool for a narrow band of high-variability, judgement-heavy, reversible, well-instrumented work, and that most of a bank's processes contain only a few such steps. Gartner's June 2025 projection that over 40% of agentic AI projects would be cancelled by 2027 [PUBLIC — Gartner, 2025; VERIFY] is largely a statement about this misfit.

## Theme 4: Business case and TCO

**Q16. What goes into the TCO of a GenAI or agentic system?**

Four buckets over at least three years. Build: data preparation and access, integration with core systems, engineering, construction of evaluation sets, security review and model-risk validation. Run: inference and token costs, platform and infrastructure, observability and monitoring, ongoing evaluation, guardrail services, and human review capacity, which is often the largest run cost and the most often omitted. Change: process redesign, training, role changes and adoption management, the 70% in BCG's heuristic [PUBLIC — BCG, 2023–24]. Refresh: model upgrades and re-validation, prompt and retrieval re-tuning when models change, and drift monitoring. The common errors are to model inference as the dominant cost, when it is typically 10–25% of three-year TCO for an operations use case [ESTIMATE — assumption], and to omit human review and re-validation. The term TCO comes from Gartner in 1987 [PUBLIC]; the principle, count every cost of owning not just acquiring, is what a CFO will hold you to.

**Q17. How do you estimate inference cost for a use case?**

Bottom up, per task. Define the workflow steps that call a model; for each, estimate input tokens (system prompt, retrieved context, conversation state) and output tokens; multiply by the model's price per million tokens; sum across steps; multiply by annual volume. Then apply the levers: prompt caching for repeated system and context prefixes, typically 75–90% off cached input [ESTIMATE — order of magnitude, VERIFY]; batch processing for non-urgent work, around 50% off; and model tiering, routing high-volume simple steps to small models. Worked [ILLUSTRATIVE — composite]: a dispute agent with six model steps averaging 6,000 input and 600 output tokens per step is 36,000 input and 3,600 output tokens per case. At USD 3 per million input and USD 15 per million output that is about USD 0.11 input plus USD 0.05 output, roughly USD 0.16 per case; at 600,000 cases a year, about USD 96,000 before caching. The point to make: inference is rarely the line that breaks the case; integration and review are.

**Q18. How does a CFO challenge an AI business case, and how do you pre-empt it?**

Eight challenges recur. Baseline inflation: pre-empt with a measured, dated baseline and source. Capacity release versus cost: CFOs discount "hours saved" at 30–70% [ESTIMATE — practitioner range] unless tied to a headcount plan, vacancy freeze or volume absorption; state the conversion mechanism. Ramp optimism: show adoption and accuracy curves, not a step change. Double counting: reconcile with other programmes claiming the same FTE. Run-cost omission: show the full TCO including human review and re-validation. Attribution: define the counterfactual and the measurement method up front. Risk-adjustment: probability-weight benefits by evidence level, for example 90% for measured pilot results, 50% for analogues, 25% for hypotheses. Terminal assumptions: no benefit beyond the horizon unless contracted. The strongest pre-emption is a benefits register with an owner, a baseline, a measurement method and a tracking cadence for each line.

**Q19. Explain NPV, IRR and payback and when each misleads.**

NPV discounts all future net cash flows to today at the organisation's hurdle rate: NPV = Σ CF_t / (1+r)^t minus initial investment. Positive means the programme creates value at that hurdle. IRR is the discount rate at which NPV is zero; it is a return measure that is intuitive but misleading when cash flows change sign more than once or when comparing projects of different scale. Payback is the time to recover the investment; it ignores everything after payback and ignores discounting, but CFOs like it because it measures exposure. For AI programmes, NPV at the bank's hurdle, often 8–12% [ESTIMATE — practitioner range], is the decision metric; payback under 24 months is usually needed for sponsorship; IRR is reported for comparison. The specific AI distortion is a benefit ramp that back-loads value, which makes payback look worse than NPV suggests; show both and explain why.

**Q20. How do you risk-adjust an AI business case?**

Three mechanisms, applied transparently. Evidence-weighted benefits: tier each benefit line by evidence quality, measured in a pilot, observed at an analogue, or hypothesis, and apply probability weights such as

0.9, 0.5 and 0.25 [ESTIMATE — practitioner practice]. Scenario analysis: build low, base and high cases that vary the three or four assumptions that drive most of the value, typically adoption rate, accuracy at autonomy threshold, volume growth and run cost, and show NPV for each. Sensitivity: a tornado chart ranking assumptions by their effect on NPV, which tells the CFO what to monitor. Add a contingency on costs, commonly 15–25% on build [ESTIMATE — practitioner range], and a delay scenario, because regulated deployments slip on validation. Present the risk-adjusted NPV as the headline and the unadjusted as context. CFOs trust an advisor who discounts their own numbers before being asked.

## Theme 5: Governance and regulation

**Q21. How does the EU AI Act affect a bank's AI programme?**

The Act [PUBLIC — Regulation (EU) 2024/1689, in force 1 Aug 2024; VERIFY current timeline] is risk-based. Prohibited practices applied from February 2025. Obligations for general-purpose AI model providers applied from August 2025. Most high-risk obligations under Annex III were scheduled for August 2026, and the Commission's November 2025 digital omnibus proposal would delay some of them, so verify the current dates. For banks, Annex III lists creditworthiness assessment of natural persons and life and health insurance risk assessment and pricing as high-risk, which brings risk management, data governance, logging, human oversight, accuracy and robustness, and conformity requirements. Most operations use cases, KYB, disputes, AML triage, are not Annex III high-risk, but they carry transparency obligations where people interact with AI, and they still sit under the bank's model risk and conduct frameworks. The consultant's job is to classify every use case against the Act early, because classification determines cost and timeline.

**Q22. Compare NIST AI RMF, ISO/IEC 42001 and PRA SS1/23.**

They operate at different levels. NIST AI RMF 1.0 [PUBLIC — Jan 2023] is a voluntary framework with four functions, Govern, Map, Measure and Manage, plus a Generative AI Profile [PUBLIC — NIST AI 600-1, July 2024]; it is a vocabulary and a checklist for risk practice. ISO/IEC 42001 [PUBLIC — Dec 2023] is a certifiable management-system standard, like ISO 27001 for security; it specifies the organisational system for AI governance and can be audited. PRA SS1/23 [PUBLIC — published May 2023, effective May 2024; VERIFY scope] is supervisory expectation for UK banks on model risk management, with five principles: model identification and risk classification, governance, development and implementation and use, independent validation, and risk mitigants. In practice a UK bank uses SS1/23 to decide what counts as a model and how to tier and validate it, ISO 42001 to structure the management system if it wants certification, and NIST to shape the risk taxonomy. GenAI systems test all three because "the model" is now a model plus prompts plus retrieval plus tools.

**Q23. Is an LLM-based agent a "model" under model risk management?**

Yes, and more than one. SS1/23 defines a model broadly as a quantitative method that processes inputs into outputs using assumptions, and a bank's inventory should treat the agentic system as a model with components: the foundation model, the prompts, the retrieval corpus and index, the tools and the orchestration logic. The practical approach is to tier by materiality and use, validate the system end to end on a golden set rather than attempting to validate the foundation model's internals, and put change control on every component, because a prompt edit or a corpus update changes behaviour as surely as a parameter change. Independent validation should test accuracy, groundedness, robustness to adversarial inputs, and the effectiveness of guardrails and human oversight. Treat vendor model updates as model changes that trigger re-validation. This is where many GenAI programmes stall in banks, so plan validation effort and timeline into the roadmap from day one.

**Q24. How do you test an AI system for bias in a lending context?**

Start from the protected characteristics and proxies relevant to the jurisdiction, then test outcomes, not intentions. Measure disparate impact: compare approval rates, pricing and error rates across groups; the four-fifths rule from US employment law [PUBLIC — EEOC Uniform Guidelines, 1978] is a common screening heuristic, where a selection rate below 80% of the most favoured group's rate triggers scrutiny. Test for

proxy discrimination by checking whether features correlate with protected attributes, including in embeddings and text features for LLM-based systems. Examine calibration and error-rate parity, since equal approval rates can hide unequal false-negative rates. For generative components, test for differential quality of explanations or recommendations across groups with templated inputs. Document the trade-offs, because fairness metrics conflict and a choice must be made and recorded. The Apple Card episode [PUBLIC — Nov 2019; NYDFS report, Mar 2021] showed that even without a finding of unlawful discrimination, inability to explain outcomes is itself a conduct and reputational failure.

**Q25. What does the RBI's FREE-AI framework ask of Indian financial institutions?**

The Reserve Bank of India constituted the FREE-AI committee in December 2024 and published its report in August 2025 [PUBLIC — RBI, 2025; VERIFY exact wording and implementation status]. It sets out seven guiding "sutras": trust is the foundation; people first; innovation over restraint; fairness and equity; accountability; understandable by design; and safety, resilience and sustainability, and makes 26 recommendations across six pillars spanning infrastructure, policy, capacity, governance, protection and assurance. The posture is notably pro-innovation compared with the EU, with proposals such as shared data and compute infrastructure and a tolerant approach to first-mover errors, alongside board-level accountability, AI policies, incident reporting and consumer protection. For a consultant, it means Indian banks will be expected to have board-approved AI policies, inventories and grievance mechanisms, and that explainability and accountability will be examined. Treat it as direction of travel and check what has been converted into binding regulation.

## Theme 6: Delivery and operating model

**Q26. Why do AI pilots stall, and what do you do about it?**

Pilots stall because they are designed to prove the technology rather than to reach production. Causes, in rough order of frequency: no production owner or budget beyond the pilot; data access granted for a sandbox but not for production; integration never scoped; model risk validation not planned; success criteria technical rather than business; and change capacity absent, so the pilot works and nobody uses it. The statistics are stark: Gartner projected at least 30% of GenAI projects abandoned after proof of concept by end 2025 [PUBLIC — Gartner, July 2024; VERIFY], and the MIT NANDA report claimed around 95% of enterprise GenAI pilots showed no measurable P&L impact [PUBLIC — MIT NANDA, Aug 2025; VERIFY, methodology contested]. The remedy is to decide the production path before the pilot starts: owner, funding, integration budget, validation plan, exit criteria including business metrics, and a kill decision date. A pilot without a kill date is a hobby.

**Q27. Compare the operating-model archetypes for AI.**

Centralised centre of excellence: one team owns strategy, platform and delivery; fast to stand up, builds deep expertise, but becomes a bottleneck and distant from the business. Hub-and-spoke: a central hub owns platform, standards, risk and scarce skills; spokes in business units own use cases and adoption; the most common target state for banks because it balances control and proximity. Federated: business units own everything with light central coordination; speeds local innovation and fragments standards, risk and spend. Embedded: AI roles sit permanently in product teams; the end state for digitally mature organisations. McKinsey and QuantumBlack research has reported hub-and-spoke as the most common high-performer configuration [PUBLIC — McKinsey; VERIFY specific publication]. The evolution is usually CoE to hub-and-spoke to embedded as maturity grows. The decision criteria are regulatory exposure, talent scarcity, platform reuse and the speed of business demand. Whatever the archetype, risk and validation must have an independent line.

**Q28. What roles does an AI programme need, and in what ratio?**

A delivery squad for an agentic use case typically needs an AI product owner from the business, an agent or prompt engineer, one or two software engineers for integration, a data engineer, an ML or evaluation engineer, a domain SME at least half-time, and access to AI risk and validation, security and change

management. Ratios [ESTIMATE — practitioner observation]: roughly one product owner and one SME per squad, two to three engineers per agent engineer, and one validation resource per two or three squads in regulated settings. Platform roles, LLMOps, observability, model gateway and guardrail services, sit centrally. The under-resourced roles are almost always evaluation engineering and change management. The mistake is to staff a GenAI programme like a data-science programme; agentic systems are integration and process-change heavy, which the 10-20-70 heuristic [PUBLIC — BCG] captures.

**Q29. How do you design KPIs for an AI deployment?**

Build a KPI tree with three layers and a causal chain between them. Model KPIs: accuracy, groundedness, false-positive and false-negative rates, latency, cost per task, human-override rate. Operational KPIs: cycle time, throughput per FTE, rework rate, queue age, exception volume. Business KPIs: unit cost, loss rate, revenue or retention, customer satisfaction, regulatory findings. Distinguish leading from lagging: override rate and adoption lead unit cost by months. Set targets at three, six and twelve months, baseline every metric before go-live, and define the measurement method and counterfactual. Watch for pitfalls: Hawthorne effects in pilots, baseline drift when volumes change, and attribution when several programmes touch the same process. The most useful single KPI for an agentic system in early life is the override rate by reason, because it tells you simultaneously about model quality, trust calibration and where to move the autonomy dial.

**Q30. How do you manage adoption and the workforce impact?**

Treat adoption as a designed outcome, not a hoped-for one. Use a change framework, Kotter's eight steps [PUBLIC — Kotter, 1996] for leadership and ADKAR from Prosci [PUBLIC — Hiatt, 2003] for individual change, and add three AI-specific elements. Trust calibration: train people to know when the system is likely wrong, because both over-trust, automation bias, and under-trust destroy value. Role redesign: specify what the human does after automation, exception handling, quality assurance, customer judgement, and pay and grade accordingly. Workforce transition: be explicit about whether the case assumes attrition, redeployment or redundancy, and plan reskilling early. The Commonwealth Bank of Australia case in 2025, where redundancies attributed to an AI voice bot were reversed after volumes did not fall as assumed [PUBLIC — Aug 2025; VERIFY], shows what happens when headcount is removed on a modelled rather than measured effect. Measure adoption weekly in the first quarter.

## Theme 7: Industry and cases

**Q31. Where does AI create the most value in a bank today?**

Three clusters, in order of proven value. Operations with high volumes of document- and text-heavy work under codified rules: KYC and KYB, AML alert triage, sanctions adjudication, disputes, payment investigations, reconciliation breaks. Here agentic and GenAI patterns compress cycle time and cost and the KPIs are measurable. Risk and fraud, where machine learning has been delivering for a decade and GenAI adds narrative generation and investigator assistance; Mastercard reported an average 20% fraud-detection uplift from generative techniques in 2024 [PUBLIC — Mastercard, Feb 2024; VERIFY]. Knowledge work and servicing: adviser assistants, contact-centre copilots and coding assistants, where value is real but diffuse and hard to convert to cost. The McKinsey Global Institute estimated USD 200–340 billion of annual value for banking from generative AI [PUBLIC — MGI, June 2023]. My view for a client: fund operations use cases for cost and cycle time, risk for loss reduction, and treat knowledge-worker tools as a productivity floor rather than a business case.

**Q32. What do you take from the Klarna case?**

Klarna announced in February 2024 that its OpenAI-based assistant handled two-thirds of customer-service chats in its first month, equivalent to about 700 full-time agents, with an estimated USD 40 million profit improvement [PUBLIC — Klarna, Feb 2024]. In 2025 the CEO publicly acknowledged the company had over-indexed on cost and would re-hire humans to restore service quality [PUBLIC — interviews, May 2025; VERIFY]. Three lessons. Scope automation by error tolerance and customer impact, not by cost ambition; the

reversal was about quality on the cases the assistant should never have fully owned. Keep human capacity as a designed tier, not a residual. And be careful with early claims: first-month resolution statistics measure deflection, not resolution quality, and the market now discounts them. For a consultant, it is the best public illustration of why human-in-the-loop design is an economic decision as well as a risk one.

**Q33. What did HSBC do with AI in anti-money laundering, and why does it matter?**

HSBC worked with Google Cloud on AI-based transaction monitoring, the "Dynamic Risk Assessment" approach later productised as Google's AML AI. Google Cloud reported that the system identified two to four times more suspicious activity while cutting alert volumes by around 60% [PUBLIC — Google Cloud, June 2023; VERIFY]. It matters for three reasons. It is a machine-learning risk-scoring case, not a GenAI case, which is a useful corrective when a client assumes AI means chatbots. It shows the dual benefit shape regulators care about: fewer false positives and more true positives at the same time, which is the only way to justify replacing rules-based monitoring to a supervisor. And it is a multi-year, bank-and-hyperscaler build, not a procurement, which tells a client what commitment looks like. In an AML conversation I use it to frame the ambition and then immediately caveat that the regulator must accept the model under the bank's MRM framework.

**Q34. Tell me about an AI failure in financial services and what it teaches.**

Two I use. Zillow Offers [PUBLIC — Nov 2021]: an algorithmic home-buying business wound down after pricing-model losses of more than USD 500 million and a roughly 25% workforce cut. The lesson is that a model's error distribution matters more than its average accuracy when the model is exposed to a market with adverse selection; sellers accepted offers when the model over-priced. The Apple Card episode [PUBLIC — Nov 2019; NYDFS Mar 2021]: allegations of gender bias in credit limits; the regulator found no unlawful discrimination but criticised the inability to explain decisions to customers. The lesson is that explainability is a conduct requirement independent of the fairness finding. For agentic systems I add Air Canada [PUBLIC — Feb 2024], where a tribunal held the airline liable for its chatbot's incorrect fare statement: the organisation owns what its AI says. Each one maps to a design control: error-cost analysis, explanation capability, and output guardrails with human accountability.

**Q35. How do you apply AI in payments operations specifically?**

Payments operations is a strong fit because the data is structured, the rules are codified and the volumes are high. Four use cases. Exceptions and repairs: ISO 20022 structured party and remittance data lets models repair and route exceptions with far higher precision than free-text MT messages allowed; agents can assemble the fix and a human releases high-value items. Investigations: cross-border inquiries are text-heavy, multi-party and SLA-bound, an ideal orchestrated-agent domain with retrieval over scheme and correspondent rules. Fraud and authorised push payment scams in real-time rails: the sub-second window excludes heavy models from the authorisation path, so ML scoring sits in-line and GenAI sits in pre-transaction customer warnings and post-transaction investigation. Reconciliation: pattern-matching for breaks with agentic root-cause narratives. The strategic point: ISO 20022 migration is the data foundation most AI assessments in payments under-weight; banks that populated the structured fields have a feasibility advantage.

## Theme 8: Personal and behavioural

**Q36. Why AI strategy consulting, and why now?**

Give a reason tied to your trajectory and the market, not enthusiasm for technology. "I have spent eight years in financial services consulting, mostly payments, and the last two watching AI move from analytics into the operating core of the processes I know: onboarding, disputes, screening. The scarce skill in this market is not building models; it is knowing a bank's processes and data well enough to decide where AI belongs and defending that to a COO, a CFO and a model risk committee. That is the intersection I sit at. Now matters because banks are moving from pilots to scaled programmes, and that transition is where strategy

consultants earn their fee: scoping, business case, operating model and governance. I want to lead that work, and this practice is where that work is being done at the level I aspire to."

**Q37. Tell me about a time you changed a client's or team's mind with analysis.**

Use the underwriting diagnostic. "The working hypothesis was that AI would draft credit memos. My SME interviews and time analysis showed the dominant pain point was information assembly across disconnected systems, not analysis. I presented the heatmap and argued that the proposition should be orchestrated assembly with the analyst owning judgement, which also kept us clear of autonomous credit decisioning and its regulatory classification. The team re-scoped. What made it land was that the recommendation reduced regulatory exposure and increased value at the same time, so it had no natural opponent. What I would do differently is quantify in the interviews rather than afterwards." Keep it to 90 seconds; name the decision, the evidence and the outcome.

**Q38. Describe a failure and what you learned.**

Use the prototype. "I built the dispute prototype before I built its evaluation set. The architecture worked on the scenarios I had in mind and then produced confident but wrong outcomes on edge cases I had not written down, including a broad agent attempting actions outside its remit. I retrofitted a golden set and narrowed each agent's tools and prompt. The lesson I carry into every engagement now is that the evaluation set is the specification: you define what correct looks like, with coverage of the ugly cases, before you build or buy anything. It also changed how I write business cases: accuracy on the golden set at the chosen autonomy level is now an explicit assumption, not an afterthought."

**Q39. How do you handle a client stakeholder who wants to automate everything?**

"I take the ambition seriously and redirect it to where it survives. First I agree the goal, usually unit cost or cycle time, and then I show the task-level scoring: which steps are deterministic and belong in rules, which are judgement-heavy and agent-fit, and which are irreversible and stay human. I put a number on the automatable share and a number on the cost of being wrong at each step. Then I show the Klarna arc [PUBLIC — 2024–25; VERIFY] as what over-reach looks like in public. Most sponsors want the outcome, not the autonomy level; once they see that a hybrid design delivers most of the value with a fraction of the risk, the conversation moves to sequencing. If they still insist, I document the risk acceptance explicitly and make sure risk and compliance are in the room."

**Q40. Where do you see agentic AI in banking in three years, and what would you tell a CEO to do now?**

"Three years out I expect agentic systems to run the evidence-gathering and drafting layers of most operations processes, with humans owning the consequential decisions, and I expect regulators to have settled what validation of an agentic system means. I do not expect autonomous decisioning in credit or adverse customer outcomes in that window; the regulatory and liability picture argues against it. I would tell a CEO three things. Fix the data and integration foundations now, because they gate everything and take longest. Build the model-risk and evaluation capability for probabilistic systems, because validation, not technology, is the bottleneck in banks. And pick two or three processes to reshape end to end rather than fifty pilots, with a production owner and a kill date for each. The value is in reshaping, not deploying tools, and the organisations that will lead are those that treat AI as a process-redesign programme with a technology component."

## Part A self-check quiz

Attempt before consulting the answer key in the appendix.

1. In one sentence each, what do ML, GenAI and agentic AI do, and what new class of risk does each add?
2. A sponsor proposes a GenAI programme budget that is 60% model licences, 30% engineering and 10% change. Which heuristic do you cite and what do you say?

3. State the four guardrail layers for an agentic system and the design principle about where guardrails are enforced.
4. A dispute agent runs six model steps averaging 5,000 input and 500 output tokens at USD 3 and USD 15 per million tokens. What is the inference cost per case and per 500,000 cases before caching?
5. Name the five PRA SS1/23 principles and say why prompts and retrieval corpora fall under model change control.

## Part B: Five AI case drills

Each drill is [ILLUSTRATIVE — composite]. Run it aloud against a 25-minute timer: two minutes of clarifying questions, three minutes to lay out a structure, fifteen minutes of analysis with the interviewer feeding numbers, five minutes for recommendation and risks. The model structures below are what a manager-level candidate produces; the "what distinguishes a manager-level answer" line is the grading criterion.

### Case drill 1: AI strategy for a regional bank

**Prompt.** "Nordbank [ILLUSTRATIVE — composite] is a EUR 50 billion asset regional bank with retail and SME banking, about 6,000 staff, a cost-income ratio of 62% and a legacy core. The CEO has been told by the board to 'have an AI strategy' by the next strategy day in ten weeks. What should it be?"

**Clarifying questions a strong candidate asks.**

- What is the strategic ambition: defend cost position, grow in SME, or both? What is the board's risk appetite?
- What exists already: data platform, cloud posture, any AI in production, a model risk function?
- What is the cost base split between operations, distribution, risk and support, and where is the biggest pain?
- Any regulatory constraints or commitments: EU AI Act readiness, supervisor expectations?
- What budget envelope is realistic for a three-year programme?

**Structure.**

1. Ambition: which of defend, extend, transform, and the one or two business outcomes the strategy must move.
2. Value scan: where the money is, by domain, using cost and revenue pools.
3. Foundations: data, technology, talent and governance maturity against the ambition.
4. Portfolio: long list to prioritised waves using value versus feasibility with an agentic-fit lens.
5. Operating model and governance: how the bank will deliver and control it.
6. Business case and roadmap: investment, benefits, NPV and the first 100 days.

**Key analyses with numbers.** Opex of about EUR 1.1 billion [ESTIMATE — 62% of roughly EUR 1.8 billion income]. Interviewer gives a split: operations and servicing 35%, distribution 30%, risk and compliance 15%, technology 12%, support 8%. Operations and servicing is EUR 385 million; the AI-addressable share of effort in document- and text-heavy operations is 25–35% [ESTIMATE — practitioner range], giving EUR 95–135 million of gross potential. Realistic capture in three years is around 40% of potential, so EUR 40–55 million of run-rate saving, which is 3.5–5.0% of opex and would move cost-income from 62% towards 58–59% with flat income. Risk and compliance adds EUR 10–15 million from AML and KYC efficiency. Revenue levers in SME, pricing and next-best-action, are plausible but lower-evidence; include at 25% probability weight. Programme investment EUR 60–80 million over three years, of which data and integration foundations about 40%, use cases 35%, change and capability 25%. Payback inside year three; five-year NPV positive at 10% in the base case.

**Recommendation.** An ambition of "defend and extend": first, reshape three operations processes end to end, onboarding and KYC, payments exceptions and investigations, and servicing, with agentic designs under

human oversight; second, deploy productivity tools firm-wide as a floor, not a business case; third, build the foundations, a data platform, a model gateway, an evaluation capability and model-risk coverage for GenAI, as a funded workstream; fourth, a hub-and-spoke operating model with a small central hub and spokes in operations and SME. Present the board a 100-day plan, three-year roadmap and a benefits register with owners.

**What distinguishes a manager-level answer.** Sizing the value from cost pools rather than listing use cases; explicitly separating deploy from reshape; naming foundations as the gating item; a risk-adjusted number and a payback; and the operating model as part of the strategy, not an afterthought.

## Case drill 2: Should a card issuer build an agentic dispute platform?

**Prompt.** "A card issuer [ILLUSTRATIVE — composite] with 20 million cards handles about one million disputes a year with 300 operations staff. A vendor has pitched an agentic dispute platform; the COO asks whether to build, buy or do nothing."

**Clarifying questions.**

- Current cost per dispute, cycle time, and write-off below investigation threshold?
- Dispute mix: fraud versus non-fraud, and share resolved by rule today?
- Existing case-management system and its integration options?
- Risk appetite for automated provisional credits and automated denials?
- The vendor's commercial model and what exactly it automates?

**Structure.** Baseline economics; where the value sits by dispute step; design options (rules uplift only, agent-assist, agentic with HITL); build versus buy versus partner; business case and risks.

**Key analyses with numbers.** Baseline: 300 FTE at USD 60,000 loaded is USD 18 million, plus USD 4 million systems, about USD 22 per dispute. Interviewer gives a time study: intake and classification 15%, evidence gathering 35%, rule application and decision 25%, customer communication 15%, quality and rework 10%. Agentic-fit scoring places evidence gathering, classification and communication drafting as agent-fit with review, rule application largely in a rules engine, and final decisions above a value threshold with humans. Automatable effort about 60%, retaining 15% of released capacity for review and exceptions, so net 45% of labour, USD 8 million a year. Goodwill write-offs below the investigation threshold are USD 15 million; better evidence and representment cut them 20%, USD 3 million. Faster cycle time cuts provisional-credit funding cost by about USD 1 million. Total steady-state benefit about USD 12 million a year. Costs: build USD 8 million, run USD 4 million a year including inference of roughly USD 0.2 per case, platform, evaluation and human review. Ramp 30%, 70%, 100%. Five-year NPV at 10% about USD 12 million; payback early in year three; IRR in the low 40s per cent.

| Year | Benefit (USD m) | Run cost (USD m) | Build (USD m) | Net (USD m) | Discounted at 10% |
|---|---|---|---|---|---|
| 0 | 0.0 | 0.0 | 8.0 | -8.0 | -8.0 |
| 1 | 3.6 | 4.0 | 0.0 | -0.4 | -0.4 |
| 2 | 8.4 | 4.0 | 0.0 | 4.4 | 3.6 |
| 3 | 12.0 | 4.0 | 0.0 | 8.0 | 6.0 |
| 4 | 12.0 | 4.0 | 0.0 | 8.0 | 5.5 |
| 5 | 12.0 | 4.0 | 0.0 | 8.0 | 5.0 |
| NPV | | | | | 11.7 |

**Recommendation.** Not "do nothing": the economics clear the hurdle even with 25% haircuts on benefits. Not a pure vendor platform either, because dispute handling under scheme rules is close to the issuer's customer promise and the vendor's model behaviour is outside the issuer's change control. Recommend "buy the plumbing, build the judgement": an orchestration and LLMOps platform from a vendor, agents and

rules specified and owned by the issuer, integrated with the existing case system, with autonomy earned step by step and human gates on write-offs and adverse decisions.

**What distinguishes a manager-level answer.** Step-level scoring rather than a process-level verdict; separating the labour saving from the write-off and funding levers; a cash-flow table with a ramp; a build-buy answer at the layer level; and naming where the human stays.

## Case drill 3: GenAI in a bank's contact centre

**Prompt.** "A retail bank [ILLUSTRATIVE — composite] runs a 3,000-agent contact centre with 25 million contacts a year. The head of servicing wants a GenAI chatbot to cut cost by 30%. Advise."

**Clarifying questions.** Channel mix and cost per contact by channel; intent distribution and share already contained by existing automation; complaint and conduct exposure; current customer satisfaction; appetite for agent-assist versus customer-facing automation.

**Structure.** Baseline by channel and intent; three levers (containment, agent assist, demand elimination); sequencing by risk; business case; conduct and adoption risks.

**Key analyses with numbers.** Cost base around USD 200 million at USD 8 per contact. Channel split: 15 million voice at USD 10, 10 million digital chat at USD 5. Intent analysis shows 40% of digital chats are low-complexity informational or transactional and 25% of those are already contained by menus. A GenAI assistant could contain a further 30 percentage points of the low-complexity digital volume, 1.2 million chats at USD 5, about USD 6 million. Agent assist, real-time knowledge retrieval, summarisation and after-call work automation, cuts average handling time 12% across voice, worth about USD 18 million, with a 60% conversion of capacity to cost through attrition and vacancy management, USD 11 million realised. Demand elimination by fixing the top five failure-driven intents, surfaced by GenAI contact analytics, removes 5% of contacts, USD 10 million. Total realistic saving USD 27 million, about 13% of cost, not 30%. Programme cost USD 12 million build and USD 6 million run.

**Recommendation.** Reset the ambition to 12–15% in two years with a path to 20%; lead with agent assist and contact analytics, which carry low conduct risk and build the knowledge base the customer-facing assistant will later need; introduce customer-facing containment only on intents with high accuracy on a golden set and an always-available human handover; measure first-contact resolution and complaints alongside cost. Cite Klarna's 2024 claims and 2025 adjustment [PUBLIC — 2024–25; VERIFY] and the Commonwealth Bank of Australia reversal [PUBLIC — Aug 2025; VERIFY] as the two public cautions on over-reach and premature headcount removal.

**What distinguishes a manager-level answer.** Challenging the 30% rather than accepting it; three distinct levers with different risk profiles; capacity-to-cost conversion stated explicitly; and conduct risk treated as a design constraint with public evidence.

## Case drill 4: AML false-positive reduction

**Prompt.** "A universal bank [ILLUSTRATIVE — composite] generates 1.2 million transaction-monitoring alerts a year with a 97% false-positive rate and employs 600 level-one analysts. The head of financial crime wants to use AI to cut the cost. What do you recommend, and what will the regulator ask?"

**Clarifying questions.** Current monitoring system (rules-based scenarios?); suspicious activity report yield and any regulatory findings; data quality on customer risk and transaction features; the bank's model risk framework and relationship with the supervisor; appetite for suppressing alerts versus re-ranking them.

**Structure.** Where the cost and risk sit in the alert funnel; two AI levers (alert scoring and investigation assistance); regulatory conditions; phased plan; economics.

**Key analyses with numbers.** Level-one cost: 600 analysts at USD 55,000 loaded is USD 33 million; handling time 45 minutes per alert. Lever one, machine-learning alert risk scoring layered on existing scenarios, as in the HSBC and Google Cloud case where Google reported two to four times more suspicious activity found

with around 60% fewer alerts [PUBLIC — Google Cloud, June 2023; VERIFY]: assume a conservative 45% reduction in alerts that need level-one review via auto-closure of lowest-risk alerts with sampling, leaving 660,000. Lever two, GenAI investigation assistance that assembles customer context, transaction history and prior alerts into a draft narrative: handling time from 45 to 30 minutes. Capacity needed: 660,000 alerts times 0.5 hours is 330,000 hours; at 1,600 productive hours per analyst that is about 206 analysts against 600 today. Release of roughly 390 FTE, USD 21 million, of which the bank should reinvest a share in level-two investigation quality. Costs: USD 10 million build including validation and USD 5 million run.

**Regulatory conditions.** The supervisor will ask for evidence that suspicious-activity yield does not fall: run the scoring model in shadow mode for at least six months, perform below-the-line testing on auto-closed alerts, demonstrate stability and explainability of risk scores, document the model under the MRM framework with independent validation, and keep an audit trail for every auto-closure. Under UK SS1/23 [PUBLIC — PRA, 2023; VERIFY] this is a high-materiality model.

**Recommendation.** Phase one, GenAI investigation assistance, low regulatory friction, immediate time saving. Phase two, risk scoring in shadow mode with regulator engagement, then re-ranking, then auto-closure of the lowest tier with sampling. Reinvest 20% of released capacity into level-two and typology development. Headline: level-one cost down about 55% in three years with higher detection, not lower.

**What distinguishes a manager-level answer.** Treating the regulator as a design stakeholder; sequencing by regulatory friction; the shadow-mode and below-the-line test as explicit steps; and framing the outcome as better detection at lower cost rather than cost alone.

## Case drill 5: Rescuing a failing AI pilot

**Prompt.** “A bank’s KYC document-extraction GenAI pilot [ILLUSTRATIVE — composite] is at month nine. Target accuracy was 95%; it is at 82%. Analysts have stopped using it. The sponsor wants to know whether to fix it, pivot or kill it.”

**Clarifying questions.** How is accuracy measured and on what sample; error distribution by document type and field; was there a production path, integration and owner from the start; what do analysts say; what has been spent and what remains?

**Structure.** Diagnose across six causes (problem fit, data, model and evaluation, integration, adoption, governance); quantify the fix; decide against explicit criteria; re-plan with gates.

**Key analyses with numbers.** The interviewer gives an error analysis: 60% of errors come from three document types, scanned and handwritten forms, which are 30% of volume; on the remaining 70% of volume, digital documents, accuracy is 96%. The pilot has been measured on a blended sample, so the headline hides a usable component. Adoption failed because outputs appeared in a separate screen and analysts re-keyed data, which doubled their work; no integration budget had been allocated. Governance: no evaluation set existed at the start, so the target was arbitrary.

**Decision.** Pivot, not kill. Re-scope to digital documents at autonomy level three, sampled review, with handwritten and scanned documents routed to analysts with the model’s extraction shown as a draft at autonomy level one. Fund integration into the case system so extracted fields pre-populate. Build a golden set of 1,000 documents stratified by type and define accuracy per field with thresholds, 97% for digital pre-population, before any further model work. Set a 90-day gate: adoption above 70% of eligible cases and field accuracy at threshold, otherwise stop. Expected value: 70% of volume at 20 minutes saved per case; for 200,000 cases a year that is about 47,000 hours, roughly 29 FTE, or USD 1.6 million a year at USD 55,000 [ESTIMATE], against USD 0.8 million to integrate and complete.

**What distinguishes a manager-level answer.** Decomposing the accuracy figure before judging it; naming the real failure as integration and governance rather than the model; a kill criterion with a date; and matching autonomy level to measured accuracy by segment.

# Part C: Three AI guesstimates

A guesstimate is scored on structure, labelled assumptions and sanity checks, not on the final number. State the assumption, state why it is reasonable, do the arithmetic aloud, and end with a check against something you know.

## Guesstimate 1: Annual inference cost of an LLM assistant for 20,000 relationship managers

**Structure.** Users times usage times tokens per use times price, then apply the cost levers.

1. Users and days: 20,000 RMs, 220 working days a year, 80% active on a given day [ESTIMATE — assumption] gives 3.52 million user-days.
2. Usage: 12 queries per active user-day [ESTIMATE — observed ranges for adviser assistants are roughly 5–20] gives about 42 million queries a year.
3. Tokens per query: system prompt and tool definitions 1,500, retrieved context 3,000, conversation history 1,000, output 500 [ESTIMATE — assumption]. That is 5,500 input and 500 output tokens.
4. Annual tokens: 42 million times 5,500 is about 231 billion input tokens; 42 million times 500 is about 21 billion output tokens.
5. Price, frontier model: USD 3 per million input and USD 15 per million output [ESTIMATE — order of magnitude, VERIFY]. Input cost USD 693,000; output cost USD 315,000; total about USD 1.0 million a year.
6. Levers: prompt caching on the system prompt and tool definitions, about 27% of input, at 90% discount saves roughly USD 170,000; routing 60% of queries to a mid-tier model at a third of the price saves a further USD 300,000 or so. Realistic range USD 0.5–1.0 million a year.

**Sanity check.** USD 25–50 per RM per year, against a loaded RM cost of USD 120,000–200,000. Inference is well under 0.1% of the cost of the people using it. The programme's real costs are integration with CRM and research systems, evaluation, and change; a plausible three-year TCO is USD 15–25 million [ESTIMATE], of which inference is under 10%.

## Guesstimate 2: KYC analysts in UK banking and the addressable automation value

**Structure.** Headcount by institution tier, loaded cost, addressable share, capture rate.

1. Institutions: five large UK banking groups, about 15 mid-sized banks and building societies, roughly 150 smaller banks, foreign branches and large payment and e-money firms [ESTIMATE — order of magnitude].
2. Headcount: large groups run financial-crime operations in the low thousands each, of which KYC and customer due diligence is perhaps 40%, say 1,200 each, giving 6,000. Mid-sized firms about 150 each, 2,250. Smaller firms about 20 each, 3,000. Onshore-equivalent total about 11,000 [ESTIMATE]. Add offshore and outsourced capacity serving UK books, commonly equal to or greater than onshore in large groups, say 9,000. Total around 20,000 FTE.
3. Cost: blended loaded cost of GBP 45,000 (onshore GBP 60,000, offshore GBP 25,000) [ESTIMATE] gives an annual cost pool of about GBP 900 million.
4. Addressable effort: document collection and extraction, registry and screening lookups, data normalisation and narrative drafting are 50–60% of analyst time [ESTIMATE — consistent with task-level studies in composite engagements]; take 55%, GBP 495 million gross.
5. Capture: 50% of addressable within three years after human review, change and regulatory caution, about GBP 250 million a year run-rate across the UK market.

**Sanity check.** GBP 250 million is about 1% of UK banks' total operating costs of roughly GBP 25–30 billion for the large groups [ESTIMATE — order of magnitude, VERIFY], which is plausible for one process family. If the number had come out at 10% of opex, an assumption would be wrong.

### Guesstimate 3: Global annual value of AI in card disputes

**Structure.** Dispute volume times cost per dispute times addressable share, plus a loss lever.

1. Volume: global card purchase transactions are of the order of 700 billion a year [ESTIMATE — order of magnitude from industry statistics, VERIFY]. Dispute rates run around 0.1–0.2% of transactions [ESTIMATE]; take 0.15%, about 1.05 billion disputes a year.
2. Handling cost: issuer and acquirer operations cost per dispute of USD 15–40 depending on market and complexity [ESTIMATE]; take USD 25 blended, giving a global handling cost pool of about USD 26 billion.
3. Addressable effort: as in the issuer case, about 60% of handling effort is agent-fit, with 15 points retained for review, net 45%, about USD 12 billion gross.
4. Capture: 40% within five years given legacy systems and scheme-rule change, about USD 4.7 billion a year.
5. Loss lever: merchant and issuer write-offs and first-party misuse losses are a multiple of handling cost; even a 10% reduction via better evidence and representment plausibly adds a similar order of magnitude, USD 3–5 billion [ESTIMATE — wide uncertainty].

**Sanity check.** A total of USD 8–10 billion a year is a small fraction of global card-industry revenue, which runs in the hundreds of billions of dollars [ESTIMATE — order of magnitude], and is consistent with the operations share of the McKinsey Global Institute banking estimate of USD 200–340 billion for generative AI overall [PUBLIC — MGI, June 2023]. Say explicitly that the loss lever carries the widest uncertainty.

## Part D: Full mock interview with grading rubrics

Have a partner read each question. Answer in 60–120 seconds. Score each with the rubric (0 = missing or wrong, 1 = weak, 2 = adequate, 3 = strong). A model strong answer follows each rubric; do not read it until you have answered.

### Question 1 (fit): “Tell me about yourself and why this practice.”

| Score | Descriptor |
|---|---|
| 1 | Chronological CV recital; enthusiasm for AI as the reason |
| 2 | Clear trajectory, mentions AI work, generic reason for the firm |
| 3 | Trajectory framed as deliberate convergence of payments depth and AI scoping work; a specific, defensible reason this practice |

**Model strong answer.** “Eight years in financial services consulting, mostly payments: ISO 20022 programmes, real-time payments, cross-border and cards. Over the last two years I moved to where the money in those processes is now going, AI: I scored a Tier-1 bank’s KYB process for agentic fit, which set the scope of a supervisor-led multi-agent design; I ran the as-is diagnostic that re-scoped an underwriting proposition; and I built a working multi-agent dispute prototype to settle design questions before engineering spend. I want this practice because it does the reshape work, not deploy work, with the risk and business-case rigour that banks require, and because my differentiator, knowing the processes and data AI is being pointed at, is most valuable here.”

### Question 2 (experience probe): “On the KYB work, what did you personally decide?”

| Score | Descriptor |
|---|---|
| 1 | Describes team activity; cannot isolate own decisions |
| 2 | Names own analysis; decisions attributed to others |

| Score | Descriptor |
|---|---|
| 3 | Names two or three decisions owned, the analysis behind each, and one that was wrong and corrected |

**Model strong answer.** “Three things. Which steps were scored as rules rather than agents: about a third, the deterministic checks, and I defended that against a vendor who wanted everything in agents. The calibration of the six dimension weights with the risk function, which is where the scoring became defensible. And the recommendation that risk rating and adverse decisions stay human, on error tolerance. One I got wrong first: our initial weights overvalued volume and let two deterministic high-volume steps score as agent-fit; risk challenged it and we re-weighted. The architecture itself was the engineering lead’s; I owned the boundary.”

## Question 3 (concept): “Why does an agent need guardrails outside the model?”

| Score | Descriptor |
|---|---|
| 1 | Generic “AI can make mistakes” |
| 2 | Lists guardrail types without the enforcement-location argument |
| 3 | Four layers, plus the argument that model-based checks share the model’s attack surface, plus an example |

**Model strong answer.** “Because a check performed by the model can be defeated by the same inputs that compromise the model: a prompt injection that gets the agent to attempt an unauthorised refund can also get it to report that the refund was authorised. So the four layers, input filtering, output validation, action controls and governance logging, must be enforced in deterministic code the model cannot influence: tool allow-lists, value limits, schema validation, approval gates. In my dispute prototype the broad agent attempted actions outside its remit until I narrowed tool access per specialist; narrowing the prompt alone was not enough.”

## Question 4 (framework): “How would you prioritise 40 use cases a bank has brainstormed?”

| Score | Descriptor |
|---|---|
| 1 | “Value versus effort” with no dimensions or process |
| 2 | Scoring dimensions named; weights and sequencing vague |
| 3 | Dimensions, weight-setting with the sponsor, agentic-fit and risk overlays, dependency mapping, and waves with a rationale |

**Model strong answer.** “First clean the list: de-duplicate, cluster into domains, write each as a problem statement with a value hypothesis; 40 usually becomes 20. Then score value, on cost, revenue, risk and customer experience, and feasibility, on data readiness, integration complexity, model maturity, regulatory exposure and change effort, with weights agreed in the room with the sponsor and risk, because weights are a political decision. Overlay a regulatory tier and a dependency map; the top scorer is often blocked by a data foundation. Then sequence into waves: two or three quick wins to fund credibility, the foundations, then the reshape bets. The output the executive committee sees is a 2x2 and a roadmap; the scoring table is the appendix they will ask for.”

## Question 5 (case): “A COO asks whether to automate sanctions alert adjudication with an agent. Walk me through it in five minutes.”

| Score | Descriptor |
|---|---|
| 1 | Jumps to a yes or no |

| Score | Descriptor |
|---|---|
| 2 | Structure present; numbers thin; regulator absent |
| 3 | Baseline, step-level fit, autonomy by alert tier, regulator conditions, economics and sequencing |

**Model strong answer.** "Baseline first: alert volume, true-match rate, handling time and cost; typically 95%-plus of name-screening alerts are false positives [ESTIMATE — practitioner range]. Then score the steps: evidence gathering, identifier comparison, narrative drafting are agent-fit; the dismissal decision is deterministic for clear non-matches and judgement-heavy for partial matches; escalation of true matches stays human. Autonomy by tier: auto-dismiss only the lowest-risk tier with sampling, agent-draft with analyst decision for the middle, human-led for potential matches. Economics: if 60% of alerts are tier one and handling time halves on the rest, level-one capacity falls by roughly 60–70% [ESTIMATE — composite]. Conditions: shadow mode, below-the-line testing, independent validation under the MRM framework, and supervisor engagement before any auto-dismissal. Sequence: assist first, auto-dismiss later, earned by evidence."

## Question 6 (numbers defence): "Your business case claims USD 12 million a year. I think half of it is hours saved that never become cost. Defend it."

| Score | Descriptor |
|---|---|
| 1 | Restates the number |
| 2 | Acknowledges the point; no mechanism offered |
| 3 | Separates levers, states the capacity-to-cost conversion mechanism and rate, offers the risk-adjusted figure unprompted |

**Model strong answer.** "You are right to separate them. Of the USD 12 million, USD 8 million is labour capacity, USD 3 million is write-off reduction and USD 1 million is funding cost; the last two are cash regardless. On the USD 8 million, the conversion mechanism is attrition and vacancy management at a 14% attrition rate on 300 staff, which releases about 40 roles a year without redundancy; we assume 70% conversion in year two and 90% in year three, and the benefits register names the head of operations as owner. If you prefer, the risk-adjusted case at 60% conversion is USD 9 million a year and still clears the hurdle with payback inside three years. I would rather you approve the conservative number and we beat it."

## Question 7 (governance): "A UK bank wants to deploy an agentic KYB system. What will the model risk function require, and how long will it take?"

| Score | Descriptor |
|---|---|
| 1 | "Governance and compliance" without specifics |
| 2 | Names SS1/23; vague on what validation means for an agentic system |
| 3 | Inventory and tiering, end-to-end validation on a golden set, change control on prompts and corpora, vendor update handling, a realistic timeline |

**Model strong answer.** "Under SS1/23 [PUBLIC — PRA, 2023; VERIFY] the system goes into the model inventory, probably as a high-materiality model given its role in onboarding decisions. Validation will be end to end on a golden set rather than of the foundation model's internals: accuracy and groundedness per task, robustness to adversarial inputs, guardrail effectiveness and the human-oversight design. Prompts, retrieval corpora and tool configurations go under change control, and vendor model updates are treated as model changes that trigger re-validation. Expect three to six months from design freeze to validation sign-off if the

evaluation set and documentation are built during development, and much longer if they are retrofitted [ESTIMATE — practitioner range]. I would put validation on the critical path from day one."

## Question 8 (behavioural): "Tell me about a time you were wrong on a client engagement."

| Score | Descriptor |
|---|---|
| 1 | A non-failure dressed as a failure |
| 2 | A real failure; learning is generic |
| 3 | A real, specific failure, the consequence, the correction, and a changed practice |

**Model strong answer.** "I built the dispute prototype before building its evaluation set. It worked on the scenarios I had in mind and produced confident, wrong outcomes on edge cases I had not written down; one agent attempted actions outside its remit. The consequence was two weeks of rework and a credibility dent with the engineering lead. I built a stratified golden set, narrowed each agent's tools and prompt, and documented failure modes. The practice I changed is that the evaluation set is now the first deliverable on any AI workstream I run, and golden-set accuracy at the chosen autonomy level is an explicit business-case assumption."

## Question 9 (curveball): "Isn't all of this just hype? Most pilots fail."

| Score | Descriptor |
|---|---|
| 1 | Defends AI uncritically or concedes entirely |
| 2 | Cites a statistic; weak explanation |
| 3 | Agrees with the evidence, explains the mechanism of failure, states where value is proven, and turns it into the firm's value proposition |

**Model strong answer.** "The failure statistics are real: Gartner projected 30% of GenAI projects abandoned after proof of concept [PUBLIC — Gartner, 2024; VERIFY], and the MIT NANDA report put the share of pilots with no measurable P&L impact at around 95%, with methodological caveats [PUBLIC — Aug 2025; VERIFY]. But the mechanism is not that the technology does not work; it is that pilots are built without a production owner, integration budget, validation plan or business metric. Where those exist, operations use cases are delivering: HSBC's AML results [PUBLIC — Google Cloud, 2023; VERIFY] and Morgan Stanley's adviser assistant adoption [PUBLIC — 2023–24; VERIFY] are measured. So the hype is real and so is the value; the difference is scoping, governance and business-case discipline, which is precisely what a strategy practice sells."

## Question 10 (close): "What questions do you have for me?"

| Score | Descriptor |
|---|---|
| 1 | None, or questions answerable from the website |
| 2 | Reasonable questions about the practice |
| 3 | Two or three questions that show commercial and delivery understanding and invite the interviewer's judgement |

**Model strong answer.** Ask: "Where in the deploy, reshape, invent spectrum is most of your banking work today, and where do you expect it in two years?" "How do your teams handle model-risk validation with clients: do you own it, or do you work alongside the client's second line?" "What has distinguished the consultants who have grown fastest in this practice?"

## Self-scoring and interpretation

Total the ten scores out of 30.

| Total | Interpretation | Action |
|---|---|---|
| Under 15 | Foundations gap | Re-read Modules 1–4 and 7–9; rebuild the three narratives in Module 19 |
| 15–21 | Practitioner | Drill Parts B and C; rehearse numbers defence (Question 6) and governance (Question 7) until both score 3 |
| 22–26 | Manager-level | Polish altitude and brevity; practise the curveball and the close |
| 27–30 | Ready | Run a cold mock with a new interviewer; focus on case speed and quantification |

Re-run the mock no sooner than a week later with different numbers fed in by your partner, so you are practising structure rather than memorising answers.

# Appendix — Answer keys

Attempt each quiz before reading its key. Each answer gives the expected response, a short explanation, and what a manager-level AI consultant would add.

## Module 0 — Diagnostic model answers

Score 0, 1 or 2 per question against these. The “benchmark” line describes what a manager-level AI consultant’s answer contains.

**Q1. Tool choice.** A rules engine for the eligibility decision itself: the logic is codified (scheme reason codes, time limits, amount thresholds), the inputs are structured, and the regulator needs reproducibility and auditability, which deterministic logic gives and an LLM does not. LLMs add value upstream and downstream: extracting facts from unstructured cardholder correspondence and merchant evidence, classifying the dispute reason, drafting the representment or customer letter, and summarising the case for the agent. Benchmark: names the rules engine first, explains why (determinism, audit), then positions the LLM on the unstructured edges, and mentions that the combination is the typical production pattern.

**Q2. Context window.** The context window is the maximum number of tokens the model can attend to in one call (prompt plus generated output). Cost scales roughly linearly with input tokens and latency rises with length; attention cost grows faster than linearly in naive implementations. Stuffing the whole file degrades retrieval quality (“lost in the middle” effects), increases hallucination risk through irrelevant context, and makes every call expensive regardless of how much of the file is relevant. Retrieval (RAG) selects the relevant passages instead. Benchmark: gives the definition, the cost and latency consequence, the quality consequence, and names RAG as the alternative design.

**Q3. Supervisor vs orchestrator-worker.** Supervisor (hierarchical): a supervising agent routes the task to specialised sub-agents, evaluates their outputs and decides next steps; sub-agents may themselves have sub-agents. Orchestrator-worker: a central orchestrator decomposes a task into sub-tasks, dispatches them to identical or interchangeable workers (often in parallel) and merges results. Use supervisor for KYB where sub-tasks differ in nature (ownership resolution, adverse media, document verification) and need judgement about what to do next. Use orchestrator-worker for parallel bulk work such as screening 2,000 names against lists or summarising 50 documents the same way. Benchmark: distinguishes heterogeneity-plus-judgement from homogeneity-plus-parallelism, and gives a BFSI example of each.

**Q4. Evaluation claims.** Ask: accurate against what reference (who labelled the golden set, how large, how representative of our alert mix)? Accurate on what unit (per narrative, per fact, per field)? Measured when and on which model version, and what is the false-positive versus false-negative split? Insist on faithfulness/groundedness (every claim traceable to source evidence) and a hallucination rate or unsupported-claim rate, plus precision and recall on the key extracted fields rather than a single accuracy figure. Benchmark: challenges the denominator and the labelling, asks for a confusion matrix style view, and names faithfulness explicitly.

**Q5. Attribution.** 10-20-70 is BCG: roughly 10% of the effort and value is in algorithms, 20% in technology and data, 70% in people, process and business change. “Rewired” (2023) is by McKinsey authors Lamarre, Smaje and Zemmel; the six capabilities are business-led digital roadmap, talent, operating model, technology, data, and adoption and scaling. Benchmark: both attributions correct, six capabilities listed, and a one-line point of view on what each framework is useful for.

**Q6. High value, low feasibility quadrant.** Do not discard them and do not force them into wave one. Decompose each into the specific feasibility blockers (data access, integration, regulatory sign-off, model capability), cost the removal of each blocker, and treat the blockers as enablers in the roadmap. Some of these use cases become wave two or three once the enabler lands; others reveal that the feasibility score was really a data problem shared across several use cases, which strengthens the case for a data investment. Junior mistake: labelling them “strategic bets” and parking them with no owner, or sliding them into wave

one because a sponsor likes them. Benchmark: explicit decomposition into blockers, blockers become roadmap enablers, and a sequencing logic.

**Q7. Cost arithmetic.** Input: 60,000 x 40,000 = 2.4 billion tokens x USD 3 per million = USD 7,200. Output: 60,000 x 4,000 = 240 million tokens x USD 15 per million = USD 3,600. Total inference roughly USD 10,800 a year [ESTIMATE]. Inference is rarely the largest TCO line for a workflow like this. Integration and data engineering, the evaluation and guardrail stack, human review capacity, and ongoing model and prompt maintenance usually dominate, often by an order of magnitude. Benchmark: arithmetic correct, states the surprise (inference is small), names what actually dominates, and notes that the pattern can reverse for high-volume conversational use cases.

**Q8. CFO challenge.** Replace year-one 40% with a benefit ramp (for example 10% in year one, 25% in year two, 40% steady state) tied to adoption milestones; convert FTE "reduction" into a mix of attrition-based reduction, redeployment and avoided hiring, each with a different cash realisation; apply a probability or haircut per benefit lever and show the P50 and P90 views; and separate hard cash benefits from capacity benefits. Present an unadjusted case, a risk-adjusted case and the break-even adoption level. Benchmark: ramp, realisation mechanics, risk-adjustment, and a sensitivity presentation.

**Q9. Regulation.** NIST AI RMF 1.0 functions: Govern, Map, Measure, Manage. Yes: Annex III of the EU AI Act lists AI systems used to evaluate the creditworthiness of natural persons or establish their credit score as high-risk (excluding fraud detection). The bank (as deployer, and as provider if it built the model) faces obligations including risk management, data governance, technical documentation, logging, human oversight, accuracy and robustness requirements, and registration, with the main high-risk obligations applying from August 2026 [VERIFY — the Commission's November 2025 omnibus proposal may shift dates]. Benchmark: four functions correct, Annex III recognised, deployer versus provider obligations distinguished, and a date with a verify flag.

**Q10. Positioning.** A strong answer uses the client-ask / what-we-delivered / key-decisions / client-benefit structure, states the personal decision the candidate owned (for example "I designed the six-dimension agentic-fit scoring and ran the SME calibration sessions"), and names the trade-off they argued for. The sentence an interviewer pushes on is any claim of ownership: expect "what did you personally decide, and what would have happened if you had decided differently". Benchmark: specific personal decision, a trade-off, a quantified or qualitative client outcome, and awareness of where the probe will land.

## Module 1 — Quiz answers

**1.** Rules engine (humans author explicit rules); classical ML (learned parameters from structured labelled data); deep learning (learned weights of multi-layer networks from large raw data); generative AI (transformer weights pre-trained on web-scale corpora by a vendor, then aligned); agentic AI (a generative model in a loop with tools, memory and planning, so logic emerges from the model plus orchestration code). Benchmark: names all five and makes the "who produces the logic" distinction explicit, since that is what drives governance.

**2.** Rules engine as the decision of record. An LLM can extract or normalise inputs from unstructured documents and draft the outcome narrative, with its outputs validated before entering the rule evaluation. Benchmark: says "decision of record", positions the LLM at the edges, and mentions validation of LLM outputs before they enter the rules.

**3.** 0.97^8 = 0.78, so about 78% end to end. Implication: narrow the agent's scope, add verification or retry steps, place human gates before irreversible actions, and measure end-to-end success rather than per-step accuracy. Benchmark: correct arithmetic plus at least two design responses.

**4.** Technical: structured bureau and application features with long labelled default history make a scorecard accurate, stable and monotonic. Regulatory: adverse-action and explainability requirements (reason codes), existing model-risk approval under SR 11-7 or SS1/23, and EU AI Act Annex III high-risk status all favour an explainable, validated scorecard. Benchmark: one of each, with a named regulation.

**5.** Kept deterministic: rule-based thresholds and the decision of record where regulators require explainability. Replaced: the alert-generation layer, where a learned risk model reduced false positives and surfaced more true suspicious activity. The split matters because the learned model's outputs could be explained with feature-level reasons and validated as a model, while the regulatory decision logic remained auditable. Benchmark: identifies the layered split and links it to model-risk governance rather than just to performance.

## Module 2 — Quiz answers

**1.** Generation is autoregressive: each output token requires a full forward pass conditioned on all previous tokens, so output tokens are produced sequentially, while input tokens are processed in parallel in one pass. Vendors price output higher to reflect this compute and latency profile. Benchmark: uses the word autoregressive and links it to both latency and price.

**2.** Retrieval is the binding constraint: accuracy is bounded by recall, so 28 points of error come from retrieval and about 11 from generation. Fix retrieval first: structural chunking, hybrid search, re-ranking and query rewriting; then address generation faithfulness. Benchmark: performs the decomposition explicitly and names at least two retrieval improvements.

**3.** Prompting controls behaviour and format within a call; RAG supplies knowledge at inference time; fine-tuning changes weights to teach style, format or a narrow skill. Fine-tuning does not reliably add factual knowledge and that knowledge goes stale. Benchmark: three distinct problems, plus the governance cost of a fine-tuned model as a bonus point.

**4.** Cached portion: 3,000 tokens at 90% discount saves 2,700 x 3.00 / 1,000,000 = USD 0.0081 per case. Over 60,000 cases with the 1.30 overhead: 60,000 x 0.0081 x 1.30 = about USD 632 a year. Not material; caching matters for multi-turn conversations with large shared prefixes, not for single-shot batch processing. Benchmark: correct arithmetic and the judgement that caching's value depends on the usage pattern.

**5.** Fixed-size with overlap; structural or semantic chunking (by heading, clause or paragraph), optionally with parent-child retrieval. Fixed-size chunks split clauses mid-sentence, separate conditions from their exceptions, and lose the heading context that tells the model which policy a passage belongs to, which degrades both retrieval and faithfulness. Benchmark: names two strategies and gives a mechanism-level reason, not just "it works better".

## Module 3 — Quiz answers

**1.** Model: which tier for which agent (frontier for judgement, mid-tier for extraction). Tools: scope and least privilege per agent, with read/write and reversibility flags. Memory: what persists, where, and for how long (data protection). Planning: which pattern (ReAct, plan-and-execute, supervisor, orchestrator-worker) and why. Benchmark: four components plus a specific, defensible design decision for each rather than a definition.

**2.** 0.96^12 = 0.61, about 61%. Responses: reduce dependent steps by making routine sub-steps deterministic, and add verification or self-correction steps that catch and fix errors before they propagate; human gates before irreversible actions also cap the consequence. Benchmark: correct arithmetic and the insight that verification steps break the independence assumption in your favour.

**3.** Approve (pre-action gate), review-sample (post-action QA), exception-only (thresholds route hard cases to humans). For refunds above a threshold, use approve: the action is irreversible and financially material, so a human reviews the evidence and rationale before execution. Below the threshold, exception-only with sampled QA. Benchmark: names all three, chooses approve for the irreversible case, and notes that thresholds are the control.

**4.** MCP standardises how an application exposes tools, resources and prompts to a model (agent-to-tool); A2A standardises how agents discover and exchange tasks with each other (agent-to-agent). Governance: for

MCP, a catalogue of approved internal servers with authentication and audit at the server; for A2A, policies on which external agents may be called and what data may cross the boundary. Benchmark: correct distinction, attribution (Anthropic 2024; Google 2025), and a governance point rather than a feature list.

**5.** Free-text sending by an agent to an external party risks disclosing customer data, hallucinating commitments and violating scheme rules; a template-only tool constrains the action space so the agent can only choose and populate approved messages. This is an action guardrail (constraining what the agent can do), supported by an output guardrail (PII filter). Benchmark: identifies the conduct and privacy risk, names the action layer, and notes the supporting output check.

## Module 4 — Quiz answers

**1.** Availability (documents in systems without APIs); quality (multiple versions of the same procedure); structure and parseability (scanned documents, tables as images); metadata (no type, date, jurisdiction tags); access control and lineage (folder-level entitlements only); privacy and residency (PII in free text); labelled examples (outcomes recorded, rationales not); freshness (no owner for corpus updates). Any five. Benchmark: dimensions paired with concrete gaps, and the observation that readiness is scored per use case.

**2.** HNSW is a graph-based approximate nearest-neighbour index with high recall and fast queries at high memory cost; IVF clusters vectors and searches only the nearest clusters, trading some recall for lower memory. Hybrid search combines vector similarity with lexical matching, which is essential because names, aliases, identifiers and codes have weak semantic signal and need exact or near-exact lexical matching. Benchmark: correct trade-off and the recognition that index choice matters less than hybrid search and metadata filtering.

**3.** Verbosity bias (prefers longer answers): use rubric-based scoring with length-neutral criteria. Position bias (prefers the first option): randomise order in pairwise comparisons. Self-preference and inconsistency: calibrate against a human-labelled subset, report agreement, and keep a permanent human sample. Benchmark: three named biases with specific mitigations and mention of calibration statistics.

**4.** A narrative that recommends dismissing a true sanctions match is a regulatory breach with potential enforcement consequences; the risk appetite for that failure is zero, so any miss blocks release. Faithfulness at 0.97 accepts a small rate of unsupported claims because every narrative is reviewed by an analyst at assist level, so the residual risk is controlled downstream; the threshold would rise if autonomy increased. Benchmark: links thresholds to risk appetite and to the autonomy level rather than to achievable performance.

**5.** Inference: 220,000 x 0.04 = USD 8,800. QA: 1.5 x 70,000 x 1.27 = USD 133,350. Tooling: USD 120,000. Total run cost about USD 262,000. Benefit: 12 x 65,000 x 1.27 = USD 990,600. Ratio roughly 3.8 to 1 benefit to run cost, before build cost and before the benefit ramp. Benchmark: correct arithmetic, the observation that inference is 3% of run cost, and the caveat that build cost and ramp are excluded.

## Module 05 — Quiz answers

**1.** Business-led digital roadmap; talent; operating model; technology; data; adoption and scaling [PUBLIC — Lamarre, Smaje, Zemmel, “Rewired”, 2023]. KYB depends most on data (entity, ownership and document data sit in several systems and must become governed data products before any agent can reason over them) and on adoption and scaling (onboarding is run by operations teams whose workflow, controls and second-line sign-off must change for benefits to appear). Manager-level benchmark: names all six without hesitation, picks two with a one-line reason each, and notes that governance (model risk) is a close third for a regulated process.

**2.** Diagnosis: the budget is inverted against the 10-20-70 heuristic [PUBLIC — BCG, 2023–24]; only 15 percent sits in the 70 percent category, so the programme is shaped to produce pilots that cannot scale. Most important reallocation: move spend from licences and platform into process redesign, controls engagement

and front-line change, targeting change at 50 to 60 percent of total. Manager-level benchmark: states the diagnosis in one sentence, names the reallocation, and adds the caveat that 10-20-70 is a heuristic and that the exact split should be derived from the specific domains.

**3.** Deploy: adopt off-the-shelf tools for broad productivity, e.g. an enterprise GenAI assistant for all staff. Reshape: redesign an end-to-end process around AI, e.g. agentic dispute resolution with human exception handling. Invent: new propositions, e.g. an AI-native embedded-finance credit product for SME platforms [PUBLIC — BCG framing, 2023–24, VERIFY wording]. Manager-level benchmark: adds that measurable P&L sits mostly in Reshape and that portfolio shape should be checked against maturity.

**4.** "Our diagnostic shows the platform is already ahead of the organisation: technology scores 3, adoption scores 1 and data scores 2. I would put the next EUR 12 million into data products for the two priority domains and into a funded change and adoption capability, and revisit the platform upgrade once production deployments are being throttled by the platform rather than by the organisation." Manager-level benchmark: references the evidence, redirects rather than refuses, and states the trigger that would change the answer.

**5.** Defend: 8 to 12 percent of EUR 400 million = EUR 32 to 48 million a year. Transform: 35 percent of EUR 90 million = EUR 31.5 million, plus 8 percent of the remaining EUR 310 million = EUR 24.8 million, total EUR 56.3 million a year (a range of roughly EUR 50 to 62 million if the "elsewhere" rate is flexed between 6 and 10 percent). Manager-level benchmark: shows the arithmetic, labels the result as run-rate at maturity rather than year-one value, and notes that a ramp and a risk adjustment (Module 9) must be applied before the figure goes to a CFO.

## Module 06 — Quiz answers

**1.** L3 tasks have a definable input, output, decision logic and error profile, which is what an AI pattern acts on; L2 stages are where benefits appear because a stage only gets faster or cheaper when its whole set of tasks changes. Example L3: "build ownership structure and identify UBOs" (fit assessed here). Example L2: "entity verification" (business case built here, covering registry lookup, extraction, UBO unwinding, ID and V and reconciliation together). Manager-level benchmark: adds that assessing fit at L2 hides judgement tasks inside automatable stages, and that building cases at L3 produces unbankable micro-benefits.

**2.** Volume, effort, error and rework, delay, customer or colleague friction. Weight Delay most heavily, because it measures each task's share of elapsed cycle time including waiting, which is the direct driver of the ambition; Effort would be weighted for a cost ambition. Manager-level benchmark: notes that heat is not fit and that weighting should be agreed with the sponsor before scoring, not adjusted afterwards.

**3.** Task 2.2 (chasing) has the higher fit at high autonomy. It is high volume, low judgement, bounded by policy (what to chase, how often, when to escalate), reversible (a wrong chase is a nuisance, not a regulatory breach), and acts through tools the agent can be given (case system, email, portal). Checker review (6.1) is a control with regulatory accountability; a copilot can flag inconsistencies but the approval decision stays with a human. Manager-level benchmark: frames the reason in terms of error tolerance and accountability, and proposes the copilot pattern for 6.1 rather than dismissing it.

**4.** Example: "Agent that classifies and extracts incoming KYB documents, checks completeness and expiry against the jurisdiction-specific requirements matrix, and returns a completed document pack to the analyst with exceptions flagged." Baseline: documents received a year, handle time per document, first-time completeness rate, average days to complete the pack, rework rate from the checker on document issues. Manager-level benchmark: the rewrite names a pattern, a task and an outcome, specifies the autonomy option, and the baseline list is specific enough to be pulled from the case system.

**5.** False positives: 48,000 x 0.94 = 45,120. Time saved per false positive: 9 - 3 = 6 minutes. Total saved: 45,120 x 6 = 270,720 minutes = 4,512 hours. FTE: 4,512 / 1,600 = 2.8 FTE. Manager-level benchmark: shows the arithmetic, then adds that the saving is capacity rather than cost until the roles are redeployed or

removed, and that the true-positive handle time may rise slightly because analysts spend more time on the cases that matter.

## Module 07 — Quiz answers

**1.** How much is it worth (value scoring); can we do it and how hard (feasibility scoring); should an agent do it and how autonomously (agentic-fit assessment), with the risk overlay as the fourth, modifying question. Agentic fit is assessed at task level because a use case bundles tasks with different variability, judgement and error profiles; the pattern and autonomy decision only makes sense for one task at a time. Manager-level benchmark: names all instruments without blending them and explains that the overlay comes after scoring so it modifies rather than contaminates.

**2.** Value = 0.35 x 4 + 0.20 x 3 + 0.25 x 2 + 0.20 x 5 = 1.40 + 0.60 + 0.50 + 1.00 = 3.50. Feasibility = 0.25 x 3 + 0.20 x 4 + 0.20 x 4 + 0.20 x 2 + 0.15 x 3 = 0.75 + 0.80 + 0.80 + 0.40 + 0.45 = 3.20. Quick-wins quadrant, but the regulatory score of 2 should trigger a look at the risk overlay before it is sequenced. Manager-level benchmark: computes correctly and spots the regulatory signal inside an otherwise favourable score.

**3.** Variability override: if task variability scores 2 or below, use a rules engine regardless of total, because a deterministic solution is cheaper, auditable and avoids model risk validation of a stochastic component; KYB example: determining the required document set by entity type and jurisdiction. Risk-tier override: if the overlay assigns Tier 1, cap autonomy at human-in-the-loop regardless of score; KYB example: adjudication of sanctions and PEP screening hits. Manager-level benchmark: states both precisely and notes that the LLM can still add value in the rules case (explaining the pack to the client) and in the Tier 1 case (drafting the disposition).

**4.** Dependencies: it needs document intake, registry reconciliation and the entity and ownership data product from Waves 1 and 2. Risk and control change: it alters the periodic review control, which requires MLRO sign-off on the design and Tier 2 validation, both of which take time. A third acceptable answer: change effort scored 2 because it changes the operating rhythm of the whole KYC team. Manager-level benchmark: gives two distinct reasons, at least one of which is a dependency rather than a difficulty.

**5.** Total = 5 + 2 + 4 + 2 + 4 + 4 = 21, the “agent with human-in-the-loop” band. Judgement scoring 2 means the decision is largely discretionary, so the agent should prepare and propose rather than decide; error tolerance scoring 2 means errors are severe or hard to reverse, so the human gate must be an approval before any action is taken, not a sampling review after the fact. Manager-level benchmark: totals correctly, names the band, and derives the gate type (pre-action approval) from the two low scores rather than defaulting to a generic “human in the loop”.

## Module 08 — Quiz answers

**1.** Consumption-based run cost (inference scales with volume and design, so it must be modelled per case, not as a licence). Human-in-the-loop cost (sampling QA, exceptions and appeals are permanent and are the control a supervisor will examine). Assurance cost (evals, guardrails, monitoring and model risk validation recur and are mandatory for higher-tier systems). Lifecycle churn (models are deprecated on 12 to 24 month cycles, forcing re-baselining and re-validation). Manager-level benchmark: names all four and notes that inference is usually the smallest of the four for bank back-office volumes.

**2.** Input tokens per case: 6 x 12,000 = 72,000; cached half: 36,000 x USD 0.30 per million = USD 0.0108; uncached half: 36,000 x USD 3.00 per million = USD 0.108; output: 6 x 400 = 2,400 x USD 15 per million = USD 0.036. Raw inference = USD 0.155 per case. Loaded at 30 percent overhead = USD 0.201 per case. Annual: 86,000 x 0.201 = about USD 17,300. Manager-level benchmark: computes correctly, labels the prices as date-sensitive estimates, and remarks that USD 17,000 a year is immaterial next to the integration and review costs, so the design decision should be driven by accuracy rather than token cost.

**3.** Cost per case counts every case handled; cost per successful outcome divides by the success rate, so re-work, reversals, escalations and re-contacts raise it. Dispute example: an agent auto-resolves a chargeback

for USD 0.40 of inference, but if 15 percent of its decisions are reversed on appeal, each reversal costs a human investigation of 25 minutes plus a provisional-credit write-off, so the cost per successful outcome is several dollars higher than the per-case figure and the autonomy share should be re-examined. Manager-level benchmark: makes the link to the autonomy share and to Klarna's 2025 walk-back as the public example of the gap.

**4.** Because inference was under 5 percent of three-year TCO at 600,000 cases; halving a small number is still a small number, and routing adds build cost and a second eval set. Routing becomes material when tokens per case or volume rise enough that inference approaches 15 to 20 percent of run cost, for example with long evidence packs, image inputs or many more agent steps, or at several million cases a year. Manager-level benchmark: states the proportion, names the condition, and recommends designing the architecture to allow routing without building it on day one.

**5.** Any three of: vendor and model lock-in (lifecycle line, model refresh and re-validation); data egress and residency charges (run, platform and infrastructure); model risk validation effort (build, validation and assurance, and lifecycle for re-validation); eval set maintenance (run, monitoring and evals); observability log storage and retention (run, platform); legal and privacy review of vendor terms (build, validation and assurance). Manager-level benchmark: places each cost in the right line and notes which are schedule risks as well as money risks (validation, legal review).

## Module 09 — Quiz answers

**1. Vendor case-study haircut.** Apply roughly 50–70%, treating it as vendor claim or benchmark-from-another-bank evidence. The other bank's baseline, volumes, process design and data quality differ from the client's, and vendors publish their best outcomes. Manager-level benchmark: names the evidence-haircut ladder, states that the haircut falls as the client runs its own PoC and pilot, and proposes the pilot measurement that would justify lifting it.

**2. Capacity release vs cash-releasing.** Cash-releasing benefit reduces an actual cost line (payroll, contractor, vendor) that an owner commits to cut. Capacity release frees hours while headcount stays flat, so it only becomes bookable if (a) roles are removed through attrition or contractor release, (b) staff are redeployed into measurable revenue or risk work, or (c) the freed capacity absorbs planned volume growth that would otherwise have required funded hires. Manager-level benchmark: insists on a signed headcount bridge by quarter and reports unbooked capacity separately.

**3. NPV and payback.** Discount factors at 10%: 1.000, 0.909, 0.826, 0.751, 0.683. PVs: −3.00, −0.45, 1.49, 1.50, 1.37. NPV ≈ GBP 0.90m. Cumulative undiscounted: −3.0, −3.5, −1.7, +0.3, so payback falls in Year 3; interpolating 1.7/2.0 = 0.85 gives about 2.85 years. Manager-level benchmark: shows the factors, rounds sensibly, and comments that payback near three years is at the edge of a typical bank CFO's comfort range.

**4. Combined ramp.** 0.50 × 0.80 × 0.75 = 0.30, so the Year 1 ramp is 30%, not 50%. A case assuming 50% in Year 1 overstates Year 1 benefit by two-thirds and will miss its first-year target, damaging credibility for later funding. Manager-level benchmark: explains that each component is individually plausible but multiplicative, and ties coverage to the IT roadmap and adoption to the change plan.

**5. CFO challenges and pre-emptions.** Any four of: double counting (reconcile across programme benefit registers and show net-of-overlap); baseline inflation (trailing twelve months, fully loaded cost with HR sign-off); ramp optimism (decompose into coverage, adoption, effectiveness tied to roadmap); run-cost omission (full TCO including inference, human review, model risk, refresh); attribution (control group or phased rollout agreed before go-live); capacity release (signed headcount bridge); terminal value (no benefit growth beyond volume, include refresh cost); downside (probability-weighted NPV and gated funding). Manager-level benchmark: gives the pre-emption as a specific artefact or method, not a general reassurance.

## Module 10 — Quiz answers

**1. Four layers and postures.** Foundation model: buy via API or open-weight, never build. Platform (gateway, orchestration, vector store, evals, observability, guardrails): buy components, assemble and own the control plane. Application (agent workflows, prompts, tools, rules, integration): build for differentiating processes, buy for commodity ones. Services: partner for surge and specialist skills with capability transfer, retain product ownership. Manager-level benchmark: explains why the posture differs per layer and gives a banking example at the application layer.

**2. Golden dataset over public benchmarks.** Public benchmarks measure general capability on tasks unlike the bank's documents and decisions, so they predict domain performance poorly; and vendors optimise for public leaderboards, so scores are inflated relative to real tasks. The bank's own golden set measures the exact task, document mix and failure modes, and doubles as the baseline for ongoing monitoring and model risk validation. Manager-level benchmark: adds that the golden set also makes the evaluation reproducible when the model version changes.

**3. Weight sensitivity.** Vendor A: 4.5×0.35 + 4.0×0.15 + 4.0×0.20 + 3.0×0.15 + 3.5×0.10 + 2.5×0.05 = 1.575 + 0.60 + 0.80 + 0.45 + 0.35 + 0.125 = 3.90. Build under the same weights: 1.40 + 0.525 + 0.90 + 0.525 + 0.30 + 0.25 = 3.90. A tie, so Vendor A does not overtake; the ranking is sensitive to the exit weight and the steering committee should decide the weight explicitly. Manager-level benchmark: shows the working and interprets the tie as a governance decision rather than a scoring artefact.

**4. Five AI-specific clauses.** Data use and training prohibition (confidentiality, regulatory data rules); model change notification and version pinning (model risk validation invalidated by silent updates); quality SLAs on the client's test set (vendor accountability for accuracy, not just uptime); output IP indemnity and ownership of prompts, fine-tunes and evaluation data (infringement risk, lock-in); exit and portability with transition assistance (concentration and substitutability under outsourcing and resilience rules). Also acceptable: regulatory cooperation and audit rights; price protection. Manager-level benchmark: links each clause to a specific regulatory expectation or model risk control.

**5. Per-alert pricing with growth.** Year 1: 180,000 × 5 = EUR 0.90m. Year 2: 207,000 × 5 = EUR 1.035m. Year 3: 238,050 × 5 = EUR 1.19m. Three-year total ≈ EUR 3.13m, against EUR 2.70m at flat volume. Protections: volume-tiered pricing with declining unit rates, a cap on annual unit-price increases, a committed-volume discount in exchange for a floor, and a clause converting to a platform fee above a threshold. Manager-level benchmark: notes that per-transaction pricing transfers volume risk to the buyer and that the bank should model TCO on its own growth forecast, not the vendor's.

## Module 11 — Quiz answers

**1. Stage questions.** PoC: can the technique do the task at acceptable quality on our data? Pilot: does it work in the real workflow with real users, and what is the measured benefit? MVP: can it run in production with controls at partial volume and hold its KPIs? Scale: does the benefit hold across the full population and can the organisation run it? A PoC cannot prove business value because it runs on a sample outside the workflow, so it measures neither user behaviour change nor benefit against a baseline. Manager-level benchmark: adds that booking PoC results in a business case is the most common source of later credibility loss.

**2. Four dimensions with pilot examples.** Technical: accuracy on live data within one percentage point of golden-set accuracy. Operational: handle time per case falls to a stated target and HITL override rate sits within a defined band. Business: benefit measured on the pilot population and extrapolated with a haircut, signed by the business owner. Risk: pilot risk assessment closed, fairness check passed, zero severity-1 incidents. Manager-level benchmark: names who owns each threshold, particularly that risk and accuracy thresholds belong to the second line.

**3. Hours and FTE.** Auto-closed: 147,000 × 0.44 = 64,680 alerts × 9 min = 582,120 min ≈ 9,700 hours. Remaining: 82,320 alerts × 3.3 min = 271,656 min ≈ 4,530 hours. Total ≈ 14,230 hours; at 1,600 productive

hours ≈ 8.9 FTE equivalents. Manager-level benchmark: distinguishes FTE equivalents (capacity) from booked FTE (headcount bridge).

**4. Low override rate.** A very low override rate can indicate automation bias: analysts accepting proposals without genuine review, which removes the human control the model risk tier depends on. It can also indicate that analysts are not using the tool at all if measured on the wrong denominator. Set a band with a floor and a ceiling and test override quality through QA sampling. Manager-level benchmark: proposes a specific control such as seeded test cases or randomised blind review to detect automation bias.

**5. Six causes and remedies.** No production path (define architecture, integration and funding in principle before PoC); no business owner (named owner signs hypothesis and criteria); data access not secured (start approvals in parallel with PoC); model risk engaged late (classify and agree evidence pack at PoC start); vague or moving criteria (quantified four-dimension gates with stopping rule); benefit not measured (pilot in workflow with control or before/after); technology before problem (use-case assessment first); run costs and team ignored (TCO and operating model at pilot exit); no change capacity (fund change in MVP); point solutions (platform after second PoC). Any six with correct pairing. Manager-level benchmark: sequences the remedies and names the first three to act on.

## Module 12 — Quiz answers

**1. Annex III points.** Point 5(b): creditworthiness assessment or credit scoring of natural persons, excluding fraud detection. Point 5(c): risk assessment and pricing for life and health insurance for natural persons. Not high-risk under Annex III: AML transaction monitoring, sanctions screening, dispute resolution, corporate lending to legal entities, internal knowledge assistants (though chatbots carry Article 50 disclosure duties). Manager-level benchmark: adds that employment-related AI (point 4) catches HR analytics and that all dates are subject to the November 2025 omnibus proposal [VERIFY].

**2. SS1/23 principles and definition.** Model identification and model risk classification; governance; model development, implementation and use; independent model validation; model risk mitigants. The model definition includes qualitative or expert-judgement inputs and qualitative outputs, which brings LLM-based systems inside model risk management, so the debate shifts from “is it a model” to “which tier and how do we validate the system in use”. Manager-level benchmark: names the SMF accountability and the self-assessment expectation, and notes the initial scope of firms with internal model permissions [VERIFY].

**3. Disparate impact.** Ratio = 47% / 62% = 0.758. Below 0.80, so it fails the four-fifths screen and warrants investigation; failing the screen is evidence of adverse impact, not proof of unlawful discrimination, and the bank must examine whether legitimate, non-proxy factors explain the gap. Manager-level benchmark: proposes a second metric such as equalised odds and notes the impossibility results when base rates differ.

**4. LLM rationale.** A generated rationale is post-hoc text produced by the same model, not a faithful trace of the factors that determined the output; research (Turpin et al., 2023) shows stated reasoning can diverge from the actual drivers. The architectural answer is to keep the decision in an explainable component (scorecard, constrained model, rules) that the agent calls as a tool, and to require the LLM’s text to cite verifiable evidence for every fact rather than to justify the decision. Manager-level benchmark: distinguishes citation grounding (checkable) from rationale generation (useful for review, not an explanation).

**5. NIST functions.** Govern, Map, Measure, Manage. GenAI Profile categories include confabulation, information integrity, information security, intellectual property, harmful bias and homogenisation, data privacy, and value chain and component integration. Manager-level benchmark: explains that Govern is cross-cutting and uses the RMF as a crosswalk to ISO/IEC 42001 and the EU AI Act rather than as a separate programme.

## Module 13 — Quiz answers

1. **Talent, money and decision rights.** A CoE holds all three centrally; hub-and-spoke splits them (hub: platform money, standards, scarce talent; spokes: use-case money, business decision rights,

embedded delivery); federated gives nearly all three to business units and keeps policy centrally. Manager-level benchmark: a strong answer also names what stays central even in a federated model (model inventory, tiering, validation standards).

2. **Four validators in total.** 14 systems at one validator per five systems is 2.8, rounded up to 3 for full validation, plus 1 for the remaining 26 lighter-touch systems, giving 4. The binding assumption is the effort norm per Tier 1 or Tier 2 validation (initial validation plus annual re-validation plus event-driven re-validation on model or prompt changes); if agentic systems change frequently, the ratio tightens and the number rises. Manager-level benchmark: a strong answer rounds up explicitly, states the assumption, and notes that validation is often the critical path on the roadmap rather than engineering.
3. **Business-funded.** Each unit funds its own use cases and no one funds cross-cutting platform and foundations. The fix is a hybrid: central funding for platform and risk functions, business funding for use cases, and a stage-gated central pool for new domains, with a benefits register so the centrally funded hub is credited for benefits landing in business P&Ls. Manager-level benchmark: mentions chargeback as a later-stage discipline once platform services are mature.
4. **Roughly 10% algorithms and models, 20% technology and data, 70% people, process and business change (BCG, 2023–24).** It implies that change, adoption, process redesign and domain SME capacity should dominate the resourcing plan; an operating model whose headcount is mostly data scientists and engineers is inverted relative to where value is created and lost. Manager-level benchmark: attributes it to BCG, treats it as a heuristic not a law, and converts it into a staffing check.
5. **Because otherwise each spoke rebuilds the same components (gateway, retrieval, evaluation, guardrails), producing duplicated spend, inconsistent controls and no central model inventory.** The hub should ship the minimum platform (gateway, retrieval service, eval harness) so spokes build on it from day one; the sequencing is what makes the target state credible. Manager-level benchmark: a strong answer ties this to regulatory consequences (fragmented standards visible to the PRA) as well as cost.

## Module 14 — Quiz answers

1. **Calibrated trust, over-trust (misuse), under-trust (disuse), abuse** (Lee and See, 2004; Parasuraman and Riley, 1997). AML symptoms: calibrated, override rate tracks measured error rate and audit finds few missed escalations; misuse, analysts clear agent-prioritised alerts in seconds and blind audit finds missed suspicious activity; disuse, analysts re-run their own checks and the agent's narratives are rarely used in SAR filings; abuse, the triage agent is used to make final close decisions on alert classes it was never validated for. Manager-level benchmark: a strong answer links misuse to a specific design counter-measure such as forced disagreement sampling.
2. **Analyst cost falls by roughly GBP 3.7m a year.** Before: 300,000 x 3.0 x 12 minutes = 10.8m minutes = 180,000 hours x GBP 50 = GBP 9.0m. After: 300,000 x (1.6 x 12 + 2) minutes = 300,000 x 21.2 = 6.36m minutes = 106,000 hours x GBP 50 = GBP 5.3m. Difference: 180,000 - 106,000 = 74,000 hours x GBP 50 = GBP 3.7m. Manager-level benchmark: a strong answer states this is capacity released, not yet a saving, and notes the agent's run cost and the review-step quality must be netted off.
3. **Desire (and, close behind, Reinforcement).** Fear about roles suppresses Desire; unchanged performance measures (cases per hour) undermine Reinforcement. The single design choice that addresses it most directly is role redesign with changed performance measures agreed by HR, Operations and Risk before go-live, with affected staff co-designing the exception taxonomy and HITL thresholds. Manager-level benchmark: attributes ADKAR to Prosci (Hiatt, 2003) and distinguishes it from Kotter's organisational-level steps.

4. **Goodhart's law: when a measure becomes a target it ceases to be a good measure.** If straight-through rate alone is targeted, thresholds get lowered and quality falls; pairing it with override accuracy or blind-audit error rate makes the trade-off visible. Example pairs: straight-through rate with audit error rate; handle time with rework rate; cost per case with complaints per 1,000 cases. Manager-level benchmark: a strong answer says the pair should carry equal weight in the steering committee's decision rules, not just appear on the dashboard.
5. **A 6 to 12 month baseline period** (to absorb seasonality and avoid measuring an unusual month), **and a control group** (a region, product line, or randomised share of cases kept on the manual process for the first six months), plus a sustainment check at month 9 to 12 to see whether pilot-period performance persisted once observation faded. Manager-level benchmark: mentions pre-agreed attribution rules with Finance for KPIs that other initiatives also move.

## Module 15 — Quiz answers

1. **Labour arbitrage** earns margin by delivering the same work from a lower-cost location with process standardisation; the unit of sale is labour and scale economics are weak. **Automation arbitrage** earns margin from the gap between a price still anchored to labour units and a cost base that has partly become software with near-zero marginal cost. Under an unchanged FTE-based contract the provider captures all of it, and has a disincentive to automate because automation reduces billable FTE. Manager-level benchmark: a strong answer says the pricing model, not goodwill, decides who captures the gap.
2. **New price EUR 18.0m (24 x 0.75); new margin EUR 5.0m (18.0 - 13.0), or 27.8% of price and 38.5% on cost.** Current margin is EUR 4.0m (24 - 20), 16.7% of price and 20% on cost. The provider's absolute and percentage margin both rise even after a 25% price cut, which shows that a 25% cut leaves a large share of the automation gain with the provider. Manager-level benchmark: the strong answer proposes a step-down schedule so later gains flow to the client, and asks whether the EUR 13m includes amortised transition cost.
3. Any four of: **decision-level accuracy and quality SLAs with measurement method and sample size; model and prompt change notification with re-validation evidence and client veto for Tier 1; audit and validation access for the client's second line and regulators; data-use restrictions (no training on client data without consent, residency, retention); liability and indemnity for decision errors reflecting regulatory exposure; exit and portability covering golden datasets, prompts, exception taxonomies and transition assistance; benefit-sharing and step-down mechanics with a frozen baseline; minimum human-oversight commitments by risk tier.** Manager-level benchmark: explains why each matters in model-risk or outsourcing-rule terms, not just as a list.
4. **Because (a) retained roles are more senior and more expensive per head** (exception adjudicators, QA auditors, agent supervisors), **and (b) new cost lines appear** (AI run cost, agent engineering and evaluation, amortised transition cost, higher risk cost). A third reason: fixed costs such as licences and management do not scale down proportionally. In the worked example, 57.5% headcount reduction gave a 35% cost reduction. Manager-level benchmark: quantifies the gap and uses it to set client expectations before the provider does.
5. **No.** The bank remains accountable for the outcomes of models used in its business. The relevant UK references are PRA SS1/23 (model risk management principles, effective May 2024) for the model itself and PRA SS2/21 (outsourcing and third-party risk management) for the arrangement, with the EBA outsourcing guidelines (2019) and DORA (from January 2025) relevant for EU-regulated entities, all `[VERIFY]`. In practice the provider's agent must be in the bank's model inventory with validation evidence, monitoring and audit access. Manager-level benchmark: distinguishes outsourcing the operation of a model from outsourcing accountability, and names the SMF accountability under SS1/23.

## Module 16 — Quiz answers

1. **Real-time authorisation fraud scoring: classical or deep-learning ML**, because decisions are needed in tens of milliseconds with deterministic, explainable, monitored behaviour; LLMs are excluded from the real-time path. **SAR narrative drafting: GenAI (RAG-grounded)**, because it is unstructured text generation from an evidence pack, with mandatory analyst sign-off. **Applying chargeback time limits: rules engine**, because scheme rules are deterministic, change on a schedule and need a named owner and audit trail. **Investigating a non-receipt claim: agentic orchestration**, because it is multi-step across systems and correspondents with unstructured messages and judgement at branches, with fund movements kept behind deterministic controls. Manager-level benchmark: a strong answer explains that most processes contain all four kinds of work and the skill is in the decomposition.
2. **Net annual benefit of roughly GBP 4.5m.** Current cost: 1.5m x GBP 8 = GBP 12.0m. Effort reduction: 45% x 12.0m = GBP 5.4m capacity released. Run cost: 1.5m x GBP 0.60 = GBP 0.9m. Net: 5.4 - 0.9 = GBP 4.5m. Manager-level benchmark: a strong answer labels the GBP 5.4m as capacity released rather than saving, notes build cost and validation effort are excluded, and adds the non-cost benefits (SLA breaches, liquidity) qualitatively.
3. **Credit underwriting.** Annex III of Regulation (EU) 2024/1689 lists AI systems used to evaluate the creditworthiness of natural persons or establish their credit score as high-risk; the distinction that limits the effect is that corporate and SME lending to legal persons falls outside that annex entry, though it remains subject to model-risk governance (SS1/23, SR 11-7) and anti-discrimination law. Both points `[VERIFY]` given the proposed digital omnibus changes to timelines. Manager-level benchmark: also notes that preparation steps (spreading, memo drafting) carry lower regulatory weight than the decision itself.
4. **The test: compare the cost of a false positive versus a false negative at the decision point, and ask whether the action is reversible.** Releasing a payment after a sanctions hit: a false negative (releasing a true match) is irreversible and a regulatory breach, so no autonomy above a conservative threshold and human decision with full logging. Issuing a GBP 20 provisional credit: the error is bounded and reversible (the credit can be reversed after review), so high autonomy with sampling review is appropriate. Manager-level benchmark: a strong answer frames the output as an autonomy level by decision class that can be presented to a regulator.
5. **Because structured fields replace free text.** ISO 20022 carries structured names and addresses, LEIs, purpose codes and the end-to-end UETR, which reduce false-positive hits at screening, give the resolution agent reliable identifiers to compare, and let an investigation agent locate and trace a payment across correspondents without parsing MT-style narrative fields. The effect is higher extraction accuracy, fewer ambiguous cases and more reliable tool calls. Manager-level benchmark: a strong answer notes that the benefit depends on counterparties actually populating the structured fields, which is uneven during the coexistence period `[VERIFY]`.

## Module 17 — Quiz answers

1. **Klarna 2024 claims sit at rung 1 to 2; HSBC AML AI at rung 3.** Klarna's two-thirds-of-chats figure was a first-month production metric (rung 2) but the USD 40m profit figure was a forecast (rung 1); the 2025 correction confirms the haircut was warranted. HSBC's results came from a system in live use across markets with regulator-visible metrics (rung 3), though no booked P&L figure has been disclosed, so it is not rung 4. Manager-level benchmark: distinguishes the volume claim from the financial claim within the same case, and says explicitly which rung each sits on and why.
2. **Commonwealth Bank of Australia (2025) is the baseline-error case.** Capacity was removed on a forecast reduction in call volumes that did not materialise. The sequencing rule: release capacity only after deflection is observed in production telemetry for at least one full demand cycle, and

book redeployment before headcount release in the business case. Manager-level benchmark: cites the case, names the driver, and states the rule as a design principle rather than a lesson learned.

3. **Correction: Erica is predominantly intent classification and rules-based fulfilment, not generative AI, and its success rests on a bounded scope with deterministic actions.** The evidence it provides is that bounded, deterministic customer automation works at scale; it says little about open-ended GenAI chat in front of customers, where the relevant public evidence is Klarna's correction and Air Canada's liability ruling. Manager-level benchmark: corrects the pattern mislabel politely and immediately offers the cases that do bear on the client's question.

4. **200,000 employees is an access or licence count: it measures deployment, not value.** Outcome metrics to track instead: weekly active usage by role (a leading indicator) and a value metric tied to a specific workflow, such as hours released per analyst in a defined process or cycle-time reduction in a named use case that rides on the platform. Manager-level benchmark: distinguishes input, adoption and outcome metrics, and ties value to vertical use cases rather than the enabler.

5. **Arithmetic: 240,000 alerts × 45 minutes = 180,000 hours = 112.5 FTE at 1,600 hours.** A 60% alert reduction removes 144,000 alerts, 108,000 hours, 67.5 FTE of capacity. First-year realised saving will be lower because the remaining alerts are richer and take longer to investigate, because validators and regulators need to accept the model before thresholds are retired, and because capacity converts to cost only through attrition or redeployment. Manager-level benchmark: shows the arithmetic cleanly, labels it an estimate, and names at least two reasons for the first-year gap without being asked.

## Module 18 — Quiz answers

1. **Revised NPV is about EUR 25m.** Year 3 net falls from 21 to 3 (42 − 18 = 24 benefit, minus 12 investment and 9 run); year 4 net falls from 34 to 28 (42 − 4 − 10). NPV = −15/1.10 − 5/1.21 + 3/1.331 + 28/1.4641 + 34/1.6105 = −13.6 − 4.1 + 2.3 + 19.1 + 21.1 = EUR 24.8m. Allowing for rounding in the module's figures, an answer between EUR 24m and 26m is acceptable; the point is that a one-year wave 2 slip removes roughly 40% of NPV. Manager-level benchmark: rebuilds the table rather than guessing, states the sensitivity as a share of base NPV, and names the mitigation (start KYC data remediation in month 1).

2. **D4, error tolerance and reversibility, decided it.** A missed beneficial owner is a regulatory failure that is hard to detect after approval and expensive to remediate, so despite a composite score on the automation boundary the task was placed in assist with mandatory human confirmation for layered structures. The most relevant SS1/23 principle is Principle 5, model risk mitigants, which expects compensating controls where model limitations are known; Principle 1 (classification) is also acceptable if argued through materiality. Manager-level benchmark: says explicitly that the score informs and the risk dimension decides, and connects the design choice to a named regulatory principle.

3. **Capacity conversion at 60% cuts NPV from USD 20.2m to 7.1m; a straight-through rate of 35% cuts it to 13.1m.** The conversion driver is larger because it scales the entire USD 12m effort benefit, which is 77% of total benefit, whereas the straight-through rate moves only part of that benefit and leaves the loss and complaint benefits intact. The message to the CFO is that the business case is a workforce case first and a technology case second. Manager-level benchmark: quotes both NPV figures, explains the mechanism rather than only the ranking, and draws the implication for where management attention goes.

4. **Per dispute: mid-tier 42,000 × 0.80/1m + 4,200 × 4/1m = USD 0.051; frontier 18,000 × 3/1m + 1,800 × 15/1m = USD 0.081; total USD 0.13, plus 30% for retries and evaluation = USD 0.17; 600,000 disputes = USD 0.10m a year against USD 4.4m run cost.** The implication is that model choice should be presented to the client on accuracy, controllability, data residency and vendor

stability, not on price per token, and that integration and people dominate TCO. Manager-level benchmark: does the arithmetic from memory of the shape (tokens × price, tiered routing, overhead), labels it an order-of-magnitude estimate, and makes the presentational point unprompted.

5. **The low-value write-off pool is a policy decision to credit disputes under a threshold rather than investigate them, so better handling does not change it; counting it would be claiming a benefit the platform cannot cause.** Excluding it signals to a CFO that the team understands causal attribution and will not inflate the case, which buys credibility for the benefits that are claimed. Manager-level benchmark: explains the causal logic, uses the word attribution, and notes that the exclusion was stated to the CFO explicitly rather than hidden.

## Module 19 — Quiz answers

**1. Altitude ladder.** Task (what you physically did), workstream output, engagement recommendation, client outcome. Project B example: I ran SME interviews and mapped the underwriting process (task); the workstream produced a pain-point heatmap showing information assembly, not analysis, consumed most analyst time (output); the proposition was re-scoped from memo drafting to orchestrated assembly with the analyst owning judgement (recommendation); the client got a proposition that fits inside model-risk constraints and targets the largest cost pool (outcome). Manager-level benchmark: all four rungs in under 90 seconds, with the decision at rung three named explicitly.

**2. Reframing the "analyst work" probe.** "The scoring itself is mechanical; the judgement was in calibrating the weights with the risk function and deciding which steps were rules, which were agents and which stayed human. Those calls set the scope, and the scope set the business case." Manager-level benchmark: no defensiveness, immediate move up the ladder, names the decision owner.

**3. Rules-vs-agent principle.** Project C, the disputes prototype: reason-code validity and chargeback timeframe checks were placed in a rules engine because they are deterministic, and an agent only adds cost and probabilistic error. Manager-level benchmark: cites the specific step and the reason (determinism, downside-only error), and connects it to the error-tolerance dimension of the agentic-fit framework in Project A.

**4. Public pairings.** Klarna [PUBLIC — Feb 2024 and May 2025; VERIFY]: automation scope must be set by error tolerance and service quality, not cost ambition alone, which is what HITL gates enforce. Commonwealth Bank of Australia [PUBLIC — Aug 2025; VERIFY]: baseline the volume effect before removing cost; the redundancy reversal came from an unvalidated volume assumption. Either HSBC AML AI or Morgan Stanley is also acceptable with a correct lesson. Manager-level benchmark: correct year, correct direction of the lesson, and a link back to a design choice in the learner's own work.

**5. Strategy vs implementation answer.** Any version that (a) owns both, (b) states that most time is spent on scoping, business case and risk posture, (c) explains implementation depth as a way of making recommendations survive model risk and engineering scrutiny, and (d) states clearly what the learner is not (a production engineering lead). Manager-level benchmark: under 80 words, no hedging, includes one concrete example of implementation knowledge changing a recommendation.

## Module 20 — Part A self-check quiz answers

**1. ML, GenAI, agentic.** ML learns a mapping from historical data to a prediction or label (risk: wrong predictions, drift, bias). GenAI produces content from a prompt and context (risk: wrong or ungrounded content, hallucination). Agentic AI wraps a model in a loop with tools, memory and planning to take multi-step actions (risk: wrong actions with real-world effect, compounding errors). Manager-level benchmark: "predicts, produces, acts" plus a one-line control implication for each layer.

**2. Inverted budget.** Cite BCG's 10-20-70 heuristic [PUBLIC — BCG, 2023–24]: roughly 10% algorithms, 20% technology and data, 70% people, process and change. A 60/30/10 budget is inverted and will under-deliver

on adoption; propose re-balancing towards integration, evaluation, validation and change, and ask what the 60% of licences is buying at expected volumes. Manager-level benchmark: names the heuristic, states it is a heuristic not a formula, and notes that regulated banking needs a larger governance share than the rule implies.

**3. Guardrail layers.** Input (injection detection, data classification, scope filters), output (schema validation, groundedness, PII and toxicity filters, confidence routing), action (tool allow-lists, value and volume limits, idempotency and rollback, approval gates), and policy or governance (logging, versioning, kill switches, re-validation). Principle: enforce guardrails in deterministic code outside the model, because a model-based check shares the model's attack surface. Manager-level benchmark: all four layers plus the enforcement-location argument with an example such as prompt injection.

**4. Inference cost.** Input: 6 × 5,000 = 30,000 tokens × USD 3 per million = USD 0.090. Output: 6 × 500 = 3,000 tokens × USD 15 per million = USD 0.045. Per case USD 0.135; for 500,000 cases about USD 67,500 a year before caching [ESTIMATE — order of magnitude pricing, VERIFY]. Manager-level benchmark: shows the arithmetic, then observes that inference is a minor line and that caching, batching and model tiering could roughly halve it; integration and human review dominate TCO.

**5. SS1/23 principles.** Model identification and model risk classification; governance; model development, implementation and use; independent model validation; model risk mitigants [PUBLIC — PRA SS1/23, 2023; VERIFY scope]. Prompts and retrieval corpora fall under change control because they alter system behaviour as materially as a parameter change: a prompt edit or a corpus update can change outputs on the golden set, so they must be versioned, tested and re-validated like any model component. Manager-level benchmark: five principles in order, plus the "the model is the whole system" argument and the point that vendor model updates also trigger re-validation.